

IELTS ACADEMIC READING

Passage 3 Collection

*A complete practice book of real and adapted IELTS Reading
Passage 3 texts, with full-text passages, original question types,
and a complete answer key*

IB

70

PASSAGES

982

QUESTIONS

Full

ANSWER KEY

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Compiled by @bekhruzposts

Introduction

Welcome to the IELTS Academic Reading — Passage 3 Collection, a practice book bringing together 70 real and adapted Passage 3 texts in one place, compiled and prepared by Teacher Ibodullaev Bekhruz.

Each passage is presented exactly as it would appear in the exam — the full text with paragraph labels, and every original question type (Matching Headings, Matching Information, Multiple Choice, Summary Completion, and more) — with blank spaces and answer circles left for you to write your own answers, just like on test day.

At the back of the book you will find a complete Answer Key for every passage, with the exact sentence from the passage that supports each correct answer, so you can check your work and understand exactly why an answer is correct.

How to use this book:

- Work through one passage at a time under timed conditions (about 20 minutes each).
- Write your answers directly in the circles and blanks provided.
- Check your answers against the key at the back, and read the supporting evidence for anything you got wrong.
- Revisit difficult passages after a few days to reinforce what you have learned.

Good luck with your preparation, and follow **@bekhruzposts** for more free IELTS materials.

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Passage 1

200 Years of Australian Landscapes at the Royal Academy in London

This exhibition promises to chart the evolution of a nation through its art, but not everyone agrees with the reasons behind the choice of artwork

This exhibition promises to chart the evolution of a nation through its art, but not everyone agrees with the reasons behind the choice of artwork. For the casual viewer, the exhibition of landscapes, Australia, selected by the Royal Academy of Art, will be a spectacular guide through Australian art history. Included in the exhibition are a range of artists and styles, dating from the earliest days of colonial art and progressing through expressionism and modernism to the greats of the 20th century, culminating with the current generation of Australian artists. It is hardly surprising, then, that this results in a flexible, wide-ranging notion of landscape.

But this landmark exhibition gives rise to some questions, and perhaps problems, regarding Britain's relationship with its former colony. By choosing a style of painting at which British artists excel, the Academy could be seen as inviting criticism of the Australian interpretation by comparison. But it is the very theme of landscape that hints at a telling attitude towards Australian art from Britain. To consider it condescending is perhaps too strong, but for Joanna Mendelsohn, an Australian critic and Associate Professor at the University of NSW College of Fine Arts (COFA), there is a suggestion that British artistic values have directed this exhibition rather than allowing Australia the freedom to demonstrate its maturity. What Mendelsohn found surprising about this exhibition was that the underlying rules for the selection of works seemed to have been so conservative. Since the landscape is a very strong British artistic theme, it appeared to her that when the British looked at the art of a former colony, there was a tendency for them to think that those colonies would continue to be like the British themselves. In reviewing Australian art, the British insisted on looking at the genre of landscape painting.

Because of colonial ties, it was inevitable during Australian art's formative years that it would reflect Britain's devotion to the beloved landscape before its own character and idiosyncrasies took shape. And while Mendelsohn's concern over the exhibition's conventional selection is valid, the Academy is nevertheless embracing the peculiarities of Australian art from the mid-19th century onward, albeit within the boundaries of landscape. Australia is curated by Kathleen Soriano, director of exhibitions at the Royal Academy. "Certainly, the influence of English, French, or German art is much more evident in the early periods, in the early 1800s to mid-1800s," she says. "What I wanted to show was how Australian art develops a real distinctiveness, associated with the landscape

and the light." The fusion of tradition of the European kind with something more specifically Australian, and often personal, is crucial to the exhibition and extends particularly to some of the more contemporary artists involved.

Sydney-born video artist Shaun Gladwell is a good example of this. Gladwell's most famous piece, which is featured in the exhibition, is *Storm Sequence* (2000), a video of Gladwell skateboarding on the Bondi seafront as one of Sydney's signature brutal storms lingers offshore. It is his acknowledgment of landscape (or seascape) tradition, coloured by Gladwell's own individualism. "To exhibit my work in this show might make some sense because I was interested in Turner and the idea of atmosphere affecting vision, something I was really interested in around the time of *Storm Sequence*," says Gladwell. "I was thinking about this tradition of Romantic landscape, but I wanted to make it personal." But he didn't want to just embark on borrowing imagery from elsewhere. He wanted to bring it to his experience and his world through skateboarding and beach culture.

So while it may seem narrow for Britain to reduce Australian art to the genre of landscape, there can be little denying that British landscape painting is still relevant to a current generation of Australian practitioners, however indirectly.

Visitors to the exhibition encounter Australian Aboriginal art first, the idea being that these works warrant a prominent position because they were first. Over the last couple of decades, London has hosted many successful exhibitions of Aboriginal art in smaller spaces, but for Soriano, Australia represents an opportunity to place such art in a new context, with new relationships to the art of the settlers and white Australia. "One of the reasons landscape struck me as being the right theme was because Aboriginal art started in and on the landscape," she says. "[The exhibition] is a beautiful meshing of the two different kinds of art that allowed me to bring them together comfortably and honestly with this theme. It was important for me to present Indigenous art to audiences, and I felt it was important that it was seen as part of Australian art history, rather than a separate exhibition on its own."

Meanwhile, Australian critic Mendelssohn also points out that London is increasingly less important to today's generation of artists, and this somewhat weakens the ceremony surrounding the exhibition in London. "China is the most important art market in the world," she says. "You've made it in Shanghai, you've made it. The world has changed. My students in Australia, who come from all over the world, really want to see Venice Biennale and Art Basel, but they're less interested in going to London. When I was growing up, London was the destination, and then when I was at university all the smart young things wanted to go to New York," she added. "Now they want to go everywhere. There's no such thing as the centre and the periphery like there used to be. It's much more complicated."

Questions 27–31

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write **YES**, **NO** or **NOT GIVEN**.

- 27 As expected, the artworks chosen for the exhibition reflect a narrow interpretation of landscape.
- 28 The Academy rejected Australian suggestions for the subject of the exhibition.
- 29 The colonial relationship meant that early Australian landscape painting followed the traditions of English landscape painting.
- 30 The exhibition reflects the fact that Australian art developed its own particular qualities.
- 31 Contemporary Australian artists have generally rejected British landscape traditions.

Questions 32–36

Choose the correct letter, A, B, C or D.

- 32 What is the writer's main point in the second paragraph?
 - A Australian landscape painting derives from the British tradition.
 - B Australian landscape painting is more highly regarded than British.
 - C Britain is still imposing its principles on Australian art.
 - D British art cannot be compared to Australian art.
- 33 What does Joanna Mendelssohn find surprising?
 - A Modern Australian landscape painting has great variety.
 - B The guidelines for the choice of work were very traditional.
 - C Landscape painting remains a popular subject for British artists.
 - D The British find the Australian landscape unsuitable as a subject.
- 34 Shaun Gladwell's work is included in the exhibition because
 - A it adopts a subjective approach to depicting the landscape.
 - B skateboarding is an inspiration to many Australian artists.
 - C storms are a significant feature in the Australian landscape.
 - D Bondi is an iconic Australian location.
- 35 What was the reason for Soriano including Aboriginal art in the exhibition?
 - A It is not well known in London art circles.
 - B Aboriginal landscape painting influenced Australian settlers.
 - C It is part of the Australian artistic tradition and not independent.
 - D Aboriginal art has been displayed in London before.

- 36 By referring to China, Mendelssohn is making the point that
- A having an exhibition in London is not as important as it used to be.
 - B young artists in Britain are not interested in Australian art.
 - C art from Shanghai is more important than Australian art.
 - D New York is still a preferred destination for young artists.

Questions 37–40

Complete each sentence with the correct ending, A–F, below. Write the correct letter in each box.

- 37 In spite of its conservatism, the Royal Academy exhibition... 37 _____
- 38 Australian art of the early to mid-1800s... 38 _____
- 39 The modern work by Gladwell chosen for the exhibition... 39 _____
- 40 Including Aboriginal art in the exhibition... 40 _____

LIST OF ENDINGS

- A reflects the mood created by the natural environment
- B demonstrates that the dominant art form in Australia is landscape painting
- C demonstrates an understanding of the historical importance of the land
- D showcases a very small number of artists
- E demonstrates a strong European flavour
- F shows an acceptance of the unique qualities of Australian art

Passage 2

A New Stage in the Study and Teaching of History

For hundreds of years, historians have relied on written or printed documents to provide the bulk of their source materials, and they have largely communicated with students and the wider public by writing books and journal articles. Today, however, the printed word is being superseded by a diversity of forms of communication, above all moving images on video or film.

A The development of this new form of communication is leading to a growing gap between the practice of professional historians based in academia, and the practice of those aiming to popularise the study of history among the general public, and to encourage people to create their own records for the future. On the one hand, there are mainstream academics who continue to use only the written word as they examine more and more fields with an ever-increasing number of sophisticated methodologies. On the other hand, film and video, especially as broadcast on television, are probably the major influence on the public's consciousness of history, as they see film of events of fifty or a hundred years ago, events they had previously only read about.

B In a related development, a great many people now document local and family events in the form of videos; many schools, too, produce video yearbooks. All these visual records may well prove to be invaluable sources of information for future historians. The glaring contradiction is that the two approaches – the academic and what we might term the popular – have intersected very little: with a few notable exceptions, professional historians have tended to avoid involvement in television programmes about history, and have even less impact on what is being captured and preserved on video. And the potential of moving images has wielded negligible influence on the academic study of history.

C This gulf can be seen as resulting from the willingness or otherwise of individual historians to accept the validity of new forms of communication in the study of history. This is not the first time that the question has arisen. The study of history, as conceived of today, began with the transition from oral to literate culture, leading to the earliest written records and the earliest historical studies. The next great shift came with the advent of printing, which transformed everything. Today, as the printed word loses its dominance, historians are faced with a variety of forms of communication, ranging from simple audiotape to the promising complexities of videodiscs linked with computers. As yet, however, the use of moving images to record current events for the benefit of future historians does not even have a commonly agreed name.

D This does not mean that mainstream historians have totally rejected the use of moving images as sources: the majority seem intrigued by the idea, and valuable research has been carried out into the history and analysis of films with a broad circulation, using them as a

source of information on the social and intellectual history of the twentieth century. Journals such as American History Review have played a significant role in this field.

E Yet the number of historians using moving images in their research or teaching is very small. The barrier seems to be that the profession is structured around the medium of the written word, and is somewhat insulated in its academic setting. The use of moving images presents a substantial challenge to this setting and its assumptions. As a result, historians have rejected the training, the institutions, the motivations and the professional structures that would be needed in order to use moving images effectively. Above all, they have rejected the necessity to learn complicated new skills.

F So why should historians make this change? Clearly, films or videos of events and people can be used as solid evidence of the past, linked to the words of the narrator (whether a television presenter/historian or a university teacher giving a lecture) but carrying information in their own right. Film has reintroduced the oral form as a mode of research and communication for documenting historical events. Now, with moving images, people are reminded that oral communication is not limited to words: it also includes body language, expression and tone, and is embedded in a context. Little of this is evident in a written transcript. A further effect of video and film is that the narrator gives up some control and has less need to give explanations, while the viewer becomes involved in the process of interpreting and understanding history.

G Film or videotape can also aid historians by simplifying the work of the interviewer. Instead of trying to carry on an interview while simultaneously making notes about setting and other unspoken data, this new kind of historian can concentrate on the interview itself, and study the film later. The many benefits of using moving images as historical evidence easily outweigh worries about cost, technical skills, or the effect of a camera on a person telling his or her story. Moving images enhance the quality of historical research, and suggest new directions for historians to explore.

Questions 1–9

Reading Passage 3 has seven paragraphs, A–G.

Which paragraph contains the following information? Write the correct letter, A–G. You may use any letter more than once.

Q	Statement	
1	an overview of the range of methods that have been used over time to document history	
2	the main reason why many historians are unwilling to use films in their work	

3	a reference to some differences between oral and written communication	
4	how most citizens today gain an understanding of history	
5	how current student events are sometimes captured for future audiences	
6	mention of the fact that the advantages of film are greater than the disadvantages	
7	the claim that there is no official title for film-based historical work	
8	reference to the active role the audience plays when watching films	
9	a list of requirements that historians see as obstacles to their use of film to record history	

Questions 10–14

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 10 The needs of students in school have led to improvements in the teaching of history.
- 11 Academic and popular historians have different attitudes towards the value of innovations in communication.
- 12 It is common for historians to play a major role in creating historical documentaries for television.
- 13 Articles in American History Review have explored aspects of modern history through popular films.
- 14 Developments in technology are influencing a range of academic subjects.

Passage 3

Are We Born Able to Imitate?

A famous claim about newborn babies copying facial expressions has, decades later, unravelled

In the late 1970s, psychologists Andrew Meltzoff and Keith Moore reported that newborn babies could copy a range of facial expressions, including tongue protrusion, mouth opening and lip pursing. Surely such an achievement could only be due to clever genes; after all, it's impossible to imagine how babies could learn to produce the expressions they see on the faces around them, just a few hours or days after birth. According to Meltzoff and Moore, their findings indicated that we must be born with a cognitive mechanism that connects 'felt but unseen movements of the self with the seen but unfelt movements of the other'.

However, worrying cracks began to appear in Meltzoff and Moore's picture as soon as their findings were published in *Science* magazine – and these cracks have been spreading ever since, in fine but disfiguring lines. From the outset, other experts on child development were unable to replicate the results. They found that newborns copied sticking out their tongues, but not other facial gestures – so perhaps instead of an elaborate imitation mechanism, it was a simple reflex in response to stimulation. Evidence that older infants don't copy any facial gestures, not even tongue protrusion, supported that interpretation.

Further doubts arose when it was discovered that adults can't imitate facial expressions without something called 'mirror experience.' Experiments have shown that people who film themselves and then try to copy their own facial expressions simply by looking at a frozen image on the screen do no better on their tenth attempt than on their first. The only way they improve is with 'mirror experience' – if they watch a reflection of their face as they are trying to recreate the expression. It seems that, in order to learn imitation, we need to see, as well as feel, what we are doing. If we were born with a cognitive mechanism that connects seeing another person's gesture with the sensation of replicating it ourselves, we should be able to improve our imitation without 'mirror experience'.

For several decades, child psychologists have watched the flaws in Meltzoff and Moore's work becoming more apparent, without ever abandoning their belief in it. One of the reasons for this is that Meltzoff portrays *Homo imitans* – a species born able to mimic – in beautiful, compelling prose. Additionally, every loving parent is drawn to the idea that if you stick out your tongue at a newborn, the baby likes to return the compliment. In the scientific world, the concept of an imitation instinct swiftly became a foundation stone. By the 1990s, it was supporting other claims about how the mind is shaped by the genes. Researchers might have realized that, if the imitation instinct was discredited, other treasured ideas would fall apart.

But there were also practical considerations. When small babies are not asleep, they are often inattentive; when they're alert, they're often absorbed in things around them, but not by the things a researcher wants them to look at. In each study of newborn imitation, only a small number of babies stayed attentive for long enough to be analyzed. This sampling, along with small variations in experimental procedure, could have produced misleading results. So, until recently, the idea of a newborn imitation instinct was still scientifically plausible.

However, a study led by Virginia Slaughter at the University of Queensland in 2016 was so large and thorough that it finally undermined this argument. Slaughter and colleagues studied an unprecedented number of babies at one, three, six and nine weeks old. They tested for imitation for nine gestures, following gold-standard procedures. They found evidence for just one: tongue protrusion. In every other case, the babies were no more likely to produce the gesture after seeing it done than they were if they saw something else. Of course, you can't prove a negative – that no newborn baby has ever been able to imitate. But in the aftermath of the Queensland study, we have no good reason to believe that newborns can copy gestures or, therefore, that humans have an instinct for imitation.

What we're left with instead is a wealth of evidence that humans *learn* to imitate in much the same way as we pick up other social skills. In a 2018 study at Birkbeck College in London, Carina de Klerk and her colleagues discovered that a baby's capacity to imitate depended on how much parents had imitated its facial expressions. The more parents copy a baby's facial gesture, the more likely the baby is to imitate it. But the effect highlighted here doesn't generalize. Parents who copy babies' smiles make the infants better at imitating smiles, but not at copying hand movements. This specificity is important because it shows that a simple learning mechanism is involved: babies associate the feel of their own mouth opening with the sight of another mouth opening. Being imitated doesn't make babies think that imitation is a good idea, or give them the capacity to imitate any action they like. Rather, parental imitation gives the baby 'mirror experience', an opportunity to connect a visual image of an action with the feeling of doing it.

In the space of a few decades, it's become clear that imitation is not a cognitive instinct and is not in our genes. Rather, children learn to imitate through interaction with other people. Adults and peers help them to do this, not only by copying what the child does, but through games, songs and rhymes that synchronize actions, by reacting in the same way and at the same time (for example, 'Ooh, what's that?'), and, in most cultures, by allowing the child to spend time in front of actual mirrors, while building up the range of movements that they can copy in future. Whether the mirror is real or metaphorical, 'mirror experience' is what makes children imitators.

Questions 1–5

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 What can be found in the first paragraph?
 - A the reasoning behind Meltzoff and Moore's view
 - B the public reactions to Meltzoff and Moore's findings
 - C the scientific background to Meltzoff and Moore's research
 - D the procedures followed during Meltzoff and Moore's studies
- 2 In the second paragraph, regarding Meltzoff and Moore's work, what is the writer referring to with the phrase 'spreading ever since, in fine but disfiguring lines'?
 - A mounting awareness of deficiencies in it
 - B rising numbers of publications covering it
 - C growing curiosity about the ideas underlying it
 - D increasing numbers of conflicting interpretations of it
- 3 According to the writer, what does evidence from older infants suggest?
 - A Older babies may not be as focused on adult faces as newborn babies
 - B Certain facial gestures could take longer than others for infants to learn
 - C Babies must lose the ability to imitate tongue protrusion at a certain stage
 - D Newborns' tongue protrusion is probably not a response to seeing this gesture
- 4 What do the experiments on 'mirror experience' suggest?
 - A We intuitively associate gestures with the emotions they correspond to
 - B We find it hard to copy an image of a gesture without sufficient practice
 - C We lack a natural capacity to reproduce gestures based solely on observation
 - D We can instinctively imitate our own gestures better than those of others
- 5 The word 'this' in the fourth paragraph refers to
 - A changing attitudes towards Meltzoff and Moore's work
 - B experts choosing to disregard clear scientific evidence
 - C new developments arising within the field of child psychology
 - D improvements occurring in the way research was conducted

Questions 6–10

Complete the summary using the list of words, **A–I**, below.

WORD LIST

- A consistency
- B challenge
- C appeal
- D obligation

- E controversy
- F reluctance
- G confidence
- H writing
- I research

HOMO IMITANS AND THE IMITATION INSTINCT

The success of the *Homo imitans* theory is, in part, due to the high quality of Meltzoff's (6) _____. Particularly for parents, there was also great (7) _____ in the notion of babies imitating their facial expressions. Then, as the concept of an imitation instinct came to form the basis of other genetic theories, scientists may have felt a certain (8) _____ to question it.

These factors, combined with the (9) _____ of carrying out studies on newborns and a certain lack of (10) _____ in how the experiments were conducted, meant that Meltzoff and Moore's idea remained influential well into the 21st century.

Questions 11–16

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- (11) Virginia Slaughter and her team used reliable methodology in their research.
- (12) Slaughter's team were surprised by evidence showing babies' ability to imitate tongue protrusion.
- (13) The findings of the Queensland study proved unconvincing.
- (14) The results of the Birkbeck study suggest that copying babies' expressions teaches them to imitate a range of physical actions.
- (15) It has become evident that verbal input can help children to imitate adults' behaviours.
- (16) The first gestures which children imitate via 'mirror experience' may vary depending on their culture.

Passage 4

Assessing the Risk

How do we judge whether it is right to go ahead with a new technology? Apply the precautionary principle properly and you won't go far wrong, says Colin Tudge.

As a title for a supposedly unprejudiced debate on scientific progress, "Panic attack: interrogating our obsession with risk" did not bode well. Held last week at the Royal Institution in London, the event brought together scientists from across the world to ask why society is so obsessed with risk and to call for a "more rational" approach. "We seem to be organising society around the grandmotherly maxim of 'better safe than sorry'," exclaimed Spiked, the online publication that organised the event. "What are the consequences of this overbearing concern with risks?"

The debate was preceded by a survey of 40 scientists who were invited to describe how awful our lives would be if the "precautionary principle" had been allowed to prevail in the past. Their response was: no heart surgery or antibiotics, and hardly any drugs at all; no aeroplanes, bicycles or high-voltage power grids; no pasteurisation, pesticides or biotechnology; no quantum mechanics; no wheel; no "discovery" of America. In short, their message was: no risk, no gain.

They have absolutely missed the point. The precautionary principle is a subtle idea. It has various forms, but all of them generally include some notion of cost-effectiveness. Thus the point is not simply to ban things that are not known to be absolutely safe. Rather, it says: "Of course you can make no progress without risk. But if there is no obvious gain from taking the risk, then don't take it."

Clearly, all the technologies listed by the 40 well-chosen savants were innately risky at their inception, as all technologies are. But all of them would have received the green light under the precautionary principle because they all had the potential to offer tremendous benefits — the solutions to very big problems — if only the snags could be overcome.

If the precautionary principle had been in place, the scientists tell us, we would not have antibiotics. But of course, we would — if the version of the principle that sensible people now understand had been applied. When penicillin was discovered in the 1920s, infective bacteria were laying waste to the world. Children died from diphtheria and whooping cough, every open-drain brought the threat of typhoid, and any wound could lead to septicaemia and even gangrene.

Penicillin was turned into a practical drug during the Second World War when the many pestilences that result from war threatened to kill more people than the bombs. Of course antibiotics were a priority. Of course, the risks, such as they could be perceived, were worth taking.

And so, with the other items on the scientists' list: electric light bulbs, blood transfusions, CAT scans, knives, the measles vaccine — the precautionary principle would have prevented all of them, they tell us. But this is just plain wrong. If the precautionary principle had been applied properly, all these creations would have passed muster, because all offered incomparable advantages compared to the risks perceived at the time.

Another issue is at stake here. Statistics are not the only concept people use when weighing up risk. Human beings, subtle and evolved creatures that we are, do not survive to three-score years and ten simply by thinking like pocket calculators. A crucial issue is the consumer's choice. In deciding whether to pursue the development of new technology, the consumer's right to choose should be considered alongside considerations of risk and benefit. Clearly, skiing is more dangerous than genetically modified tomatoes. But people who ski choose to do so; they do not have skiing thrust upon them by portentous experts of the kind who now feel they have the right to reconstruct our crops. Even with skiing, there is the matter of cost-effectiveness to consider: skiing, I am told, is exhilarating. Where is the exhilaration in GM soya?

Indeed, in contrast to all the other items on Spiked's list, GM crops stand out as an example of a technology whose benefits are far from clear. Some of the risks can at least be defined. But in the present economic climate, the benefits that might accrue from them seem dubious. Promoters of GM crops believe that the future population of the world cannot be fed without them. That is untrue. The crops that really matter are wheat and rice, and there is no GM research in the pipeline that will seriously affect the yield of either. GM is used to make production cheaper and hence more profitable, which is an extremely questionable ambition.

The precautionary principle provides the world with a very important safeguard. If it had been in place in the past it might, for example, have prevented insouciant miners from polluting major rivers with mercury. We have come to a sorry pass when scientists, who should above all be dispassionate scholars, feel they should misrepresent such a principle for the purposes of commercial and political propaganda. People at large continue to mistrust science and the high technologies it produces partly because they doubt the wisdom of scientists. On such evidence as this, these doubts are fully justified.

Questions 1–6

*Do the following statements agree with the information given in Reading Passage 3? Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 The title of the debate is not unbiased.
- 2 All the scientists invited to the debate were from the field of medicine.

- 3 The message those scientists who conducted the survey were sending was people shouldn't take risks.
- 4 All the 40 listed technologies are riskier than other technologies.
- 5 It was worth taking the risks to invent antibiotics.
- 6 All the other inventions on the list were also judged by the precautionary principle.

Questions 7–13

Complete the summary below. Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

When applying the precautionary principle to decide whether to invent a new technology, people should also take into consideration the 7 _____, along with the usual consideration of 8 _____. For example, though risky and dangerous enough, people still enjoy 9 _____ for the excitement it provides. On the other hand, experts believe that the future population desperately needs 10 _____, in spite of their undefined risks. However, the research conducted so far has not been directed towards increasing the yield of 11 _____, but to reduce the cost of 12 _____ and to bring more profit out of it. In the end, such selfish use of the precautionary principle for business and political gain has often led people to 13 _____ science, for they believe scientists are not to be trusted.

Question 14

Choose the correct letter, **A**, **B**, **C** or **D**.

- 14 What is the main theme of the passage?
- A people have the right to doubt science and technologies.
 - B the precautionary principle could have prevented the development of science and technology.
 - C there are not enough people who truly understand the precautionary principle.
 - D the precautionary principle bids us take risks at all costs.

Passage 5

Australia's Megafauna Controversy

Just how long did humans live side by side with megafauna in Australia? Barry Brook, Richard Gillespie and Paul Martin dispute previous claims of a lengthy coexistence.

Over the past 50 millennia, Australia has witnessed the extinction of many species of large animals, including rhinoceros-sized wombat and goannas the size of crocodiles. Debate about the possible cause of these extinctions has continued for more than 150 years and one of the crucial questions raised is how long humans and megafauna coexisted in Australia. We need to know the overlap of time to make an informed choice between the two main theories regarding the causes of these extinctions. If humans and megafauna coexisted for a protracted period then climate change is the more likely cause. However, if the megafauna became extinct shortly after the arrival of humans, then humans are the likely culprits.

The archaeological site at Cuddie Springs in eastern Australia appears to be well preserved. This dusty claypan holds within its sediments a rich cache of flaked stone and seed-grinding tools, and side by side with these clear signals of human culture are the bones of a dozen or more species of megafauna. Drs Judith Field and Stephen Wroe of the University of Sydney, who excavated the site, claim that it provides unequivocal evidence of a long overlap of humans and megafauna, and conclude that aridity leading up to the last Ice Age brought about their eventual demise. In the longstanding explanation of this site, artefacts such as stone tools and extinct animal remains were deposited over many thousands of years in an ephemeral lake – a body of water existing for a relatively short time – and remained in place and undisturbed until the present day.

There is no disputing the close association of bones and stones at Cuddie Springs, as both are found 1 to 1.7 metres below the modern surface. The dating of these layers is accurate: ages for the sediments were obtained through radiocarbon dating of charcoal fragments and luminescence dating of sand grains from the same levels (revealing when a sample was last exposed to sunlight). Intriguingly, some of the stone tools show surface features indicating their use for processing plants, and a few even have well-preserved blood and hair residues suggesting they were used in butchering animals.

But is the case proposed by Field and Wroe clear-cut? We carried out a reanalysis of the scientific data from Cuddie Springs that brings into question their conclusions. The amount of anthropological evidence found at the site is remarkable: we estimate there are more than 3 tonnes of charcoal and more than 300 tonnes of stone buried there. Field and Wroe estimate that there are approximately 20 million artefacts. This plethora of tools is hard to reconcile with a site that was only available for occupation when the lake was dry. Furthermore, no cultural features such as oven pits have been discovered. If the sediment

layers have remained undisturbed since being laid down, as Field and Wroe contend, then the ages of those sediments should increase with depth. However, our analysis revealed a number of inconsistencies.

First, the charcoal samples are all roughly 36,000 years old. Second, sand in the two upper levels is considerably younger than charcoal from the same levels. Third, Field and Wroe say that the tools and seed-grinding stones used for plant and animal processing are ancient, yet they are very similar to implements found elsewhere that were in use only a few thousand years ago. Also of interest is the fact that a deep drill core made a mere 60 metres from the site recovered no stone artefacts or fossil bones whatsoever. These points suggest strongly that the sediments have been moved about and some of the old charcoal has been re-deposited in younger layers. Indeed, one sample of cow bone found 1 metre below the surface came from sediments where charcoal dated at 6,000 and 23,000 years old is mixed with 17,000-year-old sand. The megafauna bones themselves have not yet been dated, although new technological developments make this a possibility in the near future.

We propose that the archaeologists have actually been sampling the debris carried by ancient flood channels beneath the site, including charcoal transported from bushfires that intermittently occurred within the catchment. Flood events more likely explain the accumulation of megafauna remains, and could have mixed old bones with fresh deposits. European graziers also disturbed the site in 1876 by constructing a well to provide water for their cattle. Given the expense of well-digging, we speculate that the graziers made sure it was protected from the damage caused by cattle hooves by lining the surface with small stones collected from further afield, including prehistoric quarries. This idea is consistent with the thin layer of stones spread over a large area, with cattle occasionally breaking through the gravel surface and forcing the stone and even cattle bones deeper into the waterlogged soil.

The lack of conclusive evidence that humans and megafauna coexisted for a lengthy period casts doubt on Field and Wroe's assertion that climate change was responsible for the extinction of Australia's megafauna. However, we do not suggest that newly arrived, well-armed hunters systematically slaughtered all the large beasts they encountered. Recent studies based on the biology of modern-day large mammals, combined with observations of people who still practise a traditional hunter-gatherer lifestyle, reveal an unexpected paradox and suggest a further possible explanation as to what happened. Using a mathematical model, it was found that a group of 10 people killing only one juvenile *Diprotodon* each year would be sufficient to bring about the extinction of that species within 1,000 years. This suggests that here, as in other parts of the world, the arrival of humans in lands previously inhabited only by animals created a volatile combination in which large animals fared badly.

Questions 27–30

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write **YES**, **NO** or **NOT GIVEN**.

- 27 Field and Wroe argue that findings at the Cuddie Springs site show that people lived in this area at the same time as megafauna.
- 28 Field and Wroe believe it is likely that smaller megafauna species survived the last Ice Age.
- 29 The writers believe that the dating of earth up to 1.7m below the present surface at Cuddie Springs is unreliable.
- 30 Some artefacts found at Cuddie Springs were preserved well enough to reveal their function.

Questions 31–35

Complete the summary below. Choose the correct letter, A–I, from the word bank below. Write the correct letter in each box.

- 31 One objection to Field and Wroe's interpretation is the large quantity of charcoal, 31 _____ and artefacts found at Cuddie Springs.
- 32 Such large numbers of artefacts would be impossible if the area had been covered with 32 _____ for a period.
- 33 There is also a complete lack of man-made structures, for instance those used for 33 _____.
- 34 While some material in the highest levels of sediment is 36,000 years old, the 34 _____ in the same level is much more recent.
- 35 Further evidence against human occupation is the absence of tools and 35 _____ a short distance away.

WORD BANK

- A seeds
- B stone
- C sand
- D cooking
- E deep drill core
- F water

- G** fossil bones
- H** sediment
- I** storage

Questions 36–40

Choose the correct letter, **A**, **B**, **C** or **D**.

- 36** What conclusions did the writers reach about the inconsistencies in the data from Cuddie Springs?
- A** The different layers of sediment have been mixed over time.
 - B** The sand evidence is unhelpful and should be disregarded.
 - C** The area needs to be re-examined when technology improves.
 - D** The charcoal found in the area cannot be dated.
- 37** According to the writers, what impact could a natural phenomenon have had on this site?
- A** Floods could have caused the death of the megafauna.
 - B** Floods could have disturbed the archaeological evidence.
 - C** Bushfires could have prevented humans from settling in the area for any length of time.
 - D** Bushfires could have destroyed much of the evidence left by megafauna and humans.
- 38** What did the writers speculate about the people who lived at this site in 1876?
- A** They bred cattle whose bones could have been confused with megafauna.
 - B** They found that the soil was too waterlogged for farming.
 - C** They allowed cattle to move around freely at the site.
 - D** They brought stones there from another area.
- 39** In the final paragraph, what suggestion do the writers make about Australia's megafauna?
- A** A rapid change in climate may have been responsible for the extinction of the megafauna.
 - B** Megafauna could have died out as a result of small numbers being killed year after year.
 - C** The population of humans at that time was probably insufficient to cause the extinction of the megafauna.
 - D** The extinction of ancient animals should not be compared to that of modern-day species.
- 40** Which of the following best represents the writer's criticism of Field and Wroe?
- A** Their methods were not well thought out.
 - B** Their excavations did not go deep enough.
 - C** Their technology failed to obtain precise data.
 - D** Their conclusions were based on inconsistent data.

Passage 6

Business Innovation

As new 'wonder products' are getting harder and harder to find, what should companies do to survive in today's ever more competitive markets?

The Oxford English Dictionary defines innovation as 'making changes to something established'. Invention, by contrast, is the act of 'coming upon or finding; discovery'. Revolutionary 'inventions' of 20th-century US firms include, for example, the safety razor, the hydraulic brake, and DNA finger-printing. However, as Adrian Slywotsky of Mercer Management Consulting says, 'In most industries, truly differentiating new-product breakthroughs are becoming increasingly rare.' And as 'blockbuster' novel products become steadily more elusive, big companies would do well to focus instead on making lots of small things better.

Unfortunately, big companies often have big problems with innovation. 'Innovators try to change the status quo,' says Bhaskar Chakravorti of the Monitor Group, 'which is why markets tend to resist them.' Innovations frequently disrupt the way companies do things (and may have been doing them for years). And it is not just markets that are nervous of innovation. Michael Hammer, co-author of 'Re-engineering the Corporation', cites the case of a PC-maker that set out to imitate Dell's famous 'Build to Order' system of computer assembly. The company found its attempts were frustrated by both its head of manufacturing, who feared for his job, and its head of marketing, who did not want to upset his existing retail outlets. So the proposal got nowhere, and Dell continued to dominate the business.

Clayton Christensen, a Harvard Business School Professor, describes how disruptive innovations — simpler, cheaper and more convenient products that seriously upset the status quo — often herald the rapid downfall of thriving, long-standing businesses. This, it is widely recognised, is because most such organisations are designed to grow through a series of sustaining innovations — the sort which simply improve on existing products for existing markets.

William Baumol, professor at New York University, argues that large companies have been learning important lessons from the history of innovation. Most have both cut back and redirected their Research and Development spending in recent years. Innovation by big companies is now based less on the discoveries of white-coated scientists, and more on incremental improvements in the processes that constitute daily operations.

According to marketing specialist Vijay Vishwanath, companies also need to adjust to the ongoing fragmentation of markets. Once-uniform mass markets are breaking up into countless 'niches', in which everything has to be customised for a small group of

consumers. Looking for blockbusters in such a world is a daunting task. Another problem, according to Chakravorti, lies in the marketing of innovations. Too many executives are still stuck with the strategies used to sell Kodak's first camera 120 years ago, when the product was so revolutionary that the company could forget about competition for at least a decade. Today, rival products can be rushed onto the market at speeds which would have been unimaginable in those days.

Companies that fail to come up with headline-hitting new products should not despair. 'There are plenty of other, albeit less glamorous, areas where innovation can take place,' says Erik Brynjolfsson of the MIT Sloan School of Management. He claims that the roots of the USA's productivity surge originate in 'a genuine revolution in how American companies are using information technology to reinvent their business processes from top to bottom'.

Hammer highlights several instances where firms have added considerably to their shareholder value through what he calls operational innovations — creativity in their business processes. Notable among these is the American retail giant Wal-Mart. While superficially mundane, their pioneering idea of 'cross-docking' — shifting merchandise off trucks from suppliers and straight onto trucks heading for stores — has been fundamental to the company's ability to offer low prices, the platform for its outstanding success.

Companies are also being encouraged to embrace strategic innovations, following the example of such firms as Southwest Airlines, a low-cost American regional carrier, whose creative strategies and shrewd timing won it a victory over its ailing competitor US Airways. The Swedish packaging company Tetra Pak also used this style of innovation, by moving away from delivering ready-made containers for customers into providing machinery and supplying dedicated materials for them to make their own packaging. This strips out many costs from the process, and also makes it very difficult for the customer to switch suppliers. Like Southwest Airlines, they achieved success through strategic innovations alone with little innovation in either the underlying technology or the product sold.

In his book 'How to Grow When Markets Don't', Slywotsky and his co-author Richard Wise recommend what they call demand innovations. A few far-sighted organisations such as the French industrial gas company Air Liquide have found success not — as in the case of strategic innovations — by meeting existing demand in a new way, but by discovering new forms of demand and adapting to supply the products or services required. Air Liquide had been a leading supplier of industrial gases until the early 1990s, when gas became a commodity and their operational income plunged. Realising the value of the skills they had gained over the decades, the company became a supplier not only of gas, but also of the management services to accompany it. Within a short time their profits rose again.

There are certain things that managers can do to make innovations like these happen within their organisations. For example, projects with potential should be rapidly hived off into independent business units, away from the smothering influence of the status quo. The ultimate outcome of any one innovation may still be unpredictable, the process from which it emerges is not.

Questions 1–5

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- ① In order to stay in the market, companies today need to concentrate their efforts on finding genuinely original products.
- ② Disruptive innovations often pose serious threats to even successful, well-established companies.
- ③ A key stage in the development of sustaining innovations involves research into the strengths and weaknesses of competitor's products.
- ④ Manufacturers today are able to compete with innovations over shorter time periods than in the past.
- ⑤ It is hard to tell what the long-term result of any innovation might be.

Questions 6–9

Look at the following statements (Questions 6–9) and the list of researchers below. Match each statement with the correct researcher, **A–E**.

LIST OF PEOPLE

- A** Bhaskar Chakravorti
- B** Michael Hammer
- C** William Baumol
- D** Vijay Vishwanath
- E** Erik Brynjolfsson

Q	Statement	
6	the need for companies to cater for ever more specialised sets of customers	
7	an explanation for the recent upward trend in the US economy	

8	an example of company personnel who resisted innovation	
9	a shift in the way many companies are organising their budgets	

Questions 10–14

Complete the note below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

TYPE OF INNOVATIONS:

Operational innovations

DEFINITION:

Being innovative in the way existing business ⑩ ————— are carried out

EXAMPLE:

Moving goods by a system known as: ⑪ —————

COMPANY:

Wal-Mart

TYPE OF INNOVATIONS:

⑫ ————— innovations

DEFINITION:

Developing new approaches to meeting needs of existing markets

EXAMPLE:

Giving customers the means to manufacture products themselves

COMPANY:

Tetra Pak

TYPE OF INNOVATIONS:

⑬ ————— innovations

DEFINITION:

Identifying and meeting new types of customer needs

EXAMPLE:

Offering customers ⑭ ————— to go with the main product

COMPANY:

Air Liquide

Passage 7

Children's Literature Studies Today

Who studies children's literature, and what makes the field so hard to define?

Who studies children's literature and what is it that they study? The answers to this question are complex and messy, because of the many confounding factors which exist in this field.

Firstly, unlike literature for adults, children's literature is not generally written by its own readers. Adults write for children, and thus adult perceptions of what children are and of what they could and should be become woven into the literature.

Furthermore, some of those who study children's literature (and those who write certain kinds of children's books) are less interested in literary values than in the kinds of lessons it can teach—either in terms of creating better children or in terms of serving a particular curriculum. The issue of how a teacher can use a children's book is often contentious, but even outside the classroom, much material for children is still didactic.

Thirdly, while almost all literature is currently promoted within a strong commercial matrix, children's literature is often especially targeted for marketing initiatives. This fact means that readers are often recruited with a message that is negligibly literary and significantly oriented to ideas of consumption. Daniel Hade (2002) has raised useful questions about whether children's experience of reading is altered when their books are part of a larger marketing framework involving the movie, the game, and the toy of a popular children's book. How children perceive and respond to their stories in this new context is an important question.

It is also important to note that texts in an ever-increasing range of new media compete with print media for the attention of the child reader, and create definitional issues for scholars. Does the term 'literature' exclusively imply a verbal text? If not, where are the limits? Could a literary computer game ever be considered a work of literature? If not, what kind of attention should be paid to it, since children themselves undoubtedly perceive their print literature as part of a broader continuum? The internet provides one forum through which children now communicate with each other. (In 2003, the internet search engine Google listed 7,920,000 sites relating to the Harry Potter novels; even allowing for duplication and dead ends, that is a number with revolutionary implications.)

Finally, in the context of the higher education institutions where the formal study of children's literature is often located, at least three disciplinary frameworks (English, education, and librarianship) fragment the focus of scholarly study of children's literature.

How is the value of the imaginative encounter with the work of literature sustained and honored among such a welter of conflicting interests? One route through this maze is to ask the child readers for help. As David Lewis (2001) has perceptively noted, what children think of reading is not usually the same as what adults think, whether teachers or parents. As Lewis points out, children ‘sometimes see more and they often see differently’. If those who study reading can explore children’s perceptions as well as those of adults, their understanding of the nature of reading will be enhanced.

Lewis makes a further valid point when he adds that exploring children’s perceptions is usually justified for educational reasons: “It is true that a better understanding of how children read and how they learn to read, is a prerequisite to improved approaches to teaching.” However, it can also be argued, as Lewis rightly does, that when children’s responses to literature are accessed and interpreted, they frequently lead to an understanding of how picture books appeal to children.

Young people’s accounts of what and how they read also enable a more sophisticated description of many of the complex processes involved in reading. All descriptions of reading run the risk of solipsism: i.e. this is how I read so this is what reading is for everyone. Asking other readers how they read, however, reduces that risk. For example, if I am a strong visualizer as I read, I may consider that visualization is a key component of successful reading and I may judge books by their capacity to evoke a vivid visual response. Other readers, however, may help me to realize that not everyone reads with mental pictures. Some readers respond to the patterns of the words, ‘hearing’ them inaudibly like a subliminal radio program. Others respond to the patterns of feelings in the story, responding with an emotional connection. Talking to competent readers, of all ages, provides a better understanding of reading experiences.

Children’s insights are even more important when it comes to understanding the significance of print literature as one aspect of literary culture. Too often adults assume that reading any book at all is a more worthwhile experience than playing a digital game of any kind. A humbler approach would include asking why the game appeals to the player. Many adults will probably never develop the automatic skills to process a game as readily as they can read a book. This does not indicate that a book is better, but that a particular set of skills is absent. Non-players must acknowledge that some fictional universes are thus closed to them, and a logical response would be to find someone who can guide them to the pleasures and challenges of the gaming world. Games need to be judged individually just as books do, and any evaluative framework needs to take this into account.

Questions 27–29

Choose the correct letter, A, B, C or D.

- 27 Which of the following best summarises the writer's argument in the second paragraph?
- A Children are portrayed as adults see them.
 - B Children are unable to write their own stories.
 - C Adults fail to stimulate children's imaginations.
 - D Adult literature is too difficult for children.
- 28 In the third paragraph, what does the writer say is the main interest of some people who study children's literature?
- A The quality of the writing.
 - B The imaginative content of stories.
 - C The instructive nature of children's books.
 - D The way children are written about in stories.
- 29 The main point of the writer's argument in the fifth paragraph is to
- A demonstrate that academics consider computer games to be a logical extension of children's literature.
 - B explore the impact of computers on the boundaries of children's literature.
 - C illustrate that literature and computer games have different origins.
 - D prove that children are using computers more than they are reading literature.

Questions 30–34

Do the following statements agree with the views of the writer in Reading Passage 3?

Write YES if the statement agrees with the views of the writer, NO if it contradicts them, or NOT GIVEN if it is impossible to say what the writer thinks about this.

- 30 Children tend to make a clear distinction between print literature and electronic media.
- 31 The study of children's literature at higher education institutions is restricted to one subject area.
- 32 Exploring children's perceptions of reading will assist parents to choose suitable books for children.
- 33 Adults may appreciate the appeal of illustrated stories better, if they have more information on how children read.
- 34 Children should be asked what features they would like digital games to include.

Questions 35–39

Complete each sentence with the correct ending, A–H, below. Write the correct letter in each box.

- 35 Young people's accounts of how they read lead to 35 _____
- 36 The risk of solipsism is reduced by 36 _____
- 37 Strong visualisers judge books on the basis that they are 37 _____

38 Children's insights are important in

38 _____

39 When adults read a book more easily than they play a digital game it simply suggests

39 _____

LIST OF ENDINGS

- A** accepting that literature exists in a variety of forms today
- B** verbalising the words in their heads
- C** discovering the reading techniques used by others
- D** the style of written stories changing over time
- E** the lack of some specific abilities
- F** a deeper knowledge of the intricacies of reading
- G** children teaching adults to play computer games
- H** creating a variety of images in their minds

Question 40

Choose the correct letter, A, B, C or D.

- 40 What was the writer's main purpose in writing this article?
- A** To evaluate how the process of reading fits into children's literature studies.
 - B** To discuss the impact of the increasing commercial influence on children's literature studies.
 - C** To review the challenges in the field of children's literature studies and suggest how to proceed.
 - D** To provide arguments in favor of including computerized forms of children's literature studies.

Passage 8

Conscious and Unconscious Thought

What the latest psychology research reveals about how the unconscious mind solves problems the conscious mind cannot

Great scientists and artists have long known about the unconscious mind. The founder of psychoanalysis, Sigmund Freud, argued that most of our real motivation lies below conscious awareness and that we require the services of a psychoanalyst to explain ourselves to ourselves. Even those celebrated scientists who believe that Freud exaggerated the importance of the unconscious argue that it is a part of our minds which we should learn to control and take advantage of.

The great mathematician Alfred North Whitehead seemed to be a supporter of unconscious mental processing when he said: 'It is a profoundly erroneous truism, repeated by all copybooks and by eminent people making speeches, that we should cultivate the habit of thinking about what we are doing. The precise opposite is the case.' What Whitehead meant by this is that some of the greatest breakthroughs in creativity occur when a problem is consciously shelved for a while – 'don't think about it, just sleep on it' – after which the unconscious mind offers a solution. This is a process which psychoanalysts call 'incubation'.

Now the latest psychological research confirms that we can all incubate a problem to our own advantage. Ap Dijksterhuis, a psychologist at the University of Amsterdam, has recently published a series of experiments in which students were presented with complex everyday problems to solve. In one experiment they were given the opportunity to think carefully about a decision; in another experiment they were distracted by an irrelevant task whilst making a decision. The surprising result, published in the *Journal of Personality and Social Psychology*, was that 'unconscious' thinkers made the better decisions.

Timothy Wilson and colleagues from the University of Virginia's psychology department conducted another experiment. Participants were presented with five posters and asked to choose one to take home. A little later, they were telephoned and asked how satisfied they were with their choice – a measure of whether they had made the right decision from a subjective point of view. Some participants had picked their poster straight away, whereas others were asked to analyse carefully the pros and cons of each poster. It turned out that people who had analysed were less satisfied with their choice than people who had not applied logic. Paradoxically, it seems that those who had consciously weighed up the various attributes had made relatively poor decisions.

The conclusion that has to be drawn from current research suggests the somewhat counter-intuitive idea that the more complex a problem is, the less likely it becomes that conscious thought can contribute much. If a dilemma is particularly difficult, a lot of information has to be taken into account and it appears that conscious thought is not good at this. The message is that when it comes to intricate problems with many variables, you should let the unconscious deal with it.

Using dreaming is one way of doing this. Dreams are the part of sleep most strongly correlated with rapid eye movement (REM) and have been associated with enhanced creativity and novel solutions. The latest research suggests that, if you want to enhance unconscious problem-solving, you should try to think about your problems immediately after dreaming. In a recent experiment, psychiatrist Matthew Walker and colleagues from the Laboratory of Neurophysiology at Harvard Medical School woke up half of their experimental subjects immediately after REM sleep, and the other half just after sleep with no dreams, and then asked them to solve crossword puzzles. REM awakenings provided a 32 per cent advantage in the number of puzzles solved, compared with non-REM awakenings.

But you don't have to be asleep to use the unconscious mind: when you are not thinking of anything in particular, which is why daydreams are so useful in the quest for creativity. Daniel Goleman, the famous psychologist who proposed the theory of 'emotional intelligence', urges us to be more aware of, and discount, a key obstacle that the conscious mind imposes on the unconscious: that of self-censorship. This is the voice that whispers to you, 'they'll think I'm foolish' or 'that will never work', keeping you imprisoned within the boundaries of what is deemed acceptable. Howard Gardner, a professor of cognition and education at Harvard University, suggests that we free ourselves from the straitjacket of always doing things the same way. In seeking to go beyond the routine and conventional, you will gain confidence in trusting your unconscious, he says.

Questions 1–4

Look at the following statements and the list of people below.

Match each statement with the correct person, A–F.

LIST OF PEOPLE

- A** Alfred North Whitehead
- B** Ap Dijksterhuis
- C** Timothy Wilson

- D Matthew Walker
- E Daniel Goleman
- F Howard Gardner

Q	Statement	
1	People find it easier to solve problems at a particular point in the sleep cycle.	
2	People should change their general behaviour patterns in order to benefit more from the unconscious.	
3	When solving problems, people should try to ignore negative thoughts.	
4	People can come up with ideas which are more innovative if they temporarily put a problem aside.	

Questions 5–7

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Many scientists argue that people should make better use of their unconscious mind. Alfred North Whitehead, for example, gave support for the idea of (5) _____ – a term for people deliberately ignoring a problem over a period of time.

Experiments in Amsterdam that required students to consider a tricky situation found them making good decisions while carrying out an unconnected (6) _____. Similarly, an experiment in Virginia which required subjects to make a selection from a number of different (7) _____ found people feeling happier with their decisions if they had made them quickly.

Questions 8–13

Do the following statements agree with the information given in Reading Passage 3? Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- (8) Our unconscious mind helps us more than our conscious mind with complex problems.
- (9) Resting before going to sleep can result in improved solutions to problems.

- ⑩ Problems are more likely to be solved after dreamless sleep.
- ⑪ Matthew Walker's team assessed their subjects using a memory test.
- ⑫ Daydreams are more helpful to the unconscious than dreams which occur during sleep.
- ⑬ People often create their own barriers to problem-solving.

@bekhruzposts

Passage 9

Designed to Last

Could Better Design Cure Our Throwaway Culture?

Jonathan Chapman, a senior lecturer at the University of Brighton, UK, is one of a new breed of 'sustainable designers'. Like many of us, they are concerned about the huge waste associated with Western consumer culture and the damage this does to the environment. Some, like Chapman, aim to create objects we will want to keep rather than discard. Others are working to create more efficient or durable consumer goods, or goods designed with recycling in mind. The waste entailed in our fleeting relationships with consumer durables is colossal.

Domestic power tools, such as electric drills, are a typical example of such waste. However, much DIY the purchaser plans to do, the truth is that these things are thrown away having been used, on average, for just ten minutes. Most will serve 'conscience time', gathering dust on a shelf in the garage; people are reluctant to admit that they have wasted their money. However, the end is inevitable: thousands of years in land-fill waste sites. In its design, manufacture, packaging, transportation and disposal, a power tool consumes many times its own weight of resources, all for a shorter active lifespan than that of the average small insect.

To understand why we have become so wasteful, we should look to the underlying motivation of consumers. "People own things to give expression to who they are, and to show what group of people they feel they belong to," Chapman says. In a world of mass production, however, that symbolism has lost much of its potency. For most of human history, people had an intimate relationship with objects they used or treasured. Often, they made the objects themselves, or family members passed them on. For more specialised objects, people relied on expert manufacturers living close by, whom they probably knew personally. Chapman points out that all these factors gave objects a history — a narrative — and an emotional connection that today's mass-produced goods cannot possibly match. Without these personal connections, consumerist culture idolizes novelty instead. People know that they cannot buy happiness, but the chance to remake themselves with glossy, box-fresh products seems irresistible. When the novelty fades, they simply renew the excitement by buying more.

Chapman's solution is what he calls 'emotionally durable design'. He says the challenge for designers is to create things we want to keep. This may sound like a tall order but it can be surprisingly straightforward. A favourite pair of old jeans, for example, just do not have the right feel until they have been worn and washed a hundred times. It is as if they are sharing the wearer's life story. The look can be faked, but it is simply not the same. Walter Stahel, visiting professor at the University of Surrey, UK, calls this 'the teddy bear factor'.

No matter how ragged and worn a favourite teddy becomes, we don't rush out and buy another one. As adults, our teddy bear connects us to our childhood and this protects it from obsolescence. Stahel argues that this is what sustainable design needs to do with more products.

The information age was supposed to lighten our economies and reduce our impact on the environment, but, in fact, the reverse seems to be happening. We have simply added information technology to the industrial era and speeded up the developed world's metabolism. The cure is hardly rocket science: minimise waste, stop moving things around so much and use people more. So, what will post-throwaway consumerism look like? It might be as simple as installing energy-saving light bulbs, more efficient washing machines or choosing locally produced groceries with less packaging. In general, we will spend less on goods and more on services. Instead of buying a second car, for example, we might buy into a car-sharing network. Rather than following our current wasteful practices, we will buy less and rent a lot more; why own things such as tools that you use infrequently, especially things that are likely to be updated all the time?

Consumer durables will increasingly be sold with plans for their disposal. Electronic goods such as mobile phones will be designed to be recyclable, with the extra cost added into the retail price. Following Chapman's notion of emotionally durable design, there will be a move away from mass production and towards tailor-made articles and products designed and manufactured with greater craftsmanship, products which will be repaired rather than replaced, in the same way as was done in our grandparents' time. Companies will replace profit from bulk sales by servicing and repairing products chosen because we want them to last.

Chapman acknowledges that it will be a challenge to persuade people to buy fewer goods, and ones that they intend to keep. At the moment, price competition between retailers makes it cheaper for consumers to replace rather than repair. Products designed to be durable and emotionally satisfying are likely to be more expensive, so how will we be persuaded to choose sustainability? Tim Cooper, from Sheffield Hallam University in the UK, points out that many people are already happy to pay a premium for quality, and that they also tend to value and care more for expensive goods. Chapman is also positive: "People are ready to keep things for longer," he says. "The problem is that a lot of industries don't know how to do that." Chapman believes that sustainable design is here to stay. "The days when large corporations were in a position to choose whether to jump on the sustainability band-wagon or not are coming to an end," he says. Whether this is also the beginning of the end of the throwaway society remains to be seen.

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 In the second paragraph, the expression 'conscience time' refers to the fact that the owners
- A wish they had not bought the power tool.
 - B want to make sure the tool is stored safely.
 - C feel that the tool will increase in value in the future.
 - D would feel guilty if they threw the tool away immediately.
- 2 Jonathan Chapman uses the word 'narrative' in the third paragraph to refer to the fact that the owner
- A told a story about how the item was bought.
 - B was aware of how the item had come into being.
 - C felt that the item became more useful over time.
 - D was told that the item had been used for a long time.
- 3 In the third paragraph, the writer suggests that mass-produced goods are
- A inferior in quality.
 - B less likely to be kept for a long time.
 - C attractive because of their lower prices.
 - D less tempting than goods which are traditionally produced.
- 4 Lack of personal connection to goods is described as producing
- A a belief that older goods are superior.
 - B an attraction to well-designed packaging.
 - C a desire to demonstrate status through belongings.
 - D a desire to purchase a constant stream of new items.
- 5 Jeans and teddy bears are given as examples of goods which
- A have been very well designed.
 - B take a long time to show wear.
 - C are valued more as they grow older.
 - D are used by the majority of the population.

Questions 6–9

Do the following statements agree with the information given in Reading Passage 3? Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 6 People often buy goods that they make little use of.
- 7 Understanding the reasons for buying goods will help to explain why waste occurs.
- 8 People already rent more goods than they buy.
- 9 Companies will charge less to repair goods in the future.

Questions 10–14

Complete the summary using the list of words, A–I, below.

WORD LIST

- A mobile phones
- B clothing
- C tools
- D laundry
- E computers
- F food
- G heating
- H cars
- I teddy bears

The writer believes that the recipe for reducing our impact on the environment is a simple one. He states that we should use less energy for things such as lighting or ⑩ _____, and buy ⑪ _____ that will not need to be moved across long distances. Some expensive items such as ⑫ _____ could be shared, and others which may be less expensive but which are not needed often, such as ⑬ _____, could be rented instead of being purchased. He believes that manufacturers will need to design high-technology items such as ⑭ _____ so that they can be recycled more easily.

Passage 10

Digital History

How web-based video-conferencing and collaborative websites are reshaping historical research

Digital communication has revolutionised the way we teach and study history as well as virtually all other academic disciplines. Students and teachers have access to huge amounts of electronic resources. Those like myself, studying the early modern period of European history, can now download a quarter of a million relevant, original, printed works published between the invention of printing and the end of the eighteenth century. Many national libraries and historical archives have very helpful, searchable, online catalogues, and historical journals are increasingly available in electronic as well as print formats. All this facilitates learning and research, and reduces the advantages once enjoyed by prestigious and well-funded international universities and centres for historical research.

Yet how far has the digital age transformed how the individual historian works? There is a stereotype of the scholar alone in his or her workshop, crafting a publication destined for a bookshop or library. Indeed, this is how many in the profession probably still think of themselves. The solitary researcher searches through an archive or library — even if this process is now made easier by technology — and then compiles the results into a book or article which probably has a severely restricted readership.

However, the communication revolution has changed all this. I want to explore the use of two pieces of technology which my colleagues and I are using in a historical research project I am currently involved in, which is examining language in early modern European history. These pieces of technology will undoubtedly have significant implications for future study and research in history — and indeed in most other academic subjects as well. The first is web-based video-conferencing. The software we are using in our project allows us to link any number of individuals together simultaneously. It is possible for us to bring together for discussions scholars who have a tremendous range of special interests, from historians with different specialities, to literary critics, to political scientists. They may be geographically far-flung. Indeed, our project connects academics in over a dozen institutions. Time zones permitting, there is no obstacle to collaboration with colleagues anywhere in the world, so that conversations, and the sharing of information, ideas and opinions, can occur more regularly. Virtual meetings can be recorded to create an archive of debates that can be viewed or listened to subsequently by others, creating a history, if you will, of a historical research project. This is especially important for future historians trying to validate or replicate the original research. The video links can project presentations, including accompanying slides and other electronic images, and participants at one site can even control a computer at another.

In addition to its use in research projects, video-conferencing can encourage interchanges among students from different universities, so that the isolation felt by many, especially PhD students, will probably be reduced if they feel part of an international community, rather than of just one institutional body. Video-conferencing can also allow several universities to combine their scarce resources to present joint classes in minor or simply very specialized subjects (perhaps a seminar on the ancient history of the Vlach people in southeast Europe) and thus keep such subjects alive.

The second technology we are employing in our historical project is an extension of the web-based teaching tools that are now common in most universities for the purpose of giving students access to course documentation. This technology is an interactive website which enables an advanced research group like ours to plan activities and share documents, including copies of important original historical documents, and other resources. The website can also use a 'wiki', which allows multiple users to edit shared documents, whilst significantly also retaining all previous versions. The wiki principle means that texts can be jointly authored, amended and revised. Moreover, it offers a different type of publication because its publishing output is not static; rather it can be altered by authors or even by readers.

Of course, the use of technology in historical research is not without its problems. Our project has had difficulties with the quality of sound and pictures in video-conferencing. Institutional technical support may be required. Conservative scholars may be uncomfortable using the website or video-conferencing and may need more guidance: a set of informal social rules is necessary when discussing issues with people with whom you cannot make direct eye contact! For some, the idea of being recorded or filmed can be disconcerting.

Other issues centre on copyright — of the resources and of the intellectual work of participants. Will such work encourage the pirating of primary resources that are currently 'owned' by companies that have invested money in making them available? To whom does the product of a collaboration belong? And who can claim to have had an idea when it emerges from a group discussion? With many scholars working together, how can a level of excellence be maintained? And what of research ideas which are controversial or incorrect or even offensive?

Yet the advantages of the new technologies for history and other academic fields surely outweigh any problems and, in any case, their development is probably inevitable. Indeed, the potential of routine technologically aided collaboration in the arts and humanities has barely been explored. But just as Wikipedia took an old idea — the print encyclopedia — and transformed it, so our old methods of research and teaching may well be reshaped by collaborative technologies.

Questions 1–4

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 The writer thinks that recent technological advances in academic research have brought about
- A more equality among institutions.
 - B more institutional publications.
 - C the publication of more specialised journals.
 - D the production of more original documents.
- 2 The writer says that the historians today tend to think of themselves as people
- A who are highly competitive.
 - B who work collaboratively with colleagues.
 - C who are socially and professionally isolated.
 - D who are different from their popular image.
- 3 The writer refers to his own particular research project as an example of
- A why historians have special technological needs.
 - B the reasons for the discovery of modern technologies.
 - C the contrast between old-style research methods and newer ones.
 - D how new technology can be used in research in many academic fields.
- 4 Why does the writer say the recording of video-conferencing meetings is useful?
- A It makes participants more careful of what they say.
 - B It allows later research into how the project was carried out.
 - C It provides evidence of the researcher's technological skills.
 - D It eliminates the need for using printed or graphic images.

Questions 5–8

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 5 Some technological advances can help reduce the loneliness of certain higher level students.
- 6 Video-conferencing is especially useful for consultations between students and teachers.
- 7 An interactive website is more suitable for students than for high level research work.
- 8 Wiki publications are not regarded by academics as very reliable.

Questions 9–14

Complete the summary using the list of words, **A–K**, below.

WORD LIST

- A cautious
- B determination
- C categories
- D transformation
- E contributions
- F qualified
- G ownership
- H continuation
- I insupportable
- J confident K guidelines

PROBLEMS OF NEW TECHNOLOGIES

New technologies may in the future result in a ⑨ _____ of the ways in which teaching and research are carried out. However, there will be difficulties, such as disagreements over the ⑩ _____ of the materials produced by a collective effort. Also, with ⑪ _____ from a number of people, it may be hard to maintain high standards. More ⑫ _____ participants in projects may not be at ease with the new technologies and may not feel ⑬ _____ certain aspects of videoconferencing. In fact, new ⑭ _____ for personal interaction might be called for.

Passage 11

Dressed to Dazzle

As high-tech materials invade high-street fashion, prepare for clothes that are cooler than silk and warmer than wool, keep insects at arm's length, and emit many pinpricks of coloured light.

The convergence of fashion and high technology is leading to new kinds of fibres, fabrics and coatings that are imbuing clothing with equally wondrous powers. Corpe Nove, an Italian fashion company, has made a prototype shirt that shortens its sleeves when room temperature rises and can be ironed with a hairdryer. And at Nexia Biotechnologies, a Canadian firm, scientists have caused a stir by manufacturing spider silk from the milk of genetically engineered goats. Not surprisingly, some industry analysts think high-tech materials may soon influence fashion more profoundly than any individual designer.

A big impact is already being made at the molecular level. Nano-Tex, a subsidiary of American textiles maker Burlington, markets a portfolio of nanotechnologies that can make fabrics more durable, comfortable, wrinkle-free and stain-resistant. The notion of this technology posing a threat to the future of the clothing industry clearly does not worry popular fashion outlets such as Gap, Levi Strauss and Lands' End, all of which employ Nano-Tex's products. Meanwhile, Schoeller Textil in Germany, whose clients include famous designers Donna Karan and Polo Ralph Lauren, uses nanotechnology to create fabrics that can store or release heat.

Sensory Perception Technologies (SPT) embodies an entirely different application of nanotechnology. Created in 2003 by Quest International, a flavour and fragrance company, and Woolmark, a wool textile organisation, SPT is a new technique of embedding chemicals into fabric. Though not the first of this type, SPT's durability (evidently the microcapsule containing the chemicals can survive up to 30 washes) suggests an interesting future. Designers could incorporate signature scents into their collections. Sportswear could be impregnated with anti-perspirant. Hayfever sufferers might find relief by pulling on a T-shirt, and so on.

The loudest buzz now surrounds polylactic acid (PLA) fibres - and, in particular, one brand-named Ingeo. Developed by Cargill Dow, it is the first man-made fibre derived from a 100% annually renewable resource. This is currently maize (corn), though in theory any fermentable plant material, even potato peelings, can be used. In performance terms, the attraction for the 30-plus clothes makers signed up to use Ingeo lies in its superiority over polyester (which it was designed to replace).

As Philippa Watkins, a textiles specialist, notes, Ingeo is not a visual trend. Unlike nanotechnology, which promises to 'transform what clothes can do, Ingeo's impact on fashion will derive instead from its emphasis on using natural sustainable resources. Could

wearing synthetic fabrics made from polluting and non-renewable fossil fuels become as uncool as slipping on a coat made from animal fur? Consumers should expect a much wider choice of 'green' fabrics. Alongside PLA fibres, firms are investigating plants such as bamboo, seaweed, nettles and banana stalks as raw materials for textiles. Soya bean fibre is also gaining ground. Harvested in China and spun in Europe, the fabric is a better absorber and ventilator than silk, and retains heat better than wool.

Clunky earlier attempts involved attaching electronic components to the fabrics after the normal weaving process. But companies such as SOFTswitch have developed electro-conductive fabrics that behave in similar ways to conventional textiles.

Could electronic garments one day change colour or pattern? A hint of what could be achieved is offered by Luminex, a joint venture between Stabio Textile and Caen. Made of woven optical fibres and powered by a small battery, Luminex fabric emits thousands of pinpricks of light, the colour of which can be varied. Costumes made of the fabric wowed audiences at a production of the opera Aida in Washington, DC, last year.

Yet this ultimate of ambitions has remained elusive in daily fashion, largely because electronic textiles capable of such wizardry are still too fragile to wear. Margaret Orth, whose firm International Fashion Machines makes a colour-changing fabric, believes the capability is a decade or two away. Accessories with this chameleon-like capacity - for instance, a handbag that alters its colour - are more likely to appear first.

Questions 1–6

Look at the following companies (Questions 1–6) and the list of new materials below.

Match each company with the correct material, **A–H**. **NB** You may use any letter more than once.

New Materials

A material that can make you warmer or cooler **B** clothing with perfume or medication added

C material that rarely needs washing **D** clothes that can change according to external heat levels

E material made from banana stalks **F** material that is environmentally-friendly

G fibres similar to those found in nature **H** clothes that can light up in the dark

Q	Statement	
1	Corpe Nove	
2	Nexia Biotechnologies	
3	Nano-Tex	
4	Schoeller Textil	
5	Quest International and Woolmark	

Questions 7–14

Complete the summary below. Write **NO MORE THAN TWO WORDS** from the passage for each answer.

MAJOR CHANGES IN FABRICS**USING PLANTS**

Nanotechnology will bring changes we can see, while the brand called (7) _____ will help the environment. Fibre made from the (8) _____ plant has better qualities than silk and wool.

ELECTRONICS

In first attempts to use electronics, companies started with a material made by a standard (9) _____ method and then they fixed (10) _____ to the material.

LUMINEX FABRIC

- needs a (11) _____ to make it work.
- has already been used to make stage (12) _____.
- is not suitable for everyday wear because it is too (13) _____.

The first products that can change colour are likely to be (14) _____.

Passage 12

Facial Expression

What our faces reveal — and hide — about our inner state

A A facial expression is one or more motions or positions of the muscles in the skin. These movements convey the emotional state of the individual to observers. Facial expressions are a form of nonverbal communication. They are a primary means of conveying social information among humans, but also occur in most other mammals and some other animal species. Facial expressions and their significance in the perceiver can, to some extent, vary between cultures with evidence from descriptions in the works of Charles Darwin.

B Humans can adopt a facial expression to read as a voluntary action. However, because expressions are closely tied to emotion, they are more often involuntary. It can be nearly impossible to avoid expressions for certain emotions, even when it would be strongly desirable to do so; a person who is trying to avoid insulting an individual he or she finds highly unattractive might, nevertheless, show a brief expression of disgust before being able to reassume a neutral expression. Microexpressions are one example of this phenomenon. The close link between emotion and expression can also work in the other direction; it has been observed that voluntarily assuming an expression can actually cause the associated emotion.

C Some expressions can be accurately interpreted even between members of different species — anger and extreme contentment being the primary examples. Others, however, are difficult to interpret even in familiar individuals. For instance, disgust and fear can be tough to tell apart. Because faces have only a limited range of movement, expressions rely upon fairly minuscule differences in the proportion and relative position of facial features, and reading them requires considerable sensitivity to same. Some faces are often falsely read as expressing some emotion, even when they are neutral, because their proportions naturally resemble those another face would temporarily assume when emoting.

D Also, a person's eyes reveal much about how they are feeling, or what they are thinking. Blink rate can reveal how nervous or at ease a person maybe. Research by Boston College professor Joe Tecce suggests that stress levels are revealed by blink rates. He supports his data with statistics on the relation between the blink rates of presidential candidates and their success in their races. Tecce claims that the faster blinker in the presidential debates has lost every election since 1980. Though Tecce's data is interesting, it is important to recognize that non-verbal communication is multichanneled, and focusing on only one aspect is reckless. Nervousness can also be measured by examining each candidates' perspiration, eye contact and stiffness.

E As Charles Darwin noted in his book *The Expression of the Emotions in Man and Animals*: the young and the old of widely different races, both with man and animals, express the same state of mind by the same movements. Still, up to the mid-20th century most anthropologists believed that facial expressions were entirely learned and could therefore differ among cultures. Studies conducted in the 1960s by Paul Ekman eventually supported Darwin's belief to a large degree.

F Ekman's work on facial expressions had its starting point in the work of psychologist Silvan Tomkins. Ekman showed that contrary to the belief of some anthropologists including Margaret Mead, facial expressions of emotion are not culturally determined, but universal across human cultures. The South Fore people of New Guinea were chosen as subjects for one such survey. The study consisted of 189 adults and 130 children from among a very isolated population, as well as twenty-three members of the culture who lived a less isolated lifestyle as a control group. Participants were told a story that described one particular emotion; they were then shown three pictures (two for children) of facial expressions and asked to match the picture which expressed the story's emotion.

G While the isolated South Fore people could identify emotions with the same accuracy as the non-isolated control group, problems associated with the study include the fact that both fear and surprise were constantly misidentified. The study concluded that certain facial expressions correspond to particular emotions and can not be covered, regardless of cultural background, and regardless of whether or not the culture has been isolated or exposed to the mainstream.

H Expressions Ekman found to be universal included those indicating anger, disgust, fear, joy, sadness, and surprise (note that none of these emotions has a definitive social component, such as shame, pride, or *schadenfreude*). Findings on contempt (which is social) are less clear, though there is at least some preliminary evidence that this emotion and its expression are universally recognized. This may suggest that the facial expressions are largely related to the mind and each part of the face can express specific emotion.

Questions 28–32

Complete the summary paragraph below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

The result of Ekman's study demonstrates that fear and surprise are persistently

(28) _____ and made a conclusion that some facial expressions have something to do with certain (29) _____. This cannot be concealed, despite of (30) _____, and whether the culture has been (31) _____ or (32) _____ to the mainstream.

Questions 33–38

The reading passage has eight paragraphs, **A–H**. Which paragraph contains the following information? Write the correct letter, **A–H**.

NB You may use any letter more than once.

Q	Statement	
33	the difficulty identifying the actual meaning of facial expressions	
34	the importance of culture on facial expressions is initially described	
35	collected data for the research on the relation between blink and the success in elections	
36	the features on sociality of several facial expressions	
37	an indicator to reflect one's extent of nervousness	
38	the relation between emotion and facial expressions	

Questions 39–40

Choose **TWO** letters, **A–E**.

Which **TWO** of the following statements are true according to Ekman's theory?

OPTIONS

- A** No evidence shows animals have their own facial expressions.
- B** The potential relationship between facial expression and state of mind exists.
- C** Facial expressions are concerning different cultures.
- D** Different areas on face convey certain state of mind.
- E** Mind controls men's facial expressions more obvious than women's.

Answer: (39) (40)

Passage 13

First Memories

Psychologist Richard Worrington reviews research into our earliest memories

People have sometimes told me they dread the question, 'What is your earliest memory?' Perhaps they find the whole notion too simplistic, like being asked to name a favourite film or book, when the choice is almost limitless. But we should welcome enquiries about our first memories - it is a compliment if others are interested in our lives and experiences. Besides that, the human memory is a fascinating area of study. Research shows that some subjects can provide rich details of a first memory very quickly while others struggle to come up with anything at all. Their range of memories is also predictably diverse, from holidays and injuries to routine activities such as walking in the park or having a family meal. Most studies show that around 25% of first memories are of a negative event, while nearly double that proportion are positive, and these findings are constant regardless of the subjects' ages or backgrounds.

Historically, the topic of human memory in general most certainly attracted a great deal of scholarly attention. But our earliest memories were largely overlooked by researchers, which turns out to have been an important omission. Scientists developed a reasonable understanding of what we now call 'semantic memory', which is the ability to recall what might loosely be called data, such as dates, times or places in which a particular incident occurred. In general terms, semantic memory is fixed and static. However, there are other forms of memory, which collectively I call 'subjective memories', and these are much more fluid. For example, numerous experiments studying the memories held by witnesses to an event have shown conclusively that recollections of what happened can vary dramatically between individuals, and even in the same person over time. Our earliest memories are another example of subjective memory.

So what do we know about when, why and how our earliest memories form? Scientists today recognise what has been termed 'childhood amnesia'. This is the finding that babies and toddlers do not form lasting memories. For the majority of adults, their earliest memories begin once the period of childhood amnesia has come to an end, at about three and a half years of age. These are the memories that will endure for the rest of their lives. However, beyond that fixed starting point much variation exists. For example, one study found that the earliest memories of American children were more likely to feature solitary play or events where they were the central character. Asian children, in contrast, tended to remember family or school events in which they might have played no central role. Based on the assumption that American society places greater emphasis on individualism, this implies that earliest memories are influenced by social priorities.

Another important idea is that of 'autobiographical memory'. This is the concept that many people construct a narrative of their lives and select memories that contribute most forcefully to that narrative. This seems to be relevant to our earliest memories in the following sense. Once people reach maturity, they often develop a strong feeling of who they are, perhaps based on their chosen line of work or their interests. Studies show that, for example, the earliest memories of an accomplished sportswoman were of herself as a child athlete, whereas a chef recalled preparing a meal as a young boy. Thus, who we are now appears to have some bearing on what we choose to recall from long ago.

A picture is emerging, then, of the unreliability of our earliest memories. Certainly, studies show that some earliest memories are simply made up. However, these seem to be a small minority. Much more common is that our earliest memories may be pieced together from different elements of reality. They may feature an item that we do remember from our childhood, such as a pram or toy, but embellished in some way to include a larger event. Or they may feature information that actually comes from a photograph, a family story, a memory that we've heard from somebody else, or some combination of all of these. Research shows that some memories stand out from the rest as much clearer and more colourful. These are usually formed during what's known as the 'reminiscence bump', which occurs between the ages of 16 and 25. But even these memories are no less likely to be created.

In fact, research shows that only so-called 'flashbulb memories' are likely to be completely true. These are memories that derive from events about which we have particularly strong feelings, perhaps of intense sadness or happiness. But most of our other early memories are, at least to some extent, chosen by us. The question is, though, does this really matter? They are still much-loved memories and perhaps we can learn about ourselves from them. Often these memories are strong indicators of who we are, or would like to be, and exchanging first memories with others fosters connection and intimacy. Such factors represent important elements of our humanity, whether or not the memory itself is completely accurate.

Questions 1–5

Choose **YES** if the statement agrees with the claims of the writer of the text, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 1 People ought to be pleased when asked, 'What is your earliest memory?'
- 2 Most research subjects have surprisingly similar first memories.
- 3 Negative first memories are more interesting than positive ones.

- 4 In the past, academics completed enough studies of earliest memories.
- 5 Experiments using witnesses to events have provided reliable results.

Questions 6–9

Choose the correct answer.

- 6 The writer refers to American and Asian children
 - A to suggest that most people's earliest memories are of themselves.
 - B to illustrate how earliest memories reflect cultural values.
 - C to contrast two methods of researching earliest memories.
 - D to explain why some earliest memories last longer than others.
- 7 In the fourth paragraph, the writer concludes that earliest memories
 - A may depend on an adult's sense of identity.
 - B might not affect later character development.
 - C could help an adult select a suitable career.
 - D should not influence decisions in later life.
- 8 What is the writer's main point in the fifth paragraph?
 - A Certain types of memories are often invented.
 - B Members of the same family have similar memories.
 - C Some people have more reliable memories than others.
 - D Memories are constructed from a variety of sources.
- 9 What is the writer doing in the final paragraph?
 - A asking why false memories are so common
 - B arguing for more research into false memories
 - C suggesting that false memories still have value
 - D comparing the value of true and false memories

Questions 10–14

Choose the correct explanation, A–G, for each term.

LIST OF EXPLANATIONS

- A It describes recollections that are associated with very strong emotions.
- B It is associated with a period of our lives that we would like to return to.
- C It describes the period after which lifelong memories begin.
- D It is associated with the memories that we recall most vividly.
- E It describes certain events that we wish we could forget.
- F It describes hard facts, such as when or where an event happened.
- G It is associated with a story we have created about ourselves.

Q	Statement	
10	semantic memory	
11	childhood amnesia	
12	autobiographical memory	
13	reminiscence bump	
14	flashbulb memory	

@bekhruzposts

Passage 14

How Green is your PlanIT Valley?

A new European company plans to build a smart, low-carbon city from scratch — and its founders say the technology already exists to do it

The first thing to say about the people running new European company Living PlanIT is that they are ambitious. Not only are they planning to build a smart green city from scratch at a site in northern Portugal, but they also hope to establish their PlanIT Valley development as both a genuine European alternative to the USA's Silicon Valley, and a working model that will inspire the next generation of low carbon cities. These will combine real environmental sustainability with a quality of IT-enhanced urban living almost unrecognisable from the crowded, polluted and disorganised reality that is city life for most people.

Many are sceptical about the company's plans for a brand-new smart city packed with cutting-edge green technology which can house 225,000 people while producing 'negligible' greenhouse gas emissions. To them, the technical challenges, combined with the \$10 billion that the company needs to raise to see the project through from beginning to end, make Living PlanIT's plans sound more like an admirable experiment rather than a viable construction project. Their scepticism is not helped by the company's use of marketing language that, in order to understand it, requires a degree in Public Relations. For example, the company's claim that its 'design and manufacturing platforms enable the convergence of computing, network and sensing technologies with the fabric of buildings and places, demonstrated at urban scale in the development and operations of PlanIT Valley' does not really explain what it does.

However, when you listen to chief executive Steve Lewis outline his plans for the company, it becomes possible to believe that they might just deliver on their absurdly ambitious promises. His rationale for the company is admirably simple. He argues that the construction industry remains the last sector of the economy to resist the IT revolution that has enhanced efficiencies across every other industry, from car manufacturing to food production. Their existing techniques are inadequate, he says, for today's technology-rich and environmentally-aware requirements. Here, therefore, is an opportunity to entirely update the building process. Taking lessons from other manufacturing industries, including aerospace, automotive and shipbuilding, project leaders identified a number of elements that feature in modern manufacturing processes and which could be applied to modern buildings from the very start of the construction process.

Living PlanIT plans to integrate IT into the fabric of the city. It is installing many thousands of sensors that allow an urban operating system to deliver intelligent buildings that are constantly optimised to enhance comfort, productivity and environmental

sustainability. Meanwhile, the latest renewable energy technologies and green building techniques will allow the city to operate with a virtually non-existent carbon footprint. Of course, such techniques are not new, but they are rarely put into practice. In fact, the IT industry has been complaining for a long time that the construction sector has failed to make adequate (if any) use of IT in its buildings.

So, what makes PlanIT Valley different? For a start, the company has considerable power and influence, both in terms of the team it has assembled and the financial backing it has already been promised. Lewis and many on the senior management team have served as senior executives with other major IT companies. This means that they not only have the right experience, but also the essential 'anything is possible' mentality that is a feature of such companies. In addition, Lewis claims that the company has already invested \$300 million in putting together its team of engineers and developing its technology portfolio. As a result, the project is establishing a degree of credibility that far exceeds that of other similar projects.

More importantly, however, almost all of the technology the company is planning to deploy in PlanIT Valley either already exists or is viable from a technical point of view. Intelligent buildings that know to turn the air-conditioning on before you even realise you are hot may sound like something out of a science-fiction novel, but we are increasingly living in a science-fiction age. You do not need to invent anything new to develop a zero-carbon smart city, you just have to put all the right technologies together in the right place.

If Living PlanIT achieve this integration, it will hopefully be able to prove the final part of Lewis' claim. Namely, that cost concerns surrounding green developments are ill-founded. A more automated approach to construction coupled with long-term efficiency gains delivered by intelligent infrastructure more than cancel out the extra money required to build the development in the first place. And if it can win the economic argument, future opportunities are enormous. As Lewis points out, projected world population growth means that the world has to deliver between 9,500 and 10,000 new cities over the next forty years to house everyone. There is no chance of avoiding dangerous levels of climate change unless this expansion is delivered in an environmentally-sustainable manner. That would provide quite a business opportunity for the company that can deliver the solution.

The PlanIT Valley project presents many problems in terms of project management and coordination, and there is a huge amount of work to be done before the first residents are able to move in. Whatever the outcome, it is hard not to admire a project that will put so many theories about smart cities to the test all in one go. First, it is challenging accepted assumptions on how cities should be designed and constructed. Secondly, it will show what

can be done if connectivity and intelligence are built into the design from the beginning. Thirdly, it will be a pilot project for a whole range of new services and, equally important, new types of collaboration.

Questions 1–5

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 What do we learn about the people running the company Living PlanIT in the first paragraph?
 - A They are going to build the world's first green smart city.
 - B They want to build a replica of Silicon Valley in Europe.
 - C They want their new city to be the first of other similar cities.
 - D They think that the quality of life in most other cities is unacceptable.
- 2 Why are a lot of people skeptical about PlanIT Valley?
 - A They do not think that the plans are practical or possible.
 - B They do not understand how the technology works.
 - C They have misunderstood the reasons why Living PlanIT is building a new city.
 - D They have not read the marketing information clearly enough.
- 3 Who or what does "Their" in "Their existing techniques" refer to?
 - A Living PlanIT
 - B The construction industry
 - C Car manufacturers
 - D Food producers
- 4 The senior management team at Living PlanIT
 - A have more power and influence than teams working on other, similar projects.
 - B have raised enough money to build PlanIT Valley.
 - C work elsewhere when they are not working on the PlanIT Valley project.
 - D possess the right attitude required to undertake such a major project.
- 5 Why, according to the author, is the PlanIT Valley project so admirable?
 - A It shows how good working practices can have a successful outcome.
 - B It presents such a challenge to its creators.
 - C It is putting several theories into practice simultaneously.
 - D It demonstrates why collaboration between builders and IT experts is so important.

Questions 6–10

Do the following statements agree with the claims of the writer in the Reading passage?

Write **YES** if the statement agrees with the claims of the writer, **NO** if it contradicts them, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 6 Living PlanIT's marketing information is unclear and confusing for many people.

- 7 Steve Lewis presents a convincing argument that suggests PlanIT Valley might succeed.
- 8 Some of the buildings in PlanIT Valley have features that can be found in aircraft, cars or ships.
- 9 Living PlanIT's idea of building a smart, environmentally-friendly city is original and unique.
- 10 Some of the technology in PlanIT Valley was probably inspired by science fiction stories.

Questions 11–14

Complete each sentence with the correct ending, A–G, below. Write the correct letter in each box.

- 11 The IT industry believes that, when it comes to putting up new buildings, IT is currently 11 _____
- 12 Creating new technology for PlanIT Valley is 12 _____
- 13 Steve Lewis believes that the economic arguments against green developments are 13 _____
- 14 Some organisational aspects of the PlanIT Valley project are 14 _____

LIST OF ENDINGS

- A technically possible
- B underused
- C difficult to avoid
- D not valid
- E inefficient
- F too ambitious
- G extremely challenging

Passage 15

How a Prehistoric Predator Took to the Skies

Is that a bird in the sky? A plane? No, it's a pterosaur. Kate Thomas meets Professor Matthew Wilkinson, who built a life-size model to find out how this prehistoric predator ever got off the ground.

Pterosaurs existed from the Triassic period, 220 million years ago, to the end of the Cretaceous period, 65 million years ago, when South America pulled away from Africa and the South Atlantic was formed. They are among the least understood of all the extinct reptiles that once spent their lives in the skies while the dinosaurs dominated the land. Pterosaurs had no feathers, but at least part of their bodies was covered in hair, not unlike bats. Some believe this is an indication they were warm-blooded. Researchers also debate whether pterosaurs travelled on the ground by walking on their hind legs, like birds, or by using all fours, relying on their three-toed front feet as well as their four-toed rear feet.

Vertebrates, pterosaurs were the first species possessing backbones to become airborne, but scientists have never quite understood their flight technique. How, they wondered, did such a heavy creature ever manage to take off? How could a wing that appears to have been supported by fine, hollow bones have carried one into the sky? Then came the discovery of a site in Brazil's Araripe basin. Here, not only were hundreds of fossils of amphibians* and other reptiles found, but archaeologists unearthed a number of very well-preserved pterosaurs. The anhanguera — a fish-eating sub-species of pterosaur that ruled the skies in the Cretaceous period — was among them. With a wingspan of up to 12 metres, they would have made an amazing sight in the sky — had any human been there to witness it. 'I've been studying pterosaurs for about eight years now,' says Dr Matthew Wilkinson, a professor of zoology at Cambridge University. With an anhanguera fossil as his model, Wilkinson began gradually reconstructing its skeletal structure in his Cambridge studio. The probability of finding three-dimensional pterosaur fossils anywhere is slim. 'That was quite a find,' he says. 'Their bones are usually crushed to dust.' Once the structure was complete, it inspired him to make a robot version as a way to understand the animal's locomotion. With a team of model-makers, he has built a remote-controlled pterosaur in his studio. 'Fossils show just how large these creatures were. I've always been interested in how they managed to launch themselves, so I thought the real test would be to actually build one and fly it.'

Wilkinson hasn't been alone in his desire to recreate a prehistoric beast. Swiss scientists recently announced they had built an amphibious robot that could walk on land and swim in water using the sort of backbone movements that must have been employed by the first creatures to crawl from the sea. But Wilkinson had the added complication of working out his beast's flight technique. Unlike those of bats or flying squirrels, pterosaur wings — soft,

stretchy membranes of skin tissue — are thought to have reached from the chest right to the ankle, reinforced by fibres that stiffened the wing and prevented tearing. Smaller subspecies flapped their wings during takeoff. That may have explained the creatures' flexibility, but it did not answer the most pressing question: how did such heavy animals manage to launch themselves into the sky? Working with researchers in London and Berlin, Wilkinson began to piece together the puzzle.

It emerged that the anhanguera had an elongated limb called the pteroid. It had previously been thought the pteroid pointed towards the shoulder of the creature and supported a soft forewing in front of the arm. But if that were the case, the forewing would have been too small and ineffectual for flight. However, to the surprise of many scientists, fossils from the Araripe basin showed the pteroid possibly faced the opposite way, creating a much greater forewing that would have caught the air, working in the same way as the flaps on the wings of an aeroplane. So, with both feet on the ground, the anhanguera might have simply faced into the wind, spread its wings and risen up into the sky. Initial trials in wind tunnels proved the point — models of pterosaurs with forward-facing pteroids were not only adept at gliding, but were agile flyers in spite of their size. 'This high-lift capability would have significantly reduced the minimum flight speed, allowing even the largest forms to take off without difficulty,' Wilkinson says. 'It would have enabled them to glide very slowly and may have been instrumental in the evolution of large size by the pterosaurs.'

Resting in the grass at the test site near Cambridge, the robot-model's wings ripple in the wind. In flight, the flexible membrane, while much stiffer than the real thing, allows for a smooth takeoff and landing. But the model is also troubled by other mechanical problems. 'Unlike an aircraft, which is stabilised by the tail wing at the back, the model is stabilised by its head, which means it can start spinning around. That's the most problematic bit as far as we're concerned,' Wilkinson says. 'We've had to take it flying without the head so far.' When it flies with its head attached, Wilkinson will finally have proved his point.

So what's next for the zoologist — perhaps a full-size Tyrannosaurus rex? 'No,' he tells me: 'We're desperate to build really big pterosaurs. I'm talking creatures with even greater wingspans, weighing a quarter of a ton. But,' he adds, just as one begins to fear for the safety and stress levels of pilots landing nearby at Cambridge City Airport, 'it's more likely we'll start off with one of the smaller, flapping pterosaurs.' This is certainly more reassuring. Let's hope he is content to leave it at that.

Questions 27–32

Complete the summary using the list of words, A–L, below.

LIST OF WORDS

- A front feet
- B fish
- C dinosaurs
- D reptiles
- E flexibility
- F hind legs
- G amphibians
- H birds
- I strength
- J weight K tail L hair

Pterosaurs are believed to have existed until the end of the Cretaceous period. They are classed as (27) _____ which were capable of flight, although, unlike modern species, they had some (28) _____, which is evidence of their having had warm blood. There are two theories as to how they moved on land: perhaps with all their feet or by using their (29) _____ only. Another mystery has concerned the ability of the pterosaur to fly despite its immense (30) _____, and the fact that the bones making up the wing did not have great (31) _____. Thanks to reptile fossils found in Brazil, we now know that the subspecies known as anhanguera had wings that were 12 metres across and that it mainly survived on (32) _____.

Questions 33–36

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write **YES**, **NO** or **NOT GIVEN**.

- (33) It is rare to find a fossil of a pterosaur that clearly shows its skeleton.
- (34) The reason for building the model was to prove pterosaurs flew for long distances.
- (35) It is possible that pterosaur species achieved their wing size as a result of the pteroid.
- (36) Wilkinson has made several unsuccessful replicas of the pterosaur's head.

Questions 37–40

Choose the correct letter, **A**, **B**, **C** or **D**.

- 37 What was Professor Wilkinson's main problem, according to the third paragraph?
- A Early amphibians had a more complex structure than pterosaurs.
 - B Pterosaur wings could easily be damaged while on the ground.
 - C Flying squirrels and bats were better adapted to flying than pterosaurs.
 - D Large pterosaurs were not able to take off like other flying animals.
- 38 What did Professor Wilkinson discover about a bone in pterosaurs called a pteroid?
- A It was in an unexpected position.
 - B It existed only in large species of pterosaurs.
 - C It allowed pterosaurs to glide rather than fly.
 - D It increased the speed pterosaurs could reach in the air.
- 39 According to the writer, the main problem with the remote-controlled 'pterosaur' is that
- A it has been unable to leave the ground so far.
 - B it cannot be controlled when its head is attached.
 - C its wing material is not flexible enough.
 - D the force of the wind may affect its test results.
- 40 What does 'it' in the last sentence refer to?
- A the information the tests have revealed
 - B Wilkinson's sense of achievement
 - C Wilkinson's desire to build models
 - D the comparison between types of models

Passage 16

Improving Patient Safety

How the design of drug packaging and patient information leaflets can make the difference between safety and tragedy

One of the most prominent design issues in pharmacy is that of drug packaging and patient information leaflets (PILs). Many letters have appeared in *The Journal's* letters pages over the years from pharmacists dismayed at the designs of packaging that are "accidents waiting to happen".

Packaging design in the pharmaceutical industry is handled by either in-house teams or design agencies. Designs for over-the-counter medicines, where characteristics such as attractiveness and distinguishability are regarded as significant, are usually commissioned from design agencies. A marketing team will prepare a brief and the designers will come up with perhaps six or seven designs. These are whittled down to two or three that might be tested on a consumer group. In contrast, most designs for prescription-only products are created in-house. In some cases, this may simply involve applying a company's house design (ie, logo, colour, font, etc). The chosen design is then handed over to design engineers who work out how the packaging will be produced.

The author of the recently published "Information design for patient safety," Thea Swayne, tracked the journey of a medicine from manufacturing plant, through distribution warehouses, pharmacies and hospital wards, to patients' homes. Her book highlights a multitude of design problems with current packaging, such as look-alikes and sound-alikes, small type sizes and glare on blister foils. Situations in which medicines are used include a parent giving a cough medicine to a child in the middle of the night and a busy pharmacist selecting one box from hundreds. It is argued that packaging should be designed for moments such as these. "Manufacturers are not aware of the complex situations into which products go. As designers, we are interested in not what is supposed to happen in hospital wards, but what happens in the real world," Ms Swayne said.

Incidents where a drug has been injected intrathecally (into the spine) instead of intravenously (into a vein) are a classic example of how poor design can contribute to harm. Investigations following these tragedies have attributed some blame to poor typescript.

Child protection is another area that gives designers opportunities to improve safety. According to the Child Accident Prevention Trust, seven out of 10 children admitted to hospital with suspected poisoning have swallowed medicines. Although child-resistant closures have reduced the number of incidents, they are not fully child-proof. The definition of such a closure is one that not more than 15 percent of children aged between

42 and 51 months can open within five minutes. There is scope for improving what is currently available, according to Richard Mawle, a freelance product designer. "Many child-resistant packs are based on strength. They do not necessarily prevent a child from access, but may prevent people with a disability," he told *The Journal*. "The legal requirements are there for a good reason, but they are not good enough in terms of the users," he said. "Older people, especially those with arthritis, may have the same level of strength as a child," he explained, and suggested that better designs could rely on cognitive skills (eg, making the opening of a container a three-step process) or be based on the physical size of hands.

Mr Mawle worked with GlaxoSmithKline on a project to improve compliance through design, which involved applying his skills to packaging and PILs. Commenting on the information presented, he said: "There can be an awful lot of junk at the beginning of PILs. For example, why are company details listed towards the beginning of a leaflet when what might be more important for the patient is that the medicine should not be taken with alcohol?"

Look-alike boxes present a potential for picking errors and an obvious solution would be to use colours to highlight different strengths. However, according to Ms Swayne, colour differentiation needs to be approached with care. Not only should strong colour contrasts be used, but designating a colour to a particular strength (colour coding) is not recommended because this could lead to the user not reading the text on a box.

Design features can provide the basis for lengthy debates. For example, one argument is that if all packaging is white with black lettering, people would have no choice but to read every box carefully. The problem is that trials of drug packaging design are few – common studies of legibility and comprehensibility concern road traffic signs and visual display units. Although some designers take results from such studies into account, proving that a particular feature is beneficial can be difficult. For example, EU legislation requires that packaging must now include the name of the medicine in Braille but, according to Karel van der Waarde, a design consultant to the pharmaceutical industry, "it is not known how much visually impaired patients will benefit nor how much the reading of visually able patients will be impaired".

More evidence might, however, soon be available. EU legislation requires PILs to reflect consultations with target patient groups to ensure they are legible, clear and easy to use. This implies that industry will have to start conducting tests. Dr van der Waarde has performed readability studies on boxes and PILs for industry. A typical study involves showing a leaflet or package to a small group and asking them questions to test understanding. Results and comments are used to modify the material, which is then tested on a larger group. A third group is used to show that any further changes made are

an improvement. Dr van der Waarde is, however, sceptical about the legal requirements and says that many regulatory authorities do not have the resources to handle packaging information properly. "They do not look at the use of packaging in a practical context – they only see one box at a time and not several together as pharmacists would do," he said.

The RCA innovation exhibition this year revealed designs for a number of innovative objects. "The popper", by Hugo Glover, aims to help arthritis sufferers remove tablets from blister packs, and "pluspoint", by James Cobb, is an adrenaline auto-injector that aims to overcome the fact that many patients do not carry their auto-injectors due to their prohibitive size. The aim of good design, according to Roger Coleman, professor of inclusive design at the RCA, is to try to make things more user-friendly as well as safer. Surely, in a patient-centred health system, that can only be a good thing. "Information design for patient safety" is not intended to be mandatory. Rather, its purpose is to create a basic design standard and to stimulate innovation. The challenge for the pharmaceutical industry, as a whole, is to adopt such a standard.

Questions 1–6

Look at the following statements and the list of people or organisation below.

Match each statement with the correct person or organisation, **A–D**.

NB You may use any letter more than once.

LIST OF PEOPLE/ORGANISATION

- A** Thea Swayne
- B** Child Accident Prevention Trust
- C** Richard Mawle
- D** Karel van der Waarde

Q	Statement	
1	Elderly people may have the same problem with children if the lids of containers require too much strength to open.	
2	Adapting packaging for the blind may disadvantage the sighted people.	
3	Specially designed lids cannot eliminate the possibility of children swallowing pills accidentally.	
4	Container design should consider situations, such as drug used at home.	
5	Governing bodies should investigate many different container cases rather than individual ones.	

6	Information on the list of a leaflet hasn't been in the right order.	
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Questions 7–11

Complete the notes using the list of words, **A–G**, below.

WORD LIST

- A** consumers
- B** marketing teams
- C** pharmaceutical industry
- D** external designers
- E** in-house designers
- F** design engineers
- G** pharmacist

PACKAGING IN PHARMACEUTICAL INDUSTRY

DESIGNS FOR OVER-THE-COUNTER MEDICINES

First, **7** ————— make the proposal, then pass them to the **8** —————.
Finally, these designs will be tested by **9** —————.

PRESCRIPTION-ONLY

First, the design is made by **10** ————— and then subjected to **11** —————.

Questions 12–14

Choose the correct letter, **A, B, C** or **D**.

- 12** What may cause the accident in "design container"?
- A** a print error
 - B** style of print
 - C** wrong label
 - D** the shape of the bottle
- 13** What do people think about the black and white only print?
- A** Consumers dislike these products.
 - B** People have to pay more attention to the information.
 - C** That makes all products looks alike.
 - D** Sighted people may feel it more helpful.

- 14 Why does the writer mention "popper" and "pluspoint"?
- A to show that container design has made some progress
 - B to illustrate an example of inappropriate design which can lead to accidents
 - C to show that the industry still needs more to improve
 - D to point out that consumers should be more informed about the information

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Passage 17

Inside the Mind of a Fan

How watching sport affects the brain

A At about the same time that the poet Homer invented the epic hero, the ancient Greeks started a festival in which men competed in a single race, about 200 metres long. The winner received a branch of wild olives. The Greeks called this celebration the Olympics. Though the ancient sprint remains, today the Olympics are far more than that. Indeed, the Games seem to celebrate the dream of progress as embodied in the human form. That the Games are intoxicating to watch is beyond question. During the Athens Olympics in 2004, 3.4 billion people, half the world, watched them on television. Certainly, being a spectator is a thrilling experience: but why?

B In 1996, three Italian neuroscientists, Giacomo Rizzolatti, Leonardo Fogassi and Vittorio Gallese, examined the premotor cortex of monkeys. They discovered that inside these primate brains there were groups of cells that 'store vocabularies of motor actions'. Just as there are grammars of movement. These networks of cells are the bodily 'sentences' we use every day, the ones our brain has chosen to retain and refine. Think, for example, about a golf swing. To those who have only watched the Master's Tournament on TV, golfing seems easy. To the novice, however, the skill of casting a smooth arc with a lop-sided metal stick is virtually impossible. This is because most novices swing with their consciousness, using an area of brain next to the premotor cortex. To the expert, on the other hand, a perfectly balanced stroke is second nature. For him, the motor action has become memorized, and the movements are embedded in the neurons of his premotor cortex. He hits the ball with the tranquility of his perfected autopilot.

C These neurons in the premotor cortex, besides explaining why certain athletes seem to possess almost unbelievable levels of skill, have an even more amazing characteristic, one that caused Rizzolatti, Fogassi and Gallese to give them the lofty title 'mirror neurons'. They note, 'The main functional characteristic of mirror neurons is that they become active both when the monkey performs a particular action (for example, grasping an object or holding it) and, astonishingly, when it sees another individual performing a similar action.' Humans have an even more elaborate mirror neuron system. These peculiar cells mirror, inside the brain, the outside world: they enable us to internalize the actions of another. In order to be activated, though, these cells require what the scientists call 'goal-orientated movements'. If we are staring at a photograph, a fixed image of a runner mid-stride, our mirror neurons are totally silent. They only fire when the runner is active: running, moving or sprinting.

D What these electrophysiological studies indicate is that when we watch a golfer or a runner in action, the mirror neurons in our own premotor cortex light up as if we were the ones competing. This phenomenon of neural mirroring was first discovered in 1954, when two French physiologists, Gastaut and Bert, found that the brains of humans vibrate with two distinct wavelengths, alpha and mu. The mu system is involved in neural mirroring. It is active when our bodies are still, and disappears whenever we do something active, like playing a sport or changing the TV channel. The surprising fact is that the mu signal is also quiet when we watch someone else being active, as on TV; these results are the effect of mirror neurons.

E Rizzolatti, Fogassi and Gallese call the idea for mirror neurons the 'direct matching hypothesis'. They believe that we only understand the movement of sports stars when we 'map the visual representation of the observed action onto our motor representation of the same action'. According to this theory, watching an Olympic athlete 'causes the motor system of the observer to resonate. The "motor knowledge" of the observer is used to understand the observed action.' But mirror neurons are more than just the neural basis for our attitude to sport. It turns out that watching a great golfer makes us better golfers, and watching a great sprinter actually makes us run faster. This ability to learn by watching is a crucial skill. From the acquisition of language as infants to learning facial expressions, mimesis (copying) is an essential part of being conscious. The best athletes are those with a premotor cortex capable of imagining the movements of victory, together with the physical properties to make those movements real.

F But how many of us regularly watch sports in order to be a better athlete? Rather, we watch sport for the feeling, the human drama. This feeling also derives from mirror neurons. By letting spectators share in the motions of victory, they also allow us to share in its feelings. This is because they are directly connected to the amygdala, one of the main brain regions involved in emotion. During the Olympics, the mirror neurons of whole nations will be electrically identical, their athletes causing spectators to feel, just for a second or two, the same thing. Watching sports brings people together. Most of us will never run a mile in under four minutes, or hit a home run. Our consolation comes in watching, when we gather around the TV, we all feel, just for a moment, what it is to do something perfectly.

Questions 1–6

Reading Passage 3 has six paragraphs, A–F. Which paragraph contains the following information? Write the correct letter, A–F, in boxes 1–6. NB You may use any letter more than once.

Q	Statement	
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1	an explanation of why watching sport may be emotionally satisfying	
2	an explanation of why beginners find sporting tasks difficult	
3	a factor that needs to combine with mirroring to attain sporting excellence	
4	a comparison of human and animal mirror neurons	
5	the first discovery of brain activity related to mirror neurons	
6	a claim linking observation to improvement in performance	

Questions 7–9

Choose the correct letter, **A**, **B**, **C** or **D**.

- 7 The writer uses the term 'grammar of movement' to mean
- A a level of sporting skill.
 - B a system of words about movement.
 - C a pattern of connected cells.
 - D a type of golf swing.
- 8 The writer states that expert players perform their actions
- A without conscious thought.
 - B by planning each phase of movement.
 - C without regular practice.
 - D by thinking about the actions of others.
- 9 The writer states that the most common motive for watching sport is to
- A improve personal performance.
 - B feel linked with people of different nationalities.
 - C experience strong positive emotions.
 - D realize what skill consists of.

Questions 10–14

Do the following statements agree with the views of the writer in Reading Passage 3? Write **YES** if the statement is true, **NO** if the statement is false, or **NOT GIVEN** if the information is not given in the passage.

- 10 Inexpert sports players are too aware of what they are doing.
- 11 Monkeys have a more complex mirror neuron system than humans.
- 12 Looking at a photograph can activate mirror neurons.
- 13 Gastaut and Bert were both researchers and sports players.
- 14 The mu system is at rest when we are engaged in an activity.

Passage 18

Kinesics

An introduction to the study of body language and non-verbal communication

① **A** Psychology is a well-established subject, but one area of it, 'kinesics' – the study of body language – was not identified until 1952, and research only began in earnest in the 1960s. Kinesics is based on the behavioural patterns of non-verbal communication. Clinical studies have revealed the extent to which body language can actually contradict verbal communications. A classic example is the young woman who told her psychiatrist that she loved her boyfriend very much while shaking her head from side to side in subconscious denial.

② **B** Body language also sheds light on the dynamics of interfamily relationships. A family sitting together can give a revealing picture of itself simply by the way its members move their arms and legs. If the mother, for example, crosses her legs first and the rest of the family then follows suit, she has set the lead for the family action, though she, as well as the rest of the family, may not be aware she is doing it. In fact, her words deny her leadership as she asks her husband or children for advice. But the unspoken, follow-the-leader clue in the actions of the family members give the family set-up away to someone knowledgeable in kinesics.

③ **C** Another kinesic signal is the unconscious widening of a person's pupils when their eyes see something pleasant. Experiments have shown that we become aware of how that person feels, although we are conscious neither of seeing the signal nor of giving it meaning.

④ **D** This kinesic principle has been used on a commercial level to detect the effect of a television advertisement. While the ad is being shown to a selected audience, their eyes are photographed. Later, the film is carefully studied to detect just when there is any widening of the eye; in other words, when there is any unconscious, positive response to the advertisement.

⑤ **E** Body language can include any voluntary or involuntary movement of a part or all of the body, used by a person to communicate an emotional message to the outside world. To understand this unspoken body language, kinesics experts often have to take into consideration cultural and environmental differences. The average person, unschooled in cultural nuances of body language, is often mistaken when decoding what he or she sees.

⑥ **F** In addition to sending and receiving messages, body language can also serve to break through defenses, if it is used skillfully. Often the swiftest and most obvious type of body language is touch. The touch of a hand, or an arm around someone's shoulder, can spell a more vivid and direct message of friendliness than dozens of words. But such a

touch must come at the right moment and in the right context, or the other person may take it as an intrusion to his or her personal space. For every situation there must be two elements to body language: the delivery of the message and the reception of the message.

⑦ **G** However, some people are 'touchers', compulsive touchers, who seem completely impervious to all messages they may get from friends and acquaintances. They are people who will touch others even though they are bombarded with body-language requests not to. There are also people who avoid touching others altogether.

⑧ **H** We act out our state of being with non-verbal language. We lift one eyebrow for disbelief, shrug our shoulders for indifference, tap our fingers for impatience. The gestures are numerous, and while some are deliberate and other are almost deliberate, there are some, such as rubbing under noses for puzzlement or claspings our arms to protect ourselves, that are mostly unconscious. Kinesics is a study of the mixture of all body movements, from the very deliberate to the completely unconscious, from those that apply only in one culture to those that cut across cultural barriers.

Questions 1–8

The reading passage has eight paragraphs, A–H.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i How the study of body language has changed
- ii A possible business application of body language
- iii Using body language as a tool to deceive others
- iv Communicating a wide range of messages
- v A branch of an older academic field
- vi The need for skill when interpreting body language
- vii Recognizing a positive attitude without realizing it
- viii How power is linked with certain family roles
- ix A form of body language that can be misrepresented
- x Imitating the chief person in a group
- xi Ignoring signals from other people

Questions 9–13

Do the following statements agree with the claims of the writer in the reading passage? Write **YES** if the statement reflects the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 9 Little work was done in the field of kinesics for the first few years after the subject had been identified.
- 10 Family leaders consistently show their dominance through speech and body language.
- 11 The use of kinesics in connection with television advertising has increased sales of products.
- 12 Touching may be regarded as an unwelcome gesture.
- 13 Most signals with the same meaning in all cultures are unconscious.

Question 14

Choose the correct letter, **A**, **B**, **C** or **D**.

- 14 The writer's intention is
 - A to present recent findings in kinesics to specialists.
 - B to introduce kinesics to general readers.
 - C to examine weaknesses in kinesics.
 - D to identify the scientific basis of kinesics.

Passage 19

Learning to be Bilingual

Dr Chisato Danjo, Lecturer in Japanese and Linguistics, examines the bilingual family home

Few people would consider mastering more than one language a bad idea. Consequently, parents who speak different languages from each other are generally keen for their children to learn both. They understand that the family setting they create is central to this, and seek ways to ensure their children thrive bilingually. One of the best-known approaches is the 'one-parent-one-language' strategy (OPOL). Each parent uses their native language when communicating with their children, so the children learn both simultaneously. OPOL emphasises consistency – each parent sticking to one language – as key to its approach. But this creates the myth that mixing languages should always be avoided. My recent study, part of a new wave of multilingualism studies, would suggest that this received wisdom is just that: a myth. My research looked at Japanese-British families living in the UK with pre- and early school-age children who were following a more-or-less strict OPOL language policy. I was particularly interested in examining the impact of OPOL in the family home – how does this unique language environment affect the way children use languages?

Most of the Japanese mothers who participated were fluent in Japanese and English, while the fathers possessed merely an elementary grasp of Japanese. This made English the primary language of communication between the parents and outside the home. For this reason, the mothers were careful to carve out additional space for more sustained Japanese language learning with their children. In other words, this dedicated space for communicating in Japanese was time the children spent exclusively with their mother. This seemed to create a connection between 'Japanese language' and 'motherhood' in the children's perception. This link became apparent in the way the children used Japanese as a means of emotional bonding with their mother. In addition, they adopted a broader behavioural repertoire than that which is usually associated with a language. For example, switching to Japanese could sometimes serve as a method to appease Mum when she seemed unhappy. At other times, refusing to communicate in Japanese was a useful means of defiance, even when the dispute was not related to language.

The OPOL approach emphasises the need for parents to monitor children's language closely and correct them if they mix the two languages. In practice, many parents speaking the minority language are bilingual themselves – so they understand what their children are saying even when they do mix the two. In addition, parents feel it's problematic to keep correcting children when they mix languages. This is particularly the case when children show annoyance at being corrected. The parents believe it is a lot more important to have a meaningful conversation with their children.

But what if a child uses language that cannot readily be categorised as either one language or the other? An example from my study involved the use of English words adapted to accommodate Japanese pronunciation. One of these borrowed words, 'ice cream', is usually pronounced 'aisukurimu'. The distinction between singular and plural does not exist in Japanese nouns in the English language sense, so whether using singular or plural, even in a borrowed word, 'aisukurimu' is the form normally used. But one of my child participants showed his mother a drawing of two cones of ice cream and described them as 'aisukurimuzu', with a Japanese pronunciation but in English plural form. The child had created something in between.

Another example from my study focused on the interaction between Japanese-English bilingual siblings. In one case, a six-year-old girl was trying to convince her four-year-old brother to let her play with his toys. Following firm rejections by her brother, the girl drew on her communicative repertoire to convince him. To start with, she shifted from an authoritative demand to a softer and humbler appeal. She rephrased the question by using various polite forms. Then, her voice became more nasal, suggesting she was about to burst into tears. Even more interestingly, while the negotiation had begun in English, in the middle she shifted to Japanese. Although this may give the impression of language mixing, a considerably more complex process was taking place. The shift was accompanied by the incorporation of Japanese cultural elements, such as honorific titles that emphasise emotional attachment, a relationship of dependence between sister and brother, and an assumed obligation to care on the part of the brother. She succeeded.

These examples show how creatively and strategically human beings use language in their daily communication. Whether bilingual or not, we all constantly select from our repertoire anything that will best serve our purpose. For instance, imagine you want to ask a neighbour a favour. You would use polite language in a friendly voice. But what about your facial expression? Your body language? It is likely you would make appropriate choices. For bilinguals, shifting between languages is all part of their repertoire.

Our language repertoires are shaped by meaning, based on knowledge gathered throughout our lives, and the ways we use language also shape its meaning. The use of OPOL in the family brings specific meaning to language used at home, and children make full use of emergent meaning in their own interactions.

The popularity of OPOL amongst parents rests on the simplicity of its message, which is that it should be applied consistently. But when we see a child actively using, adapting and negotiating their repertoire, it casts doubt on the belief that it's bad for children to mix languages. What they could actually be doing is demonstrating high-level flexibility and interpersonal skills.

Being bilingual isn't simply about an ability to speak two languages. Rigidly policing consistency in the OPOL approach could actually inhibit bilingual children's linguistic ability and creativity. And in the same way, it could also limit their parents' ability to reveal their own bilingual skills, using their own repertoires.

Questions 1–6

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 What does the writer say about families in which the parents speak different languages?
 - A They constantly research ways to help their children learn both languages.
 - B They realise the importance of their role in their children's language learning.
 - C They prioritise their children's language learning over their own.
 - D They find the OPOL approach the easiest to adopt.
- 2 The writer uses the term 'received wisdom' in the first paragraph to illustrate that
 - A the success of OPOL depends on the languages spoken by the parents.
 - B OPOL fails if the parents are not consistent in its use.
 - C one generally accepted principle of OPOL is incorrect.
 - D there has been a lack of research into OPOL.
- 3 The mothers who took part in the writer's research
 - A compensated for the fact that their children were exposed to more English than Japanese.
 - B took their children's language learning more seriously than their partners did.
 - C used language learning to increase the time they spent with their children.
 - D were reluctant for their partners to speak Japanese to their children.
- 4 The writer says that the children in her study
 - A preferred to use Japanese exclusively for communicating with their mothers.
 - B understood their mothers better as their knowledge of Japanese increased.
 - C argued about things that would be unimportant for monolingual children.
 - D used language in a way that is uncommon.
- 5 What does the writer suggest about correcting children who are learning to be bilingual?
 - A It is usually avoided if it causes bad feeling.
 - B Parents tend to prioritise it over maintaining a conversation.
 - C It is unnecessary with the OPOL method of learning.
 - D Parents are less likely to do it if they speak both languages.
- 6 The writer gives the 'ice cream' example to show that
 - A bilingual children find the use of borrowed words less confusing than other children.
 - B bilingual children like to test their parents with words they know to be wrong.
 - C knowledge of two languages leads bilingual children to invent new word forms.
 - D it may be difficult to distinguish which language a bilingual child is using.

Questions 7–10

Complete the summary using the list of words, A–I, below.

A BILINGUAL CHILD NEGOTIATES WITH HER BROTHER

Faced with her little brother's (7) _____ to let her play with his toys, a young bilingual child used all of her communication (8) _____ to change his mind. First, she expressed her (9) _____ in gentler, more polite language, and pretended to be about to cry. Having switched from English to Japanese, she exploited aspects of Japanese culture, such as reminding him of his (10) _____ to her as her brother. She succeeded in her attempts to convince him.

WORD LIST

- A request
- B attempts
- C strategies
- D misunderstandings
- E responsibility
- F agreement
- G judgement
- H refusal
- I behaviour

Questions 11–14

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write **YES** if the statement agrees with the claims of the writer, **NO** if it contradicts them, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- (11) Even people who speak only one language know how to adapt their way of communicating according to the situation.
- (12) People who grow up in OPOL families will go on to use this approach with their own children.
- (13) The reason the OPOL approach is favoured by many parents is that it encourages them to switch languages when necessary.
- (14) Bilingual children who mix languages eventually develop a preference for one of the languages.

Passage 20

Left, or Right?

An overview of some research into lateralisation: the dominance of one side of the body over the other

A Creatures across the animal kingdom have a preference for one foot, eye or even antenna. The cause of this trait, called lateralisation, is fairly simple: one side of the brain, which generally controls the opposite side of the body, is more dominant than the other when processing certain tasks. This does, on some occasions, let the animal down: such as when a toad fails to escape from a snake approaching from the right, just because its right eye is worse at spotting danger than its left. So why would animals evolve a characteristic that seems to endanger them?

B For many years it was assumed that lateralisation was a uniquely human trait, but this notion rapidly fell apart as researchers started uncovering evidence of lateralisation in all sorts of animals. For example, in the 1970s, Lesley Rogers, now at the University of New England in Australia, was studying memory and learning in chicks. She had been injecting a chemical into chicks' brains to stop them learning how to spot grains of food among distracting pebbles, and was surprised to observe that the chemical only worked when applied to the left hemisphere of the brain. That strongly suggested that the right side of the chick's brain played little or no role in the learning of such behaviours. Similar evidence appeared in songbirds and rats around the same time, and since then, researchers have built up an impressive catalogue of animal lateralisation.

C In some animals, lateralisation is simply a preference for a single paw or foot, while in others it appears in more general patterns of behaviour. The left side of most vertebrate brains, for example, seems to process and control feeding. Since the left hemisphere processes input from the right side of the body, that means animals as diverse as fish, toads and birds are more likely to attack prey or food items viewed with their right eye. Even humpback whales prefer to use the right side of their jaws to scrape sand eels from the ocean floor.

D Genetics plays a part in determining lateralisation, but environmental factors have an impact too. Rogers found that a chick's lateralisation depends on whether it is exposed to light before hatching from its egg – if it is kept in the dark during this period, neither hemisphere becomes dominant. In 2004, Rogers used this observation to test the advantages of brain bias in chicks faced with the challenge of multitasking. She hatched chicks with either strong or weak lateralisation, then presented the two groups with food hidden among small pebbles and the threatening shape of a fake predator flying overhead.

As predicted, the birds incubated in the light looked for food mainly with their right eye, while using the other to check out the predator. The weakly-lateralised chicks, meanwhile, had difficulty performing these two activities simultaneously.

E Similar results probably hold true for many other animals. In 2006, Angelo Bisazza at the University of Padua set out to observe the differences in feeding behaviour between strongly-lateralised and weakly-lateralised fish. He found that strongly-lateralised individuals were able to feed twice as fast as weakly-lateralised ones when there was a threat of a predator looming above them. Assigning different jobs to different brain halves may be especially advantageous for animals such as birds or fish, whose eyes are placed on the sides of their heads. This enables them to process input from each side separately, with different tasks in mind.

F And what of those animals who favour a specific side for almost all tasks? In 2009, Maria Magat and Culum Brown at Macquarie University in Australia wanted to see if there was general cognitive advantage in lateralisation. To investigate, they turned to parrots, which can be either strongly right- or left-footed, or ambidextrous (without dominance). The parrots were given the intellectually demanding task of pulling a snack on a string up to their beaks, using a co-ordinated combination of claws and beak. The results showed that the parrots with the strongest foot preferences worked out the puzzle far more quickly than their ambidextrous peers.

G A further puzzle is why are there always a few exceptions, like left-handed humans, who are wired differently from the majority of the population? Giorgio Vallortigara and Stefano Ghirlanda of Stockholm University seem to have found the answer via mathematical models. These have shown that a group of fish is likely to survive a shark attack with the fewest casualties if the majority turn together in one direction while a very small proportion of the group escape in the direction that the predator is not expecting.

H This imbalance of lateralisation within populations may also have advantages for individuals. Whereas most co-operative interactions require participants to react similarly, there are some situations – such as aggressive interactions – where it can benefit an individual to launch an attack from an unexpected quarter. Perhaps this can partly explain the existence of left-handers in human societies. It has been suggested that when it comes to hand-to-hand fighting, left-handers may have the advantage over the right-handed majority. Where survival depends on the element of surprise, it may indeed pay to be different.

Questions 1–4

Complete each sentence with the correct ending, **A–F**, below. Write the correct letter in each box.

- 1 In the 1970s, Lesley Rogers discovered that
- 2 Angelo Bisazza's experiments revealed that
- 3 Magat and Brown's studies show that
- 4 Vallortigara and Ghirlanda's research findings suggest that

- 1 _____
- 2 _____
- 3 _____
- 4 _____

LIST OF ENDINGS

- A lateralisation is more common in some species than in others
- B it benefits a population if some members have a different lateralisation than the majority
- C lateralisation helps animals do two things at the same time
- D lateralisation is not confined to human beings
- E the greater an animal's lateralisation, the better it is at problem-solving
- F strong lateralisation may sometimes put groups of animals in danger

Questions 5–9

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

LESLEY ROGERS' 2004 EXPERIMENT

Lateralisation is determined by both genetic and (5) _____ influences. Rogers found that chicks whose eggs are given (6) _____ during the incubation period tend to have a stronger lateralisation. Her 2004 experiment set out to prove that these chicks were better at (7) _____ than weakly lateralised chicks. As expected, the strongly lateralised birds in the experiment were more able to locate (8) _____ using their right eye, while using their left eye to monitor an imitation (9) _____ located above them.

Questions 10–14

Reading Passage 3 has eight paragraphs, A–H. Which paragraph contains the following information?

NB You may use any letter more than once.

Q	Statement	
10	description of a study which supports another scientist's findings	
11	the suggestion that a person could gain from having an opposing lateralisation to most of the population	

12	reference to the large amount of knowledge of animal lateralisation that has accumulated	
13	research findings that were among the first to contradict a previous belief	
14	a suggestion that lateralisation would seem to disadvantage animals	

Passage 21

Let's All Be Friends

How the science of social networks could help governments change behaviour

A From healthcare to climate change, governments today face a range of problems in which they must persuade people to change their behaviour. But maybe instead of relying on their powers of persuasion, governments should consider using the science of social networks as a tool to do this. True, many claims for the power of these are based on the hype surrounding major social networking websites. But the basic idea is simple: people can join together in groups with particular patterns of social links, and these patterns then have important effects on the way they behave.

B The shape of these networks has surprising effects. Take an unlikely example: stage musicals. Brian Uzzi is a sociologist at Northwestern University in Chicago. He is also a big fan of musicals. Musicals have been big business for decades, but investors have to guess which shows will be a hit. *Bye Bye Birdie*, a profitable 1960 production, ran for 607 nights. *Bring Back Birdie*, its 1981 sequel, closed after just four. Intrigued, Uzzi used network science to find out why.

C He investigated 321 American musicals that were launched between 1945 and 1989, paying particular attention to whether the team of director, producers, choreographers and writers had worked together before. After crunching the statistics, he discovered something remarkable. Teams who had never worked together, perhaps unsurprisingly, fared poorly: their 'weak' networks meant a lack of creative vision, and lots of failures. And at the other extreme, teams that had worked together successfully also tended to produce flops. Sometimes, lacking outside creative input, the team just rehashed the same ideas that worked the last time; sometimes, lacking newcomers, they 'developed' their vision in unwise ways. Either way, lightning rarely struck twice. But, in between, Uzzi found a point of balance. Groups with exactly the right mix of new and old participants reliably produced hits. This variation in the 'density' of the ties allowed easy communication and fostered greater creativity; new ideas from the outsiders meshed with the experience of the insiders. It made no difference whether a musical was about cats or roller-skating, or who starred in it – what mattered was the nature of the network binding its team together.

D These insights have some fascinating consequences for policymaking – particularly in the area of health. According to network science, teenagers are more likely to adopt unhealthy eating habits if their friends' friends consume junk food, even if they don't personally know those individuals. Conversely, they may be encouraged to eat more healthily if their immediate friends do. This opens up opportunities for governments to reduce spending. For example, suppose a health authority has £100 and wants to improve the diets of ten people. If it invests £10 in each individual separately, it might only

influence one or two. However, by investing the entire sum in persuading one or two key individuals who are central to a social network, the authority could trigger a cascade effect, influencing the entire group for a fraction of the cost. The same principle could be applied to other areas, such as encouraging physical exercise or quitting smoking.

E However, the influence of networks is not always positive. The same mechanisms that can spread healthy behaviours can also amplify harmful ones. Financial panic, political extremism, and the rapid proliferation of unhealthy fads can all be accelerated by tightly-knit social networks. This presents a significant challenge for policymakers: how to harness the power of social connectivity for good while mitigating its potential for harm. Understanding the precise structure of these networks – identifying the key influencers and the strength of the connections between people – is therefore critical.

Questions 27–31

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

BRIAN UZZI'S MUSICALS RESEARCH

Brian Uzzi's research into Broadway musicals revealed that the success of a production was not determined by its subject matter or cast, but by the (27) _____ of its creative team. He found that groups composed entirely of newcomers were often unsuccessful due to a lack of (28) _____. Conversely, teams that had a history of working together also frequently produced (29) _____ because they tended to reuse old ideas. The most successful teams had a (30) _____ of experienced and new members. This balance in network (31) _____ promoted effective communication and creativity.

Questions 32–36

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write **YES** if the statement agrees with the claims of the writer, **NO** if it contradicts them, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- (32) Governments are currently very effective at using social network science.
- (33) Bring Back Birdie was financially successful.
- (34) Uzzi's study analysed musicals from the first half of the 20th century.
- (35) The eating habits of a teenager can be affected by someone they have never met.
- (36) Investing money in central figures is guaranteed to be more cost-effective.

Questions 37–40

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

THE IMPLICATIONS OF NETWORK SCIENCE

- **POSITIVE USE FOR GOVERNMENTS:**

can be used to improve public health initiatives, e.g., diet and 37 _____.

- Targeting key people in a network can create a 38 _____ effect, influencing many others.

- **RISKS AND CHALLENGES:**

networks can also spread 39 _____ behaviours, like financial panic or extreme views.

- To use networks effectively, policymakers must understand their 40 _____.

Passage 22

Life on Mars?

Terraforming may sound like something out of science fiction, but some believe it is possible to turn that fiction into fact

As plans are slowly being drawn up for the first manned mission to Mars, many space travel sceptics are asking one vital question: why go there? Mars is a barren, desolate planet, and with its thin atmosphere and bitterly cold climate, it would appear to be completely unsuitable for human life. Above all, it is a very distant place, and getting there would be an enormous challenge. However, the planet might just hold the key to long-term human survival. With the Earth's population currently at more than seven billion and climbing, we may eventually be forced to look elsewhere in the solar system for somewhere to live. It is just possible that, contrary to photographic evidence, Mars may be more promising than it appears.

Today, Mars is a viciously cold, dry place. However, it does have some things in common with our own planet. For example, it has a daily rotation rate of 24 hours 37 minutes, compared with 23 hours 56 minutes on Earth. It also has an axial tilt of 24 degrees, which is just half a degree more than Earth's, and a gravitational pull one third of Earth's. Furthermore, it holds many of the elements that are required to support life, including carbon and oxygen (in the form of carbon dioxide), nitrogen, and frozen water at its polar ice caps. In fact, if you were to travel back in time several billions years, you would notice some remarkable parallels between the atmosphere on Earth then and Mars today. Back then, Earth was also a lifeless planet; until photosynthetic bacteria developed and began to produce enough oxygen to allow for the development of animal and plant life, our atmosphere also consisted entirely of carbon dioxide and nitrogen.

It comes as no surprise to learn, therefore, that some scientists believe the same process which turned Earth's atmosphere from mostly carbon dioxide into breathable air could be repeated on Mars, but by using technology rather than by letting nature and evolution take its natural course. Terraforming, as this process is known, would initially create a greenhouse effect that would heat the planet, which in turn would create other conditions necessary to provide a suitable living environment for plants and animals. However, it would be a highly challenging undertaking, and the process of terraforming the entire planet into an Earth-like habitat could still take many thousands of years.

Three terraforming methods have been suggested, with the first already under development, albeit for a different purpose. At present, the American space agency NASA is working on a system that will use large mirrors to capture the sun's radiation. This radiation will be used to propel spacecraft through space, removing the need for heavy and expensive rocket fuel. With a few changes, it might be possible to use similar mirrors to

reflect the sun's radiation and heat the surface of Mars. Aimed at the planet from a distance of two hundred thousand miles, these enormous mirrors would raise the surface temperature by a few degrees. If they were concentrated on the polar ice caps, they would provide enough heat to melt the polar ice caps and release the carbon dioxide that is believed to be trapped there. Gradually, as the temperature rose, greenhouse gases would be released, and this would create a form of Martian global warming, the first stage in making the planet sustainable for life.

The second method would be to set up greenhouse gas 'factories' in order to raise the temperature of the planet. It is generally accepted that greenhouse gases produced by heavy industry are raising the Earth's temperature. Therefore, by building hundreds of greenhouse-gas emitting factories on Mars, a similar effect could be achieved. Carbon dioxide, methane and other greenhouse gases would be pumped into the Martian atmosphere. The same factories would then produce oxygen by mimicking the natural process of plant photosynthesis: they would inhale the carbon dioxide they produce, and then emit oxygen. The process could be accelerated by 'sowing the planet's surface with photosynthetic bacteria, which would increase the rate at which oxygen is produced. Eventually, there would be enough oxygen on the planet for humans to breathe using only special apparatus similar to that used by mountain climbers.

The third, and by far the most extreme, method has been proposed by space scientists Robert Zubrin and Christopher McKay. They believe that it would be possible to produce greenhouse gases and water by firing large, ammonia-bearing asteroids at the planet. Each asteroid would weigh about ten billion tons, and would be powered by huge rocket engines which would move it towards Mars at over 10,000 miles per hour. At this speed, it would take each asteroid about ten years to reach its destination. The energy produced by one asteroid slamming into Mars' surface, say Zubrin and McKay, would raise the temperature of the planet by three degrees Celsius and melt about one thousand billion tons of ice at the polar caps. They believe it would take many of these asteroids, and at least fifty years, in order to create a temperate climate and enough water to cover a quarter of the planet's surface.

Terraforming Mars, if it is ever attempted, will be neither cheap nor easy. And it certainly won't be quick: although optimists like Zubrin and McKay say it could be achieved in five or six decades, the reality is that terraforming is more likely to take hundreds or even thousands of years. Furthermore, it will stretch human ingenuity to its limits, and will require levels of will and commitment that have rarely been seen before. The challenge of developing a habitable environment and bringing life to the cold, dry world of Mars is fraught with challenges, but it might just be one that saves the human race.

Questions 1–5

Do the following statements agree with the views of the writer in the Reading Passage? Write **YES** if the statement agrees with the writer's view, **NO** if the statement contradicts the writer's view, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 1 Pictures of Mars suggest it might make a good place for people to settle.
- 2 Modern Mars and ancient Earth looked remarkably similar.
- 3 One method of terraforming could involve adapting technology that is already under development.
- 4 Greenhouse gas factories would provide enough oxygen for people to breathe without special equipment.
- 5 Terraforming Mars would be an extreme test of human skill and intelligence.

Questions 6–9

Choose the correct letter, **A**, **B**, **C** or **D**.

- 6 Which one of these factors suggests that Mars might be a good place for people to settle?
 - A It is not too far from Earth.
 - B It has no other life forms living there.
 - C It has a cool, dry climate.
 - D It has some similarities with Earth.
- 7 The first step in terraforming Mars would be to
 - A make the planet warmer.
 - B create a breathable atmosphere.
 - C find a suitable source of water.
 - D create a habitat for living organisms.
- 8 Special factories on Mars could be used to
 - A control the level of greenhouse gases.
 - B absorb excess levels of carbon dioxide.
 - C produce oxygen in a manner similar to plants.
 - D help grow essential bacteria.
- 9 What is the writer's main purpose in the passage?
 - A To explain why we need to terraform Mars.
 - B To illustrate the three processes required to terraform a planet like Mars.
 - C To consider how and why Mars might be terraformed.
 - D To demonstrate how straightforward it would be to terraform a planet.

Questions 10–14

Complete the summary using the list of words, A–I, below.

WORD LIST

- A cover
- B create
- C hit
- D increase
- E land
- F drive
- G power
- H rise
- I shoot

ZUBRIN AND MCKAY'S ASTEROID METHOD

One method of terraforming Mars would be to ⑩ _____ asteroids at the planet. Rockets attached to an enormous asteroid would propel it towards Mars, taking ten years to ⑪ _____ the enormous distances required. The asteroid would ⑫ _____ the planet with incredible force and ⑬ _____ enough energy to ⑭ _____ the planet's temperature. The result would be a temperate climate and lots of water from melting ice caps.

Passage 23

Life on Mars?

A century of speculation, disappointment, and renewed hope about life on the Red Planet

A When scientists began observing Mars with their first primitive telescopes, they assumed from the start that the planet was peopled. The 18th-century astronomer William Herschel, noting the seasons and atmosphere of Mars, declared that 'its inhabitants enjoy a situation similar to our own'. A century later, astronomers asserted that they could detect canals running across the Martian landscape. These canals, argued US scientist Percival Lowell, had been built by technologically advanced Martians to transport water from the poles to the equator of their drying world. 'Martian folk are possessed of inventions of which we have not dreamed,' he argued.

B The vision of an advanced culture trapped on a drying world was taken up by the British writer H.G. Wells in 1898 in his masterpiece *The War of the Worlds*, in which Earth is subjected to a pitiless invasion from a race of hideous Martians armed with poison gas and heat rays. Martians became synonymous with grotesque, threatening aliens, at least as far as the public was concerned. Scientists were more restrained, but still supported the idea that the planet might support relatively complex life forms.

C Then, in the late 1960s, rocket science changed our image of Mars forever. The US Mariner probes swept past Mars and sent back pictures, not of a world covered in vegetation, as many scientists still expected, but of vistas of cratered, arid deserts. Finally, in 1975, two huge US probes settled on the Martian surface and began a series of experiments aimed at pinpointing, once and for all, the existence of air-breathing organisms. None were found, for the Viking probes landed in areas as dead as the Sahara Desert. All that was sent back were images of a bleak, boulder-strewn terrain. Mars was declared officially dead – for a while.

D Since that time, two critical pieces of evidence have been found on Earth which support the idea that Mars has life. The first is known as ALH84001, a meteorite that was blasted into space when a large body struck Mars 15 million years ago and which spiralled across the solar system to crash into the Antarctic. Analysis carried out by NASA scientist Imre Friedmann showed that the meteorite contained elements identical to those found on Mars by the Viking probes, and revealed chains of crystals similar to those created by earthbound bacteria. Friedmann believes that these could only have been created by living organisms. Others, however, argue that the crystals could have been laid down by simple chemical processes...

E In the past few years, astronomers have become convinced that the Martian surface, although hostile and arid today, was once lush and welcoming. Images sent back by recent probes such as the Mars Global Surveyor have revealed dried-up riverbeds and old estuaries. Rivers poured into ancient seas, it appears, and rain swept the sodden terrain. Then catastrophe struck. Mars lost its magnetic field several billion years ago, explains British astronomer Paul Murdin. 'Earth retained one, and it protects our atmosphere. Without that protection, Mars's gases and water boiled off.' That is why they are so different today.

F The discovery of signs of water on Mars has changed everything. Some researchers believe that water from underground reservoirs still gushes through Martian valleys as occasional flash floods bubble to the surface. In the 1990s, the Mars Odyssey probe began to map the chemical make-up of the Mars soil, and discovered that much of the ground in the southern hemisphere of Mars is mixed with large quantities of ice. And where there is ice, there is likely to be liquid water, probably underneath...

G Current projects are hoping to clarify the question of the presence of water once and for all. These are aimed at conducting experiments to detect the presence of a substance called carbon 13 in soil and rock samples on Mars; this would be, according to some researchers, an unequivocal indicator of the presence of living organisms, past or present.

H But would the presence of a few Martian organisms alter our world view in any significant way? Professor Juan Pérez Mercader of Madrid's Centre for Astrobiology in Spain believes that it would. 'For a start, we will be able to find out if Martian organisms have DNA, the golden molecule that controls the replication of earthly creatures.' If there is no DNA, that would show that Martian biology is quite unlike Earth's, and that on two planets, side by side in the same solar system, life evolved independently. If DNA is located, it would mean that life on Mars and on Earth may spring from the same source...

Questions 27–32

Reading Passage 3 has eight paragraphs, A–H.

Which paragraph contains the following information?

Write the correct letter, A–H, in boxes 27–32.

Q	Statement	
27	evidence that a rock found on Earth originated on Mars	
28	a plan to analyse material on Mars	
29	the reason Mars is no longer similar to Earth	

30	an observation suggesting that there is currently water on Mars	
31	a claim that construction had taken place on Mars	
32	a possibility that humans and Martians could share the same origins	

Questions 33–36

Choose the correct letter, **A**, **B**, **C** or **D**.

- 33** The 18th-century astronomer William Herschel believed that
- A** the Martians were physically similar to humans.
 - B** the Martian environment resembled that of Earth.
 - C** members of the Martian race were extremely primitive.
 - D** the climatic conditions on Mars were very dry.
- 34** At the end of the 19th century, the public's ideas about Martians changed because of
- A** scientific discussion.
 - B** media reports.
 - C** a work of literature.
 - D** a magazine illustration.
- 35** What does the writer say happened in the 1960s?
- A** The appearance of some regions of Mars changed.
 - B** Evidence of organisms was found on Mars.
 - C** The Martian atmosphere was tested scientifically.
 - D** Earlier beliefs about Mars were disproven.
- 36** What do scientists disagree about regarding the ALH84001 meteorite?
- A** Whether it really originated on Mars.
 - B** Whether its crystal formations were created by living organisms.
 - C** How long ago it struck the Antarctic.
 - D** Whether it should be studied further.

Questions 37–40

Choose **YES** if the statement agrees with the claims of the writer,

choose **NO** if the statement contradicts the claims of the writer,

or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 37** Lowell believed that Martian technology was more developed than human technology.
- 38** Information gained from Viking probes remains unchallenged.
- 39** There is evidence that there has been significant climate change on Mars.
- 40** Spanish scientists plan to launch space probes if DNA is found on Mars.

Passage 24

Literacy and Rural Communities in Australia

How low literacy skills present different, often hidden, social costs in the Australian countryside

A Little is known about the incidence of literacy in rural areas of Australia, but anecdotal evidence suggests that lack of literacy skills may present particular problems and have different social costs in rural areas compared to urban areas. Social costs, of course, differ at different life stages. For young people, inadequate literacy skills can affect their transition from school to work; for older people lack of literacy skills can reduce their employment options as well as reducing their effectiveness or ability in the workplace. Assuming life-stage differences remain much the same in both city and country areas, it is still possible to make some general comments about differences between urban and rural areas in Australia. It is important, however, to remind the reader of the considerable diversity of rural areas.

B It is not unusual for males in rural areas to have left school early, or to have left with inadequate literacy skills, and then to move onto farms. This can be due to a number of causes, among which are family pressure or desire to follow the father into the family business, learning difficulties or unhappiness at school, or even peer-group pressure from other young males who have decided to work on their family farms. There are numerous areas where literacy skills are required in farming, such as reading directions for (often dangerous) sprays, the mixing of animal feed, the mixing of fuel for machinery, care of machinery, care of animals, and understanding letters from government authorities. In addition, there are the numerous tasks associated with running a business. Although the responsibility for the business side of the farm may be carried by a female partner with more adequate literacy skills, there remain a number of areas where inadequate literacy skills in males can lead to difficult or dangerous situations.

C It is difficult to estimate the social costs of low literacy levels for women in non-metropolitan areas. Some research in this area has shown that women leave school early and travel to the city to find work; women's employment options are limited generally, but for women with low literacy skills there is often no alternative to dependence — on family, partner or government benefits. Also, according to a 1989 survey, about ten per cent of women living in rural areas were born outside Australia. We can assume that a fair proportion of these have literacy needs.

D It is suggested that in times of hardship in farming areas, when there is the threat or actuality of banks foreclosing on mortgages, inadequate literacy skills add to the normal stresses. Farmers and others with inadequate literacy skills affected by a rural down-turn

have fewer alternatives for employment, less flexibility, perhaps fewer support mechanisms, and therefore more strains both personally and with a spouse or partner. It is not inconceivable that the additional stress contributes to marriage breakdown or suicide.

E Problems can become acute in regional towns or areas which are based on one or two major industries. A proportion of manual and construction workers and trade assistants in such towns have low levels of literacy, but are able to find employment so long as new projects and development continue. Shift work and relatively high wages make such work relatively attractive. Once the development is completed, however, there are few alternative options.

F In some small communities, there is a tendency for the problem of poor literacy skills to be continued with the next generation, thereby repeating the problem. Economic hardship, the need for children to travel (or be driven) long distances to school, and the low value placed on literacy may mean that helping with farm work is seen as more important than regular attendance at school. More children in rural areas than urban areas tend to leave school without completing a full secondary education, despite increasing school retention rates generally, although there is considerable variation between rural areas.

G The closeness of some rural communities and country towns sometimes means that people are reluctant to admit their literacy problems. While this is generally the case in both city and country locations, cities at least do offer some anonymity for people who make the decision to start a literacy class. In a small country town or rural area, going to a class becomes much more public: tutors in country areas have reported that some students travelled long distances to another town in order to avoid having to reveal their 'problem' in the local area. Obviously, a change in public perception of literacy problems is needed, if people with low literacy skills in rural areas are to avoid the social stigma which is attached to them in the minds of most people.

Questions 1–6

*Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- 1 There is a lack of factual information on literacy levels in rural Australia.
- 2 The rural areas of Australia are very similar to each other.
- 3 Most businesses in rural areas are family-run.
- 4 Women who can read and write have a wide choice of jobs in rural areas.
- 5 In some rural areas, industrial jobs are popular.

- 6 Children in urban areas stay on at school longer than children in rural areas.

Questions 7–11

Reading Passage 3 has seven paragraphs, A–G. Which paragraph contains the following information?

Write the correct letter, A–G, in boxes 7–11.

Q	Statement	
7	how an economic recession may aggravate the problems caused by low levels of literacy	
8	measures which individuals may take to hide the fact that they have poor literacy skills	
9	examples of things that rural workers may need to read	
10	the ways literacy problems can affect people over the course of their working life	
11	a reference to youngsters encouraging each other to leave school early	

Questions 12–14

Choose **THREE** letters, A–G.

Which **THREE** of the following areas are explored by the writer of this article?

OPTIONS

- A the effectiveness of tuition for literacy problems
- B reasons why employment prospects can be limited for workers with literacy problems
- C the effect of literacy problems on family mobility
- D factors that worsen the difficulties caused by literacy problems
- E how families have a low regard for literacy
- F how literacy levels among men compare with those of women
- G the probability of literacy levels rising in the future

Answer: 12 13 14

Passage 25

Living Dunes

The strange, restless life of the desert's giant sand dunes

27 **A** Armies of giant sand dunes are advancing across the world's deserts, engulfing anything that crosses their path. They are tens of metres tall and hundreds of metres long. Fortunately, they aren't going very fast. Even the smallest, speediest dunes only travel about 100 metres over the course of a year, while the bigger ones, which weigh something in the order of 10,000 tonnes, barely move one metre in that time. However, their insidious creep can have serious consequences if there is an oil installation or a railway line in their path.

28 **B** About 47% of the world's land mass, including Antarctica, most of Australia and large areas of Africa, is classified as arid or semi-arid desert. Only around 20% of that is sand-covered, however, and over half of that is classified as 'linear' sand dunes. These form in a long curving wave, as a result of wind blowing strongly from several quarters, flipping them from side to side. Although linear dunes are static, sand blowing off them can cause problems for desert villages, burying crops and buildings.

29 **C** Moving dunes make up just a small percentage of the rest, but they are of the most interest to scientists. They are known as 'barchans': heavy, crescent-shaped sand piles with a ridged crest and two elongated arms, one curving away to either side. 'Barchan dunes only tend to form where you have one-directional winds on the edge of sandy deserts near coastal areas,' says Giles Wiggs, a geomorphologist at Oxford University, who has been studying the formation and movement of sand dunes for more than a decade.

30 **D** But even with strong winds, how can entire barchans move while retaining their form? That question was first answered in the mid-20th century by British explorer Ralph Alger Bagnold, and his answer hinges on the fact that dunes aren't solid, but granular. Bagnold figured out how barchan dunes are able to move grain by grain. Imagine a single grain of sand being blown up the back of a dune by the wind and deposited on the top. More grains follow the same pattern, until the accumulated weight of piled-up sand finally pushes the top down the dune face. The grain tumbles, then stops on the face until subsequent mini-avalanches bury it. Eventually, it reappears at the back of the dune, ready to repeat the process. As this happens to every grain of sand in the dune, the whole thing creeps in the direction of the prevailing wind.

31 **E** The relationship between the wind and barchan dunes is complex. As a dune grows, it modifies the speed and course of the wind, which in turn alters how that dune and its neighbours evolve. 'Interestingly, the dune can regulate its own shape, and maintain it as it

moves,' says Dr Stéphane Douady, a physicist at Ecole Normale Supérieure (ENS) in France. 'Even when two dunes collide, they quickly take on their distinctive shapes again. It's like a living organism.'

32 F Douady and his colleagues have also been studying an even odder phenomenon than moving dunes: some barchans actually sing. Local legends attributed the sounds to dangerous spirits which were trying to trap unwary travellers. Douady is more pragmatic. 'It's a strong booming noise with a low frequency,' he explains, making a noise like a foghorn to demonstrate. 'It can last for a long time — up to several minutes. It's a very loud sound and you don't understand where it's coming from when you first hear it.' There are about 50 dunes distributed across 35 deserts round the world that are known to sing. Douady says the sound is caused by the way sand avalanches down the faces of particular dunes. Rather than tumbling randomly, the sand grains flow in synchrony and set each other vibrating like the membrane on a gigantic loudspeaker. The synchronisation causes the air to move in and out between the grains, creating a powerful sound wave.

33 G What really surprised the scientists, however, was that they were able to take samples of the singing sand back to France and replicate the sound at ENS, proving that it's the sand, not the dune shape, that causes the sound. Their studies show the grains are a uniform shape, well-rounded from years of striking each other, and that the variations in size affect the tone. Crucially, the grains are coated with a special veneer, which Douady calls 'desert glaze', made from a precise combination of minerals from surrounding rocks including iron, aluminium, manganese, silicon and calcium. The team found that after a month or so, the veneer wore off and the grains lost their 'voice'. 'We managed to reproduce the desert glaze and then the grains started to sing again,' says Douady. 'We tried putting the coating onto different grains, but they weren't round enough and it didn't work. But some American colleagues made some artificial grains and managed to make them sing, after covering them in desert glaze.' Douady has now made recordings of dunesong from all over the world which are to be made into a CD.

Questions 27–33

Reading Passage 3 has seven paragraphs, A–G.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Moving rapidly across the desert
- ii Recreating the process in a laboratory
- iii Strange music created by human movement

- iv A potential threat to industry and communications
- v Dunes coming together and re-forming
- vi Needing a specific combination of conditions
- vii A continuous cycle
- viii The commonest type of dune
- ix Old superstitions demystified

Questions 34–36

Choose the correct letter, **A**, **B**, **C** or **D**.

- 34 What are we told about linear dunes?
 - A They are formed by strong winds blowing from one direction.
 - B They can move up to 100 metres in a twelve-month period.
 - C They develop in a recognisable shape.
 - D They make up about 20% of the world's deserts.
- 35 Bagnold discovered that movement in barchans was caused by
 - A the long straight shape of the dunes.
 - B the particular composition of the dunes.
 - C the exceptionally heavy nature of the sand grains.
 - D the unusual strength of the wind in certain seasons.
- 36 Why does Dr Douady compare a barchan dune to a living organism?
 - A It starts small and then increases in size.
 - B It has an effect on its immediate surroundings.
 - C Its relations with desert organisms are quite developed.
 - D Its outline stays the same even after a period of movement.

Questions 37–40

Complete the summary below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

SINGING DUNES

Singing dunes, which belong to the type of dunes known as 37 ————, produce a very loud sound which is transmitted at a low frequency. Researchers have worked out that sand grains fall down the dune and start vibrating against other grains, forming a sound wave.

Research proves that the individual grains have a similar (38) _____, but the differences in dimensions alter the (39) _____ of the 'song'. Each grain is covered with a mixture of different (40) _____ and this is vital to the sound production.

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Passage 26

Memory Champions...or just good at tricks?

What separates the world's top mental athletes from the rest of us — natural gift, or learnable technique?

A The Examinations School at Oxford University is where, last August, 34 contestants gathered at the World Memory Championships. It is an impressive historical venue, known for its grandiose setting of oak-panelled rooms, large Gothic windows, and looming portraits of eminent dukes and earls. It is here that generations of Oxford students have been tested in their final exams on their memory and knowledge, and it is here that these contestants were challenged to demonstrate their exceptional mnemonic abilities. Reciting poems, memorizing digit sequences, and recalling faces from photographs are just a few of the tasks that test the limits of human memory in this competitive setting.

B They are using a technique known as the loci method, reportedly originating in 477 BC with the Greek poet Simonides of Ceos. He was able to memorize the guest list at a dinner by closing his eyes and recalling each individual around the table. The loci method, which was rediscovered and popularized by Simonides, has become an integral part of historical and modern memory training. This method involves associating items to be remembered with specific physical locations, creating a mental map that facilitates recall. After Simonides's discovery, the loci method became widely embraced across ancient Greece, serving as a fundamental technique for orators and scholars. It was so effective that it was incorporated into educational curricula and became a cornerstone of rhetorical training. Aristotle wrote favorably about it, recognizing its value and effectiveness. Later, a number of treatises on the art of memory were published in Rome, further cementing its importance and utility in memory training. This method not only aids in memorization but also enhances the ability to retrieve information by navigating through these mental loci, effectively walking through a structured memory palace.

C One memory champion, Ed Cooke, can memorize the order of ten decks of playing cards in less than one hour or one deck of cards in less than one minute. Psychologists Elizabeth Valentine and John Wilding have teamed up with Eleanor Maguire to study how memory champions like Cooke operate differently from average individuals. Their collaboration aims to explore the neural underpinnings that might explain the exceptional memory skills exhibited by competitors in high-pressure environments such as memory championships. When the researchers analyzed the brain scans from the number memorization, they found that the memory champions were activating some brain regions that were different from those the control subjects were using. These regions are known to be involved in visual memory and spatial navigation.

D To enable himself to memorize quickly, Ed Cooke has memorized a specific human being, verb, and object that he associates with every card in the deck. This detailed and highly personalized system allows him to rapidly encode and decode information during competitions. When Cooke commits a deck to memory, he does it three cards in one go, with every three-card group forming a composite picture of a person doing something to an object. This method not only enhances the memorization process but also makes the recall process faster and more efficient. He then places those images along a specific familiar path in his mind, creating a vivid mental journey that he can easily traverse. When it comes to recall, Cooke takes a mental walk along his route and translates the images into cards, effectively and efficiently converting his vivid mental images back into the specific order of the deck. This innovative approach to memory exercises showcases the extraordinary capabilities of mental athletes and the sophisticated strategies they employ to achieve their remarkable feats.

E The term 'photographic memory' is frequently used to describe people with exceptional recall. Yet, most researchers agree there is no proof that such a memory truly exists. If it is indeed a myth, it highlights how learned strategies are often more effective than natural talent. The belief in 'photographic memory' remains a point of contention among the public and scientists alike. Despite the absence of solid scientific evidence, this myth endures, reflecting how society's fascination with remarkable memory tends to ignore the practical techniques and training that genuinely drive mnemonic success.

F The most renowned of the naturals was the Russian journalist S.V. Shereshevski, who could remember long sequences of numbers learned decades before, as well as poems, strings of meaningless syllables, and nearly anything else he was asked to recall. Shereshevski also experienced synesthesia, a rare sensory condition where the senses are interconnected, which significantly improved his memorization ability. His skills were not limited to these accomplishments; his remarkable memory covered an almost endless range of information, making him a figure of great interest and study. This synesthetic trait not only made his memorization tasks easier but also more vivid and long-lasting. His case has been extensively examined, as it offers deep insights into the potential of human memory when enhanced by neurological variations.

G K. Anders Ericsson, a well-known psychologist, claims that anyone can develop memory abilities comparable to those of memory champions through consistent practice and proper techniques. During two years of experimentation, S.V. increased his ability to recall numbers from seven to over eighty. This study led Ericsson to conclude that there is no such thing as an innately superior memory. When he revisited original case reports of natural memorizers, he discovered that their remarkable abilities were actually based on the use of specific techniques — sometimes unconsciously — combined with extensive

practice. Strategists are usually outstanding at remembering only certain kinds of material, whereas naturals exhibit strong memory across many types of tasks and can retain those memories more easily.

Questions 27–32

Reading Passage 3 has seven paragraphs, A–G. Which paragraph contains the following information? Write the correct letter, A–G.

Q	Statement	
27	a description of how a contestant remembers complex sequences	
28	an example of how one person's memory was improved over time	
29	a description of a memorizing competition	
30	a commonly held belief about memory that many experts say is unproven	
31	a reference to the first recorded use of memorizing through location	
32	a reference to an unusual condition which results in improved memory	

Questions 33–36

Complete the summary below. Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Using visual imagery and spatial navigation to remember numbers are investigated and explained. A (33) _____, an action, and a thing are assigned to consecutive cards in the pack. Combining these images allows Cooke to memorise (34) _____ at a time. The mind pictures are then positioned on a (35) _____. The pack of cards is remembered by going for a (36) _____, while the pictures are converted back into cards.

Questions 37–40

Complete each sentence with the correct ending, A–F, below. Write the correct letter in each box.

37 Valentine, Wilding and Maguire's experiment shows that the memory contestants (37) _____

38 After Simonides discovered the loci method in 477 BC, it was (38) _____

39 Ericsson finds out that natural memorisers are better than strategic memorisers at

39 _____

40 According to K. Anders Ericsson, Shereshevski's skills were

40 _____

LIST OF ENDINGS

- A** used mainly for educational purposes.
- B** activated brain areas associated with visual and spatial memory.
- C** became widespread in ancient Greece for memorizing speeches and texts.
- D** the result of techniques and extensive practice.
- E** retaining a wide variety of tasks.
- F** benefited from additional cognitive abilities.

Passage 27

Metropolis

A classic science-fiction film — why Fritz Lang's 1927 vision of a mechanised future still speaks to the modern global economy

When the German director Fritz Lang visited the United States in 1924, his first glimpse of the country was a night-time view of the New York skyline from the deck of an ocean liner. This, he later recalled, was the direct inspiration for what is still probably the most innovative and influential science-fiction film ever made — Metropolis. Metropolis is a bleak vision of the 21st century that is at once both chilling and exhilarating. This spectacular city of the future is a technological marvel of high-rise buildings connected by elevated railways and airships. It's also a world of extreme inequality and social division. The workers live below ground and exist solely to operate the city's machines in an endless routine of mind-numbing 10-hour shifts, while the city's elite lead lives of luxury high above. Presiding over them all is the Master of Metropolis, John Fredersen, whose sole satisfaction seems to lie in the exercise of power.

Lang's graphic depiction of the future is conceived in almost totally abstract terms. The function of the individual machines is never defined. Instead this mass of dials, levers and gauges symbolically stands for all machines and all industry, with the workers as slave-like extensions of the equipment they have to operate. Lang emphasises this idea in the famous shift-change sequence at the start of the movie, when the workers walk in zombielike geometric ranks, all dressed in the same dark overalls and all exhibiting the same bowed head and dead-eyed stare. An extraordinary fantasy sequence sees one machine transformed into a huge open-jawed statue which then literally swallows them up.

On one level, the machines and the exploited workers simply provide the wealth and services which allow the elite to live their lives of leisure, but on a more profound level the purpose of all this crazed industry is to serve itself. Power, control, and the continuance of the system from one 10-hour shift to the next, are all that counts. The city consumes people and their labour and, in the process, becomes a perverse parody of a living being.

It is enlightening, I think, to relate the film to the modern global economy in which multinational corporations now routinely close their factories in one continent so that they can take advantage of cheap labour in another. Like the industry in Metropolis, these corporations' goals of increased efficiency and profits have little to do with the welfare of the majority of their employees or that of the population at large. Instead their aims are to sustain the momentum of their own growth and to increase the monetary rewards to a tiny elite — their executives and shareholders. Fredersen himself is the essence of the big company boss: most modern industrialists would probably feel perfectly at home in his huge skyscraper office with its panoramic view of the city below. And it is important that

there is never any mention of government in Metropolis — the whole concept is by implication obsolete. The only people who have power are the supreme industrialist, Fredersen, and his magician/scientist henchman Rotwang.

So far so good: when the images are allowed to speak for themselves, the film is impeccable both in its symbolism and in its cynicism. The problem with Metropolis is its sentimental storyline, which sees Freder, Fredersen's son, falling in love with the visionary Maria. Maria leads an underground pseudo-religious movement and preaches that the workers should not rebel but should await the arrival of a 'Mediator' between the 'Head' (capital) and the 'Hands' (labour). The mediator is the 'Heart' — love; as embodied, finally, by Freder's love of Maria and his father's love of him.

Lang wrote the screenplay in collaboration with his then wife, Thea von Harbou. In 1933 he fled Germany (and continued a very successful career in Hollywood). She stayed in Germany and continued to make films. There is a constant tension within the film between the too-tidy platitudes of von Harbou's script and the uncompromisingly caustic vigour of Lang's imagery.

All the same, Metropolis is probably more accurate now as a representation of industrial social relations than it has been at any time since its original release. And Fredersen is certainly still the most potent movie symbol of the handful of elusive corporate figureheads who increasingly treat the world as a Metropolis-like global village.

Questions 1–4

Choose the correct letter, A, B, C or D.

① In the writer's view, the fact that 'the function of the individual machines is never defined' (paragraph 2)

- A emphasises the fantasy element of the film.
- B makes the film too complicated.
- C strengthens the main idea of the film.
- D weakens the plot of the film.

② According to the writer, in what is the industry in Metropolis like the multinational corporations of the present?

- A both have gone totally out of control.
- B both exploit science and new technologies.
- C one person governs both systems.
- D their main concern is the continuation of the system.

- 3 The writer's main purpose in writing this article is
- A to connect the film to present-day situations.
 - B to criticise the behaviour of film-makers during the 1930s.
 - C to discuss recent critical views of the film.
 - D to recommend the film Metropolis to viewers.
- 4 Which statement best summarises the writer's attitude to the film Metropolis?
- A the film is a perfect work of art.
 - B the film is too cynical.
 - C it is a great film which could have been even better.
 - D it is a weak film with some good parts.

Questions 5–7

*Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- 5 The character of Fredersen was based on an industrialist Lang met in America in 1924.
- 6 The imagery of Metropolis is superior to its storyline.
- 7 Thea von Harbou and Fritz Lang divorced because of von Harbou's decision to remain in Germany.

Questions 8–13

*Complete the summary using the list of words, **A–K**, below.*

WORD LIST

- A machines
- B governments
- C functions
- D abstract
- E graphic
- F social vision
- G storyline
- H animals
- I power
- J purpose K organisation

The writer of the article values the film Metropolis for its ⑧ _____. It is relevant to our times mainly because it portrays people who have become like ⑨ _____. The people in the film are controlled by a dictatorial ⑩ _____, which is only interested in money and in ⑪ _____. The writer claims that the machinery in the film should take into account the fact that because the ⑫ _____ of the machines are never fully described, Lang's vision of the future is highly ⑬ _____.

Passage 28

The Value of Research into Mite Harvestmen

How a tiny, overlooked forest creature is helping scientists trace hundreds of millions of years of continental drift

Few people have heard of the mite harvestman, and fewer still would recognize it at close range. The insect is a relative of the far more familiar daddy longlegs. But its legs are stubby rather than long, and its body is only as big as a sesame seed. To find mite harvestmen, scientists go to dark, humid forests and sift through the leaf litter. The animals respond by turning motionless, making them impossible for even a trained eye to pick out. 'They look like grains of dirt,' said Gonzalo Giribet, an invertebrate biologist at Harvard University.

Dr Giribet and his colleagues have spent six years searching for mite harvestmen on five continents. The animals have an extraordinary story to tell — they carry a record of hundreds of millions of years of geological history, chronicling the journeys that continents have made around the Earth. The Earth's landmasses have slowly collided and broken apart again several times, carrying animals and plants with them. These species have provided clues to the continents' paths. The notion of continental drift originally came from such clues. In 1911, the German scientist Alfred Wegener was struck by the fact that fossils of similar animals and plants could be found on either side of the Atlantic. The ocean was too big for the species to have traveled across it on their own. Wegener speculated correctly, as it turned out, that the surrounding continents had originally been welded together in a single landmass, which he called Pangea.

Continental drift, or plate tectonics as it is scientifically known, helped move species around the world. Armadillos and their relatives are found in South America and Africa today because their ancestors evolved when the continents were joined. When South America and North America connected a few million years ago, armadillos spread north, too.

Biogeographers can learn clues about continental drift by comparing related species. However, they must also recognize cases where species have spread for other reasons, such as by crossing great stretches of water. The island of Hawaii, for example, was home to a giant flightless goose that has become extinct. Studies on DNA extracted from its bones show that it evolved from the Canada goose. Having colonized Hawaii, it branched off from that species, losing its ability to fly. This evolution occurred half a million years ago, when geologists estimate that Hawaii emerged from the Pacific.

When species jump around the planet, their histories blur. It is difficult to say much about where cockroaches evolved, for example, because they can move quickly from continent to continent. This process, known as dispersal, limits many studies. 'Most of them tend to concentrate on particular parts of the world,' Dr Giribet said. 'I wanted to find a new system for studying biogeography on a global scale.'

Dr Giribet realized that mite harvestmen might be that system. The 5,000 or so mite harvestmen species can be found on every continent except Antarctica. Unlike creatures found around the world like cockroaches, mite harvestmen cannot disperse well. The typical harvestman species has a range of fewer than 50 miles. Harvestmen are not found on young islands like Hawaii, as these types of islands emerged long after the break-up of Pangea.

According to Assistant Professor Sarah Boyer, a former student of Dr Giribet, 'It's really hard to find a group of species that is distributed all over the world but that also doesn't disperse very far.' What mite harvestmen lack in mobility, they make up in age. Their ancestors were among the first land animals, and fossils of daddy longlegs have been found in 400 million-year-old rocks. Mite harvestmen evolved long before Pangea broke up and have been carried along by continental drift ever since — they've managed to get themselves around the world only because they've been around for hundreds of millions of years, Dr Boyer said. Dr Boyer, Dr Giribet and their colleagues have gathered thousands of mite harvestmen from around the world, from which they extracted DNA. Variations in the genes helped the scientists build an evolutionary tree. By calculating how quickly the DNA mutated, the scientists could estimate when lineages branched off. They then compared the harvestmen's evolution to the movements of the continents. 'The patterns are remarkably clear,' Dr Boyer said.

The scientists found that they could trace mite harvestmen from their ancestors on Pangea. One lineage includes species in Chile, South Africa, Sri Lanka and other places separated by thousands of miles of ocean. But 150 million years ago, all those sites were in Gondwana, which was a region of Pangea.

The harvestmen preserve smaller patterns of continental drift, as well as bigger ones. After analyzing the DNA of a Florida harvestman, *Metasiro americanus*, the scientists were surprised to find that it was not related to other North American species. Its closest relatives live in West Africa. Dr Boyer then began investigating the geological history of Florida and found recent research to explain the mystery. Florida started out welded to West Africa near Senegal. North America then collided into them as Pangea was forming. About 170 million years ago, North America ripped away from West Africa, taking Florida with it. The African ancestors of Florida's harvestmen came along for the ride.

Dr Giribet now hopes to study dozens or even hundreds of species, to find clues about plate tectonics that a single animal could not show.

Questions 1–6

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 Why is it difficult to find mite harvestmen?
 - A they are too small to see with the naked eye.
 - B they can easily be confused with daddy longlegs.
 - C they are hard to distinguish from their surroundings.
 - D they do not exist in large numbers in any one place.
- 2 Why are mite harvestmen of interest to Dr Giribet and his colleagues?
 - A they have been studied far less than most other species.
 - B they show the effects of climate on the evolution of animals.
 - C they have an unusual relationship with plants and other animals.
 - D they provide evidence relating to a field of study other than insects.
- 3 What factor contributed to Wegener's idea that present-day continents used to form a single landmass?
 - A changes in the level of the ocean.
 - B the distance that species could travel.
 - C the lack of certain fossils on one side of the Atlantic.
 - D similarities in living conditions on both sides of the Atlantic.
- 4 What point is made by the reference to armadillos?
 - A regions have both separated and become connected.
 - B certain animals could travel longer distances than others.
 - C the oldest species of animals are likely to be found in Africa.
 - D there is a tendency for animals to spread in a particular direction.
- 5 Which of the following is stated about the giant flightless goose of Hawaii?
 - A Hawaii is a habitat that cannot support large birds.
 - B Hawaii is an attractive habitat for certain species of birds.
 - C flightless birds are more likely to become extinct than others.
 - D the Hawaiian goose became flightless after it had reached Hawaii.
- 6 Why is evidence from cockroaches of limited value?
 - A they spread too fast.
 - B they multiply too quickly.
 - C they are found in too few places.
 - D they have divided into too many species.

Questions 7–10

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 7 The colonization of Hawaii by geese provides evidence of continental drift.
- 8 The reason why mite harvestmen don't exist on Hawaii can be explained.
- 9 The DNA of certain species has evolved more quickly than that of others.
- 10 Dr Boyer's theory concerning the origins of Florida is widely accepted.

Questions 11–14

Complete the summary using the list of words, **A–I**, below.

WORD LIST

- A** branches
- B** fossils
- C** drift
- D** DNA
- E** evolution
- F** Pangea
- G** dispersal
- H** ancestors
- I** continents

Some of the first creatures to live on land were the **11** _____ of mite harvestmen. Boyer, Giribet and others study differences in the **12** _____ of these insects, and trace the development of a number of **13** _____ of the species. Their evolution appears to reflect changes in the location of **14** _____. For example, the same type of mite harvestman is found in places that are now far apart but used to form Gondwana, part of a huge landmass.

Passage 29

Musical Maladies

Norman M. Weinberger reviews the latest work of Oliver Sacks on music

Music and the brain are both endlessly fascinating subjects, and as a neuroscientist specialising in auditory learning and memory, I find them especially intriguing. So I had high expectations of *Musicophilia*, the latest offering from neurologist and prolific author Oliver Sacks. And I confess to feeling a little guilty reporting that my reactions to the book are mixed.

Sacks himself is the best part of *Musicophilia*. He richly documents his own life in the book and reveals highly personal experiences. The photograph of him on the cover of the book - which shows him wearing headphones, eyes closed, clearly enchanted as he listens to Alfred Brendel perform Beethoven's *Pathétique Sonata* - makes a positive impression that is borne out by the contents of the book. Sacks's voice throughout is steady and erudite but never pontifical. He is neither self-conscious nor self-promoting.

The preface gives a good idea of what the book will deliver. In it Sacks explains that he wants to convey the insights gleaned from the "enormous and rapidly growing body of work on the neural underpinnings of musical perception and imagery, and the complex and often bizarre disorders to which these are prone." He also stresses the importance of "the simple art of observation" and "the richness of the human context." He wants to combine "observation and description with the latest in technology," he says, and to imaginatively enter into the experience of his patients and subjects. The reader can see that Sacks, who has been practicing neurology for 40 years, is torn between the "old-fashioned" path of observation and the new-fangled, high-tech approach: he knows that he needs to take heed of the latter, but his heart lies with the former.

The book consists mainly of detailed descriptions of cases, most of them involving patients whom Sacks has seen in his practice. Brief discussions of contemporary neuroscientific reports are sprinkled liberally throughout the text. Part I, "Haunted by Music," begins with the strange case of Tony Cicoria, a nonmusical, middle-aged surgeon who was consumed by a love of music after being hit by lightning. He suddenly began to crave listening to piano music, which he had never cared for in the past. He started to play the piano and then to compose music, which arose spontaneously in his mind in a "torrent" of notes. How could this happen? Was the cause psychological? (He had had a near-death experience when the lightning struck him.) Or was it the direct result of a change in the auditory regions of his cerebral cortex? Electroencephalography (EEG) showed his brain waves to be normal in the mid-1990s, just after his trauma and subsequent "conversion" to music. There are now more sensitive tests, but Cicoria has declined to undergo them; he does not want to delve into the causes of his musicality. What a shame!

Part II, "A Range of Musicality," covers a wider variety of topics, but unfortunately, some of the chapters offer little or nothing that is new. For example, chapter 13, which is five pages long, merely notes that the blind often have better hearing than the sighted. The most interesting chapters are those that present the strangest cases. Chapter 8 is about "amusia," an inability to hear sounds as music, and "dysharmonia," a highly specific impairment of the ability to hear harmony, with the ability to understand melody left intact. Such specific "dissociations" are found throughout the cases Sacks recounts.

To Sacks's credit, part III, "Memory, Movement and Music," brings us into the underappreciated realm of music therapy. Chapter 16 explains how "melodic intonation therapy" is being used to help expressive aphasic patients (those unable to express their thoughts verbally following a stroke or other cerebral incident) once again become capable of fluent speech. In chapter 20, Sacks demonstrates the near-miraculous power of music to animate Parkinson's patients and other people with severe movement disorders, even those who are frozen into odd postures. Scientists cannot yet explain how music achieves this effect.

To readers who are unfamiliar with neuroscience and music behavior, *Musicophilia* may be something of a revelation. But the book will not satisfy those seeking the causes and implications of the phenomena Sacks describes. For one thing, Sacks appears to be more at ease discussing patients than discussing experiments. And he tends to be rather uncritical in accepting scientific findings and theories.

It's true that the causes of music-brain oddities remain poorly understood. However, Sacks could have done more to draw out some of the implications of the careful observations that he and other neurologists have made and of the treatments that have been successful. For example, he might have noted that the many specific dissociations among components of music comprehension, such as loss of the ability to perceive harmony but not melody, indicate that there is no music center in the brain. Because many people who read the book are likely to believe in the brain localisation of all mental functions, this was a missed educational opportunity.

Another conclusion one could draw is that there seem to be no "cures" for neurological problems involving music. A drug can alleviate a symptom in one patient and aggravate it in another, or can have both positive and negative effects in the same patient. Treatments mentioned seem to be almost exclusively antiepileptic medications, which "damp down" the excitability of the brain in general; their effectiveness varies widely.

Finally, in many of the cases described here the patient with music-brain symptoms is reported to have "normal" EEG results. Although Sacks recognises the existence of new technologies, among them far more sensitive ways to analyze brain waves than the standard neurological EEG test, he does not call for their use. In fact, although he exhibits

the greatest compassion for patients, he conveys no sense of urgency about the pursuit of new avenues in the diagnosis and treatment of music-brain disorders. This absence echoes the book's preface, in which Sacks expresses fear that "the simple art of observation may be lost" if we rely too much on new technologies. He does call for both approaches, though, and we can only hope that the neurological community will respond.

Questions 1–4

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 Why does the writer have a mixed feeling about the book?
 - A The guilty feeling made him so.
 - B The writer expected it to be better than it was.
 - C Sacks failed to include his personal stories in the book.
 - D This is the only book written by Sacks.
- 2 What is the best part of the book?
 - A the photo of Sacks listening to music
 - B the tone of voice of the book
 - C the autobiographical description in the book
 - D the description of Sacks's wealth
- 3 In the preface, what did Sacks try to achieve?
 - A make terms with the new technologies
 - B give detailed description of various musical disorders
 - C explain how people understand music
 - D explain why he needs to do away with simple observation
- 4 What is disappointing about Tony Cicoria's case?
 - A He refuses to have further tests.
 - B He can't determine the cause of his sudden musicality.
 - C He nearly died because of the lightening.
 - D His brain waves were too normal to show anything.

Questions 5–10

Do the following statements agree with the views of the writer in Reading Passage 3?

Write **YES** if the statement agrees with the views of the writer, **NO** if the statement contradicts the views of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 5 It is difficult to give a well-reputable writer a less than favorable review.
- 6 Beethoven's Pathetique Sonata is a good treatment for musical disorders.
- 7 Sacks believes technological methods is not important compared with observation when studying his patients.

- 8 It is difficult to understand why music therapy is undervalued.
- 9 Sacks should have more skepticism about other theories and findings.
- 10 Sacks is impatient to use new testing methods.

Questions 11–14

Complete each sentence with the correct ending, A–F, below. Write the correct letter in each box.

- 11 The dissociations between harmony and melody... 11 _____
- 12 The study of treating musical disorders... 12 _____
- 13 The EEG scans of Sacks's patients... 13 _____
- 14 Sacks believes testing based on new technologies... 14 _____

LIST OF ENDINGS

- A show no music-brain disorders
- B indicates that medication can have varied results
- C is key for the neurological community to unravel the mysteries
- D should not be used in isolation
- E indicate that not everyone can receive good education
- F show that music is not localised in the brain

Passage 30

Neanderthal Technology

What archaeologists have learned about how Neanderthal people made their homes and used fire

27 **A** We think of our prehistoric ancestors as people of the ice and snow, living in caves, and for many of the west European Neanderthals that is a just picture of their life. But where there were no caves, further to the east on the Russian steppe, for example, open-air sites with some sort of constructed shelter were the only option.

28 **B** We now know much more about the cave sites than the open-air ones because, historically, it was the cave sites of Western Europe that were first explored by archaeologists and also because open-air sites are harder to find — many of them have disappeared under deep mud deposits or under the rising postglacial seas. Caves, moreover, aid the survival of archaeological material and can preserve the records of remote millennia.

29 **C** In south-west France, the limestone caves of the Périgord region made ideal homes for the Neanderthal people. There were good supplies of flint to hand for axes and the like, and the caves were often sited in small river valleys that offered protection against the worst of the weather. The Neanderthals liked south-facing caves, for obvious reasons of sunshine and wind avoidance, and caves at some height above the valley floor offered refuge from floods and good game-watching vantage points.

30 **D** The Périgord region during the last ice age was, in fact, an exceptionally benign habitat for humans. It enjoyed a rather maritime climate with cooler summers that permitted the extension of tundra and steppe over its higher plateaux, and its year-round high levels of sunshine favoured the growth of the ground plants needed by reindeer, bison and horse. Winters were mildish for the ice age, animals never needed to migrate far from summer to winter, and men never needed to travel far from home to find abundant supplies of meat.

31 **E** In Central and Eastern Europe, where caves were unavailable, such open-air sites as have been discovered were mostly located near water — both because this was a good area to be for people and animals, and also because the sedimentation potential of lakes and stream courses has aided archaeological preservation — whereas erosion has presumably blown away sites which were out in the open. Some of the open-air sites in Germany, Central Europe and Russia have provided valuable information about Neanderthal man and his way of life. From Moldova, for example, comes evidence that has been interpreted as the remains of wind-break structures, or even a large tent: a ring, up to about 8 x 5m in size, of mainly mammoth bones enclosing a dense concentration of stone tools, animal bones and ash.

F From the west European caves more evidence of built structures is available, and some of it goes back a long way in time. In the Grotte du Lazaret, near Nice, at a date during the last ice age but one, claims for some sort of skin tent within the cave have been advanced, on the basis of arrangements of large stones out from the cave wall that might have supported timber struts for a covering of skins up to the rock face above. At Lazaret, what might be openings in the hypothesised tents seem to point away from the cave mouth, and finds of wolf and fox foot bones, without the rest of the skeletons, inside these 'tents' have been thought to indicate the use of animal pelts as bed coverings. The two patches of ash at Lazaret that mark ancient fires, with stone tools around them evidently made and used on the spot, are edged with small marine mollusc shells, prompting the excavator to suggest that seaweed had been used as bedding around the fires. The cave of Baume-Bonne in the Basses-Alpes region of France, another early site, boasts ten square metres of cobbles brought up from the local river and laid down, as though to take care of a puddle area in the cave, with the smoothest and roundest surfaces of the stones uppermost, and there are other similar cases.

G The ash encountered in concentrations at some sites testifies to the Neanderthal people's use of fire: not surprising, since use of fire was, by Neanderthal times, an already ancient accomplishment of evolving humanity, and survival in the sub-arctic conditions faced by the Neanderthals is inconceivable without control of fire. Fire gave warmth, light, heat for cooking and defence against predatory animals. A charred piece of birch from Krapina in Croatia, is thought to be the remains of a fire-making twirl stick. But Neanderthal hearths, in the sense of specially constructed places for fire, are fewer and harder to identify with certainty than the mere ash piles that are a regular feature of their sites. They seem often to have just lit a small fire (40-50cm across) on the existing ground surface of the cave, without preparation. Judging from the shallow penetration of heat effects under the ash, this fire was only of a short duration. Sometimes the fires were larger in size, up to one metre across, and quite irregular in shape. It is not always easy to decide how much additional structure some fires possessed: claims of stone circles to contain the fire run up against the fact that stones tend to litter the cave floors everywhere and those around a fire can quite accidentally look as though they were arranged in a circle.

Questions 27–31

Reading Passage 3 has five sections, A–E.

Choose the correct heading for each section from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Evidence of outdoor dwellings
- ii Learning to make fire

- iii A perfect place to live
- iv Examining the cave contents
- v Contrasting two types of home
- vi A vital source of power

Questions 32–36

Look at the following findings and the list of places below. Match each finding with the correct place, A–E.

NB You may use any letter more than once.

LIST OF PLACES

- A The Périgord region
- B Moldova
- C The Grotte du Lazaret
- D The cave of Baume-Bonne
- E Krapina

Q	Statement	
32	a burnt piece of wood	
33	evidence of efforts to prevent pools of water forming	
34	the remains of sea creatures	
35	a circular arrangement of animal bones	
36	evidence suggesting the use of animal fur for warmth	

Questions 37–40

Complete the summary below. Write **NO MORE THAN TWO WORDS** from the passage for each answer.

THE USE OF FIRE

Neanderthals could not have survived without fire because the conditions they lived in were (37) _____. Most evidence of purpose-built fires takes the form of ash piles, features of which suggest that the fires lasted a (38) _____ time. It is hard to be certain about the

size and structure of the fires, though they were certainly needed to protect the occupants from dangerous **39** _____, among other things.

- 40** The purpose of the writer of this article is to
- A** argue that Neanderthal homes were bigger than originally thought.
 - B** explain why Neanderthal people migrated to Western Europe.
 - C** discuss what is known about Neanderthal settlements.
 - D** track the progress of early Neanderthal development.

Passage 31

Pacific Navigation and Voyaging

How people migrated to the Pacific islands

The many tiny islands of the Pacific Ocean had no human population until ancestors of today's islanders sailed from Southeast Asia in ocean-going canoes approximately 2,000 years ago. At the present time, the debate continues about exactly how they migrated such vast distances across the ocean, without any of the modern technologies we take for granted.

Although the romantic vision of some early twentieth-century writers of fleets of heroic navigators simultaneously setting sail had come to be considered by later investigators to be exaggerated, no considered assessment of Pacific voyaging was forthcoming until 1956 when the American historian Andrew Sharp published his research. Sharp challenged the 'heroic vision' by asserting that the expertise of the navigators was limited, and that the settlement of the islands was not systematic, being more dependent on good fortune by drifting canoes. Sharp's theory was widely challenged, and deservedly so. If nothing else, however, it did spark renewed interest in the topic and precipitated valuable new research.

Since the 1960s a wealth of investigations has been conducted, and most of them, thankfully, have been of the 'non-armchair' variety. While it would be wrong to denigrate all 'armchair' research — that based on an examination of available published materials — it has turned out that so little progress had been made in the area of Pacific voyaging because most writers relied on the same old sources — travelers' journals or missionary narratives compiled by unskilled observers. After Sharp, this began to change, and researchers conducted most of their investigations not in libraries, but in the field.

In 1965, David Lewis, a physician and experienced yachtsman, set to work using his own unique philosophy: he took the yacht he had owned for many years and navigated through the islands in order to contact those men who still find their way at sea using traditional methods. He then accompanied these men, in their traditional canoes, on test voyages from which all modern instruments were banished from sight, though Lewis secretly used them to confirm the navigator's calculations. His most famous such voyage was a return trip of around 1,000 nautical miles between two islands in mid-ocean. Far from drifting, as proposed by Sharp, Lewis found that ancient navigators would have known which course to steer by memorizing which stars rose and set in certain positions along the horizon and this gave them fixed directions by which to steer their boats.

The geographer Edwin Doran followed a quite different approach. He was interested in obtaining exact data on canoe sailing performance, and to that end employed the latest electronic instrumentation. Doran traveled on board traditional sailing canoes in some of

the most remote parts of the Pacific, all the while using his instruments to record canoe speeds in different wind strengths — from gales to calms — the angle canoes could sail relative to the wind. In the process, he provided the first really precise attributes of traditional sailing canoes.

A further contribution was made by Steven Horvath. As a physiologist, Horvath's interest was not in navigation techniques or in canoes, but in the physical capabilities of the men themselves. By adapting standard physiological techniques, Horvath was able to calculate the energy expenditure required to paddle canoes of this sort at times when there was no wind to fill the sails, or when the wind was contrary. He concluded that paddles, or perhaps long oars, could indeed have propelled for long distances what were primarily sailing vessels.

Finally, a team led by P Wall Garrard conducted important research, in this case by making investigations while remaining safely in the laboratory. Wall Garrard's unusual method was to use the findings of linguists who had studied the languages of the Pacific islands, many of which are remarkably similar although the islands where they are spoken are sometimes thousands of kilometres apart. Clever adaptation of computer simulation techniques pioneered in other disciplines allowed him to produce convincing models suggesting the migrations were indeed systematic, but not simultaneous. Wall Garrard proposed the migrations should be seen not as a single journey made by a massed fleet of canoes, but as a series of ever more ambitious voyages, each pushing further into the unknown ocean.

What do we learn about Pacific navigation and voyaging from this research? Quite correctly, none of the researchers tried to use their findings to prove one theory or another; experiments such as these cannot categorically confirm or negate a hypothesis. The strength of this research lay in the range of methodologies employed. When we splice together these findings we can propose that traditional navigators used a variety of canoe types, sources of water and navigation techniques, and it was this adaptability which was their greatest accomplishment. These navigators observed the conditions prevailing at sea at the time a voyage was made and altered their techniques accordingly. Furthermore, the canoes of the navigators were not drifting helplessly at sea but were most likely part of a systematic migration; as such, the Pacific peoples were able to view the ocean as an avenue, not a barrier, to communication before any other race on Earth. Finally, one unexpected but most welcome consequence of this research has been a renaissance in the practice of traditional voyaging. In some groups of islands in the Pacific today young people are resurrecting the skills of their ancestors, when a few decades ago it seemed they would be lost forever.

Questions 1–5

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 1 The Pacific islands were uninhabited when migrants arrived by sea from Southeast Asia.
- 2 Andrew Sharp was the first person to write about the migrants to islanders.
- 3 Andrew Sharp believed migratory voyages were based more on luck than skill.
- 4 Despite being controversial, Andrew Sharp's research had positive results.
- 5 Edwin Doran disagreed with the findings of Lewis's research.

Questions 6–10

Choose the correct letter, **A**, **B**, **C** or **D**.

- 6 David Lewis's research was different because
 - A he observed traditional navigators at work.
 - B he conducted test voyages using his own yacht.
 - C he carried no modern instruments on test voyages.
 - D he spoke the same language as the islanders he sailed with.
- 7 What did David Lewis's research discover about traditional navigators?
 - A They used the sun and moon to find their position.
 - B They could not sail further than about 1,000 nautical miles.
 - C They knew which direction they were sailing in.
 - D They were able to drift for long distances.
- 8 What are we told about Edwin Doran's research?
 - A Data were collected after the canoes had returned to land.
 - B Canoe characteristics were recorded using modern instruments.
 - C Research was conducted in the most densely populated regions.
 - D Navigators were not allowed to see the instruments Doran used.
- 9 Which of the following did Steven Horvath discover during his research?
 - A Canoe design was less important than human strength.
 - B New research methods had to be developed for use in canoes.
 - C Navigators became very tired on the longest voyages.
 - D Human energy may have been used to assist sailing canoes.

- 10 What is the writer's opinion of P Wall Garrard's research?
- A He is disappointed it was conducted in the laboratory.
 - B He is impressed by the originality of the techniques used.
 - C He is surprised it was used to help linguists with their research.
 - D He is concerned that the islands studied are long distances apart.

Questions 11–14

Complete each sentence with the correct ending, **A–F**, below. Write the correct letter in each box.

- 11 One limitation in the information produced by all of this research is that it 11 _____
- 12 The best thing about this type of research 12 _____
- 13 The most important achievement of traditional navigators 13 _____
- 14 The migration of people from Asia to the Pacific 14 _____

LIST OF ENDINGS

- A was the variety of experimental techniques used
- B was not of interest to young islanders today
- C was not conclusive evidence in support of a single theory
- D was being able to change their practices when necessary
- E was the first time humans intentionally crossed an ocean
- F was the speed with which it was conducted

Passage 32

Playing Psychological Games

An analysis of the fixed, script-like patterns people fall into when interacting with each other

① **A** 'Psychological games' is an approach to relationships that developed around the 1960s. It is a way of looking at the interaction between people – identifying what seem to be fixed scripts in a seemingly spontaneous conversation. One person says something which seems to elicit a certain type of response from the other person, and the response seems to demand yet another particular response from the first person. And on it goes, as if the two people were following a script that someone had written.

② **B** Games fall into a number of categories, ranging from the harmless to the destructive. Some harmless games are even essential to social interaction – such as the 'Greeting Game' (Hello, how are you? I'm fine, how are you?) and the 'Thanks Game' (Thank you for inviting me. I had a great time.). Game playing is expected in some situations. Everyone involved knows that it is a game, and what is expected. A sales person plays a game of pleasing the prospective customer. Children play games with parents. In cases like these, the game player creates an impression, saying things which are not sincere but are ways and means of getting what they want. Other games, however, keep a relationship from developing to a more real and important level. Still others can actually be destructive, as they are played by people with deeper psychological needs and motivations for power, control or manipulation.

③ **C** Some people set out to manipulate others for their own reasons. But others may not realize that they are being manipulative. They are acting rather from an emotional script. Like a child that wants something, and does all sorts of things to get it, some game players act from their own internal desires, not realizing the effect their words and actions have on others.

④ **D** A number of potentially damaging games have been identified. In the 'Corner Game', the manipulator backs the other person into a corner – places them in a situation where anything they do is wrong. A parent complains that their son or daughter's room is never clean. Yet when the child tidies the room, the parent says, 'Why did it take you so long?' or 'You haven't tidied up inside the cupboard.' The 'It's Your Decision Game' is played by people who want to escape the responsibility of making a decision: 'I don't mind. You decide.' Although actually very much concerned about the outcome of the decision, by insisting they are not the game player forces the other person to take all responsibility for the consequences of the decision.

⑤ **E** Games may indicate a lack of confidence in the other person, an unwillingness to communicate with them directly. In the most innocent cases, they are played in an attempt at politeness, or genuine concern for the other's feelings (trying not to hurt them). However, even these well-intentioned games don't always have a good end. They can make it impossible for an atmosphere of trust to be created.

⑥ **F** At their worst, games are a way for an individual to retain power in a relationship, because their own personal feelings are not revealed. The person who uses games to their own advantage needs to win a game in order to have a sense of self-esteem – by harming someone else's self-confidence. Manipulators range from Dictator (who always has to be in charge) to Nice Guy (who exaggerates care and love for others, in order to get what he or she wants), to Protector (who is over-supportive or over-protective).

⑦ **G** Some game players have so many psychological needs that fulfilling their desires overshadows everything else in a relationship. For example, a person who needs to be the centre of attention may play games in which they consistently take the role of someone who needs help, someone who is dependent. Sometimes people fall into games in a relationship because of the roles that they think they should be playing. A young couple that accepts that the husband needs to defend his wife against criticism by his family, or that he will automatically make the decisions about minor repairs on her car, even though in both cases the wife is perfectly capable of looking after herself. Their exchange will fall into a kind of game, because they have restricted their roles to what they believe is expected of them.

Questions 1–7

The reading passage has seven paragraphs, A–G.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Towards a classification of games
- ii How the theory of games was developed
- iii Feeling good by making others have doubts about themselves
- iv Being cautious towards other people
- v Games that create permanent relationships between people
- vi Game-playing – conscious or unconscious
- vii How a relationship can be dominated by games
- viii The type of people that game players look for

- ix Some examples of harmful games
- x A tool for understanding communication

Questions 8–11

*Do the following statements agree with the claims of the writer in the reading passage? Write **YES** if the statement reflects the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- 8 When people want a certain decision they will always make it themselves.
- 9 Games that show regard for other people can prevent trust from developing in the relationship.
- 10 Giving another person too much help may be a form of manipulation.
- 11 Avoiding social pressure to behave in certain ways is difficult for a young married couple.

Questions 12–13

*Choose the correct letter, **A**, **B**, **C** or **D**.*

- 12 What point does the writer make about the 'Corner Game'?
 - A It is most commonly played in the parent-child relationship.
 - B In it, one person will be blamed whatever they do.
 - C The manipulator is normally aware of what he or she is doing.
 - D Both participants are responsible for playing the game.
- 13 Which statement best sums up the writer's opinion of the theory of psychological games?
 - A Identifying games gives useful insight into interaction between people.
 - B Relationships in which games are played are likely to end unhappily.
 - C Games are played to some extent in all relationships.
 - D It is difficult to help some people to stop playing games.

Passage 33

Rebranding Art Museums

How the 2003 redevelopment of Melbourne's National Gallery of Victoria reflects a global shift in museum design

A As anyone with even a passing interest in cultural institutions will know, art museums are adapting to keep up with changes in society. Do any of these newly developed spaces indicate a fundamental shift in the ways we engage with art at the beginning of the twenty-first century? The answer is a qualified Yes and also, it must be said, No. A clear sense of this can be gained by considering the 2003 redevelopment of the National Gallery of Victoria (NGV) in Melbourne, Australia's second largest city, in the light of the aspirations embodied in the original building that served as its template.

B The first building was considered a thoroughly modern museum when it opened to the public in 1968. We should not forget that it was also bold, entrepreneurial and innovative in ways that have done great credit to the vision and foresight of the architect, Roy Grounds. For one thing, the 1968 building signalled a strong sense of engagement with Asian and Pacific cultures, both in its architecture and through the layout of its collections. Its entrance lobby was designed to facilitate the rapid and efficient circulation of visitors. It also contained a series of greatly expanded temporary exhibition galleries and a vast Great Hall, which represented an unprecedented emphasis on the ancillary roles of corporate functions and public performances in the contemporary museum.

C This last feature might have seemed excessive to some, yet how wise it has proven over the years as museums have come to place increasing emphasis on corporate sponsorship, and on the need to generate additional funds via room hire. Furthermore, the Great Hall has for forty years provided an area for children to roam at large before being escorted through the much less spatially accommodating galleries themselves. As an unintimidating introduction to the lifelong rigours of visiting art museums, its impact on attendance cannot be overestimated.

D At the same time, other aspects of the 1968 NGV have not withstood the test of time so well. Foremost among these are the huge expanse of the exterior's windowless wall and the surrounding moat, which creates a rather unwelcoming expression of the museum as a temple of culture set apart from the everyday world. Over time, this idea has been questioned, and has been superseded by the growing impetus towards the alternative model of the museum as a more open-ended and visitor-friendly forum that engages the public more effectively.

E With these considerations in mind, Mario Bellini's redeveloped NGV opened in 2003 and his design reflects what the new, improved, twenty-first century global museum feels it should be emphasising to its visitors. Here the institution's energies are focused, not so much on the technical subtleties of how to design the galleries themselves, as on presenting to the public a spectacular image of the museum itself as a welcoming yet efficient facilitator of social interaction, popular entertainment and public knowledge. All of this is certainly impressive, but where has the art gone? We are not allowed to see the exhibitions until we have completed the necessary inductions. We are greeted by a hugely expanded cloakroom screening off our vision to the right, followed by a large visitor information office leading to the ticket booths. Above us is the cafe and to our left we cannot escape the NGV shop set alongside the exit.

F And what of the redesigned galleries themselves? In fact, they represent a wonderfully elegant reframing of the permanent collections in ways that should offer first-time visitors and seasoned members alike many new avenues for engagement in the years to come. But herein lies the conundrum posed by the NGV renovation and by the global sweep of new museum projects more generally. The mainstay elements of a permanent collection can become all too easily lost in the increased prominence that these rebranding exercises tend to place on the more glamorously spectacular aspects of the institution itself. At its most extreme, this results in the construction of new buildings that supplant the artworks inside them to become the major attractions themselves.

G These challenges are most keenly felt in the area of the temporary exhibition. Entrance to the NGV permanent collection is now free, so the NGV needs something to keep the paying public coming back. The answer lies in the international 'blockbuster' exhibition. The Impressionists, for example, netted a total of 380,000 visitors, making it one of the most popular exhibitions in Australian history. There has also been, it needs to be said, a commendable attempt to leave the international masterpieces with selected highlights from the permanent collection. Overall, though, the prevalence of these 'rental' blockbusters cannot but help introduce a certain 'off the rack' feeling to the institution's exhibition programme. More importantly perhaps, they clearly narrow the options for other worthy but less glamorous projects, and they take energy away from the curators' potential to do further work on and around the permanent collection itself.

H Yet the show must go on, and shows can't exist without the public. Even the most adroit organisation, as the NGV assuredly is, needs to be able to juggle numerous, often conflicting priorities: the need to advance scholarship, on the one hand, versus the imperative of access on the other; the need to introduce audiences to new areas, versus the necessity of luring them in with instant brand recognition. Nobody said it was going to be easy, and the effect of too much innovation in museums can be disastrous.

Questions 27–30

Choose the correct letter, **A**, **B**, **C** or **D**.

- 27 What is the writer doing in paragraph 1?
- A comparing the old NGV with the new NGV
 - B describing the role of art museums in Australian culture
 - C considering the influence of gallery design on art appreciation
 - D suggesting there has been little recent change in gallery design
- 28 The writer criticises the old NGV because
- A the permanent collection was poorly displayed.
 - B the building tended to make art seem remote.
 - C Asian and Pacific art was neglected.
 - D the foyer resembled a shop.
- 29 What does the writer notice about the new NGV?
- A He thinks the ticket booths are well positioned.
 - B He is upset that he is made to move so quickly.
 - C He is frustrated because the art is obscured by the entrance.
 - D He thinks the redevelopment fails to keep up with international trends.
- 30 The writer argues that rental 'blockbuster' exhibitions
- A should be cheaper.
 - B are a financial necessity.
 - C should be increased in number.
 - D allow the museum to present an original programme.

Questions 31–35

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write **YES**, **NO** or **NOT GIVEN**.

- 31 Roy Grounds's design failed to recognise the importance of functions and performances in museums.
- 32 Mario Bellini's new NGV rejects international museum design trends.
- 33 Mario Bellini's work on subsequent museum projects has been less successful than that on the NGV.
- 34 The NGV monitors the number of first-time visitors to permanent collections.
- 35 Too much change may have negative impacts on museums.

Questions 36–40

Complete each sentence with the correct ending, **A–G**, below. Write the correct letter in each box.

- | | | |
|-----------|--|-----------------|
| 36 | If a large space is available in the museum,... | 36 _____ |
| 37 | If children are allowed to move freely in parts of the gallery,... | 37 _____ |
| 38 | If too much emphasis is placed on the museum building itself,... | 38 _____ |
| 39 | If there is an over-reliance on 'blockbuster' exhibitions,... | 39 _____ |
| 40 | If the NGV wants to continue to be successful,... | 40 _____ |

LIST OF ENDINGS

- A** the museum programme will lose its individuality
- B** the museum will lose credibility
- C** the art loses its importance
- D** it will have to balance opposing demands
- E** this will encourage museum visitors in the future
- F** it will have the capability of increasing revenue
- G** the architect's reputation might suffer

Passage 34

Robert Louis Stevenson

A Scottish novelist, poet, essayist, and travel writer, Robert Louis Stevenson was born at 8 Howard Place, Edinburgh, Scotland, on 13 November 1850. It has been more than 100 years since his death. Stevenson was a writer who caused conflicting opinions about his works. On one hand, he was often highly praised for his expert prose and style by many English-language critics. On the other hand, others criticised the religious themes in his works, often misunderstanding Stevenson's own religious beliefs. Since his death a century before, critics and biographers have disagreed on the legacy of Stevenson's writing. Two biographers, KF and CP, wrote a biography about Stevenson with a clear focus. They chose not to criticise aspects of Stevenson's personal life. Instead, they focused on his writing, and gave high praise to his writing style and skill.

The literary pendulum has swung these days. Different critics have different opinions towards Robert Louis Stevenson's works. Though today, Stevenson is one of the most translated authors in the world, his works have sustained a wide variety of negative criticism throughout his life. It was like a complete reversal of polarity — from highly positive to slightly less positive to clearly negative; after being highly praised as a great writer, he became an example of an author with corrupt ethics and lack of moral. Many literary critics passed his works off as children's stories or horror stories, and thought to have little social value in an educational setting. Stevenson's works were often excluded from literature curriculum because of its controversial nature. These debates remain, and many critics still assert that despite his skill, his literary works still lack moral value.

One of the main reasons why Stevenson's literary works attracted so much criticism was due to the genre of his writing. Stevenson mainly wrote adventure stories, which was part of a popular and entertaining writing fad at the time. Many of us believe adventure stories are exciting, offers engaging characters, action, and mystery but ultimately can't teach moral principles. The plot points are one-dimensional and rarely offer a deeper moral meaning, instead focusing on exciting and shocking plot twists and thrilling events. His works were even criticised by fellow authors. Though Stevenson's works have deeply influenced Oscar Wilde, Wilde often joked that Stevenson would have written better works if he wasn't born in Scotland. Other authors came to Stevenson's defence, including Galsworthy who claimed that Stevenson is a greater writer than Thomas Hardy.

Despite Wilde's criticism, Stevenson's Scottish identity was an integral part of his written works. Although Stevenson's works were not popular in Scotland when he was alive, many modern Scottish literary critics claim that Sir Walter Scott and Robert Louis Stevenson are the most influential writers in the history of Scotland. While many critics exalt Sir Walter Scott as a literary genius because of his technical ability, others argue that Stevenson

deserves the same recognition for his natural ability to capture stories and characters in words. Many of Scott's works were taken more seriously as literature for their depth due to their tragic themes, but fans of Stevenson praise his unique style of story-telling and capture of human nature. Stevenson's works, unlike other British authors, captured the unique day to day life of average Scottish people. Many literary critics point to this as a flaw of his works. According to the critics, truly important literature should transcend local culture and stories. However, many critics praise the local taste of his literature. To this day, Stevenson's works provide valuable insight to life in Scotland during the 19th century.

Despite much debate of Stevenson's writing topics, his writing was not the only source of attention for critics. Stevenson's personal life often attracted a lot of attention from his fans and critics alike. Some even argue that his personal life eventually outshone his writing. Stevenson had been plagued with health problems his whole life, and often had to live in much warmer climates than the cold, dreary weather of Scotland in order to recover. So he took his family to a south pacific island Samoa, which was a controversial decision at that time. However, Stevenson did not regret the decision. The sea air and thrill of adventure complimented the themes of his writing, and for a time restored his health. From there, Stevenson gained a love of travelling, and for nearly three years he wandered the eastern and central Pacific. Much of his works reflected this love of travel and adventure that Stevenson experienced in the Pacific islands. It was as a result of this biographical attention that the feeling grew that interest in Stevenson's life had taken the place of interest in his works. Whether critics focus on his writing subjects, his religious beliefs, or his eccentric lifestyle of travel and adventure, people from the past and present have different opinions about Stevenson as an author. Today, he remains a controversial yet widely popular figure in Western literature.

Questions 27–31

Choose the correct letter, **A**, **B**, **C** or **D**.

- 27 Stevenson's biographers KF and CP
- A underestimated the role of family played in Stevenson's life.
 - B overestimated the writer's works in the literature history.
 - C exaggerated Stevenson's religious belief in his works.
 - D elevated Stevenson's role as a writer.
- 28 The main point of the second paragraph is
- A the public give a more fair criticism to Stevenson's works.
 - B recent criticism has been justified.
 - C the style of Stevenson's works overweigh his faults in his life.
 - D Stevenson's works' drawback is lack of ethical nature.

- 29 According to the author, adventure stories
- A do not provide plot twists well.
 - B cannot be used by writers to show moral values.
 - C are more fashionable art form.
 - D can be found in other's works but not in Stevenson's.

- 30 What does the author say about Stevenson's works?
- A They describe the life of people in Scotland.
 - B They are commonly regarded as real literature.
 - C They were popular during Stevenson's life.
 - D They transcend the local culture and stories.

- 31 The lifestyle of Stevenson
- A made his family envy him so much.
 - B should be responsible for his death.
 - C gained more attention from the public than his works.
 - D didn't well prepare his life in Samoa.

Questions 32–35

Do the following statements agree with the information given in Reading Passage 3?

Write **TRUE** / **FALSE** / **NOT GIVEN**

- 32 Although Oscar Wilde admired Robert Louis Stevenson very much, he believed Stevenson could have written greater works.
- 33 Robert Louis Stevenson encouraged Oscar Wilde to start writing at first.
- 34 Galsworthy thought Hardy is a greater writer than Stevenson is.
- 35 Critics only paid attention to Robert Louis Stevenson's writing topics.

Questions 36–40

Complete the notes using the list of words, A–I, below. Drag a word into a box, or tap a word then tap a box.

Sir Walter Scott and Robert Louis Stevenson

WORD LIST

- A natural ability
- B romance
- C colorful language
- D critical acclaim
- E humor

- F** technical control
- G** storytelling
- H** depth
- I** human nature

A lot of people believe that Sir Walter Scott and Robert Louis Stevenson are the most influential writers in the history of Scotland, but Sir Walter Scott is more proficient in **36** **36** _____ while Stevenson has better **37** **37** _____. Scott's books illustrate **38** **38** _____ especially in terms of tragedy, but a lot of readers prefer Stevenson's **39** **39** _____. What's more, Stevenson's understanding of **40** **40** _____ made his works have the most unique expression of Scottish people.

Passage 35

Science and Filmmaking

Academics are now working more with filmmakers who are impressed by the results of their research in computer-generated imagery (CGI)

Every year the film academy in the USA celebrates the outstanding achievements of the year in a ceremony known as the Oscars. An increasingly important component of the ceremony is the presentation of the Scientific and Technical awards. In 2004 a notable event took place: the academic world met the cinematographic world when researchers from Stanford University in the USA were awarded an Oscar. These researchers, led by Steve Marschner, were from the field of Computer Graphics at Stanford. They were part of a growing cohort of computer scientists that has become fundamental to moviemaking.

Films have shown that it is possible to use CGI to make actors look younger, older, weaker or stronger than they actually are in a surprisingly realistic manner. At least, it is possible if the altered actors are not filmed too closely. This is because the difficulty of recreating the textures of both skin and fabric means that the effect is less convincing when seen close up. The work of Marschner and his colleagues has greatly improved the accurate and realistic modeling of both skin and fabric. They recognized that one of the difficulties of creating lifelike characters in the computer world is that in CGI the characters' skin is opaque (two-dimensional) whereas real skin is in fact translucent (three-dimensional), that is to say it is semi-transparent.

Marschner and his colleagues received the Oscar for their work in successfully producing a CGI model that simulates translucency; this is when light penetrates skin and then scatters below the skin's surface before re-emerging. This is called subsurface scattering, and the mathematics for the model goes back many decades to the time when it was used in astrophysics. Because human skin is naturally translucent, it was necessary to be able to create this artificially in order to simulate the soft appearance of real skin. Previous CGI models, which assumed that skin was entirely opaque, resulted in characters with a plastic appearance. The scientists' new model of CGI was so important in bringing digital characters to life that, within two years of their original research paper, all the major special-effects studios had incorporated it into their digital rendering systems.

However, despite their award, the scientists, with admirable tenacity, continued their search for perfection, as they still did not feel that they had yet satisfactorily recreated the subtle ways light is reflected. To do this, they began to look in detail at the way skins and fabrics reflect light differently according to their make-up: the exact arrangement of fibers in fabric and the network of fibers in skin. Marschner and his team sought to do this by using computerized tomography – which is most familiar as a medical technique for examining people's internal organs. Like classical radiology it uses X-rays, but because the

image is constructed inside a computer using exposures taken from many different positions, rather than a single exposure on photographic film, it can capture fine details that are invisible in classical radiography.

Unfortunately, the scientists understood that at this point in time they could not use computer tomography on skin, because a very high intensity X-ray is needed to show the kind of detail they wanted and this would be very dangerous for human skin. They have, however, had some success with fabric. Using this new method of imaging, they are able to accurately record the three-dimensional structure of all the fibers in a number of small pieces of fabric. These same pieces of fabric, through the use of CGI, can then be patched together to form an entire garment inside a computer, in the same way that a small group of actors can be made to look like hundreds of people gathered together. A garment created through CGI is therefore made up of pieces of fabric whose internal structure has been pre-recorded. This means that the way light is reflected by the garment can be calculated far more realistically than if the scientists just made a computer model of what they thought the interior of the fabric looked like.

Cinematography will benefit from this because, although it may take some years to be able to use computerized tomographic imaging of skin, for the moment the movement of a virtual cloak or the lifting of a computerized hat should look far more realistic.

In the meantime, according to Marschner's colleague Kavita Bala, the technology might have an application in online retailing. At the moment people buying clothes over the internet have only a standard photograph to help them choose their purchases. It is hoped that if online shoppers can view items which have been presented through the use of computerized tomography graphics, they will have a much better understanding of what the material the item is made of is really like.

Marschner is now working on the way light is scattered from individual hairs. He says, 'I feel lucky to be working in this niche. I'm a visual person and to be able to spend my time scrutinizing the world around me, trying to understand why it looks the way it does, is very rewarding'.

Questions 27–32

Choose the correct letter, **A**, **B**, **C** or **D**.

- 27 What is the writer's main point in the first paragraph?
- A** Computer scientists are rarely represented at the Oscars.
 - B** Film industry awards hold little interest for computer scientists.
 - C** The USA rewards film actors more than computer scientists.
 - D** Computer scientists are becoming a vital part of the film industry.

- 28 When describing the way CGI changes actors' appearance, what does the writer suggest?
- A CGI looks best from a distance.
 - B CGI interferes with actors' facial expressions.
 - C Audiences expect too much of CGI.
 - D The scientists had hoped for more convincing results.
- 29 What does the writer suggest about the scientists' attitude to their work in the fourth paragraph?
- A They were motivated to get as near to reality as possible.
 - B They were interested in gaining recognition for their work.
 - C They thought it could have medical applications.
 - D They hoped to receive further funding for their research.
- 30 What are we told about computerized tomography?
- A It has only recently been used by doctors.
 - B It is similar to X-rays in the way it works.
 - C Filmmakers have used it for many years for special lighting effects.
 - D It can be administered using traditional radiography machines.
- 31 Which of these advantages does the writer attribute to the current use of computerized tomography?
- A It is most effective when used to create images of skin.
 - B It can make a few people seem like a crowd.
 - C It allows clothes designers to create new designs.
 - D It is practical because of the time it takes.
- 32 The writer mentions Kavita Bala in order to
- A comment on the success of CGI in commercial contexts.
 - B highlight the link between CGI and photography.
 - C show that Marschner's team are uncertain about the future of CGI.
 - D demonstrate a use for CGI outside the film industry.

Questions 33–36

*Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- 33 It used to be unusual for university researchers to receive a cinematography award.
- 34 CGI is popular among ageing actors.
- 35 The scientists' success in generating images of complete CGI garments has won them many awards.
- 36 It will be a long time before computerised tomographic fabric is used by filmmakers.

Questions 37–40

Complete the summary below. Choose the correct word, **A–I**, from the box below.

THE WORK OF MARSCHNER AND HIS COLLEAGUES

For many years, CGI characters did not appear entirely lifelike as their skin appeared plastic.

Marschner and his colleagues were the first to apply an understanding of how

37 _____ interacts with human skin. Their CGI model is based on a novel application

of principles of 38 _____, which had previously been applied in other scientific

research. The importance of CGI to the film industry has led to the 39 _____ of

Marschner's model by special-effects studios. Marschner's model has led to the

40 _____ of cinematography.

WORD LIST

- A light
- B transparency
- C age
- D use
- E astrophysics
- F mathematics
- G improvement
- H colour
- I translucency

Passage 36

Scientific Research Reveals Hidden Benefits of Regular Exercise

Exercising for up to an hour a day can improve memory and learning for both young and old, scientists have found

Concerns that children in developed countries are leading increasingly sedentary lifestyles are growing. Recent research suggests that almost nine in ten children fail to get the 60 minutes of daily exercise which is the minimum recommended for good health, and a third completed less than an hour each week. In most cases, this is because they are spending hours every day glued to televisions, the Internet and games consoles. Alarmingly, there is evidence to suggest that this lack of exercise is not only having a negative physiological effect on them, but is also adversely affecting their academic performance at school.

Psychologist Dr Aric Sigman believes that regular exercise can significantly improve pupils' academic ability, and suggests that access to high-quality PE lessons is just as likely to have a long-term impact on children's education as time spent in conventional classrooms. He also supports the long-held conviction that vigorous physical activity is much better than moderate activity. 'Children should spend at least an hour a day doing some form of vigorous exercise,' he says. And his message to schools and parents is obvious. 'Schools and parents should devise ways of increasing physical activity in and out of school time.' This, he believes, is the key to improved academic performance.

For those who are sceptical about this, and no doubt there are many, he quotes two pieces of research that underline the link between physical activity and brain capacity. One study compared brain capacity and test scores among two groups of nine- and ten-year olds, one with higher levels of physical fitness than the other. It revealed that fitter pupils had a twelve percent larger brain capacity than their peers, which was associated with better performance in cognitive tests. They were able to complete the test more quickly and got more answers correct. A second study of 1.2 million male teenagers in Sweden was perhaps even more revealing. It found that those who were fit were more likely to have a high IQ and go on to university.

Dr Sigman says, 'Physical activity is thought to help a child's cognitive processes by increasing blood and oxygen flow to the brain. This increases levels of chemicals like endorphin in the brain, which decrease stress and improve mood. It also increases growth factors that help create new nerve cells and support the connections between brain cell synapses that are the basis of learning.'

According to other researchers, there is also evidence that certain parts of the brain become larger, and that children who are fit also tend to be better at multi-tasking and performing difficult mental tasks than their unfit friends. Professor Art Kramer, director of the Beckman Institute for Advanced Science and Technology at the University of Illinois, who led the research, said their findings could have important implications for improving children's performance at school. He said it could also be used to help people combat memory loss and retain problem-solving skills in old age.

'It is a sad fact of ageing that our brain function decreases as we get older,' says Kramer. 'Increasingly, people are also living more sedentary lifestyles. While we know that exercise can have positive effects on cardiovascular disease and diabetes, we have found it can also bring about improvements in cognition and brain function. Aerobic exercise is best for this, so by starting off doing 15 minutes a day and working up to 45 minutes to an hour of continuous exercising, we can see some real improvements in cognition after six months to a year.'

Professor Kramer's team did a lot of neuroimaging work alongside their studies, which provided visual evidence to show that brain networks and structures actually change with exercise. This, they say, is the reason why their aerobically-fit test subjects were found to exhibit superior cognitive control to those who were less fit, and that regular exercise helped to improve memory, attention and an increased ability to multi-task. The hippocampus, that part of the brain involved in memory, of elderly people who exercised regularly for more than six months increased by two percent, effectively reversing brain ageing by one to two years.

Tests carried out on children also yielded some interesting results. One test involved them crossing a 'street' using a virtual reality simulation. Fitter children were better at crossing the street when distracted by music or holding a conversation on a hands-free mobile phone compared to those who were less fit. While both groups tended to walk at the same speed, the children who were less fit often misjudged the speed and distance of the computer-generated vehicles. 'The low fitness kids were just as good at crossing the street when it was the only thing they were doing,' says Kramer. 'If they were listening to music or talking on the headset, they performed badly. They often ended up with the screen going red to show they had been hit. One way to look at it is that fit children think more efficiently and so are better at multi-tasking.'

Professor Kramer presented his findings at the American Association for the Advancement of Science annual meeting in Vancouver, where other research presented showed that reducing the number of calories we consume could help to prevent brain disorders, especially in the elderly. Dr Mark Mattson, a neuroscientist from the National Institute of Ageing in Baltimore found that restricting people's diets to just 500 calories every other

day increased production of proteins that are known to protect neurons from damage. 'There is considerable evidence that doing this is not only good for your heart, but is also good for your brain', he said.

Questions 1–5

Do the following statements agree with the claims of the writer?

*Write **YES** if the statement agrees with the claims of the writer, **NO** if it contradicts them, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- ① The possible impact of a sedentary lifestyle on the way children perform in the classroom is very worrying.
- ② It is only recently that people have discovered exercise is more beneficial for you when you put more effort into it.
- ③ It is unclear what Dr Aric Sigman thinks schools and parents should do.
- ④ There are probably a lot of people who disbelieve Dr Sigman's theory.
- ⑤ Researchers were surprised to discover a link between levels of fitness in Swedish teenagers and their IQ.

Questions 6–10

Complete the summary using the list of words, A–I, below.

BENEFITS OF EXERCISE ON THE BRAIN

Dr Sigman believes that when children do physical exercise, they experience less

- ⑥ _____ as a result of a chemical change caused by increased blood and oxygen flow to the brain. Other researchers, such as Professor Art Kramer, think there is
- ⑦ _____ that some parts of the brain become bigger, and fit children are better than unfit children at doing complicated ⑧ _____ that require mental thought. Among other things, these scientific ⑨ _____ could be used to benefit the
- ⑩ _____ when it comes to fighting memory loss and keeping the skills that allow them to solve problems.

WORD LIST

- A ancients
- B anxiety
- C discoveries

- D** elderly
- E** exercises
- F** anger
- G** performances
- H** possibility
- I** proof

Questions 11–13

Choose the correct letter, **A**, **B**, **C** or **D**.

- 11 Which of the following is true, according to Professor Art Kramer?
- A** People become less active as they grow older.
 - B** Exercising is an effective way of preventing diseases like diabetes.
 - C** One particular type of exercise is more effective than others.
 - D** When exercising, you should take short breaks during the exercise period.
- 12 What did Professor Kramer's neuroimaging work show?
- A** Exercising physically alters the brain.
 - B** Unfit people have poor memories.
 - C** A particular part of the brain becomes more active during exercise.
 - D** Exercise can reduce ageing of the brain for up to two years.
- 13 What does "doing this" refer to in the final paragraph?
- A** changing the kinds of food we eat
 - B** eating no more than 500 calories a day
 - C** eating foods that are high in protein
 - D** eating less on certain days

Passage 37

Sea Change for Salinity

One of the most serious problems facing Australian farmers is an increase in the salt content in the soil – but new weapons are emerging in the fight against salinity

A Beneath the flat, impassive surface of Australia lie hidden mountains, valleys and gorges – ancient traps and channels for the deadly salt that is stealthily killing so much of the Australian landscape. The war on salt is calling forth new weapons. A suite of high technologies used by geologists to see underground and prospect for gold and minerals is now being used to pinpoint the presence of salt beneath the landscape, and predict where it might move.

B Unless this process is clearly understood, warns Chief of Exploration and Mining Dr Neil Phillips, the hard work now underway of planning and tree-planting on the surface may be rendered ineffective: salt can still sneak past and erupt, following one of the ancient river channels formed millions of years ago. The use of airborne electromagnetic to detect salt hidden beneath the landscape has been around for a decade, but the past two years have seen a major development in its precision and powers of detection. Like the use of radar in battles, it has the potential to turn the tide of the struggle in favour of the defence by helping to pinpoint, plot and predict the movements of the foe.

C Angus Howell, who farms near Warrenbayne, in Southeast Australia, saw his first outbreak of salt in 1948. Over the ensuing decades the patches spread and multiplied until they consumed almost 100 hectares. By the late 1970s, Howell and his fellow farmers had decided it was time for action and established a government-funded 'Landcare' group in a bid to save Australia's farmland. But despite a mounting effort by scientists, farmers and governments, the 'white death' continued to encroach. Small successes were eclipsed by larger defeats and fresh outbreaks.

D 'The technical solutions just aren't there yet for dealing with broadacre salinity, nor are the social and economic solutions. How do you introduce the land-use changes that are needed when people still need to make a living?' Howell asks. There is no satisfactory solution yet. Part of the problem has lain in salt's ability to mount ambushes, emerging somewhere new, sometimes unexpected and unexplained, beating plans to intercept it. Only now are scientists starting to really disclose its secret subterranean stores and passages.

E The need for such knowledge is pressing. Salt has already afflicted six million hectares of once-productive country. At present rates it is predicted that, by 2050, it will have sterilised a total of 17 million hectares and the waters of Australia's Murray River will

regularly exceed the World Health Organisation's salt limits for drinking water. Defeating this assault may take centuries, not decades.

F Electromagnetic surveys measure the electrical conductivity of soil to reveal the distribution of salt and the nature and variability of the regolith – the weathered rock and sediment that may lie above the bedrock. Magnetic surveys measure small differences in the Earth's magnetic field, enabling scientists to probe the deep past and reconstruct ancient landscapes – rivers, basins and faults now buried under tens of metres of sediments. These features help to reveal where groundwater is stored, dictate the direction of groundwater movement, and are critical to predicting or ruling out salinity hot-spots.

G Radiometric analysis is based on the detection of radiation emitted by elements contained in rocks and soils, allowing scientists to delineate landforms. These factors influence the mobility of salt through the soil profile and help determine where to plant particular crop species to tackle the problem. Using data from the Murray River region, scientists have revealed a network of ancient drainage channels buried metres beneath the current landscape. These buried channels may carry salt and sometimes run at right angles to channels on the surface. This implies that the salt could move underground in quite a different direction to what one would expect by looking at surface slope and drainage.

H One of the biggest advances in detection, says Professor Neil Phillips, has come with the integration of different techniques such as magnetics, electromagnetics and radiomagnetism, and ground mapping. Individually, these technologies only gave clues to what was going on underground. Together they provide a far more revealing picture of the subsurface landscape, several hundred metres deep. Advanced airborne electromagnetics, in particular, enables scientists to take 'slices' of the landscape at depths of five metres, ten metres, fifteen metres and so on, to determine where salt may be stored at depth. This is building up a four-dimensional picture of the subsurface landscape, enabling researchers to understand movements of salt in width, depth, breadth and time. From such technologies it will be possible to locate salt stores, identify how saline they are, look at man-made and natural changes to the landscape that may cause it to mobilise, and then predict where it will head to and over what time span. This in turn will give the salt warriors time to model various ways of containing or curbing the menace, see what works best and then try it out on the ground.

Questions 1–7

Reading Passage 3 has eight sections, A–H. Which section contains the following information?

Write the correct letter, A–H, in boxes 1–7. NB You may use any letter more than once.

Q	Statement	
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1	a prediction of the future risk of salt to water supplies	
2	the reason why technologies must be combined to be effective	
3	a reference to the recent improvements in the accuracy of airborne electromagnetics	
4	the organisation of concerned farmers into an official body	
5	the estimated length of time salinity is likely to be a problem	
6	a summary of stages in a proposed plan of action to combat the salt problem	
7	the possibility that current re-vegetation practices are a waste of time	

Questions 8–10

Look at the list of techniques (Questions 8–10) and the list of uses below. Match each technique with the correct use, **A**, **B**, **C** or **D**.

LIST OF USES

- A** can help farmers choose the best location for plants
- B** can show the composition of the top layer of the ground
- C** can detect how far below ground the salt is
- D** can determine how old the salt is in a particular area

Q	Statement	
8	Electromagnetic surveys	
9	Radiometric analysis	
10	Airborne electromagnetics	

Questions 11–14

Choose the correct letter, **A**, **B**, **C** or **D**.

- 11** What link does the writer make between salt and gold?
- A** They can both be found in the same locations.
 - B** Both have been found to have an impact on the landscape.
 - C** The same techniques can be used to find both.
 - D** Neither are present in mountainous areas.

- 12 What is the 'process' referred to in Section B?
- A the killing of vegetation by salt
 - B salt's ability to travel below ground
 - C the ability of trees to decrease salt levels
 - D the detection of salt by tracing other minerals
- 13 According to Angus Howell, one problem in the fight against salinity is that
- A not enough farmers are concerned about the fight.
 - B farmers' requests for help have been ignored.
 - C some possible measures may cause farmers to lose income.
 - D the government has not provided farmers with sufficient financial support.
- 14 Which of the following best describes the writer's view of the salinity problem in Australia?
- A Farmers are fighting an enemy that moves secretly and hides well.
 - B Farmers have been able to contain this enemy in a small area.
 - C Farmers have already had significant success in fighting this problem.
 - D Farmers need to form more organised groups to solve this problem.

Passage 38

Seeing the Colour of Sounds, Hearing the Colour of Numbers

When the senses mix together

A What is the colour of five? What is the sound of blue? To most of us such questions are either meaningless or suitable only for poetry. But for some people these are questions to which very precise answers can be given. Five, for example, for some people is green, while others say the sound of a guitar is like someone blowing on their ankles. People who 'see' colour in numbers or letters of the alphabet and 'feel' sensation in sound have synesthesia — meaning literally 'joined sensation' — an extraordinary condition that causes certain senses to 'leak' into one another.

B People whose senses behave in this way are called 'synesthetes'. Some synesthetes take pleasure in it. 'To me it's like other people see the world in black and white,' says one, who sees every letter, number, sound and pain in colour. Others learn to keep it a secret for fear of people laughing at them. But to neurologists investigating the brain it is of great interest. 'When scientists study normal perception,' says Daniel Smilek of the University of Waterloo in Ontario, Canada, 'there are lots of things we don't question because most of us perceive in the same way. Synesthesia, because it's abnormal, can give us new insights into normal perception.'

C Seventeenth-century English philosopher John Locke was the first westerner to describe synesthesia. He wrote about a man who experienced bright red as the sound of the trumpet. Later, the condition excited the imagination of nineteenth-century European painters, such as the non-synesthete Wassily Kandinsky, who believed synesthetes were like good, much-played violins, which vibrate in all their parts and fibres. But it proved impossible to research and people lost interest. Recently, however, advances in brain imaging have sparked renewed interest.

D Early claims that the multi-sensory experiences of synesthesia were linked with the hallucinations of mental illness have long been disproved. Until 1993, many researchers dismissed it as another name for a vivid imagination, but then an experiment by Simon Baron-Cohen of Cambridge University in the UK showed that synesthetes who, when tested, had linked particular colours or shapes to letters, gave the same answers in 92 per cent of instances when tested again a week later. Nonsynesthetes, given similar examples to imagine, returned the same answers in only 37 per cent of instances.

E Although there is some overlap between synesthetes as to what colour is linked to what — 56 per cent see the letter 'o' as white — most of the responses are individual. There are 30 possible sensory combinations, but links between sounds and colours are the most

common. Women are between two and eight times more likely than men to have the condition.

F As yet there is no explanation for any of this, but small pieces of the jigsaw are emerging. A brain scanning experiment by Baron-Cohen in 1995 found that when synesthetes were listening to words, areas of the brain lit up that are normally only active in response to vision and colour. From this comes the notion that we may all have synesthesia at birth, when many parts of the brain are linked, but, as we develop, connections are pruned, so our senses become separated and the synesthetic mechanism is no longer intact. Somehow, synesthetes have kept their synaptic connections intact. It's an idea that has been challenged, however, on the grounds that if you give people certain drugs they will have synesthetic experiences, which suggests the mechanism is intact in adults but repressed.

G There are many more unanswered questions about synesthesia. For example, is the synesthete's colour response to a number five triggered by actually seeing the number five written on a page, or does the synesthete 'see' the colour just by thinking of the number five? The neurologist Vilayanur Ramachandran came up with a test for synesthesia which seemed to suggest it was the sight of the number that was important. He asked people to look at a page made up of specially drawn twos and fives that were a mirror image of each other. The fives were placed at random on the page but the twos were placed to form shapes such as circles or triangles. To most people the numbers just looked like a jumble without order, but to synesthetes the patterns made by the twos were very obvious as a different colour from the fives. 'This shows they were really sensing colour in the numbers they saw,' says Ramachandran. 'Ideas don't form these kinds of patterns.'

H However, Smilek subsequently came up with evidence that what really matters is the idea of a number rather than the sight of it on a page. He asked a synesthete to do some simple mental arithmetic while looking at different coloured papers. The subject did not write her answers, but only thought of them and said them aloud. Smilek found that when the colour of the paper clashed with the colour of the answer, the subject's response was slower than when the colours were the same. An actual colour (the colour of the paper) could interfere with the colour of a number that existed only in her head (the answer to the mental arithmetic question). 'My research suggests that colour experiences coincide with the processing of meaning,' says Smilek. 'It's the concept of a number that's coloured.'

I So which is it? At present, we don't know, but in the future neurologists may be able to explain what's going on in the brains of people like the novelist Nabokov, who perceived the English 'a' as dark brown and the French 'e' as black.

Questions 27–30

Look at the following statements (Questions 27–30) and the list of people below.

Match each finding with the correct academic, **A, B, C** or **D**. **NB** You may use any letter more than once.

List of People

A Daniel Smilek

B John Locke

C Simon Baron-Cohen

D Vilyanur Ramachandran

Q	Statement	
27	For synesthetes, hearing the names of things affects the brain in the same way that looking at things does.	
28	Synesthetes are consistent in their association of a certain colour with a certain written symbol.	
29	One synesthete heard a particular musical instrument when he saw a certain colour.	
30	Because synesthetes experience things differently to other people, studying synesthesia can give fresh perspectives on how the senses usually work.	

Questions 31–35

Reading Passage 3 has nine paragraphs, **A–I**. Which paragraph contains the following information? Write the correct letter, **A–I**.

NB You may use any letter more than once.

Q	Statement	
31	an investigation into synesthesia using modern medical technology	
32	a mention of a writer who was a synesthete	
33	a definition of synesthesia	
34	two different explanations of why synesthesia is uncommon in adults	
35	a reason why some synesthetes do not tell people about their experiences	

Questions 36–40

Do the following statements agree with the information given in the passage?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 36 The painter Kandinsky admired synesthetes.
- 37 It is no longer believed that synesthesia is associated with mental illness.
- 38 Equal numbers of men and women have synesthesia.
- 39 Synesthetes have greater powers of logic than people with normal perception.
- 40 It is possible that we are all born with synesthesia.

@bekhruzposts

Passage 39

Shoemaker-Levy 9

The discovery, structure and Jovian collision of a remarkable "string of pearls" comet

A The last half of July 1994 witnessed much interest among astronomers and the wider public in the collision of comet Shoemaker-Levy (SL9) with Jupiter. The comet was discovered on 25 March 1993 by Eugene and Carolyn Shoemaker and David Levy, using the 450 mm Schmidt camera at the Mount Palomar Observatory. The discovery was based on a photographic plate exposed two days earlier. The Shoemakers are particularly experienced comet hunters with 61 discoveries to their credit. Their technique relies on the proper motion of a comet to identify the object as a non-stellar body. They photograph large areas of the sky, typically with an eight-minute exposure, and repeat the photograph 45 minutes later. Comparison of the two photographs with a stereo-microscope reveals any bodies which have moved against the background of fixed stars.

① **B** As so often in science, serendipity played a large part in the discovery of Shoemaker-Levy 9! The weather on the night of 23 March was so poor that the observers would not normally have bothered putting film into their camera. However, they had a box of old film to hand which had been partially exposed by accident some days previously, so decided to insert it into the camera rather than waste good film. Fortunately, two of the film plates, despite being fogged round the edges, captured the first image of a very strange, bar-shaped object. This object, which Carolyn Shoemaker first described as a squashed comet, later became known as comet Shoemaker-Levy 9.

② **C** Other, more powerful, telescopes revealed that the comet was, in fact, composed of 21 cometary fragments, strung out in a line, which accounted for the unusual shape. The term string of pearls was soon coined. Figure 1 is a mosaic taken by the Hubble Space Telescope (HST) during 24-27 January 1994. It shows the main fragments which at that time spanned a linear distance of approximately 600,000 km. Initially, the fragments were surrounded by extensive dust clouds in the line of the nuclei but these later disappeared. Some of the nuclei also faded out (presumably due to disintegration), while others split into multiple fragments.

③ **D** The size of the original comet and each of the fragments was, and still is, something of a mystery. The first analysis of the orbital dynamics of the fragments suggested that the comet was originally some 2.5 km in diameter with an average fragment diameter of 0.75 km. Later work gave corresponding diameters of approximately 10 km and 2 km and these values are now considered more likely. There was considerable variation in the diameters of different fragments.

④ **E** Further calculations revealed that the cometary fragments were on course to collide with Jupiter during July 1994, and that each fragment could deliver an energy equivalent to approximately 500,000 million tonnes of TNT. The prospect of celestial fireworks on such a grand scale immediately captured the attention of astronomers worldwide!

⑤ **F** Each fragment was assigned an identity letter A-W (letters "I" and "O" were not used to avoid potential confusion with numbers "1" and "0") and a coordinated program of observations was put in place world-wide to track their progress towards impact with Jupiter. As the cometary fragments reached the cloud tops of Jupiter, they were travelling at approximately 60 km/s and the chain of fragments had spread out to cover approximately 30 million km. The impacts occurred during 16-22 July. All took place at a latitude of approximately 48°S, which nominally placed them in the South Southern Temperate Region; however, visually, they appeared close to the Jovian polar region. Although the impacts all occurred some 10-15° round the limb on the far side of the planet as seen from Earth, the rapid rotation of the planet (a Jovian day is only some 10 hours long) soon carried them into the view of Earth-based telescopes. The collisions lived up to all but the wildest expectations and provided a truly impressive spectacle.

G Jupiter is composed of a relatively small core of iron and silicates surrounded by hydrogen. In the depths of the planet (approximately 1000 km and more below the visible cloud tops) the hydrogen is so compressed that it is metallic in form; further from the centre, the pressure is lower and the hydrogen is in its normal molecular form. The Jovian cloud tops visible from Earth consist primarily of methane and ammonia, with relatively small amounts of other elements and compounds which are thought to be responsible for the colours seen in the atmosphere.

H The smaller cometary fragments plunged into Jupiter, rapidly disintegrated and left little trace; three of the smallest fragments, namely T, U and V, left no discernible traces whatsoever. However, many of the cometary fragments were sufficiently large to produce a spectacular display. Each large fragment punched through the cloud tops, heated the surrounding gases to some 20,000 K on the way, and caused a massive plume or fireball up to 2000 km in diameter to rise. Before encountering thicker layers of the atmosphere and disintegrating in a mammoth shock wave, the large fragments raised dark dust particles and ultra-violet (UV) absorbing gases high into the Jovian cloud tops; in visible light, this material manifested itself as a dark scar surrounding the impact site.

I Some days after collision the impact sites began to evolve and fade as they became subject to the dynamics of Jupiter's atmosphere. No-one knows how long they will remain visible from Earth, but it is thought that the larger scars may persist for a year or more. The interest of professional astronomers in Jupiter is now waning and valuable work can therefore be performed by amateurs in tracking the evolution of the collision scars. The

scars are easily visible in a modest telescope, and a large reflector will show them in some detail. There is scope for valuable observing work from now until Jupiter reaches conjunction with the Sun in November 2004.

J Astronomers and archivists are now searching old records for possible previously unrecognised impacts on Jupiter. Several spots were reported from 1690 to 1872 by observers including William Herschel and Giovanni Cassini. The records of the BAA in 1927 and 1948 contain drawings of Jupiter with black dots or spots visible. It is possible that comet impacts have been observed before, without their identity being realised, but no-one can be sure.

Questions 1–5

Choose the most suitable headings for paragraphs **B–F** from the list of headings below.

- i Camera settings for observation
- ii Collisions on stage
- iii Size of the comet
- iv String of pearls
- v Scientific explanations
- vi Hubble Space Telescope
- vii First discovery of the squashed comet
- viii Power generated from the collisions
- ix Calculations, expectations and predictions
- x Change of the fragment's shape

Questions 6–9

Reading Passage 3 contains 10 paragraphs, **A–J**. Which paragraphs state the following information?

Write the correct letter, **A–J**, in boxes 6–9 on your answer sheet.

Q	Statement	
6	Shoemaker-Levy 9 comets had been accidentally detected.	
7	The collision caused a spectacular vision on Jupiter.	

8	Every single element of Shoemaker-Levy 9 was labeled.	
9	Visual evidence explains the structure of Shoemaker-Levy 9.	

Questions 10–14

Complete the summary below. Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

The core of Jupiter, which is enclosed by hydrogen, consists of ⑩ _____ and ⑪ _____. Hydrogen is in metallic form as it is squeezed by pressure generated from the depths of the planet. The pressure is gradually reduced from the centre to the outside layers, where hydrogen is in normal form of ⑫ _____.

Far from the ground, methane and ammonia structure the ⑬ _____, which can be observed from Earth. Colours seen in the atmosphere are largely due to other particles — ⑭ _____ — in the cloud.

Passage 40

Some views on the use of headphones

Whether wearing headphones at work, or in other areas of everyday life, is a good thing or a bad thing has generated a lot of research and opinion

To visit a typical modern office today is to walk into a room with possibly a dozen songs playing simultaneously but to hear none of them. Up to half of younger workers listen to music on their headphones, and nearly all of them think it makes them better at their jobs. In survey after survey, people report with confidence that music makes them happier, better at concentrating, and more productive.

Scientists do not share this belief; they maintain that listening to music hurts people's ability to recall other things they should be doing, and any pop song, loud or soft, reduces overall performance for both extroverts and introverts. A Taiwanese study linked music that has lyrics to lower marks on concentration tests for college students, and other research has shown music with lyrics scrambles our brains' verbal-processing skills. 'As silence has the best overall performance, it would still be advisable that people work in silence,' another reporter dryly concluded.

The question is therefore: if headphones are so bad for productivity, why do so many people at work have them? One factor to consider is that countries like the USA have moved from a farming and manufacturing economy to a service economy, with an emphasis on jobs in offices that require higher levels of concentration, reflection and creativity. As an estimated 70 percent of office workers work in open-plan office spaces, it is more important to create one's own enclosing bubble of sound. Lending strength to the argument for headphones at work is evidence that music relaxes our muscles, improves our mood, and may even moderately reduce blood pressure, heart rate and anxiety.

The story of headphones began in 1910, when the US Navy received an odd letter written in purple ink on blue-and-pink paper. The letter writer, an eccentric inventor and repairman named Nathaniel Baldwin, from the USA state of Utah, made what at the time was an astonishing claim: he had built, in his kitchen, a new kind of headset that could amplify sound. This was an opportune invention for the Navy, who asked for a sound test and then enthusiastically adopted the headsets, later called headphones, and used them in World War I for naval radio communication.

The purpose of headphones is to concentrate a quiet and private sound in the ear of the listener, which is a radical departure from music's social purpose in history. 'Music, together with dance, co-evolved biologically and culturally to serve as a technology of social bonding,' Nills L Wallin and Bjorn Merker wrote in *The Origins of Music*. Songs don't leave behind fossils, but evidence of musical notation dates back to Sumeria, 3,500

years ago, and in 1995 archaeologists discovered a bone flute in southern Europe estimated to be 44,000 years old. If music evolved as a social glue for the species, as a way to make groups and keep them together, headphones have done what writing and literacy did for language — they made music private.

Author and columnist Stephen Marche wrote that separation from other people is one of the first things ordinary Americans spend their money achieving. It is 'a by-product of a long-standing national appetite for independence,' he said. Americans are not alone in their desire for personal independence and privacy. Marche is right; wealth can buy — and modern technology can deliver — personal independence, and it is this that people have always sought.

Dr Michael Bull, an expert on personal music devices from the University of Sussex in the UK, has repeatedly made the larger point that personal music devices change how we relate to public spaces. Controlling our public spaces is more important now that more people are moving from the edges of cities to live in urban centres. 'With the urban space, the more it's inhabited, the safer you feel,' Bull says. 'You feel safe if you can feel people there, but you don't want to interact with them.' Headphones create shields for wearers, separating them from other people and their surroundings.

Headphones have their own rules of good manners; they are like wearing a 'Do not disturb' sign. We assume that people wearing them are busy and we should respect their privacy, so now people wear them to appear busy. In fact, it is now becoming quite common for people not to listen to anything at all, but just to wear headphones.

However, as pointed out at the beginning of this piece, although scientists have stated that headphones are bad for productivity, people still wear them at work. It is not just that headphones create privacy out of public areas, but also that music causes people to relax and reflect and pause. The outcome of relaxation, reflection and pausing at work won't be captured in minute-to-minute productivity metrics. What must be considered is that in moments of extreme focus, our attention radiates outward, toward the problem, rather than inward, on how to solve the problem. However, with music 'When our minds are at ease, we're more likely to direct the spotlight of attention inward,' Jonah Lehrer wrote in his book *Imagine: How Creativity Works*. 'The answers have been there all along. We just weren't listening.' In a crowded world, real estate is the ultimate scarce resource, and a headphone is a small invisible fence around our minds — making space, creating separation, and helping us listen to ourselves.

Questions 27–31

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write **YES, NO** or **NOT GIVEN**.

- 27 Young people are easily persuaded by surveys that listening to music is beneficial.
- 28 Different studies share the same conclusions about the desirability of working in silence.
- 29 Some doctors recommend wearing headphones to lower blood pressure.
- 30 Nathaniel Baldwin was a respected government researcher.
- 31 The effect of the invention of headphones is comparable to the effect of the invention of writing.

Questions 32–36

Choose the correct letter, **A, B, C** or **D**.

- 32 What does the writer suggest about a service economy?
 - A The work is mentally demanding.
 - B It provides employment for younger workers.
 - C It is a small part of a country's economy.
 - D Workers have to live in urban centres.
- 33 When the writer mentions the historical evidence for early music he is
 - A emphasizing the diversity of musical forms.
 - B expressing his frustration with the limited archaeological evidence uncovered.
 - C lending support to the view that music has been important in human history.
 - D creating a geographical map of the evolution of music.
- 34 What does the writer say about the social effects of listening to music through headphones?
 - A It has caused a reduction in the number of people who listen to music.
 - B It has increased people's participation in music events.
 - C It has reduced the global variation of music styles.
 - D It has changed the traditional role of music in society.
- 35 What does the writer say about personal independence?
 - A Americans are unique in their desire for personal independence.
 - B Personal independence is something that can be purchased.
 - C Striving for personal independence is a recent phenomenon.
 - D Personal independence destroys social connections.
- 36 Why does the writer quote Jonah Lehrer in the last paragraph?
 - A to support the writer's own view.
 - B to draw attention to an authoritative book about music.
 - C to raise awareness of people's loss of listening skills.
 - D to illustrate how music brings people closer to each other.

Questions 37–40

Complete the summary using the list of words, **A–I**, below. Drag a word into a box, or tap a word then tap a box. Tap the × to clear a box.

WORD LIST

- A** courtesy
- B** relationship
- C** difficulty
- D** countryside
- E** suburbs
- F** language
- G** barriers
- H** obstacles
- I** disapproval

Headphones and city living Dr Michael Bull believes that listening to music through headphones has changed the **37** (37) ————— the wearers of headphones have with public spaces. Living in the centre of cities is becoming popular, as people become less keen on living in the **38** (38) —————. In densely populated city centres, headphones form **39** (39) ————— that isolate people from fellow citizens and from their environment. Wearers of headphones are treated with **40** (40) ————— that other people do not receive. This is because if we see someone wearing headphones, we believe they must be occupied in some way and should not be interrupted.

Passage 41

Strictly English

British newspaper columnist Simon Heffer talks about his new book, 'Strictly English: the Correct Way to Write ... and Why It Matters', aimed at native speakers

For the last couple of years, I have sent a round-robin email to my colleagues at this newspaper every few weeks pointing out to them mistakes that we make in our use of the English language. Happily, these are reasonably rare. The emails have been circulated on the Internet – and are now available on the paper's website – and one of them ended up in the inbox of a publisher at Random House about this time last year. He asked me whether I would write a book not just on what constituted correct English, but also why it matters. The former is relatively easy to do, once one has armed oneself with the Oxford English Dictionary (OED) and some reputable grammar books by way of research materials. The latter, being a matter for debate, is less straightforward.

I suppose my own interest in language started at school. Having studied French, Latin and Greek, I saw clearly how those languages had exported words into our own. When I studied German later on, I could see even more clearly why it was the sister tongue and what an enormous impact it had had on English. I saw that words had specific meanings and that, for the avoidance of doubt, it was best to use them in the correct way. Most of all, I became fascinated by grammar, and especially by the logic that drove it and that was common to all the other languages I knew. I did not intend in those days to earn a living by writing; but I was keen to ensure that my use of English was, as far as possible, correct.

Studying English at university forced me to focus even more intently on what words actually meant: why would a writer choose that noun rather than another and why that adjective – or, in George Orwell's case, often no adjective at all. Was the ambiguity in a certain order of words deliberate or accidental? The whole question of communication is rooted in such things. For the second part of my degree, I specialized in the history of the English language, studying how words had changed their meaning and how grammar had evolved. Language had become not just a tool for me, but something of a hobby.

Can English, though, ever be fixed? Of course not: if you read a passage from Chaucer you will see that the meaning of words and the framework of grammar has shifted over the centuries, and both will continue to evolve. But we have had a standard dictionary now ever since the OED was completed in 1928, and learned men, many of whom contributed to the OED, wrote grammars a century ago that settled a pattern of language that was logical and free from the danger of ambiguity.

It is to these standards that I hope Strictly English is looking. Our language is to a great extent settled and codified, and to a standard that people recognize and are comfortable with. All my book does is describe and commend that standard, and help people towards a capable grasp of the English tongue. We shall always need new words to describe new things; but we don't need the wrong word to describe the right thing, when the right word exists. Also, English grammar shouldn't be a matter for debate. It has a coherent and logical structure and we should stick to it.

Some groups of people – state officials, academics, lawyers, certain breeds of scientist – talk to each other in a private language. Some official documents make little sense to lay people because they have to be written in a language that combines avoidance of the politically incorrect with constant use of the contemporary jargon of the profession. Some articles written by academics in particular are almost incomprehensible to those outside their circle. This is not because the outsiders are stupid. It is because the academics feel they have to write in a certain stilted, dense way in order to be taken seriously by their peers.

Many officials seem to have lost the knack of communicating with people outside their closed world. Some academics, however, are bilingual. If asked to write for a publication outside the circle – such as a newspaper – they can rediscover the knack of writing reasonably plain English. They do not indulge themselves in such a fashion when they write for learned journals. It is almost as though the purpose of such writing is not to be clear: that the writer is recording research in order to prove to peers or superiors that he has discovered something. It does not seem to bother such people that their style is considered ugly and barbaric by anyone of discernment. It is repetitious, long-winded, abstract and abstruse. Those who write in such a way probably will not easily be discouraged, unless what is considered acceptable within their discipline's changes.

The ideal style is one comprehensible to any intelligent person. If you make a conscious decision to communicate with a select group, so be it: but in trying to appeal to a large audience, or even a small one that you wish to be sure will understand your meaning, writing of the sort mentioned above will not do. This sort of writing used to be kept from the general public thanks to the need to find someone to publish it. The advent of the Internet means that it is now much more widespread than it used to be; and the fact that it is now so common and so accessible means that this sort of writing is having a harmful effect on the language and causing it to be corrupted.

Questions 1–4

Do the following statements agree with the views of the writer in the Reading Passage? Write **YES** if the statement agrees with the writer's view, **NO** if the statement contradicts the writer's view, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 1 The mistakes made by his colleagues are minor ones.
- 2 It is difficult to explain why using correct English is important.
- 3 English grammar has a different function from the grammar of other languages.
- 4 Word order may be as important as the choice of words used.

Questions 5–9

Complete the summary using the list of words, **A–H**, below.

WORD LIST

- A simple
- B general
- C controversial
- D permanent
- E knowledgeable
- F compulsory
- G historic
- H useful

THE RULES OF ENGLISH

According to the writer, the English language should not be considered something

5 ———, and this will always be the case. However, there have been accepted reference books for over a century that were produced by 6 ——— people, and these have established a system for the language that enables people to express themselves in a completely clear way. In his own book, the writer aims to describe and support the established rules of the Language that are in 7 ——— use and that people are accustomed to. He also wants his book to be 8 ——— as a way of improving people's ability at the language. He believes that there is no reason why someone's use of vocabulary should not be correct and that grammar should not be a 9 ——— subject. In his view, a system of grammar rules exists and people should always obey those rules.

Questions 10–14

Choose the correct letter, **A, B, C** or **D**.

- 10 The writer says that some groups of people use a 'private language' because
- A they do not want outsiders to be able to understand them.
 - B they want to show their superiority over other groups.
 - C they want to impress other members of their group.
 - D they do not want to use the same language as other groups.
- 11 According to the writer, some academics are capable of
- A making sense to people outside their group.
 - B writing very clearly for learned journals.
 - C changing the way they communicate within their own group.
 - D explaining other people's work to the general public.
- 12 When discussing the writing of academics about this research, the writer emphasizes
- A his own lack of knowledge of the academic world.
 - B his desire to understand what they describe.
 - C his sympathy for some of the academics.
 - D his dislike for the style used in their writing.
- 13 The writer says that the kind of language used by academics in journals
- A is becoming more widely understood by non-academics.
 - B is attracting a lot of criticism from other academics.
 - C will only change if they are forced to change it.
 - D appeals only to highly intelligent people.
- 14 The writer's opinion of the internet is that
- A it is making people more aware of the poor use of language.
 - B it is encouraging standards of language use to fall.
 - C it is enabling people to compare good and poor use of language.
 - D it is making it harder for good writing to get published.

Passage 42

Sun, Sand and Seawater

A lighting engineer's chance discovery on a Moroccan bus becomes a working greenhouse that turns desert, sun and seawater into fresh water and crops

A Sometimes ideas just pop up out of the blue. Or in Charlie Paton's case, out of the rain. 'I was on a bus in Morocco travelling through the desert,' he remembers. 'It had been raining and the bus was full of hot, wet people. The windows steamed up and I went to sleep with a towel against the glass. When I woke, the thing was soaking wet. I had to wring it out. And it set me thinking. Why was it so wet?'

B The answer, of course, was condensation. Back home in London, a physicist friend, Philip Davies, explained that the glass, chilled by the rain outside, had cooled the hot humid air inside the bus below its dew point, causing droplets of water to form on the inside of the window. Intrigued, Paton – a lighting engineer by profession – started rigging up his own equipment. 'I made my own solar stills. It occurred to me that you might be able to produce water in this way in the desert, simply by cooling the air. I wondered whether you could make enough to irrigate fields and grow crops.'

C Today, a decade on, his dream has taken shape as a giant greenhouse on a desert island off Abu Dhabi in the Persian Gulf – the first commercially viable version of his 'seawater greenhouse'. Local scientists, working with Paton, are watering the desert and growing vegetables in what is basically a giant dew-making machine that produces freshwater and cool air from sun and seawater. In awarding Paton first prize in a design competition two years ago, Marco Goldschmied, president of the Royal Institute of British Architects, called it 'a truly original idea which has the potential to impact on the lives of millions of people living in coastal water-starved areas around the world'.

D The seawater greenhouse as developed by Paton has three main parts. Two of them air-condition the greenhouse and provide water for irrigation. The front of the greenhouse faces into the prevailing wind so that hot dry air blows in through a front wall. The wall is made of perforated cardboard kept moist by a constant trickle of seawater pumped up from the ocean. The purpose is to cool and moisten the incoming desert air. The cool moist air allows the plants to grow faster. And, crucially, because much less water evaporates from the leaves, the plants need much less moisture to grow than if they were being irrigated in the hot dry desert air outside the greenhouse.

E The air-conditioning of the interior of the greenhouse is completed by the second feature: the roof. It has two layers: an outer layer of clear polyethylene and an inner coated layer that reflects infrared radiation. This combination ensures that visible light can

stream through to the plants, maximising the rate of plant growth through photosynthesis, but at the same time heat from the infrared radiation is trapped in the space between the layers, and kept away from the plants. This helps keep the air around the plants cool.

F At the back of the greenhouse sits the third element. This is the main water production unit. Here, the air hits a second moist cardboard wall that increases its humidity as it reaches the condenser, which finally collects from the hot humid air the moisture for irrigating the plants. The condenser is a metal surface kept cool by still more seawater. It is the equivalent of the window on Paton's Moroccan bus. Drops of pure distilled water form on the condenser and flow into a tank for irrigating the crops.

G The Abu Dhabi greenhouse more or less runs itself. Sensors switch everything on when the sun rises and alter flows of air and seawater through the day in response to changes in temperature, humidity, and sunlight. On windless days, fans ensure a constant flow of air through the greenhouse. 'Once it is tuned to the local environment, you don't need anyone there for it to work,' says Paton. 'We can run the entire operation off one 13-amp plug, and in the future, we could make it entirely independent of the grid, powered from a few solar panels.'

H Critics point out that construction costs of around \$4 a square foot are quite high. By way of illustration, however, Paton points out that it can cool as efficiently as a 500-kilowatt air conditioner while using less than 3 kilowatts of electricity. Thus, the plants need only an eighth of the volume of water used by those grown conventionally. And so, the effective cost of the desalinated water in the greenhouse is only a quarter that of water from a standard desalinator, which is good economics. Besides, it really suggests an environmentally-friendly way of providing air conditioning on a scale large enough to cool large greenhouses where crops can be grown despite the high outside temperatures.

Questions 1–5

*Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

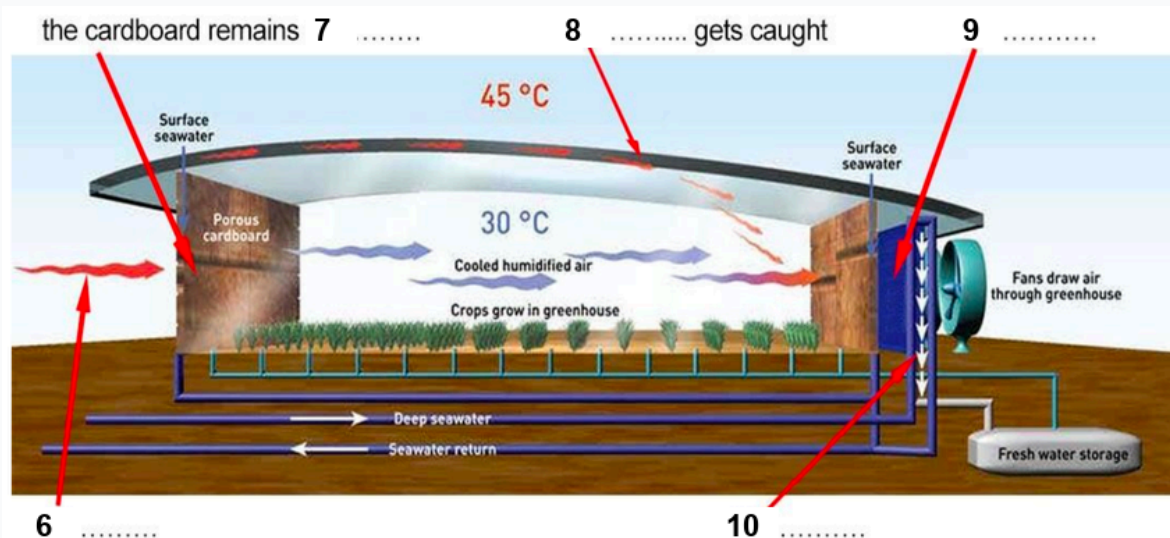
- 1 The idea just came to Charlie Paton by accident.
- 2 The bus was well ventilated.
- 3 After waking up, Paton found his towel was wet.
- 4 The fan on the bus did not work well.
- 5 Paton immediately operated his own business in the Persian Gulf after talking with Philip Davies.

Questions 6–10

Label the diagram below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

SEAWATER GREENHOUSE DIAGRAM



- ⑥ ————— blows in from outside and reaches the wall where the cardboard remains ⑦ —————.
- In the roof, ⑧ ————— gets caught between the two layers, keeping the air around the plants cool.
- At the back, the humid air reaches the ⑨ —————, which is kept cool by seawater.
- Drops of ⑩ ————— form on the condenser and flow into the fresh water storage tank.

Questions 11–14

Complete the summary below.

Using **NO MORE THAN TWO WORDS** from the Reading Passage for each answer.

To some extent, the Abu Dhabi greenhouse functions automatically. When the day is sunny, the equipment can respond to the changes in several natural elements.

- When there is no wind, ⑪ ————— help to retain the flow of air.

- Even in the future, there is an ideal plan to power the greenhouse from (12) _____.
- However, there are still some critics who argue that (13) _____ are not good economics.
- To justify himself, Paton presents favourable arguments against these critics and suggests that it is an (14) _____ approach to provide air conditioning on a large scale.

Passage 43

Telepathy

Why the scientific community remains divided over whether human beings can communicate by thought alone

Can human beings communicate by thought alone? For more than a century the issue of telepathy has divided the scientific community, and even today it still sparks bitter controversy among top academics.

Since the 1970s, parapsychologists at leading universities and research institutes around the world have risked the derision of sceptical colleagues by putting the various claims for telepathy to the test in dozens of rigorous scientific studies. The results and their implications are dividing even the researchers who uncovered them.

Some researchers say the results constitute compelling evidence that telepathy is genuine. Other parapsychologists believe the field is on the brink of collapse, having tried to produce definitive scientific proof and failed. Sceptics and advocates alike do concur on one issue, however: that the most impressive evidence so far has come from the so-called 'ganzfeld' experiments, a German term that means 'whole field'. Reports of telepathic experiences had by people during meditation led parapsychologists to suspect that telepathy might involve 'signals' passing between people that were so faint that they were usually swamped by normal brain activity. In this case, such signals might be more easily detected by those experiencing meditation-like tranquillity in a relaxing 'whole field' of light, sound and warmth.

The ganzfeld experiment tries to recreate these conditions with participants sitting in soft reclining chairs in a sealed room, listening to relaxing sounds while their eyes are covered with special filters letting in only soft pink light. In early ganzfeld experiments, the telepathy test involved identification of a picture chosen from a random selection of four taken from a large image bank. The idea was that a person acting as a 'sender' would attempt to beam the image over to the 'receiver' relaxing in the sealed room. Once the session was over, this person was asked to identify which of the four images had been used. Random guessing would give a hit-rate of 25 per cent; if telepathy is real, however, the hit-rate would be higher. In 1982, the results from the first ganzfeld studies were analysed by one of its pioneers, the American parapsychologist Charles Honorton. They pointed to typical hit-rates of better than 30 per cent – a small effect, but one which statistical tests suggested could not be put down to chance.

The implication was that the ganzfeld method had revealed real evidence for telepathy. But there was a crucial flaw in this argument – one routinely overlooked in more conventional areas of science. Just because chance had been ruled out as an explanation did not prove

telepathy must exist; there were many other ways of getting positive results. These ranged from 'sensory leakage' – where clues about the pictures accidentally reach the receiver – to outright fraud. In response, the researchers issued a review of all the ganzfeld studies done up to 1985 to show that 80 per cent had found statistically significant evidence. However, they also agreed that there were still too many problems in the experiments which could lead to positive results, and they drew up a list demanding new standards for future research.

After this, many researchers switched to autoganzfeld tests – an automated variant of the technique which used computers to perform many of the key tasks such as the random selection of images. By minimising human involvement, the idea was to minimise the risk of flawed results. In 1987, results from hundreds of autoganzfeld tests were studied by Honorton in a 'meta-analysis', a statistical technique for finding the overall results from a set of studies. Though less compelling than before, the outcome was still impressive.

Yet some parapsychologists remain disturbed by the lack of consistency between individual ganzfeld studies. Defenders of telepathy point out that demanding impressive evidence from every study ignores one basic statistical fact: it takes large samples to detect small effects. If, as current results suggest, telepathy produces hit-rates only marginally above the 25 per cent expected by chance, it's unlikely to be detected by a typical ganzfeld study involving around 40 people: the group is just not big enough. Only when many studies are combined in a meta-analysis will the faint signal of telepathy really become apparent. And that is what researchers do seem to be finding.

What they are certainly not finding, however, is any change in attitude of mainstream scientists: most still totally reject the very idea of telepathy. The problem stems at least in part from the lack of any plausible mechanism for telepathy.

Various theories have been put forward, many focusing on esoteric ideas from theoretical physics. They include 'quantum entanglement', in which events affecting one group of atoms instantly affect another group, no matter how far apart they may be. While physicists have demonstrated entanglement with specially prepared atoms, no-one knows if it also exists between atoms making up human minds. Answering such questions would transform parapsychology. This has prompted some researchers to argue that the future lies not in collecting more evidence for telepathy, but in probing possible mechanisms. Some work has begun already, with researchers trying to identify people who are particularly successful in autoganzfeld trials. Early results show that creative and artistic people do much better than average: in one study at the University of Edinburgh, musicians achieved a hit-rate of 56 per cent. Perhaps more tests like these will eventually give the researchers the evidence they are seeking and strengthen the case for the existence of telepathy.

Questions 27–30

Complete each sentence with the correct ending, **A–G**, below. Write the correct letter in each box.

- 27 Researchers with differing attitudes towards telepathy agree on (27) _____
- 28 Reports of experiences during meditation indicated (28) _____
- 29 Attitudes to parapsychology would alter drastically with (29) _____
- 30 Recent autoganzfeld trials suggest that success rates will improve with (30) _____

LIST OF ENDINGS

- A** the discovery of a mechanism for telepathy.
- B** the need to create a suitable environment for telepathy.
- C** their claims of a high success rate.
- D** a solution to the problem posed by random guessing.
- E** the significance of the ganzfeld experiments.
- F** a more careful selection of subjects.
- G** a need to keep altering conditions.

Questions 31–35

Complete the table below. Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Name / Date	Description	Result	Flaw
Ganzfeld studies 1982	Involved a person acting as a (31) _____, who picked out one (32) _____ from a random selection of four, and a (33) _____, who then tried to identify it.	Hit-rates were higher than with random guessing.	Positive results could be produced by factors such as (34) _____ or (35) _____.

Questions 36–40

Complete the table below. Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Name / Date	Description	Result	Flaw
Autoganzfeld studies 1987	_____ were _____ used for key tasks to limit the amount of _____ in carrying out the tests.	The results were then subjected to a _____.	The _____ between different test results was put down to the fact that sample groups were not _____ (as with most ganzfeld studies).

Passage 44

Termite Mounds

Could the vast towers of mud constructed by insects in sub-Saharan Africa hold the key to our energy-efficient building of the future?

27 1 To most of us, termites are destructive insects which can cause damage on a devastating scale. But according to Dr Rupert Soar of Loughborough University's School of Mechanical and Manufacturing Engineering, these pests may serve a useful purpose for us after all. His multi-disciplinary team of British and American engineers and biologists have set out to investigate the giant mounds built by termites in Namibia, in sub-Saharan Africa, as part of the most extensive study of these structures ever taken.

28 2 Termite mounds are impressive for their size alone; typically they are three metres high, and some as tall as eight metres have been found. They also reach far into the earth, where the insects 'mine' their building materials, carefully selecting each grain of sand they use. The termite's nest is contained in the central cavity of the mound, safely protected from the harsh environment outside. The mound itself is formed of an intricate lattice of tunnels, which split into smaller and smaller tunnels, much like a person's blood vessels.

29 3 This complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound. It also serves to expel spent respiratory gases from the nest to prevent the termites from suffocating, so ensuring them a continuous provision of fresh, breathable air. So detailed is the design that the nest stays within three degrees of a constant temperature, despite variations on the outside of up to 50°C, from blistering heat in the daytime to below freezing on the coldest nights. The mound also automatically regulates moisture in the air, by means of both its underground 'cellar', and evaporation from the top of the mound. Some colonies even had 'chimneys' at a height of 20cm to control moisture loss in the hottest regions of sub-Saharan Africa.

30 4 Furthermore, the termites have evolved in such a way as to outsource some of their biological functions. Part of their digestive process is carried out by a fungus, which they farm inside the mound. This fungus, which is found nowhere else on earth, thrives in the constant and optimum environment of the mound. The termites feed the fungus with slightly chewed wood pulp, which the fungus then breaks down into a digestible sugary food to provide the insects with energy, and cellulose which they use for building. And, although the termites must generate waste, none ever leaves the structure, indicating that there is also some kind of internal waste-recycling system.

31 5 Scientists are so excited by the mounds that they have labelled them a 'super organism' because, in Soar's words: 'They dance on the edge of what we would perceive to be alive,' he explains. 'If you're too hot you need to cool down, or if you're too cold you

need to warm up: that's called homeostasis. What the termites have done is to move homeostatic function away from their body, into the structure in which they live.' As more information comes to light about the unique features of termite mounds, we may ultimately need to redefine our understanding of what constitutes a 'living' organism.

③② 6 To reveal the structure of the mounds, Soar's team begins by filling and covering them with plaster of Paris, a chalky white paste based on the mineral gypsum, which becomes rock-solid when dry. The research team then carves the plaster of Paris into half-millimetre-thick slices, and photographs them sequentially. Once the pictures are digitally scanned, computer technology is able to recreate complex three-dimensional images of the mounds. These models have enabled the team to map termite architecture at a level of detail never before attained.

③③ 7 Soar hopes that the models will explain how termite mounds create a self-regulating living environment which manages to respond to changing internal and external conditions without drawing on any outside source of power. If they do, the findings could be invaluable in informing future architectural design, and could inspire buildings that are self-sufficient, environmentally, and cheap to run. 'As we head into a world of climate change, with temperatures continuing to rise,' he explains, 'there will not be enough fuel to drive air conditioners around the world.' 'It is hoped,' says Soar, 'that the findings will provide clues that aid the ultimate development of new kinds of human habitats, suitable for a variety of arid, hostile environments not only on the earth but maybe one day on the moon and beyond.'

Questions 27–33

Reading Passage 3 has seven paragraphs, numbered 1–7.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

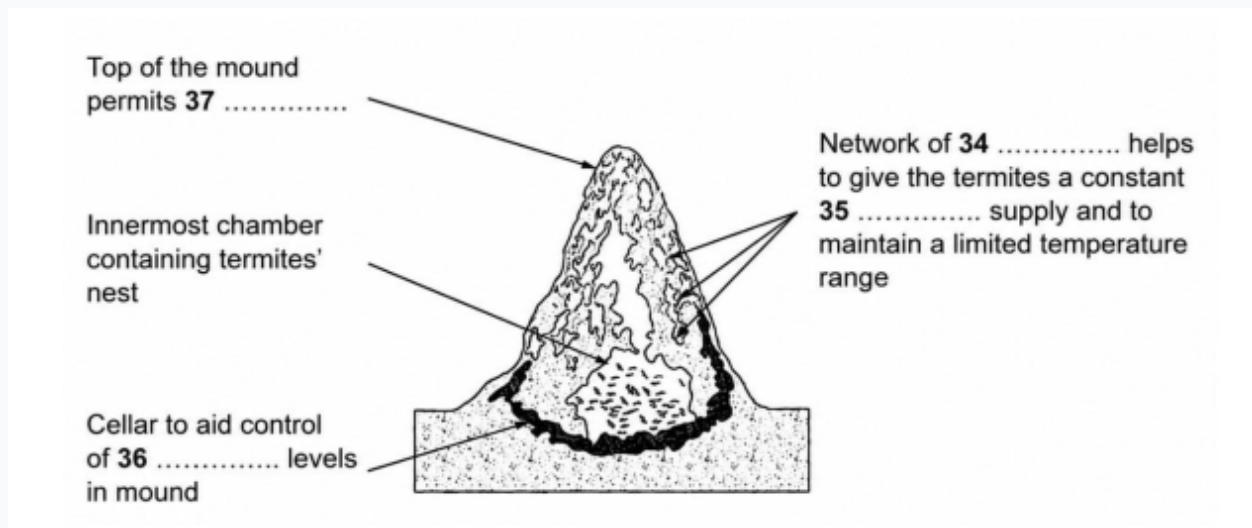
- i Methods used to investigate termite mound formation
- ii Challenging our assumptions about the nature of life
- iii Reconsidering the termite's reputation
- iv Principal functions of the termite mound
- ix A mutually beneficial relationship
- v Distribution of termite mounds in sub-Saharan Africa
- vi Some potential benefits of understanding termite architecture
- vii The astonishing physical dimensions of the termite mound

viii Termite mounds under threat from global climate change

Questions 34–37

Complete the diagram below. Write **ONE WORD ONLY** from the passage for each answer.

TERMITE MOUND CROSS-SECTION



Top of the mound permits (37) _____.

Innermost chamber contains the termites' nest.

Network of (34) _____ helps give the termites a constant (35) _____ supply and helps maintain a limited temperature range.

Cellar aids control of (36) _____ levels in the mound.

Questions 38–40

Do the following statements agree with the claims of the writer?

Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- (38) The termite mound appears to process its refuse material internally.
- (39) Dr Soar's reconstruction involves scanning a single photograph of a complete mound into a computer.
- (40) New information about termite architecture could help people deal with future energy crises.

Passage 45

The Algebra Debate

Is algebra really necessary for everyone? Educators, researchers and policymakers disagree

It is widely agreed upon that mathematical skills and knowledge are crucial for future scientists, computer programmers, and engineers, but for the rest of us, it is unclear what level of mastery is needed in this field. In recent years, the necessity of education in mathematics has been brought into question by those in the education field, social scientists, and even politicians. However, some teachers feel that mathematical skills will be an integral part of America's future success. It is believed by many that competence in the STEM fields – science, technology, engineering, and mathematics – will keep the United States economically competitive in the ever-developing and accelerating world economy. Some educators believe that if students do not take algebra classes by ninth grade, they will fall behind and not have the requisite skills and knowledge for higher education and a number of trades. Researchers, on the other hand, see a corporate agenda for forcing students to take algebra, and as many of these students are not ready for it and inevitably fail the course, the result could be higher dropout rates. So, the question remains: is algebra really that important?

First, it is important to define what algebra is and why it is so difficult for many students. To answer this question, the root of the problem must be identified, which is the difficulty young people find in learning math in general. Arithmetic, the most basic branch of mathematics, is challenging to many but most people are able to grasp the major tenets, thanks to the fact that numbers occur naturally around us in our everyday lives. Numbers are abstract, but are tied to concrete things that surround us, from taking measurements in a clothes department to weighing fresh produce at the supermarket. Past research relating to the teaching and learning of algebra was mostly concentrated on the issues of secondary school students, but these problems can be traced back to a lack of algebraic reasoning in early childhood years. Errors in school-level algebra stem from a fundamental difference in the approach to algebraic and arithmetic problem solving, as a student who has learned to solve a problem through arithmetic will need to adopt a whole new way of thinking.

In-depth research related to this issue was conducted with a group of 120 high school students in Sri Lanka, coming from three different schools in urban, semi-urban, and rural areas. A pre-screened algebra test with 30 questions was given to all students, and a random sample of 32 students was selected for interviewing following the test. The interview transcripts, and the students' written assessments were analyzed, and a number of recurring errors were prevalent. Most errors occurred during the processing stage (57.8%), followed by comprehension error (21.9%), encoding error (15.6%), and

verification error (4.7%), with these figures revealing that the majority of errors occurred during processing and comprehension. This research showed that symbolic interpretation is difficult for many students.

Both arithmetic and algebra have their own set of practical applications in life. Arithmetic is more useful in everyday activities that involve whole numbers, fractions, and percentages, such as calculating monthly budgets and taking measurements. It is indeed possible to succeed in adulthood without the use of algebra, and adults may have never learned about its practical applications. Algebra's value may not be as readily apparent as arithmetic's, but without it, access to a number of professions or university courses is impossible. Algebra may seem abstract, but that is part of why it is so useful in a number of situations and settings. It can be argued that due to its abstract nature, it should be taught from an early age, much as language already is, which is also an abstract concept.

The emphasis placed on algebra in school curricula varies significantly from country to country, as was highlighted by PISA, an international comparative study that looked into various profiles of mathematical education around the world. A consistent result of these studies was that Nordic and English-speaking countries place far less importance on the application of algebraic problem-solving than East Asian and East European countries. This was due to the fact that real-world mathematics, namely arithmetic, was the main focus of school mathematics in the former countries, and there has been a shift away from the more formal aspects of mathematics, such as algebra. There also seems to be some problems with the organization of the curricula, as students in Norway take algebra classes late in their schooling career, with teachers stressing that the abstract nature of algebra makes it difficult to teach, and therefore, difficult for students to learn. Not only is algebra difficult to teach, it is also difficult for many teachers to understand themselves, so school boards in the Nordic and English-speaking countries do not stress training their student teachers in this. For this reason, if teachers feel that they are not competent in algebra, it makes sense that they do not make a point of teaching it, or at least put off including it in their syllabi for as long as possible.

Given the vast number of difficulties experienced by both teachers and students when it comes to algebra, a central question in determining the importance of algebra is how to find a balance between what students need in their everyday lives and their requirements for this knowledge in pursuing higher education and careers later in life. Another equally important question is how arithmetic and algebra can be effectively implemented in the classroom. These questions are not easily answered, but are incredibly important and deserve more attention from both educators and researchers.

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 What can be said about knowledge of the STEM fields?
- A It will help the United States become the global economic leader.
 - B It will increase the rate of growth in the world economy.
 - C It will help the United States' economy maintain its competitive advantage.
 - D It will help students secure high-paying jobs in the future.
- 2 Why does the author mention the study in Sri Lanka?
- A To provide evidence as to why algebra is challenging for many students.
 - B To show the types of errors students make regarding algebra.
 - C To explain why students have trouble comprehending symbols.
 - D To compare the abilities of students of varying backgrounds.
- 3 How do arithmetic and algebra differ, as mentioned in paragraph four?
- A Arithmetic is more useful for gaining employment.
 - B Algebra cannot be used in practical situations.
 - C The usefulness of algebra is less obvious than arithmetic.
 - D Algebra should be taught earlier than arithmetic.
- 4 How do Nordic and English-speaking countries differ from East Asian and East European countries in their approaches to teaching mathematics?
- A Nordic and English-speaking countries do not focus on mathematics.
 - B The former countries' curricula place more emphasis on arithmetic than algebra.
 - C Algebra is not the main focus of teacher training programs in East Asian and East European countries.
 - D The latter countries teach algebra later in their curricula than the former.
- 5 How does the author feel about how both mathematical methods should be handled?
- A Higher importance should be given to algebra in the classroom.
 - B Arithmetic should be the primary focus in schools.
 - C Teachers need to improve their ability to teach both methods equally.
 - D Further research is needed on how to teach both methods effectively.

Questions 6–10

Complete the summary using the list of words, **A–H**, below.

WORD LIST

- A objects
- B source
- C beliefs
- D divergence
- E difficulties

- F uncertainty
- G concepts
- H outcomes

Why algebra is so hard

To understand why algebra is so hard, it is important to find the ⑥ _____ of the issue: a general difficulty in learning math. Arithmetic can be difficult but most people are able to figure out the basic ⑦ _____ because we are surrounded by numbers, and while they are abstract, numbers are connected to tangible ⑧ _____ in our world.

Research on the teaching and learning of algebra in the past mainly focused on ⑨ _____ faced by secondary-school students, but these can be traced back to childhood years. In fact, it turns out that errors in algebra can be traced to a/an ⑩ _____ between the two methodologies of solving math problems.

Questions 11–14

*Do the following statements agree with the claims of the writer in the passage? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- ⑪ The attention given to more practical maths in the Nordic and English speaking countries reduces their emphasis on algebra.
- ⑫ East European and East Asian students are better at algebra than those in the Nordic and English speaking countries.
- ⑬ Teachers' levels of expertise in algebra do not determine how much emphasis they put on it.
- ⑭ Scientists and educators are trying to strike a balance between arithmetic and algebra in the curriculum.

Passage 46

The Art of Deception

Do tiny changes of facial expression show whether someone is telling lies?

Forty years ago, the research psychologist Dr Paul Ekman was addressing a group of young psychiatrists in training when he was asked a question, the answer to which has kept him busy ever since. Suppose, the group wanted to know, a particular patient swears they are telling the truth. They look and sound sincere. So here is the question: is there any way you can be sure they are telling the truth? Ekman did not know the answer then, but wanted to find out.

As part of his research, he had already filmed a series of 12-minute interviews with psychiatric patients. In a subsequent conversation, one of the patients told him that she had lied to him. So, Ekman sat and looked at the film but saw nothing noteworthy. Then he slowed it down and looked again. Then he slowed it even further. And suddenly, there, across just two frames of the film, he saw it: an intense expression of extreme anguish. It lasted less than a 15th of a second, but once he had spotted the first expression, he soon found three more examples in that same interview. He termed his discovery 'micro-expression': very rapid intense demonstrations of emotion that the subject intended to be concealed.

Over the course of the next four decades, Ekman successfully demonstrated a proposition first suggested by Charles Darwin: that the ways in which we express rage, disgust, contempt, fear, surprise, happiness and sadness are universal. The facial muscles triggered by those seven basic emotions are, he has shown, essentially standard, regardless of language and culture, from the US to Japan and Brazil to Papua New Guinea. What is more, expressions of emotion are impossible to suppress and, particularly when we are lying, micro-expressions of powerfully felt emotions will inevitably flit across our face before we get the chance to stop them.

Fortunately for liars, most people will fail to spot these fleeting signals of inner torment. Of the 15,000 Ekman has tested, only 50 people, whom he calls 'naturals', have been able to do it. But given a little more training, Ekman says, almost anyone can develop the skill. He should know; since these tests were completed in the mid-1980s and the first publication of his research, he has been called in by the FBI and CIA (among countless more law-enforcement and other agencies around the world), not just to solve cases, but to teach them how to use his technique for themselves. He has held workshops for defense and prosecution lawyers, health professionals, even jealous spouses, all of them wanting to know exactly when someone is not being 100 percent candid.

Most recently, Ekman's research has resulted in a new television series about the exploits of the fictional Dr Cal Lightman, a scientist who studies involuntary body language to discover not only if you are lying, but why you might have been motivated to do so. According to the publicity blurb, Lightman is a human lie detector, even more accurate than a polygraph test. Ekman concedes he was skeptical when the producer first approached him with the idea of turning his life's work into a TV series, and initially would have stopped the project if he could. In particular, he was fearful that the show would exaggerate the effectiveness of his techniques and create the quite inaccurate impression among audiences that criminals could no longer hope to get away with lying. In the worst-case scenario, he was concerned about unfair convictions, that one day someone not properly trained in his techniques might be sitting on a jury and wrongly find someone guilty of a crime simply on the basis of a television programme.

In the end though, he was won over because the series is unusual in several respects. It is the first time, as far as Ekman is aware, that a commercial TV drama has been based on the work of just one scientist. That scientist is also deeply involved in the project, talking through plot ideas and checking five successive drafts of each script to ensure details are correct. He was also impressed with the producer's manifestly serious and well-intentioned reasons for making the programme. Now that the first series has been completed, he believes probably 80–90 per cent of the show is based on fact and that's good enough for what it is, after all, a drama, not a documentary.

Ekman, incidentally, professes to have been a terrible liar ever since he was a small boy and observes that the ability to detect a lie and the ability to lie successfully are completely unrelated. He has been asked by people running for high office if he could teach them to become more credible with the public but has always refused to use his skills in that way on ethical grounds. He also insists that there are various kinds of lies. A 'true' lie can be identified by having two essential characteristics: there must be a deliberate intent to mislead and there must be no notification that this is what is occurring. This means that an actor or a poker player isn't a true liar. They are supposed to deceive you, it's part of the game, and the same is true of flattery. He prefers to focus on the kinds of lies where the liar would be in grave trouble if they were found out, and where the target would feel properly aggrieved if they knew.

Questions 27–31

Choose the correct letter, **A, B, C or D**.

- 27 According to the writer, Ekman became interested in lying after a question from his
- A peers.
 - B patients.
 - C students.
 - D teachers.
- 28 The writer refers to the 12-minute interviews in order to
- A illustrate how frequently patients lie.
 - B describe the origins of Ekman's theories.
 - C compare Ekman's research to previous studies.
 - D show how patients' behaviour is affected by filming.
- 29 What is the writer's point in the third paragraph?
- A Micro-expressions are common to all people.
 - B Recent research has refuted an old idea.
 - C With practice we can learn to control our micro-expressions.
 - D Human society is too complex to allow for generalisations.
- 30 What are we told about Ekman's conclusions from his tests?
- A It's natural for people to lie.
 - B Few untrained people can detect lying.
 - C Most liars suffer from periods of depression.
 - D All of his subjects were trained to identify micro-expressions.
- 31 What point does the writer make about Ekman's techniques in the fourth paragraph?
- A They take a decade to teach.
 - B They have been in great demand.
 - C They have aroused the suspicions of some agencies.
 - D They can be used by a limited range of occupations.

Questions 32–36

Complete the summary using the list of words, A–I, below.

LIST OF WORDS

- A consequences
- B crimes
- C false beliefs
- D motives
- E justice
- F accuracy
- G acting
- H research
- I ratings

THE TELEVISION SERIES BASED ON EKMAN'S WORK

A new TV series based on Ekman's work features a hero named Lightman, who detects lies. Initially, Ekman was unenthusiastic about the TV project because he feared the possibility of encouraging viewers' (32) _____. For example, he was worried that one day the programme could result in (33) _____ not being carried out. Ultimately though, he has given the show his blessing because he is not aware of any other comparable programme based on a single person's (34) _____. The (35) _____ of the show's producer have been another pleasant surprise considering the genre of the programme. Ekman is happy with the show's overall (36) _____.

Questions 37–40

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write YES, NO or NOT GIVEN.

- (37) Ekman regrets the lies he told as a child.
- (38) People who are good at lying tend to be good at detecting lies.
- (39) Ekman has worked with poker players to help them lie more convincingly.
- (40) Ekman is more interested in the types of lies with serious consequences.

Passage 47

The Benefits of Being Bilingual

How growing up speaking two languages changes the way the brain works

A According to the latest figures, the majority of the world's population is now bilingual or multilingual, having grown up speaking two or more languages. In the past, such children were considered to be at a disadvantage compared with their monolingual peers. Over the past few decades, however, technological advances have allowed researchers to look more deeply at how bilingualism interacts with and changes the cognitive and neurological systems, thereby identifying several clear benefits of being bilingual.

B Research shows that when a bilingual person uses one language, the other is active at the same time. When we hear a word, we don't hear the entire word all at once: the sounds arrive in sequential order. Long before the word is finished, the brain's language system begins to guess what that word might be. If you hear 'can', you will likely activate words like 'candy' and 'candle' as well, at least during the earlier stages of word recognition. For bilingual people, this activation is not limited to a single language; auditory input activates corresponding words regardless of the language to which they belong. Some of the most compelling evidence for this phenomenon, called 'language co-activation', comes from studying eye movements. A Russian-English bilingual asked to 'pick up a marker' from a set of objects would look more at a stamp than someone who doesn't know Russian, because the Russian word for 'stamp', marka, sounds like the English word he or she heard, 'marker'. In cases like this, language co-activation occurs because what the listener hears could map onto words in either language.

C Having to deal with this persistent linguistic competition can result in difficulties, however. For instance, knowing more than one language can cause speakers to name pictures more slowly, and can increase 'tip-of-the-tongue states', when you can almost, but not quite, bring a word to mind. As a result, the constant juggling of two languages creates a need to control how much a person accesses a language at any given time. For this reason, bilingual people often perform better on tasks that require conflict management. In the classic Stroop Task, people see a word and are asked to name the colour of the word's font. When the colour and the word match (i.e. the word 'red' printed in red), people correctly name the colour more quickly than when the colour and the word don't match (i.e. the word 'red' printed in blue). This occurs because the word itself ('red') and its font colour (blue) conflict. Bilingual people often excel at tasks such as this, which tap into the ability to ignore competing perceptual information and focus on the relevant aspects of the input. Bilinguals are also better at switching between two tasks; for example, when

bilinguals have to switch from categorizing objects by colour (red or green) to categorizing them by shape (circle or triangle), they do so more quickly than monolingual people, reflecting better cognitive control when having to make rapid changes of strategy.

D It also seems that the neurological roots of the bilingual advantage extend to brain areas more traditionally associated with sensory processing. When monolingual and bilingual adolescents listen to simple speech sounds without any intervening background noise, they show highly similar brain stem responses. When researchers play the same sound to both groups in the presence of background noise, however, the bilingual listeners' neural response is considerably larger, reflecting better encoding of the sound's fundamental frequency, a feature of sound closely related to pitch perception.

E Such improvements in cognitive and sensory processing may help a bilingual person to process information in the environment, and help explain why bilingual adults acquire a third language better than monolingual adults master a second language. This advantage may be rooted in the skill of focussing on information about the new language while reducing interference from the languages they already know.

F Research also indicates that bilingual experience may help to keep the cognitive mechanisms sharp by recruiting alternate brain networks to compensate for those that become damaged during aging. Older bilinguals enjoy improved memory relative to monolingual people, which can lead to real-world health benefits. In a study of over 200 patients with Alzheimer's disease, a degenerative brain disease, bilingual patients reported showing initial symptoms of the disease an average of five years later than monolingual patients. In a follow-up study, researchers compared the brains of bilingual and monolingual patients matched on the severity of Alzheimer's symptoms. Surprisingly, the bilinguals' brains had more physical signs of disease than their monolingual counterparts, even though their outward behaviour and abilities were the same. If the brain is an engine, bilingualism may help it to go farther on the same amount of fuel.

G Furthermore, the benefits associated with bilingual experience seem to start very early. In one study, researchers taught seven-month-old babies growing up in monolingual or bilingual homes that when they heard a tinkling sound, a puppet appeared on one side of a screen. Halfway through the study, the puppet began appearing on the opposite side of the screen. In order to get a reward, the infants had to adjust the rule they'd learned; only the bilingual babies were able to successfully learn the new rule. This suggests that for very young children, as well as for older people, navigating a multilingual environment imparts advantages that transfer far beyond language.

Complete the table below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Test	Findings
Observing the (1) _____ of Russian-English bilingual people when asked to select certain objects	Bilingual people engage both languages simultaneously: a mechanism known as (2) _____
A test called the (3) _____ focusing on naming colours	Bilingual people are more able to handle tasks involving a skill called (4) _____
A test involving switching between tasks	When changing strategies, bilingual people have superior (5) _____.

Questions 6–10

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- (6) Attitudes towards bilingualism have changed in recent years.
- (7) Bilingual people are better than monolingual people at guessing correctly what words are before they are finished.
- (8) Bilingual people consistently name images faster than monolingual people.
- (9) Bilingual people's brains process single sounds more efficiently than monolingual people in all situations.
- (10) Fewer bilingual people than monolingual people suffer from brain disease in old age.

Questions 11–14

Reading Passage 3 has seven paragraphs, A–G.

Which paragraph contains the following information?

Write the correct letter, A–G, in boxes 11–14.

Q	Statement	
11	an example of how bilingual and monolingual people's brains respond differently to a certain type of non-verbal auditory input	
12	a demonstration of how a bilingual upbringing has benefits even before we learn to speak	

13	a description of the process by which people identify words that they hear	
14	reference to some negative consequences of being bilingual	

@bekhruzposts

The Cause of Linguistic Change

Why does pronunciation drift over generations — contact, fashion, or simple laziness?

All living languages undergo changes of various kinds, for which there are various reasons. The changes that have caused most dispute are those in pronunciation. We have various sources of evidence for the pronunciation of earlier times, like the evidence of spellings, the treatment of words borrowed from other languages or borrowed by them, the descriptions of contemporary grammarians, and the modern pronunciations in all the languages and dialects concerned. These, combined with a knowledge of the mechanism of speech production, can often give us a very good idea of the pronunciation of an earlier age, though absolute certainty is never possible. When we study the pronunciation of a language over any period of a few generations or more, we find that it is subject to change. Moreover, there are always large-scale regularities in the changes: for example, over a period of time, all the long a sounds in a language may change into long o sounds, or all the b sounds in a certain position (for example, at the end of a word) may change into p sounds. Such regular changes are often called sound laws. There are no universal sound laws, but simply particular sound laws for one given language at one given period.

One cause of change is the influence of one language on another. We learn our mother tongue very thoroughly, and acquire a whole set of speech habits which become second nature to us. When later we learn a foreign language, we inevitably carry over some of these speech habits into it, and so do not speak it exactly like a native. Consequently, in bilingual situations the second language tends to be modified. Such modifications may not persist, of course: an isolated Polish or German immigrant to Britain will usually have grandchildren who speak English like natives, because the influence of the general speech environment (playmates, school, work) is stronger than that of the home. But if a large and closely knit group of people adopt a new language, then the modifications they make in it may persist among their descendants, even if the latter no longer speak the original language that caused the changes. This can be seen in Wales, where the influence of Welsh has affected the pronunciation of English, and the very characteristic melody of Welsh English has been carried over from Welsh.

It is also possible that fashion plays a part in the process of change. It certainly plays a part in the spread of change: one person imitates another, and people with the most prestige are most likely to be imitated, so that a change that takes place in one social group may be imitated (more or less accurately) by speakers in another group. When a social group goes up or down in the world, its pronunciation will gain or lose prestige. Some of the changes in accepted English pronunciation in the seventeenth and eighteenth centuries have been shown to consist of the replacement of one style of pronunciation by another style already

existing, and it is likely that such substitutions were a result of the great social changes of the period: the increased power and wealth of the middle classes, and their steady infiltration upwards into the ranks of the landed gentry, probably carried elements of middle class dialect into upper class speech.

However, besides spreading changes that have already taken place, fashion may actually cause changes in pronunciation. In a stratified society, the important thing about a fashion is that it is exclusive: as soon as the fashion has penetrated to a lower social group, it is time to move on. This can be seen in clothes: fashionable people may find it flattering to be imitated, but as soon as the new fashion has really caught on they must change to something else, to mark themselves off as different. It may be the same with language, for social groups use characteristic styles of language to mark themselves off from other groups. A group that has high prestige may find that its style of speech is being imitated by other groups, and then its members may (perhaps subconsciously) begin to change it, perhaps by exaggerating its distinguishing characteristics.

However, while interaction between different languages, and between the dialects of a single language, can explain some changes in pronunciation, it cannot explain them all. Another cause that has been suggested is the fact that the imitation of children is imperfect: they copy their parents' speech, but never reproduce it exactly. This seems to be true enough, and it moreover seems to be true that even adults show a certain amount of random variation in their pronunciation. However, these facts cannot explain changes in pronunciation, unless it can be shown that there is some systematic trend in the failures in imitation: if they are merely random deviations they will cancel one another out and there will be no net change in the language. For some of these random variations to be selected at the expense of others, there must be further forces at work.

One such force, which is often suggested, is known as the principle of ease. We are all naturally lazy, it is argued, and so we tend to take short cuts in the movements of our speech organs, to replace movements calling for great accuracy or energy by less demanding ones, to omit sounds if they are not essential for understanding, and so on. For example, the word *scant* was once *skamt*, but the *m* has been changed to *n* under the influence of the following *t*. An economy of effort has thereby been achieved, because *n* and *t* are articulated in the same place (with the tip of the tongue against the teeth-ridge), whereas *m* is articulated elsewhere (with the two lips). The change from *skamt* to *scant* would be an example of assimilation, which is a very common kind of change. Assimilation is the changing of a sound under the influence of a neighbouring sound.

Assimilation is not the only way in which we change our pronunciation in order to reduce effort. A sound may come to be pronounced less energetically, wherever it occurs: the English *r* and *h* sounds are undoubtedly pronounced much less vigorously now than they

were a thousand years ago. In some positions a sound may disappear altogether: at one time the t was pronounced in words like castle and Christmas, and the g in words like gnat and gnaw. Sometimes a whole syllable is dropped out if it occurs twice running: a modern example is temporary, which in Britain is pronounced as if it were tempory.

Questions 27–30

Complete the summary below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

THE CAUSES OF LINGUISTIC CHANGE

It is known that the pronunciation of language changes over time. The changes include large-scale regularities known as (27) _____. There are several possible causes for change.

Firstly, there is the effect of one language on another. Secondly, there is the role of

(28) _____. The speech of certain groups has prestige, and is imitated by others. It has also been suggested that some changes are introduced by (29) _____, who are unable to imitate speech in an exact way. Finally, there is a factor referred to as the (30) _____, by which the effort necessary to produce words is reduced.

Questions 31–37

Do the following statements agree with the information given in Reading Passage 3?

Write **TRUE** if the statement agrees with the information, **FALSE** if it contradicts it, or **NOT GIVEN** if there is no information on this.

- (31) It is impossible to know how words used to be pronounced.
- (32) Some languages appear to have changed more than others.
- (33) Some changes in English were probably led by the middle classes.
- (34) Certain English sounds are difficult for all children to pronounce.
- (35) Scant is generally easier to pronounce than skamt.
- (36) The spelling of gnat is likely to change in the future.
- (37) The spelling of temporary no longer reflects its pronunciation.

Questions 38–40

Look at the following statements and the list of options below.

Match each statement with the correct option, **A, B, C** or **D**.

NB You may use any letter more than once.

LIST OF OPTIONS

- A** large-scale change
- B** influence of one language on another
- C** minimisation of effort
- D** imitation of prestigious group

Q	Statement	
38	the b sound changing to a p sound at the end of words	
39	the melodic pronunciation of Welsh English	
40	the disappearance of the t sound in Christmas	

Passage 49

The Conservation of Rare Species

Why "rare" means different things for different species — and why the red-cockaded woodpecker shows what it takes to rescue one

When a naturalist says that a bird or a plant is rare, he or she may mean one of several different things (Harper 1981, Rabinowitz 1981). The concept of rarity can refer to one of three characteristics: geographic range, habitat specificity, or local population size, and a classification based on the interplay between these variables yields seven different types of rarity. For example, certain species may be locally abundant over a wide geographic range, but found in only a very specific habitat, whilst others may be found in several habitats, but only in small numbers and in a geographically restricted area. We must therefore recognise that the kinds of management which will be appropriate for protecting species threatened with extinction will vary.

Classic rare species are often those which have a small geographic range and narrow habitat specificity. Many plants of this type are restricted endemics, and are often endangered or threatened (Rabinowitz 1981). Other rare species have very large geographic ranges and occur widely in different habitats but are always at low density. These species are ecologically interesting but almost never appear on lists of endangered species. So, the important point is that not all rare species are problems for conservationists.

The reasons why a particular species is rare vary. In some cases we can observe a species declining over time: for example, the African elephant population since 1950. This is a direct result of ivory poaching (Caughley et al. 1990). But not all species that have declined to rarity are so well understood. Some plant and animal species undergo bursts of colonisation and decay so that they persist as a mosaic of increasing and declining populations (Harper 1981).

For species with small geographic ranges or narrow habitat specificity we must be concerned with the spatial distribution of the population. The number or size of habitable sites may be too small, and this could be one reason the species is rare. Or, if there are many habitable sites that are not occupied, a species may be rare because of its limited dispersal powers. Within habitable sites, competition from other species, predation, disease or social interactions may restrict abundance. One example that illustrates some of these factors is the red-cockaded woodpecker.

The red-cockaded woodpecker is an endangered species of bird endemic to eastern parts of the United States of America. It was once abundant from New Jersey in the north to Texas in the south, and from the coast inland as far as Missouri, but it is now nearly extinct in the

northern and inland parts of its geographic range. The red-cockaded woodpecker is adapted to pine savannas, but most of this woodland has been destroyed for agriculture and timber production. The birds feed on insects and nest in cavities in old pine trees, and because old pines have been mostly cut down, their available habitats have been reduced (Walters 1991).

Designing a recovery programme for the red-cockaded woodpecker has been complicated by the social organisation of the species. The birds live in groups of a breeding pair and up to four helpers, nearly all males. Helpers do not breed but assist in incubation and feeding. Young birds have a choice of dispersing or staying to help in a breeding group. If they stay, they become breeders by inheriting breeding status on the death of older birds. Helpers may wait many years before they acquire breeding status. From a conservation viewpoint, the problem is that red-cockaded woodpeckers compete for breeding vacancies in existing groups, rather than forming new groups and occupying abandoned territories, or starting at a new site. The key problem is the excavation of new breeding cavities. Because of the energy and time needed, typically several years, birds are better off competing for existing territories than building new ones.

To test this idea, Walters (1991) and his colleagues artificially constructed cavities in trees at a number of sites across several forests. As predicted, breeding groups were formed only in areas where artificial cavities were drilled. This experiment showed clearly that much suitable habitat is not occupied by this woodpecker because of a shortage of cavities. Therefore, management of this endangered species was not directed at reducing mortality of the birds, but instead focused on physical characteristics of their chosen habitat.

The rescue of the red-cockaded woodpecker is a good example of how successful conservation biology depends on identifying and alleviating limiting factors. However, there can be no general prescription for rescuing rare species. Detailed information on resource requirements, social organisation, and dispersal powers are required before recovery plans can be developed.

Questions 1–4

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 What does the writer say about rarity?
- A** There is more than one criterion for defining it.
 - B** The term is a cause of dispute amongst naturalists.
 - C** It applies to only a very small number of species.
 - D** It is a phenomenon which has been widely researched.

- 2 What does the writer say about the threat of extinction?
- A All types of rarity present a problem.
 - B Few plant species are known to be vulnerable.
 - C Low density by itself does not endanger species.
 - D Lists of endangered species are not reliable.
- 3 Which of the following makes it difficult for the red-cockaded woodpecker to increase in number?
- A A proportion of male birds are sterile.
 - B Birds leave the parental nest too early.
 - C There is too much competition for food.
 - D Individuals may not breed for several years.
- 4 What was the basis of the recovery programme for the red-cockaded woodpecker?
- A Birds were bred in captivity.
 - B Nesting holes were created.
 - C Pine forests were planted.
 - D Food supplements were provided.

Questions 5–11

Do the following statements agree with the information given in Reading Passage 3? Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 5 Species with various habitat types are not necessarily widely distributed.
- 6 The rate of decline of the African elephant has slowed down.
- 7 The causes of rarity are generally easy to establish.
- 8 The traditional habitat of the red-cockaded woodpecker has been destroyed by human activity.
- 9 The red-cockaded woodpecker prefers to leave its nest before breeding.
- 10 It usually takes years for the red-cockaded woodpecker to make a nesting hole.
- 11 The project to rescue the red-cockaded woodpecker is ongoing.

Questions 12 and 13

Answer the questions below. Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

12. What does the red-cockaded woodpecker feed on?

12 _____

13. Which TWO parental tasks do the helper woodpeckers perform?

13 _____

Question 14

Choose the correct letter, **A**, **B**, **C** or **D**.

- 14 In this article, the writer's purpose is to
- A** alert the reader to the plight of the red-cockaded woodpecker.
 - B** explain how Walters's recovery programme can be applied to other species.
 - C** illustrate the factors that must be considered when dealing with rarity.
 - D** persuade readers that rarity is a neglected ecological issue.

The Dark Side of the Technological Boom

What are the effects on the individual of working in modern technological workplaces?

Changes in the way we work and how our offices are structured come at us faster and faster. Waves of state-of-the-art information technology and instant telecommunications let us reach anyone, anywhere, and speed is the key. Most of us are too busy struggling to keep pace with ongoing innovations to question the implications of our new electronic authority figures. According to a number of psychologists, however, the need to stay on top of the information flow and the extent to which we remain in touch with our offices exact a profound toll on us as individuals.

Mass exposure to technological innovations in the workplace has come too recently for psychologists to reach a consensus on its societal implications. Many agree, however, that one of the first signs of the struggle to adapt to the electronic office is often 'technostress', a cognitive shift that results from an over-identification with information systems. Psychologist Craig Brod says people become accustomed to the patterns set by electronic tools – accelerated time and yes/no logic – and internalize these patterns. When they leave the office or go home, Brod says, they need complete isolation to recover from the effects of the technology.

Brod warns that over-reliance on electronic tools could also have serious repercussions on our ability to think creatively and develop new ideas. Because we don't create in a vacuum, he points out, we need to avoid the temptation to replace informal gatherings for bouncing ideas off colleagues with electronic networking. It's also more difficult to spot errors or even evaluate the shape of a project displayed in a flat, two-dimensional way on a screen.

Electronically networked offices can also make it increasingly difficult to convince ourselves that we're doing an adequate job and accumulating enough information to make informed decisions. Philosopher Daniel Dennett points out that modern technology eliminates the possibility of unavoidable ignorance. As the opportunity to amass information grows larger, the obligation to make accurate predictions – the right decisions – becomes more onerous. Instead of consoling ourselves that we're doing as good a job as we can, we are tormented by the knowledge that the world of information is limitless.

For executives near the top of the office pyramid, the benefits of the electronic revolution – like telecommuting and flexible scheduling – may outweigh the disadvantages of being continuously on call. But in *Workplace 2000*, authors Joseph Boyett and Henry Conn describe a future in which millions of people now charged with analyzing information and

making routine decisions will be replaced by less skilled workers using 'intelligent' software to make decisions for them. They predict that a cult of performance excellence will engulf most businesses.

The millions of people on the bottom levels of electronic hierarchies are increasingly likely to spend their days in an isolated no-man's land, subservient to intelligent information systems that report their progress to unseen supervisors far away. Because computers measure quantity, not quality, such systems tend to reward employees who work faster more than those who work better.

Service people on the telephone or at a cash register curtly terminate attempts at idle conversation because their performance is being electronically monitored. Once judged on their ability to troubleshoot unexpected situations, they're now evaluated by the number of transactions they complete in a shift or the number of keystrokes required to draft a memo. In these new office 'sweatshops', the computers are running the people, not the other way around.

"I think people are going to feel an increased fragmentation of self. They won't be able to hold the pieces together," human resources consultant Philip Nicholson says. "How do you keep a coherent space if you're going in and out of spaces that don't exist?" He likens the psychic numbing of electronic information overload to symptoms of post-traumatic stress syndrome (a mental disorder following a horrific event). In office 'wars', people become overwhelmed by the sheer amount of information available, internalize the diversity of the world outside, and fear losing control of their own lives.

If we are to survive the challenges of information-driven, hardwired offices, says Nicholson, we need to provide psychological support systems. As no one has yet measured the social cost of the workplace revolution, some psychologists are mobilizing efforts to pool information as it is derived. Nicholson started the Technostress International Information Network in Massachusetts to foster an exchange of data and ideas on the effects of computerization and information technology. Meanwhile, Brod wants to examine the parallels between electronic work environments and 'sealed-cabin ecologies' like space capsules or submarines, both totally automated artificial worlds in which people live in highly confined circumstances surrounded by technology that dictates the tenor of their days as well as their survival. He is petitioning other psychologists to convince the American Psychological Association to form a specialized study group.

In addition, Brod suggests that we re-examine our value systems and that we make greater allowances for privacy in order to circumvent potential revolts against technology. "We need to coevolve with technology," he says. "These are wonderful tools, but if we exploit them without imposing appropriate values on their use, they become alienating and dangerous."

Questions 1–3

Complete each sentence with the correct ending, **A–E**, below. Write the correct letter in each box.

- | | | | |
|---|--|---|-------|
| 1 | The speed of technological changes | 1 | _____ |
| 2 | A dependency on technology and computers | 2 | _____ |
| 3 | A deterioration in personal service | 3 | _____ |

LIST OF ENDINGS

- A** requires more detailed study by psychologists
- B** means people have no time to challenge the significance of the new technology
- C** may reduce inventiveness and innovation
- D** suggests computers will take over the workplace
- E** results from increased electronic supervision

Questions 4–9

Look at the following statements and the list of people below. Match each statement with the correct person or people, **A, B, C** or **D**.

NB You may use any letter more than once.

LIST OF PEOPLE

- A** Craig Brod
- B** Daniel Dennett
- C** Joseph Boyett and Henry Conn
- D** Philip Nicholson

Q	Statement	
4	Technology has placed greater expectations on workers not to make mistakes.	
5	People will need time away from technology to reduce the frustrations caused by it.	
6	Interacting with others at work contributes to creative thinking.	
7	The psychological effect of working with technology is similar to the anxiety felt after surviving a major ordeal.	
8	Technology will ultimately increase unemployment for more highly qualified personnel.	

9	More counselling is required to help people cope with the demands of the modern workplace.	
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Questions 10–14

Do the following statements agree with the information given in Reading Passage 3? Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 10 Our knowledge of the effects of technology on workers is still limited.
- 11 An early indicator of technological anxiety is a tendency to adopt machinelike thinking.
- 12 We have now started to doubt our ability to perform well at work.
- 13 Top level managers may be more negatively affected by changes in the electronic workplace than junior workers.
- 14 Employees who learn to use new technology quickly will get promoted.

Passage 51

The Fluoridation Controversy

Examining the science and the social values behind the long-running debate over adding fluoride to public water supplies

The long-standing debate about whether to fluoridate our drinking water continues. Fluoridation is the addition of fluorine to public water supplies with the aim of reducing tooth decay. The fluorine, when mixed with water, becomes fluoride and the desired concentration of fluoride in public water is approximately one part per million, depending on the regional temperature and hence the amount of water people are likely to drink. Many studies, such as those by McClure in 1970 through to Burt in 1983, have shown that when children drink fluoridated water, their average rate of tooth decay seems greatly reduced. A typical figure claimed is 50 percent reduction. This apparently enormous benefit for children's teeth is the major argument in favor of fluoridation.

Three main grounds for opposition to fluoridation have been expressed. First, opponents claim the benefits are exaggerated or not established. Second, there are claims of health risks to parts of the population, for example, allergic reactions. It is also accepted that high levels of fluoride can cause discoloration of otherwise healthy teeth. Proponents do not consider this to be a problem in such small concentrations, whereas opponents disagree — especially because some people drink more water and obtain much more than the standard 1 milligram of fluoride per day. Third, fluoridation is thought to be an infringement on individual rights because it is compulsory medication of all members of a community.

An understanding of the fluoridation issue has important implications. If, according to the experts, fluoridation is unquestionably a beneficial and non-hazardous measure, then the wisdom of allowing the public to vote on, and reject it, must be questioned.

Almost all studies that have been done have assumed that the scientific aspects of the controversy are unproblematic, and they have excluded science from sociological examination. The traditional view is that science is a special kind of knowledge, which is established through scientific methods and objectively applied by members of a scientific community. However, in recent years there has been a major challenge to this picture by a sociology of science that shows how scientific knowledge is socially negotiated, and inevitably linked to the values of the relevant parties, both scientists and nonscientists. These challengers do not see scientific knowledge as exempt from social inquiry.

Kuhn (1970) argued that scientific knowledge does not always develop as an orderly process, but is characterized by periodic revolutions, in which the methods of study and the assessment criteria change in a fragmented way. According to Kuhn, the shift from one scientific way of thinking to another is not made solely on the basis of clear rules of formal

scientific practice, but can include social factors, though Kuhn has never developed a full analysis of what these might be. Collins (1975) took this concept further when he asserted that the outcome of experiments was not something whose meaning could be immediately comprehended, but rather something for interpretation, discussion between scientists, and reinterpretation in the light of other experiments.

One interpretation of this analysis of science is that traditional distinctions between facts and theories, and between scientific knowledge and values, can no longer be justified. Because social processes are involved at all stages of the creation, evaluation, and establishing of scientific knowledge, social values may also be involved.

In the same way as many scientists who study fluoridation have overlooked social values, sociologists have also downplayed an important part of the debate by ignoring the number of eminent scientists who have questioned aspects of fluoridation. An example is the study by Sutton in 1960, which analyzed the classic North American studies of the effect of fluoridation on tooth decay, and found that each showed significant methodological shortcomings. Sutton's detailed study throws doubt as to the extent of reductions in tooth decay from fluoridation. Yet Sutton's book is not cited in a single analysis of the fluoridation issue by any sociologist. In a situation of some scientific uncertainty, differences in values are highlighted. A supporter of fluoridation might argue, 'The evidence for the benefits of fluoridation is quite substantial, while the evidence for harm is limited and dubious. I think the likely benefits outweigh the possible dangers; hence I support fluoridation because it is the cheapest and easiest way to make sure every child reaps the benefits.' An opponent might argue, 'Though the evidence for the benefits of fluoridation is substantial, there is some doubt about it. Since fluoridation is not necessary for good teeth, we should forego the benefits if there is some slight chance of harm. Some scientists claim that a small percentage of the population could be harmed by fluoride. Therefore, I oppose fluoridation of water supplies and favor the voluntary use of fluoride tablets by those who want to take them.'

Both arguments consider the scientific evidence concerning fluoridation, but differ in their assessments of the social benefits and costs. This difference is not between rationality and irrationality but is a legitimate difference in values, for example, the positive value placed on good teeth, the negative value placed on possible health risks, and the social benefits or costs of compulsory or voluntary intake of fluorides.

From the sociological point of view, opposition to fluoridation is not necessarily irrational. Rather, claims to rationality and to scientific authority are better seen as part of a strategy to promote fluoridation than as incontrovertible statements of fact. Second, social values are likely to be bound up in any decision about fluoridation, so this is not an issue on which declarations by scientific experts ought to be considered the final word.

Questions 27–31

Choose the correct letter, **A, B, C or D**.

- 27 The optimum amount of fluorine in fluoridated water is calculated partly according to
- A how hot the area is.
 - B how warm the water is.
 - C how many dental problems there are in the community.
 - D how much fluorine the community chooses to have in its water.
- 28 One reason given by the writer for opposing fluoridation is that
- A it may contribute to tooth decay.
 - B it will be unacceptably expensive for the public.
 - C obligatory fluoridation takes away personal freedom.
 - D excessive fluoride could be added to the water by mistake.
- 29 The writer mentions Kuhn in order to
- A provide a contrast with the view of Collins.
 - B support the rational nature of scientific inquiry.
 - C demonstrate that Kuhn did not argue his case adequately.
 - D show that science can be influenced by non-scientific considerations.
- 30 What did Sutton's research discover about earlier studies in North America?
- A There were failings in the way they were carried out.
 - B The scientists involved had achieved unique results.
 - C Proponents of fluoridation had not understood its long-term effects.
 - D Fluoridation had a greater effect on tooth decay than previously believed.
- 31 In the last paragraph, what does the writer say about scientists?
- A They should reveal their true motivations.
 - B They should not decide the fluoridation policy.
 - C They are solely concerned with scientific truths.
 - D They cannot reach agreement on the fluoridation issue.

Questions 32–35

Do the following statements agree with the views of the writer in the passage?

Write **YES**, **NO** or **NOT GIVEN**.

- 32 Scientific knowledge should be kept separate from social values.
- 33 Many sociologists have disregarded the doubts that some scientists have concerning fluoridation.
- 34 Sutton's findings have been given insufficient attention by scientists outside of North America.

- 35 There are valid arguments on both sides of the fluoridation debate.

Questions 36–40

Complete each sentence with the correct ending, **A–G**, below. Write the correct letter in each box.

- | | |
|---|----------|
| 36 The traditional view of science is that | 36 _____ |
| 37 A sociological view of science argues that | 37 _____ |
| 38 Collins is of the opinion that | 38 _____ |
| 39 The writer suggests that a supporter of fluoridation may conclude that | 39 _____ |
| 40 The writer suggests that an opponent of fluoridation may conclude that | 40 _____ |

LIST OF ENDINGS

- A** the results of scientific research are not always understood at first
- B** scientific knowledge is established objectively through proper scientific methods
- C** people should be able to choose whether they want fluoride
- D** there is insufficient proof to support a cautious approach
- E** the serious damage fluoride causes far outweighs any positive effects
- F** children are not the only ones who benefit from fluoridation
- G** scientific knowledge is affected by the beliefs of everyone concerned

The Ingenuity Gap

In this book introduction the author explains what he means by 'ingenuity' and discusses the factors that influence the requirement for and provision of new ideas in today's society.

Ingenuity, as I define it here, consists not only of ideas for new technologies like computers or drought-resistant crops but, more fundamentally, of ideas for better institutions and social arrangements, like efficient markets and competent governments.

How much and what kinds of ingenuity a society requires depends on a range of factors, including the society's goals and the circumstances within which it must achieve those goals—whether it has a young population or an ageing one, an abundance of natural resources or a scarcity of them, an easy climate or a punishing one, whatever the case may be.

How much and what kinds of ingenuity a society supplies also depends on many factors, such as the nature of human inventiveness and understanding, the rewards an economy gives to the producers of useful knowledge, and the strength of political opposition to social and institutional reforms.

A good supply of the right kinds of ingenuity is essential, but it isn't, of course, enough by itself. We know that the creation of wealth, for example, depends not only on an adequate supply of useful ideas but also on the availability of other, more conventional factors of production, like capital and labor. Similarly, prosperity, stability and justice usually depend on the resolution, or at least the containment, of major political struggles over wealth and power.

The past century's countless incremental changes in our societies around the planet, in our technologies and our interactions with our surrounding natural environment, have accumulated to create a qualitatively new world.

Because these changes have accumulated slowly, it's often hard for us to recognize how profound and sweeping they've been. They include far larger and denser populations; much higher per capita consumption of natural resources; and far better and more widely available technologies for the movement of people, materials, and especially information.

In combination, these changes have sharply increased the density, intensity, and pace of our interactions with each other; they have greatly increased the burden we place on our natural environment; and they have helped shift power from national and international institutions to individuals in subgroups, such as political special interests and ethnic factions. The management of our relationship with the new world requires immense and ever-increasing amounts of social and technical ingenuity.

When we enhance the performance of any system, from our cars to the planet's network of financial institutions, we tend to make it more complex. Many of the natural systems critical to our well-being, like the global climate and the oceans, are extraordinarily complex, to begin with. We often can't predict or manage the behavior of complex systems with much precision, because they are often very sensitive to the smallest of changes and perturbations, and their behavior can flip from one mode to another suddenly and dramatically. Over the last 100 years as the human-made and natural systems we depend upon have become more complex, and as our demands on them have increased, the institutions and technologies we use to manage them must become more complex too, which further boosts our requirement for ingenuity.

However, we should not jump to the conclusion that the supply of ingenuity always increases in lockstep with our ingenuity requirement: while it's true that necessity is often the mother of invention, we can't always rely on the right kind of ingenuity appearing when and where we need it. In many cases, the complexity and speed of operation of today's vital economic, social, and ecological systems exceed the human brain's grasp. Not many of us have more than a rudimentary understanding of how these systems work. They remain fraught with countless "unknown unknowns," which makes it hard to supply the ingenuity we need to solve problems associated with these systems.

In this book, I explore a wide range of other factors that will limit our ability to supply the ingenuity required in the coming century. For example, the crush of information in our everyday lives is shortening our attention span, limiting the time we have to reflect on critical matters of public policy, and making policy arguments more superficial.

Modern markets and science are an important part of the story of how we supply ingenuity. Markets are critically important because they give entrepreneurs an incentive to produce knowledge. As for science, although it seems to face no theoretical limits, at least in the foreseeable future, practical constraints often slow its progress. The cost of scientific research tends to increase as it delves deeper into nature.

And science's rate of advance depends on the characteristics of the natural phenomena it investigates, simply because some phenomena are intrinsically harder to understand than others, so the production of useful new knowledge in these areas can be very slow.

Consequently, there is often a critical time lag between the recognition between a problem and the delivery of sufficient ingenuity, in the form of technologies, to solve that problem. Progress in the social sciences is especially slow, for reasons we don't yet fully understand; but we desperately need better social scientific knowledge to build the sophisticated institutions today's world demands.

Questions 1–4

Complete each sentence with the correct ending, **A–F**, below. Write the correct letter in each box.

- | | | | |
|---|---|---|-------|
| 1 | The author's definition of ingenuity | 1 | _____ |
| 2 | The type of ingenuity required by a society | 2 | _____ |
| 3 | The creation of wealth | 3 | _____ |
| 4 | The stability of a society | 4 | _____ |

LIST OF ENDINGS

- A** does not depend on ingenuity alone
- B** depends on the successful management of certain disputes
- C** has often been misunderstood
- D** is not limited to the creation of new inventions
- E** frequently increases in accordance with the material successes achieved
- F** is linked to factors such as the weather

Questions 5–7

Choose the correct letter, **A, B, C** or **D**.

- 5 What point does the author make about the incremental changes of the last century?
- A** Their effect on the environment has been positive.
 - B** They have not affected all parts of the world.
 - C** Their significance may not be noticed.
 - D** They have had less impact than those of previous centuries.
- 6 According to the author, one effect of the combined changes is that life has become
- A** easier.
 - B** faster.
 - C** more interesting.
 - D** more enjoyable.
- 7 What observation does the author make about complex natural systems?
- A** They can be greatly affected by minor alterations.
 - B** They cannot be compared to human-made systems.
 - C** Their performance cannot be improved by human intervention.
 - D** Their behavior is better understood than ever before.

Questions 8–14

Do the following statements agree with the views of the writer in Reading Passage 3? Write **YES** if the statement agrees with the views of the writer, **NO** if the statement contradicts the views of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 8 Changes in the last 100 years have increased the need for human ingenuity.
- 9 The amount of ingenuity available is strictly related to the demand which exists for it.
- 10 Although ingenuity may be available, it may be inappropriate for the tasks that need solutions at the time.
- 11 Few people today truly understand the way the modern world works.
- 12 Access to more and more information is improving our grasp of current affairs.
- 13 Future generations will be critical of the way today's governments have conducted themselves.
- 14 It is inevitable that some areas of scientific study advance more quickly than others.

Passage 53

The Knowledge Society

Peter Drucker on how "knowledge technologists" are reshaping work, education, and the meaning of success

A century ago, the overwhelming majority of people in developed countries worked with their hands: on farms, in domestic service, in small craft shops and in factories. There was not even a word for people who made their living other than by manual work. These days, the fastest-growing group in the developed world are "knowledge workers" — people whose jobs require formal and advanced schooling.

At present, this term is widely used to describe people with considerable theoretical knowledge and learning: doctors, lawyers, teachers, accountants, chemical engineers. But the most striking growth in the coming years will be in 'knowledge technologists': computer technicians, software designers, analysts in clinical labs, manufacturing technologists, and so on. These people are as much manual workers as they are knowledge workers; in fact, they usually spend far more time working with their hands than with their brains. But their manual work is based on a substantial amount of theoretical knowledge which can be acquired only through formal education. They are not, as a rule, much better paid than traditional skilled workers, but they see themselves as professionals. Just as unskilled manual workers in manufacturing were the dominant social and political force in the twentieth century, knowledge technologists are likely to become the dominant social — and perhaps also political — force over the next decades.

Such workers have two main needs: formal education that enables them to enter knowledge work in the first place, and continuing education throughout their working lives to keep their knowledge up to date. For the old high-knowledge professionals such as doctors, clerics and lawyers, formal education has been available for many centuries. But for knowledge technologists, only a few countries so far provide systematic and organised preparation. Over the next few decades, educational institutions to prepare knowledge technologists will grow rapidly in all developed and emerging countries, just as new institutions to meet new requirements have always appeared in the past.

What is different this time is the need for the continuing education of already well-trained and highly knowledgeable adults. Schooling traditionally stopped when work began. In the knowledge society it never stops. Continuing education of already highly educated adults will therefore become a big growth area in the next society. But most of it will be delivered in non-traditional ways, ranging from weekend seminars to online training programmes, and in any number of places, from a traditional university to the student's home. The information revolution, which is expected to have an enormous impact on education and

on traditional schools and universities, will probably have an even greater effect on the continuing education of knowledge workers, allowing knowledge to spread near-instantly, and making it accessible to everyone.

All this has implications for the role of women in the labour force. Although women have always worked, since time immemorial the jobs they have done have been different from men's. Knowledge work, on the other hand, is 'unisex', not because of feminist pressure, but because it can be done equally well by both sexes. Knowledge workers, whatever their sex, are professionals, applying the same knowledge, doing the same work, governed by the same standards and judged by the same results.

The knowledge society is the first human society where upward mobility is potentially unlimited. Knowledge differs from all other means of production in that it cannot be inherited or bequeathed from one generation to another. It has to be acquired anew by every individual, and everyone starts out with the same total ignorance. And nowadays it is assumed that everybody will be a 'success' — an idea that would have seemed ludicrous to earlier generations. Naturally, only a tiny number of people can reach outstanding levels of achievement, but a very large number of people assume they will reach adequate levels.

The upward mobility of the knowledge society, however, comes at a high price: the psychological pressures and emotional traumas of the rat race. Schoolchildren in some countries may suffer sleep deprivation because they spend their evenings at a crammer to help them pass their exams. Otherwise they will not get into the prestige university of their choice, and thus into a good job. In many different parts of the world, schools are becoming viciously competitive. That this has happened over such a short time — no more than 30 or 40 years — indicates how much the fear of failure has already permeated the knowledge society.

Given this competitive struggle, a growing number of highly successful knowledge workers of both sexes — business managers, university teachers, museum directors, doctors — 'plateau' in their 40s. They know they have achieved all they will achieve. If their work is all they have, they are in trouble. Knowledge workers therefore need to develop, preferably while they are still young, a non-competitive life and community of their own, and some serious outside interest — be it working as a volunteer in the community, playing in a local orchestra or taking an active part in a small town's local government. This outside interest will give them the opportunity for personal contribution and achievement.

Questions 1–5

Choose the correct letter, A, B, C or D.

- 1 According to the writer, a hundred years ago in the developed world, manual workers
- A were mainly located in rural areas.
 - B were not provided with sufficient education.
 - C were the largest single group of workers.
 - D were the fastest growing group in society.
- 2 The writer suggests that the most significant difference between knowledge technologists and manual workers is
- A their educational background.
 - B the pay they can expect.
 - C their skill with their hands.
 - D their attitudes to society.
- 3 He predicts that in the coming years, knowledge technologists
- A will have access to the same educational facilities as professional people.
 - B will have more employment opportunities in educational institutions.
 - C will require increasing mobility in order to find suitable education.
 - D will be provided with appropriate education for their needs.
- 4 According to the writer, the most important change in education this century will be
- A the way in which people learn.
 - B the sorts of things people learn about.
 - C the use people make of their education.
 - D the type of people who provide education.
- 5 The writer says that changes in women's roles
- A mean women are now judged by higher standards.
 - B have led to greater equality with men in the workplace.
 - C are allowing women to use their traditional skills in new ways.
 - D may allow women to out-perform men for the first time.

Questions 6–13

*Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- 6 In the knowledge society, knowledge can be passed down from parents to children.
- 7 Everyone is expected to be successful in the knowledge society.
- 8 The knowledge society means that some people may become successful by accident.
- 9 The knowledge society has both good and bad points.
- 10 Schoolchildren should not study so hard that they risk becoming ill.
- 11 It is right for schools to encourage a high degree of competition between their students.

- ⑫ When choosing outside interests, knowledge workers should avoid the need to try to do better than other people.
- ⑬ Outside interests are more fulfilling if they involve helping other people.

@bekhruzposts

The Long Road to Artificial Intelligence

Half a century after the field began, why has genuine machine intelligence proved so hard to build?

In 1950, the computing pioneer Alan Turing published a paper in the journal *Mind* entitled 'Computing Machinery and Intelligence'. It was the first serious, scholarly treatment of the concept of Artificial Intelligence (AI). It was the first serious, scholarly treatment of the concept of Artificial Intelligence (AI). Turing predicted that by the year 2000 the majority of people would be able to speak of machines thinking without expecting to be contradicted, envisaging a world very different from the one he lived in.

Despite Turing's prediction, human-level artificial intelligence had not been achieved by the year 2000. Nevertheless, one significant milestone in artificial intelligence had recently been attained. In 1997 'Deep Blue', a computer developed by IBM, defeated the world chess champion Garry Kasparov. Kasparov had beaten previous computer chess programs, which to him seemed predictable and mechanical, but he allegedly said he sensed an 'alien intelligence' on the other side of the board when he played against Deep Blue.

It is instructive to stand back and ponder this moment in the history of AI. The field had accomplished something that half a century beforehand might have been considered its crowning achievement. Humanity had been outstripped by a machine. Of course, a car can move faster than the fastest human sprinter, and a crane can carry far more than a champion weight-lifter. But intellectual prowess is what sets human beings apart from the rest of the animals, and chess is a supremely intellectual pursuit.

Yet somehow we seemed no nearer to human-level AI than in Turing's time. How could this be? The problem with Deep Blue was that it was a specialist. All it could do was play chess. Contrast this with a typical human adult. Take the office worker who has just walked past the window of the café where I am sitting with my laptop. Her day has no doubt been a busy patchwork of activities — making a packed lunch, reviewing the children's homework, driving to work, composing emails, fixing the photocopier, and so on. Each of these activities requires the exercise of multiple sensorimotor skills. Consider the task of making a packed lunch. This involves retrieving utensils and ingredients from various places, opening packets, chopping, cutting, spreading, and so on.

In short, a human being is a generalist. A human chess champion can do a whole lot more than just play chess. Moreover, a human being is adaptive. Fixing photocopiers is not an innate capability. It is learned. Had the office worker been born in a different century or a different culture, she would have acquired a different set of skills. And if she has the misfortune to lose her present job, she can re-train for another one. The achievements of AI research in a variety of specialist domains (chess being just one among many success

stories) contrast starkly with the field's failure to produce a machine with general purpose, adaptive intelligence. So our basic question is, how could we produce artificial general intelligence?

An essential feature of biological intelligence is embodiment. Unlike Deep Blue, a human being is an animal with a body, and its brain is part of that body. The brain of an animal has evolved to prevent any harm from coming to that body and to perpetuate the genes that it carries. The body has muscles, enabling it to move, and senses, so that its actions can be made to react to the state of the environment, the better to achieve its goal. The brain shapes the animal's actions according to what it perceives. Human intelligence, for all its glorious achievements, is fundamentally an extension of animal intelligence, and the human capacities for language, reason, and creativity all rest on a sensorimotor foundation.

So while the endeavor to create artificial general intelligence might succeed without much that is essential to biological life, such as metabolism and reproduction, perhaps embodiment is a methodological necessity. Perhaps the need to engage with a messy, dynamic, physical environment full of complex and varied objects, both animate and inanimate, is at the root of intelligence. The only way to form a reliable judgment of the intelligence of an artifact is to observe its behavior in an environment like our own. And the only way to achieve human-level AI, according to this way of thinking, is through robotics. Our basic question can then be reformulated: How can we endow a robot with general intelligence?

One possibility is that general intelligence is simply the sum of many specialist sensorimotor skills, and up to now, AI hasn't replicated enough of them. So when robots have been given a certain critical mass of skills, general intelligence will somehow emerge. But even if we gloss over the many engineering questions this proposal begs, it remains unconvincing. The products of such an approach might briefly give the appearance of general intelligence. But nobody would be fooled for very long. The multi-specialist robot is going to get stuck as soon as it has to face a problem that is outside its areas of expertise, an inevitable occurrence in an ever-changing world.

Perhaps the capacity to learn is enough to plug the gap here. In an unfamiliar situation, a new skill can be learned. Indeed, learning in its various forms is the backdrop to all intelligence. But learning is time-consuming and risky. The hallmark of general intelligence in a robot is therefore the ability to adapt an existing set of behaviors to new challenges, and to do so without recourse to trial and error or to training by a third party. Only when that can be achieved will artificial intelligence truly match human intelligence.

Questions 27–30

Choose the correct letter, **A, B, C or D**.

- 27 When playing chess against the computer Deep Blue, Kasparov felt that it
- A had been programmed to predict his moves.
 - B was able to see into his own thought processes.
 - C used powers that were potentially dangerous to humans.
 - D was fundamentally different from others he had encountered.
- 28 What is the writer's main idea in the third paragraph?
- A In the last 50 years machines have advanced more quickly than expected.
 - B There are certain features of chess that machines cannot replicate.
 - C Skill at chess is more significant than other machine achievements.
 - D Machines will always be more successful at some tasks than at others.
- 29 The writer describes the case of an office worker to illustrate the point that
- A everyday human actions involve a complex combination of skills.
 - B some types of mechanical activities require little conscious thought.
 - C some people are able to carry out multiple tasks at the same time.
 - D certain human tasks are more intellectually demanding than others.
- 30 In the fifth paragraph, the writer claims that
- A ideas of what success means vary across cultures.
 - B machines are most effective at game-like activities.
 - C machines are better at specialization than at adaptation.
 - D it may be impossible to isolate different skills involved in an activity.

Questions 31–36

Complete the summary using the list of words, **A–J**, below.

LIST OF WORDS

- A results
- B nourishment
- C basis
- D qualities
- E protection
- F evolution
- G surroundings
- H values
- I perception
- J movement

EMBODIMENT: AN ESSENTIAL FEATURE

Unlike machines, animals (including humans) have a body, including a brain which has evolved to provide (31) _____ for that body and allow its genes to be passed on. An animal's senses provide it with information about its (32) _____, and then it can react physically to what it (33) _____ through senses. The interaction of the body and brain therefore allows the creation of a link between (34) _____ and action. The (35) _____ which make us human and which distinguish us from animals have their (36) _____ in this link between the brain and the body, and have developed from it.

Questions 37–40

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write YES, NO or NOT GIVEN.

- (37) Any sort of artificial general intelligence must have the capacity for metabolism and reproduction.
- (38) Intelligence may depend on the ability to interact with a complex and unpredictable environment.
- (39) It is dangerous to rely too much on multi-specialist robots which could develop unfixable faults.
- (40) Training by experts is essential for acquiring certain types of skills.

Passage 55

The New Zealand Writer Margaret Mahy

A profile of one of New Zealand's most original storytellers

A In a career spanning some fifty years, Margaret Mahy has come to occupy a unique place in New Zealand writing with innovative fiction and original characterisation. In our solidly realist tradition, the linguistic fireworks of Mahy's children's fiction and the explorations of human behaviour at the heart of her supernatural teenage fiction stand gloriously alone. Mahy's clearest heir, I have always thought, is Elizabeth Knox – and maybe the true inheritance there is sheer singularity. Just as there is no one else like Knox in New Zealand writing, Mahy, too, has ventured into imaginative territory unknown to other local writers. So it was with great pleasure that I visited Mahy recently at her Governor's Bay home to talk about her new book. As with all journalists, she is generous to a fault with her time and attention; and, as always – despite her careful consideration of questions and thoughtful answers – I'm reminded she isn't truly comfortable making herself or her life the heart of any conversation.

B Certainly, any consideration of the style of Mahy's novels and picture books throws up some irresistible theories. As a writer committed to supporting herself through her art, she has seldom had the time for formal research. Rather, it has been a matter of going out and finding inspiration from her immediate environment: the writing on the side of a bus; a spelling mistake in a note to herself; the similarity between a cat and a fur hat. But despite these sometimes mundane origins, the settings for her stories are delightfully varied, as these books celebrate the dramatic plot twists and unpredictability of adventures on the high seas or in Antarctica, and also in quite unassuming places like the library or even down the back of a chair at home. Mahy has a lifelong affection for characters who are agents of upheaval and disturbance. Her junior and picture books are peppered with pirates, robbers and lions, though they appear alongside librarians, mothers and children, working against comfortable stereotype. Her fictions often have at their heart a young adult burdened with special powers, such as the ability to cast spells in order to transform their world in supernatural and fantastical ways. Another common feature is that, while the conclusions of her tales are usually predictable, they leave the reader feeling absolutely complete, the moral questions resulting from our hero's powers having been resolved.

C Many of these themes can be found in her new novel, *The Magician of Hoad*. The book was begun more than 15 years ago and envisioned as an 'entire' fantasy – one set in a fully imagined world with detailed history and complex tribal interrelationships, a classical hero quest at its heart. The story ballooned at one point to 800 pages and has been through at least two substantial rewrites. Now half its original size, it is a fascinating read – an adventure, a romance, and a gold mine of Mahy literary preoccupations.

D The other splendid Mahy publication this year is a re-issue of *Bubble Trouble* in a now beautifully illustrated edition. This tongue-twister tale first appeared in 1991, was included in *100 New Zealand Poems* (1993) and has been recited by Mahy at countless private and public functions. Perhaps more than any other work, *Bubble Trouble* is the Mahy that New Zealand children and their parents know so well, the rollicking story of a clown who serves up a joyous torrent of word play and unexpected rhyme.

E Those connections are four or five deep now. I read *The Lion in the Meadow* in the *School Journal* in the late 1960s; my stepdaughter listened to *The Boy Who Was Followed Home* over and over in the 1970s; my own children sat very still mouthing *The Great White Man-Eating Shark* in the 80s and 90s; now in the 21st century, my grandchildren have heard *Down the Dragon's Tongue*, *A Summery Saturday Morning* and *Dashing D* many times.

F Of course, readers are important to any working writer, but Mahy's espousal of the act of reading goes beyond that: a book is not properly finished, she has often said, until it has been read, because a reader brings something important to the book. So, doubtless out of need to build a market – she's not ignorant of her popularity – but also out of genuine care for that other dynamic part of the author-reader relationship, Mahy has, until recently, kept up a punishing schedule of public appearances.

G Private conversation with Mahy has always been a wild ride – marvellous, in the true sense of the word, the product of a hungry head and an infinite capacity to be astonished. She races away at one stage to consult an encyclopaedia for L M Montgomery's date of death, and speculates about the 'real' Montgomery, creator of the ever-popular *Anne of Green Gables*. Although I'd come to talk about her new book, I couldn't help but be captivated by her infectious curiosity.

H A few years ago, the writer David Hill told a funny story. Though his writing was, he conceded, very different from Mahy's, he had been affected by her peculiarly alert way of looking at the world, particularly the mad, slippery life of language. Once, Hill said, in a motel room, a sign on a door caught his eye: 'This door is alarmed.' Mahy would like that, thought Hill. She would enjoy the comedy just below the surface of the formal warning; she would leap immediately to the possibilities for story and language play: 'Yes, and this window is concerned, this light fitting is irritated.' His story was a wonderful comment on Mahy's vision.

Questions 27–31

Do the following statements agree with the views of the writer?

Write **YES** if the statement agrees with the views of the writer, **NO** if the statement contradicts the views of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 27 Mahy and Knox have sometimes criticised each other's work.
- 28 Mahy explores the traditional themes of New Zealand literature.
- 29 Mahy's children's books have been more popular than her teenage books.
- 30 Mahy is welcoming to interviewers.
- 31 Mahy is at ease speaking about her experiences.

Questions 32–35

Complete the summary using the list of words and phrases, **A–I**, below.

Mahy's determination to earn a living from writing has encouraged her to find the ideas for her work in 32 ————— places. The stories themselves are set in locations that are 33 —————. In terms of characterisation, almost all of her books feature unruly people, and very often an adolescent who possesses 34 ————— abilities. Finally, the endings of the stories tend to be 35 —————.

WORD BANK

- A disruptive and unpredictable
- B everyday
- C loveable
- D controversial
- E exotic and ordinary
- F unsatisfying but surprising
- G unconfident
- H unsurprising but satisfying
- I magical

Questions 36–40

Choose the correct letter, **A**, **B**, **C** or **D**.

- 36 What does the writer say about *The Magician of Hoad*?
 - A It is partly based on true history.
 - B It was restructured in the writing process.
 - C It is quite different from conventional stories.
 - D It does not reflect the usual themes of Mahy's work.

- 37 According to the writer, *Bubble Trouble*
- A was first illustrated in 1991.
 - B is written in plain language.
 - C is an old favourite for many families.
 - D has not been read aloud by Mahy before.
- 38 The writer's purpose in the fifth paragraph is to
- A show how Mahy's style has changed through generations.
 - B criticise Mahy's children's stories for being repetitive.
 - C describe how new media have changed entertainment.
 - D illustrate Mahy's popularity with generations of children.
- 39 According to the sixth paragraph, which of the following is true?
- A Readers sometimes do not finish Mahy's books.
 - B Mahy actively encourages feedback from readers.
 - C Readers are necessary in order to complete a book.
 - D Mahy does not realise how important she is to some readers.
- 40 What is the writer doing in the final paragraph?
- A illustrating Mahy's view of the world
 - B comparing Mahy with another writer
 - C correcting a misconception about Mahy
 - D suggesting reasons for Mahy's approach

Passage 56

The Peopling of Patagonia

Anthropologists continue to investigate human migration to Patagonia at the southern tip of South America

A The human settlement of the southern extremity of the Americas has always fascinated prehistorians. Viewed from a global perspective, this was the last major continental land mass to be reached by human beings. The earliest occupation of Patagonia carries obvious implications for understanding when the North and South American continents were peopled, because it gives a baseline that all calculations regarding the rate of dispersion of humans throughout both continents must take into account.

B For many years the human settlement of North and South America has been conceived of as beginning in the far north and travelling progressively southwards to Patagonia. However, fundamental disagreements developed concerning the length of time involved. Some scholars accepted a human presence in the Americas as early as 20,000 years ago, while others proposed that it could date no earlier than 8,000 years ago, and the debate is still with us today.

C The idea of a relatively 'late' settlement of the Americas (around 8,000 years ago) implies that a rapid process of migration took place. Herein lies a second debate which revolves around the question of how migration is to be understood. The 'late' model demands a hypothetical migration conceived of as a single, continually advancing wave of settlement. This has always been difficult to take seriously and many scholars now support the idea of an 'early' model that sees the migration as a less ordered migration, and this is surely the most realistic scenario as migrants slowly adapted to the diverse natural habitats they would have met while travelling through the continent.

D Those who argue for an earlier settlement, however, must contend with the lack of unequivocal evidence for archaeological sites older than around 14,000 years. Nevertheless, evidence for human occupation of the centre of South America is now securely dated to around 12,500 years ago at the Monte Verde site, which casts doubt on the 'late model'. The lack of archaeological evidence further south for this time period may be explained by the obstacle to humans on foot posed by the huge glacial streams that were present at that time.

E We can speculate then that the retreat of the Patagonian glaciers around 14,000 years ago allowed the initial human intrusion into a pristine environment, which was similar to that of early post-glacial Europe. Human settlement of the vast horizontal expanse of treeless high country must have been tenuous at best, and the evidence for this occupation remains relatively scant, most of it coming from rock shelters in Argentina and Chile.

F There is, however, reliable evidence from these sites to confirm the presence of humans by around 11,000 years ago in different habitats, and some hints of an even older occupation. However, some other sites where evidence for even earlier human occupation was initially posited, have recently come under fresh scrutiny. This is because anthropologists have come to recognise that bones or other evidence may be deposited in caves by natural agency, in other words by other forces such as floods or predators, and not necessarily by human action.

G We shall turn now to a more detailed discussion of the archaeological evidence found in various parts of Patagonia. At the site located beside Chinchihuapi Creek, excavations have produced convincing evidence of human occupation, including hut foundations and wooden artefacts. They were buried in layers of peat, which has the property of preserving wood remarkably well, and as a result radiocarbon dating* tests have shown these artefacts to date from around 12,500 years ago. One of the most famous Patagonian sites is a cave known as Los Toldos. However, the evidence from this site has recently been called into question, because dispersed flecks of carbon used in the test process were taken unsystematically from many different places in the site. As a result, the association of this material with the artefacts is not at all clear. About 150 kilometres south is the site called El Ceibo, where a similar collection of artefacts to that found at Los Toldos has been discovered from the lowest levels of the dig, but as yet no radiocarbon dates are available and this sort of analysis of the existing evidence is required before the site's value can be confirmed.

H The Arroyo Feo site is located very close to the high plateau. The artefacts from the earliest occupations were found at the same depth and have the same origins as those from Los Toldos, and have been securely dated to around 9,000 years ago. Another site that is mentioned in the debate is at Las Buitreras, where a number of stone flakes associated with bone remains of various animals have been discovered. However, anthropologists now believe that presumed cut marks on the bones are somewhat dubious, and despite detailed testing there is no way of securely relating any of these remains with human occupation. Finally, some 50 kilometres to the south is the site at Cueva Fell, which was the first Patagonian site to be systematically studied by modern archaeological methods. However, it is now recognised that the utility of this site must be restricted to its direct vicinity, given changes to the nearby area caused by flooding, and findings cannot be freely extrapolated further afield.

I In conclusion, based on the evidence from a number of reliable sites, it seems probable that human populations reached Patagonia around 11,000 years ago.

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 In the first paragraph, what is the writer's main point about migration to Patagonia?
 - A It started earlier than previously thought.
 - B Historians have overlooked its importance.
 - C It impacts on research into the wider region.
 - D Researchers have calculated its effects on the environment.
- 2 In the second paragraph, what is the writer's purpose?
 - A to challenge previous research
 - B to propose new areas to investigate
 - C To summarize a scholarly debate
 - D To suggest reasons for human migration
- 3 The writer refers to the 'late' model to
 - A compare it with another theory of migration.
 - B evaluate the success of American migration.
 - C Criticize the speed of research into migration.
 - D compare migration in different parts of the world
- 4 What is the writer's main point about the 'early' model?
 - A Scholars support the idea of fast migrations.
 - B It is too random to be a convincing theory.
 - C South America was more habitable at an earlier time.
 - D It is more consistent with the physical conditions of the land.
- 5 What does the writer suggest about the Monte Verde site?
 - A It is much younger than researchers once estimated.
 - B It provides supporting evidence for relatively early settlement.
 - C Archaeologists believe the site is of questionable value.
 - D Streams exposed the site, making new research possible.

Questions 6–9

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims, **NO** if the statement contradicts the claims, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 6 The conditions encountered by the first migrants to Patagonia were unique.
- 7 In the high country the first migrants hunted wild animals for food.
- 8 Archaeologists have failed to conclude from the evidence found at rock shelters in Argentina and Chile.
- 9 Archaeological evidence can be moved from place to place in a variety of ways.

Questions 10–14

Complete the summary using the list of words and phrases, **A–J**, below.

WORD LIST

- A** fixed date
- B** random collection
- C** similar properties
- D** good condition
- E** scientific evaluation
- F** huge quantities
- G** new samples
- H** reliable connection
- I** skilled preservation
- J** immediate surroundings

The archaeological evidence from Patagonia

Building remains and other evidence have been found in (10) _____ at the Chinchihuapi Creek site, and because of this it has been possible to date them to around 12,500 years ago. However, the (11) _____ of the samples taken from Los Toldos means that this site is of doubtful value. El Ceibo is a more promising dig, where the examination of artefacts would be beneficial in order to confirm the usefulness of discoveries there. The remains found at the Arroyo Feo site show (12) _____ and date from around 9,000 years ago. Unfortunately no (13) _____ can be made between the samples taken from Las Buitreras and human presence. The findings of the work carried out at Cueva Fell cannot provide useful information beyond the (14) _____. In conclusion, though the evidence is mixed, it is believed that human population of Patagonia began about 11,000 years ago.

Passage 57

The Placebo Effect

The Power of Nothing — how a treatment with no direct medical effect can still heal, and why it matters

Want to devise a new form of alternative medicine? No problem. Here is the recipe. Be warm, sympathetic, reassuring and enthusiastic. Your treatment should involve physical contact, and each session with your patients should last at least half an hour. Encourage your patients to take an active part in their treatment and understand how their disorders relate to the rest of their lives. Tell them that their own bodies possess the true power to heal. Make them pay you out of their own pockets. Describe your treatment in familiar words, but embroidered with a hint of mysticism: energy fields, energy flows, energy blocks, meridians, forces, auras, rhythms and the like. Refer to the knowledge of an earlier age: wisdom carelessly swept aside by the rise and rise of blind, mechanistic science. Oh, come off it, you are saying. Something invented off the top of your head could not possibly work, could it?

Well yes, it could — and often well enough to earn you a living. A good living if you are sufficiently convincing, or better still, really believe in your therapy. Many illnesses get better on their own, so if you are lucky and administer your treatment at just the right time you will get the credit. But that's only part of it. Some of the improvement really would be down to you. Your healing power would be the outcome of a paradoxical force that conventional medicine recognises but remains oddly ambivalent about: the placebo effect.

Placebos are treatments that have no direct effect on the body, yet still work because the patient has faith in their power to heal. Most often the term refers to a dummy pill, but it applies just as much to any device or procedure, from a sticking plaster to a crystal to an operation. The existence of the placebo effect implies that even quackery may confer real benefits, which is why any mention of placebo is a touchy subject for many practitioners of complementary and alternative medicine, who are likely to regard it as tantamount to a charge of charlatanism. In fact, the placebo effect is a powerful part of all medical care, orthodox or otherwise, though its role is often neglected or misunderstood.

One of the great strengths of CAM may be its practitioners' skill in deploying the placebo effect to accomplish real healing. "Complementary practitioners are miles better at producing non-specific effects and good therapeutic relationships," says Edzard Ernst, professor of CAM at Exeter University. The question is whether CAM could be integrated into conventional medicines, as some would like, without losing much of this power.

At one level, it should come as no surprise that our state of mind can influence our physiology: anger opens the superficial blood vessels of the face; sadness pumps the tear glands. But exactly how placebos work their medical magic is still largely unknown. Most of the scant research done so far has focused on the control of pain because it's one of the commonest complaints and lends itself to experimental study. Here, attention has turned to the endorphins, morphine-like neurochemicals known to help control pain.

"Any of the neurochemicals involved in transmitting pain impulses or modulating them might also be involved in generating the placebo response," says Don Price, an oral surgeon at the University of Florida who studies the placebo effect in dental pain. "But endorphins are still out in front."

That case has been strengthened by the recent work of Fabrizio Benedetti of the University of Turin, who showed that the placebo effect can be abolished by a drug, naloxone, which blocks the effects of endorphins. Benedetti induced pain in human volunteers by inflating a blood-pressure cuff on the forearm. He did this several times a day for several days, using morphine each time to control the pain. On the final day, without saying anything, he replaced the morphine with a saline solution. This still relieved the subjects' pain: a placebo effect. But when he added naloxone to the saline the pain relief disappeared. Here was direct proof that placebo analgesia is mediated, at least in part, by these natural opiates.

Still, no one knows how belief triggers endorphin release, or why most people can't achieve placebo pain relief simply by willing it. Though scientists don't know exactly how placebos work, they have accumulated a fair bit of knowledge about how to trigger the effect. A London rheumatologist found, for example, that red dummy capsules made more effective painkillers than blue, green or yellow ones. Research on American students revealed that blue pills make better sedatives than pink, a colour more suitable for stimulants. Even branding can make a difference: if Aspro or Tylenol is what you like to take for a headache, their chemically identical generic equivalents may be less effective.

It matters, too, how the treatment is delivered. Decades ago, when the major tranquilliser chlorpromazine was being introduced, a doctor in Kansas categorised his colleagues according to whether they were keen on it, openly skeptical of its benefits, or took a "let's try and see" attitude. His conclusion: the more enthusiastic the doctor, the better the drug performed. And this year Ernst surveyed published studies that compared doctors' bedside manners. The studies turned up one consistent finding: "Physicians who adopt a warm, friendly and reassuring manner are more effective than those whose consultations are formal and do not offer reassurance."

Warm, friendly and reassuring are precisely CAM's strong suits, of course. Many of the ingredients of that opening recipe — the physical contact, the generous swathes of time, the strong hints of supernormal healing power — are just the kind of thing likely to impress patients. It's hardly surprising, then, that complementary practitioners are generally best at mobilising the placebo effect, says Arthur Kleinman, professor of social anthropology at Harvard University.

Questions 1–6

Complete each sentence with the correct ending, **A–H**, below. Write the correct letter in each box.

NB You may use any letter more than once.

- | | | | |
|---|--|---|-------|
| 1 | Appointments with an alternative practitioner | 1 | _____ |
| 2 | An alternative practitioner's description of the treatment | 2 | _____ |
| 3 | An alternative practitioner who has faith in what he does | 3 | _____ |
| 4 | The illness of patients convinced of alternative practice | 4 | _____ |
| 5 | Improvements of patients receiving alternative practice | 5 | _____ |
| 6 | Conventional medical doctors (who are aware of placebo) | 6 | _____ |

LIST OF ENDINGS

- A** should easily be understood
- B** should improve by itself
- C** should not involve any mysticism
- D** ought to last a minimum length of time
- E** needs to be treated at the right time
- F** should give more recognition
- G** can earn valuable money
- H** do not rely on any specific treatment

Questions 7–9

Choose the correct letter, **A, B, C** or **D**.

- 7 In the fifth paragraph, the writer uses the example of anger and sadness to illustrate that
- A** people's feelings could affect their physical behaviour.
 - B** scientists don't understand how the mind influences the body.
 - C** research on the placebo effect is very limited.
 - D** how placebo achieves its effect is yet to be understood.

- 8 Research on pain control attracts most of the attention because
- A scientists have discovered that endorphins can help to reduce pain.
 - B only a limited number of researchers gain relevant experience.
 - C pain reducing agents might also be involved in the placebo effect.
 - D patients often experience pain and like to complain about it.
- 9 Fabrizio Benedetti's research on endorphins indicates that
- A they are widely used to regulate pain.
 - B they can be produced by willful thoughts.
 - C they can be neutralized by introducing naloxone.
 - D their pain-relieving effects do not last long enough.

Questions 10–14

*Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- 10 There is enough information for scientists to fully understand the placebo effect.
- 11 A London based researcher discovered that red pills should be taken off the market.
- 12 People's preference for brands would also have an effect on their healing.
- 13 Medical doctors have a range of views of the newly introduced drug chlorpromazine.
- 14 Alternative practitioners are seldom known for applying the placebo effect.

Passage 58

The Role of Accidents in Business

From cornflakes to Post-it Notes, some of the biggest business breakthroughs were never planned at all

In 1894 Dr John Kellogg and his brother, Will, were supervising a hospital and health spa in Michigan. The patients were on a restricted diet. One day, the brothers left cooked wheat untended for more than 24 hours. When they returned, they saw what they had done. It was no good to eat, but they decided to run the stale wheat through rollers, just to see how it would turn out. Normally, the process produced long sheets, but they were surprised to discover that this time the rollers created flat flakes. They baked them, and then tried the same thing with corn. From this accidental discovery came the cornflakes that generations have now been eating for breakfast.

Accidents happen; there is nothing predictable and orderly about innovation. Nobel laureate Sir Alan Hodgkin, who discovered how nerve cells transmit electrical impulses between the skin and the brain, commented: "I believe that the record of my published papers conveys an impression of directedness and planning which does not at all coincide with the actual sequence of events."

The same rule applies in business. The mistake that gave us cornflakes keeps repeating itself in the history of disruptive innovation, the kind that transforms markets. Louis Daguerre, for instance, discovered the technique that gave us photography in the 1830s, when drops of mercury from a shattered thermometer produced a photographic image. The microwave was discovered when Percy Spencer, a scientist with Raytheon, was testing a new vacuum tube and discovered that the sweet in his pocket had melted. The artificial sweetener, saccharin, was the unintentional result of a medical scientist's work on a chemical treatment for gastric ulcers. While working for the firm 3M, researcher Art Fry had no idea he was taking the first steps towards Post-it Notes when he used bits of adhesive office paper that could be easily lifted off the page to replace the scrap paper bookmarks that kept falling out of his hymn book.

Breakthrough and disruptive innovation are rarely driven by orderly process. Usually they come out of a chaotic, haphazard mess, which is why big companies, full of managers schooled in business programmes designed to eliminate random variation and mistakes, struggle with them. In these sorts of environments, accidents are called failures and are discouraged.

It is no surprise then that research from the late British economist Paul Geroski and London Business School's Constantinos Markides found that companies that were skilled at innovation were usually not that skilled when it came to commercialisation, and vice

versa. Their book, *Fast Second*, divides businesses into "colonists and consolidators." Small and nimble, colonists are adept at creating market niches but are terrible institution builders. Consolidators, with their strong cultures of discipline and cost control, know how to take clever ideas from other firms and turn them into mass-market items. Microsoft is a prime instance of this.

With companies spending hundreds of billions of dollars on research and development, US academics Robert Austin and Lee Devin examined how managers can encourage productive slip-ups. In their article *Accident, Intention and Expectation in the Innovation Process*, they argue that business processes actually prevent helpful mis-steps from occurring. According to their catalogue of accidents, not all false steps and mishaps are equal. Accidents, they say, come from unlikely mental associations such as memories and vague connections, looking for something and finding it in an unexpected way, looking for one thing and finding something else, and not looking for anything but finding something valuable.

Accident-prone innovation, they say, requires companies to get outside the "cone of expectation." It means throwing together groups from diverse backgrounds, and combining ideas in unpredictable ways. Other strategies also include having systems that watch out for accidents and examine them for value, generating them when they don't happen often enough, seizing on the useful ones, capturing their valuable features, and building on them to add value and give potential for useful accidents.

All this, however, requires thinking that is often counter-intuitive to the way businesses operate. In other words, it is the kind of thinking that goes against the beliefs of most business managers. It runs counter to the notion frequently pushed by consultants that you can "harness" creativity and direct it to line up with intention. "The cost of accidents that do not prove valuable are often of concern to people in business," they write. "In business, people tend to call such efforts failure."

There are tentative signs that more companies are starting to realise that failure can lead to commercial gain, and that this is part of the risk-taking that underpins innovation. Australia's largest brewing company, for example, made a bad error when it launched a new beer called *Empire Lager*, pitched at younger consumers. Having spent a fortune creating a beer with a sweeter taste, designing a great-looking bottle and a television campaign, Foster's was left with a drink that no-one wanted to buy. The target market was more interested in brands built up by word of mouth.

Instead of wiping the unsuccessful product launch, Foster's used this lesson learned to go on and develop other brands instead. One of them, *Pure Blonde*, is now ranked as Australia's fifth-largest beer brand. Unlike *Empire Lager*, there has been almost no promotion and its sales are generated more by word of mouth.

Other companies are taking similar steps to study their own slip-ups. Intuit, the company behind financial tools such as Quicken, holds regular "When Learning Hurts" sessions. But this sort of transformation is never easy. In a market that focuses on the short-term, convincing employees and shareholders to tolerate failure and not play it safe is a big thing to ask.

Questions 1–5

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 1 The delay in the process used by the Kellogg brothers affected the final product.
- 2 Sir Alan Hodgkin is an example of someone whose work proceeded in a logical and systematic way.
- 3 Daguerre is an exception to the general rule of innovation.
- 4 The discovery of saccharin occurred by accident during drug research.
- 5 The company 3M should have supported Art Fry by funding his idea of Post-it Notes.

Questions 6–9

Complete each sentence with the correct ending, **A–H**, below. Write the correct letter in each box.

- | | |
|---|---------|
| 6 The usual business environment... | 6 _____ |
| 7 Geroski and Markides's book... | 7 _____ |
| 8 Microsoft is an example of a company which... | 8 _____ |
| 9 The origin of useful accidents... | 9 _____ |

LIST OF ENDINGS

- A can be found in unusual thoughts and chance events
- B can be taught in business schools
- C has made a success from someone else's invention
- D is designed to nurture differences
- E is unlikely to lead to creative innovation
- F says that all mistakes are the same
- G shows that businesses are good at either inventing or selling
- H suggests ways of increasing the number of mistakes

Questions 10–14

Choose the correct letter, **A**, **B**, **C** or **D**.

- 10 How do Austin and Devin advise companies to get out of the 'cone of expectation'?
- A by decreasing the number of company systems
 - B by forming teams of different types of people
 - C by hiring new and creative people
 - D by holding regular brainstorming meetings
- 11 In recommending 'counter-intuitive' thinking, what do Austin and Devin imply?
- A that failing at business is bad for staff morale
 - B that innovation cannot be planned for
 - C that most businesses should be devoted to avoiding mistakes
 - D that the cost of mistakes is an important consideration
- 12 The writer describes the Empire Lager disaster in order to show that
- A success can come out of a business failure.
 - B the majority of companies now value risk-taking.
 - C TV advertising works better on older people.
 - D young beer drinkers do not like a sweet taste.
- 13 Pure Blonde has been more successful than Empire Lager because
- A digital media other than TV were used.
 - B it was advertised under a different brand name.
 - C it was launched with very little advertising.
 - D the advertising budget was larger.
- 14 The writer concludes that creating a culture that learns from mistakes
- A brings short-term financial gains
 - B can be very difficult for some companies
 - C holds no risk for workers
 - D is a popular move with shareholders

Passage 59

The Significant Role of Mother Tongue in Education

Why keeping children's first language alive is not a distraction from school success, but the foundation for it

One consequence of population mobility is an increasing diversity within schools. To illustrate, in the city of Toronto in Canada, 58% of kindergarten pupils come from homes where English is not the usual language of communication. Schools in Europe and North America have experienced this diversity for years, and educational policies and practices vary widely between countries and even within countries. Some political parties and groups search for ways to solve the problem of diverse communities and their integration in schools and society. However, they see few positive consequences for the host society and worry that this diversity threatens the identity of the host society. Consequently, they promote unfortunate educational policies that will make the 'problem' disappear. If students retain their culture and language, they are viewed as less capable of identifying with the mainstream culture and learning the mainstream language of the society.

The challenge for educators and policy-makers is to shape the evolution of national identity in such a way that rights of all citizens (including school children) are respected, and the cultural, linguistic, and economic resources of the nation are maximised. To waste the resources of the nation by discouraging children from developing their mother tongues is quite simply unintelligent from the point of view of national self-interest. A first step in providing an appropriate education for culturally and linguistically diverse children is to examine what the existing research says about the role of children's mother tongues in their educational development.

In fact, the research is very clear. When children continue to develop their abilities in two or more languages throughout their primary school, they gain a deeper understanding of language and how to use it effectively. They have more practice in processing language, especially when they develop literacy in both. More than 150 research studies conducted during the past 25 years strongly support what Goethe, the famous eighteenth-century German philosopher, once said: the person who knows only one language does not truly know that language. Research suggests that bilingual children may also develop more flexibility in their thinking as a result of processing information through two different languages.

The level of development of children's mother tongue is a strong predictor of their second language development. Children who come to school with a solid foundation in their mother tongue develop stronger literacy abilities in the school language. When parents and other caregivers (e.g. grandparents) are able to spend time with their children and tell stories or discuss issues with them in a way that develops their mother tongue, children

come to school well-prepared to learn the school language and succeed educationally. Children's knowledge and skills transfer across languages from the mother tongue to the school language. Transfer across languages can be two-way: both languages nurture each other when the educational environment permits children access to both languages.

Some educators and parents are suspicious of mother tongue-based teaching programmes because they worry that they take time away from the majority language. For example, in a bilingual program when 50% of the time is spent teaching through children's home language and 50% through the majority language, surely children won't progress as far in the latter? One of the most strongly established findings of educational research, however, is that well-implemented bilingual programmes can promote literacy and subject-matter knowledge in a minority language without any negative effects on children's development in the majority language. Within Europe, the Foyer program in Belgium, which develops children's speaking and literacy abilities in three languages (their mother tongue, Dutch and French), most clearly illustrates the benefits of bilingual and trilingual education (see Cummins, 2000).

It is easy to understand how this happens. When children are learning through a minority language, they are learning concepts and intellectual skills too. Pupils who know how to tell the time in their mother tongue understand the concept of telling time. In order to tell time in the majority language, they do not need to re-learn the concept. Similarly, at more advanced stages, there is transfer across languages in other skills such as knowing how to distinguish the main idea from the supporting details of a written passage or story, and distinguishing fact from opinion. Studies of secondary school pupils are providing interesting findings in this area, and it would be worth extending this research.

Many people marvel at how quickly bilingual children seem to 'pick up' conversational skills in the majority language at school (although it takes much longer for them to catch up with native speakers in academic language skills). However, educators are often much less aware of how quickly children can lose their ability to use their mother tongue, even in the home context. The extent and rapidity of language loss will vary according to the concentration of families from a particular linguistic group in the neighbourhood. Where the mother tongue is used extensively in the community, then language loss among young children will be less. However, where language communities are not concentrated in particular neighbourhoods, children can lose their ability to communicate in their mother tongue within 2-3 years of starting school. They may retain receptive skills in the language but they will use the majority language, in speaking with their peers and siblings and in responding to their parents. By the time children become adolescents, the linguistic division between parents and children has become an emotional chasm. Pupils frequently become alienated from the cultures of both home and school with predictable results.

Questions 1–4

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 What point did the writer make in the second paragraph?
 - A Some present studies on children's mother tongues are misleading.
 - B A culturally rich education programme benefits some children more than others.
 - C Bilingual children can make a valuable contribution to the wealth of a country.
 - D The law on mother tongue use at school should be strengthened.
- 2 Why does the writer refer to something that Goethe said?
 - A to lend weight to his argument
 - B to contradict some research
 - C to introduce a new concept
 - D to update current thinking
- 3 The writer believes that when young children have a firm grasp of their mother tongue
 - A They can teach older family members what they learnt at school.
 - B They go on to do much better throughout their time at school.
 - C They can read stories about their cultural background.
 - D They develop stronger relationships with their family than with their peers.
- 4 Why are some people suspicious about mother tongue-based teaching programmes?
 - A They worry that children will be slow to learn to read in either language.
 - B They think that children will confuse words in the two languages.
 - C They believe that the programmes will make children less interested in their lessons.
 - D They fear that the programmes will use up valuable time in the school day.

Questions 5–9

Complete the summary using the list of words, **A–J**, below.

WORD LIST

- A teachers
- B school
- C dislocation
- D rate
- E time
- F family
- G communication
- H type
- I ability
- J area

BILINGUAL CHILDREN

It was often recorded that bilingual children acquire the (5) _____ to converse in the majority language remarkably quickly. The fact that the mother tongue can disappear at a similar (6) _____ is less well understood. This phenomenon depends, to a certain extent, on the proportion of people with the same linguistic background that have settled in a particular (7) _____. If this is limited, children are likely to lose the active use of their mother tongue. And thus, no longer employ it even with (8) _____, although they may still understand it. It follows that teenage children in these circumstances experience a sense of (9) _____ in relation to all aspects of their lives.

Questions 10–14

*Do the following statements agree with the information given in Reading Passage 3? Write **YES** if the statement agrees with the views of the writer, **NO** if the statement contradicts the views of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- (10) Less than half of the children who attend kindergarten in Toronto have English as their mother tongue.
- (11) Research proves that learning the host country language at school can have an adverse effect on a child's mother tongue.
- (12) The Foyer program is accepted by the French education system.
- (13) Bilingual children are taught to tell the time earlier than monolingual children.
- (14) Bilingual children can apply reading comprehension strategies acquired in one language when reading in the other.

Passage 60

The Strange World of Sight

Seeing is believing, it is said. But, asks Richard Gregory, could it be the other way round?

Two of the great British men of the 17th century, the philosopher John Locke and the physicist Isaac Newton, were both aware that objects are not coloured, and that against all appearances light is not coloured either. This is still not generally recognized even now, 400 years later, because it seems so implausible. Yet it tells us something very important – that perceptions are not identical with what we perceive, and may be very different.

The most accurate historical account of perception is that of the 19th-century German scientist Hermann von Helmholtz. However, it was ridiculed at the time. Von Helmholtz thought that perceptions are unconscious inferences we make based on a combination of clues provided by the eyes and other senses, and knowledge of the world. This idea of unconscious inference for perception preceded, by several years, the psychoanalyst Freud's notion of the unconscious, which was also initially treated with derision because it undermined the notion of humans as pre-eminently rational beings who could be held responsible for their actions and awarded blame or praise accordingly.

Crucially, perception of the present depends on rich, though of course not always correct or appropriate, knowledge from the past. We interpret sense data (what we hear, touch, taste, see and smell) from the present according to what we already know. This raises the question: if we see the present through memory, why aren't past and present confused? The pioneering Russian neurologist Alexander Luria described the case of Mr. S, who had a remarkable memory. However, he was prone to just such confusions, for example mistaking seeing his clock for remembering it, and so failing to get up in the morning. This suggests that perhaps an important function of perception is to underline the present. Individual perceptions have a vividness that is rare for memories, which might be how we are able to separate them. Try this: look at something for a few seconds, and then shut your eyes and visualize it in memory. You will almost certainly find that the memory is pale by comparison with the perception. Perhaps this is why past and present are not normally confused, Luria's Mr S had exceptionally vivid memories, and rich synesthesia (experiencing perceptions from another sense as well as the one being stimulated, such as musical notes experienced as colours), which may be why he confused seeing with having seen.

The complexity of processes involved in how we see first impressed itself on me 45 years ago. With my colleague Jean Wallace, I studied the rare case of Sydney Bradford, a man who had been born blind but, through a corneal graft at the age of 52, suddenly found himself able to see. Almost immediately after the operation he was able to "see" but he could only see those things that he already knew about, having experienced them through

touch. It was his touch memories that enabled him to perceive them with his eyes. When Bradford was first taken to the zoo, he proved utterly unable to see an elephant as he had no knowledge to make sense of his perceptions.

The more recent case in California of Mike May, who was also born blind, is similar. Since his operation, his sight has gradually improved as he learns to see, for example, by understanding how shadows represent depth and tell us about the shape of things. Some of the consequences of May's new-found vision were less happy. He had been a champion blind skier, but following the operation, he would have to shut his eyes while skiing to block out what he now found was a terrifying sight.

But acceptance of this intimate connection between memory and perception, even though it was first noticed in the 17th century, has been slow in brain science. Despite the fact that state-of-the-art brain imaging shows that perception animates parts of the brain associated with both present information and memory, most research on memory and perception is still undertaken as if these were separate processes. Seeing used to be thought of as taking place only in the eyes, and in quite specialized brain regions: but now it seems that half the brain is occupied with seeing, requiring a lot of energy. Perhaps this is why we shut our eyes for a rest.

It is not just extreme cases like Mike May, but also much more common errors of seeing – illusions – that can reveal the crucial role of memory in governing what we (think we) see. Perception depends on specific knowledge and probabilities. Our brains calculate the likelihood of what is out there, and when too far-fetched, perceptions are rejected.

A dramatic and discomfiting example is looking at the two sides of a face-mask. From the front it is a convex shape with the nose sticking out. Then if the mask is rotated, the back of the mask will be seen as convex, though we know that it must be concave. It is almost, if not quite, impossible to sketch the back of a hollow mask to look as it is – hollow. Science often learns from what does not happen: people not seeing a hollow face as hollow is the most revealing experiment on perception. The unsettling truth from brain science is that even people with no visual impairment see what, at some level, they expect to see, and often miss things as they really are.

Questions 27–30

Choose the correct letter, **A**, **B**, **C** or **D**.

- 27 Why does the writer refer to Locke and Newton in the first paragraph?
- A** to indicate that his article will cover several scientific fields.
 - B** to stress how much physics has changed in 400 years.
 - C** to persuade the reader to take him seriously.
 - D** to point out that his notions are not new.

- 28 According to the writer, why was Freud's theory of the unconscious mocked?
- A It was too complex for his contemporaries to understand.
 - B It involved criticism of the way people behaved in society.
 - C People felt that it devalued the accepted concept of humanity.
 - D People assumed that it was intended as a joke.
- 29 The writer describes Mr S failing to get up in order to demonstrate
- A how realistic most people's memories are.
 - B how hard it is to tell dreaming and waking apart.
 - C how unusual it is to mistake a perception for a memory.
 - D how valuable knowledge of the past can be.
- 30 What point is the writer making in the text as a whole?
- A Perception involves much more than the data collected by the eyes.
 - B Learning to see as an adult can be a time-consuming process.
 - C Science is failing to devote enough attention to sight.
 - D Human perception is remarkably reliable.

Questions 31–36

Do the following statements agree with the views of the writer in Reading Passage 3? Write **YES** if the statement agrees with the views of the writer, **NO** if the statement contradicts the views of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 31 Sydney Bradford relied on recollections of objects he had been told about to help him see after his operation.
- 32 People who only start to see as adults can learn to see as other people do in time.
- 33 People who have gained their sight as adults find certain activities harder to do than before.
- 34 It is evident now that sight involves the eyes and one particular area of the brain.
- 35 The mask experiment is particularly useful in training people who are regaining their sight.
- 36 People with perfect vision can fail to interpret objects correctly under certain circumstances.

Questions 37–40

Complete the summary using the list of words, **A–J**, below.

WORD LIST

- A back
- B brain
- C view

- D** convex
- E** sight
- F** nose
- G** round
- H** hollow
- I** drawing
- J** preconception

THE MASK EXPERIMENT

In this experiment, having looked at the front of a simple face-mask, subjects look at the reverse. However, the subjects are convinced that they are still looking at a mask which is (37) _____ in shape. They believe that the (38) _____ is poking out in the normal manner because that is what they are used to seeing. Attempting to make a (39) _____ of the mask in this orientation leads to the same problem. The subjects fail to see a concave form because of the (40) _____ they have that the features of a face stick out.

Passage 61

The Tuatara – Past and Future

How New Zealand's ancient living fossil nearly disappeared, and how conservation science is bringing it back

A The New Zealand species of lizard, the tuatara, is firmly embedded in the national psyche: an icon for today which dates from the age of dinosaurs; an ancient reptile commemorated on the back of the five cent coin. New Zealanders feel an affinity with the tuatara, and accept that active conservation management is required to ensure it will be among the legacies left to future generations.

B When European explorers reached New Zealand in 1769 they found two large islands, which together they called the 'mainland', and many tiny offshore islands around the coast. The naturalists who came with the explorers disregarded the tuatara, though it is improbable none were seen. Only several decades later did a tuatara specimen reach the British Museum, where it was eventually classified as just another type of lizard. One of the first scientists who realized that aspects of tuatara anatomy were odd – unchanged for tens of thousands of years – was Albert Gunther in 1876. Gunther believed the tuatara was one of the most valuable objects in zoological anatomical collections, and also noted, in passing, the reptile was likely to become extinct. From today's perspective, it is striking that Gunther expressed no concern about the probable demise of the tuatara. He and his contemporaries were products of their age, strongly influenced by Charles Darwin's theory, which had only recently been published. Their views were something like this: 'Extinction is a natural process. It is sad that species disappear, but that is part of nature.'

C There is a second important aspect of Gunther's work. He recorded, correctly, that some of the mammals introduced by Europeans were predators of the tuatara – particularly rats. But what he did not realise was that New Zealand has two species of rat, both introduced, both with an appetite for tuatara: the ship's rat came with European explorers and settlers; but the kiore rat had already been in the country for hundreds of years, brought by Polynesians from the Pacific Islands. Gunther failed to recognise the distinction, believing all rats to be a relatively recent introduction.

D Little further research was conducted until Ian Crook of the NZ Wildlife Service published his findings in 1973, which can be summarized as follows. Tuatara thrive on offshore islands with no rats. Tuatara never survived on islands with ship's rats. On a few islands, small and declining populations of tuatara occur with the kiore. This should not be seen, however, as evidence that tuatara and kiore can coexist. Rather, Crook proposed, kiore probably only arrived recently on such islands, and thus the small populations represent extinctions in progress.

E Throughout the 1990s, Richard Holdaway and his colleagues at Victoria University in Wellington documented the surprising discovery that kiore probably arrived about 1800 years ago, although the human population of New Zealand is thought to be no older than 800 years. How is this possible? Presumably, Holdaway argued, the kiore were brought by Polynesian explorers who visited the country but did not settle. Thereafter, the rats were agents of ecological warfare, exterminating perhaps 1000-3000 species. Thus, tuatara and many other species were already rare or extinct when permanent human inhabitants – the Maori – arrived around 1300. This hypothesis is still being debated, but the evidence continues to accumulate in its favour.

F Conservation practice has changed dramatically since Crook's findings were published in 1973. Eradication of rats from any given environment was believed to be virtually impossible until about 1980, but since then has become routine. Enormous conservation benefits are accruing as newly rat-free offshore islands are providing sanctuaries for the country's rarest species. In 1995, for example, Nicola Nelson of the Department of Conservation established 68 tuatara on Titi Island. Since then, four more populations of tuatara have been established elsewhere under similar conditions. Today, numbers of tuatara are still a fraction of what they once were, but for the first time in 1800 years the decline has been reversed.

G While the recovery of rare species is itself a good thing, the truly significant outcome of this research is that it liberates the imagination. If we can remove predatory introduced mammals from islands, why not from the mainland too? Perhaps the questions we ask should demonstrate even more visionary ambition. Can non-mammalian pests also be removed from the mainland? Our rivers, for example, are full of surrogate rats, in the form of introduced species of fish called trout. Some day more people will understand that trout have replaced a whole native fauna in our waterways, just as rats replaced tuatara on the mainland. Will such knowledge lead to the creation of mainland 'aquatic islands' where we can once again establish those species of indigenous fish that used to live in our rivers? Similarly, can bellbirds and tuis replace birds like starlings and mynahs?

H The answers to such questions are uncertain, and opposing sides will doubtless be fiercely debated. But the role of scientific knowledge in illuminating the past will be crucial. Just as we now no longer tolerate extinction, in the future we may no longer accept a mainland devoid of the biological wonders of our past such as tuatara. Conservation is thus not primarily about the past but about imagining and then creating the future we wish for our children and ourselves. For 80 million years until humans arrived, tuatara occurred throughout New Zealand – might they do so again?

Choose the correct letter, **A**, **B**, **C** or **D**.

- 27 What are we told about the Europeans who arrived in 1769?
- A They thought there was only one large island.
 - B They had not come to study natural history.
 - C They had no interest in the tuatara.
 - D They sent a tuatara to the British Museum.
- 28 What does the author say about Albert Gunther in paragraph 3?
- A He believed the tuatara could fetch a high price.
 - B He was typical of his generation of scientists.
 - C He disagreed with Charles Darwin's theory.
 - D He wanted to stop the tuatara becoming extinct.
- 29 What did Albert Gunther think about the rats in New Zealand?
- A They did not eat the tuatara.
 - B There was one species of rat.
 - C There had always been rats in New Zealand.
 - D They were killed by Polynesians.
- 30 What did Ian Crook conclude from his research?
- A Tuatara are safe on small islands.
 - B Ship's rats kill more tuatara than kiore.
 - C Kiore cannot swim to offshore islands.
 - D Rats and tuatara cannot live together.
- 31 What were the findings of Richard Holdaway's research?
- A Maori settled more recently than previously thought.
 - B The first Polynesian explorers formed permanent settlements.
 - C Ship's rats are the oldest rat species in the country.
 - D Rats caused extinctions before any humans settled.

Questions 32–35

Do the following statements agree with the claims of the writer in Reading Passage 3?

Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 32 The available research supports Holdaway's theory but it has not been proved.
- 33 Nowadays, it is possible to totally destroy a population of rats on a small island.
- 34 Crook was the first person to recognize the potential of offshore islands as sanctuaries.
- 35 Tuatara numbers are continuing to fall.

Questions 36–40

Complete the summary using the list of words, A–H, below.

What conclusions can we draw?

The most important result of the tuatara research is that it frees our (36) _____.
For example, there are many similarities between rats and (37) _____. Should we
now go further and consider reintroducing (38) _____ to our mainland rivers?
Perhaps our children will come to believe in the (39) _____ of species, in the same
way that our generation refuses to accept (40) _____.

WORD BANK

- A natural evolution
- B creative thought
- C indigenous plants
- D trout
- E pollution
- F restoration
- G native fish
- H extinction

Passage 62

The Work of Caravaggio

Inside a High Court battle over whether a £42,000 card-game painting is really a lost Caravaggio worth £10 million

A Every once in a while, a controversy takes the art world by storm. One such example is the case of a painting of a group of three men playing cards, which may or may not be by the Italian painter Caravaggio (1571 to 1610) and which has been at the centre of a case at the High Court in London. The painting was owned by one Mr Lancelot William Thwaytes, who, back in 2006, sold the painting through the London auction house Sotheby's for £42,000. The painting was bought on behalf of the art collector and Caravaggio expert Sir Denis Mahon. After carrying out extensive research into and restoration of the painting, Sir Denis announced that the painting was in fact an original Caravaggio. It has since been valued at £10 million. Mr Thwaytes proceeded to sue the auctioneers for professional negligence, arguing that they should have consulted more experts when assessing the painting, and advised him of its potential value. Had they done so, he insists, he could have sold the painting for millions.

B Authenticating a work of art is often difficult, especially when it is, as in this case, several hundred years old, and at least one tool for the expert, namely records of all prior owners, is limited or non-existent. In some cases, these records can be traced right back to the artist himself, but this is rare. Most judges, at least in the English-speaking world, are reluctant to rule on whether an artwork should or should not be attributed to a particular artist, as this question lies outside their field of expertise.

C In civil legal cases, when a decision is challenged in court, a judge must decide if the experts are right or wrong. The standard of proof is 'more likely than not', or 'on the balance of probabilities'. And yet, in the art world, the degree of proof required is more similar to that needed in criminal trials, which require 'proof beyond all reasonable doubt'. No one would pay full price for a painting that was more-likely-than-not, on-the-balance-of-probabilities, by the legendary artist Picasso.

D An additional difficulty in attributing a work to a particular artist arises when the artist had a studio, where pupils may have been engaged to make copies of works by the master himself. This was the case with artists such as Guido Reni, but not with Caravaggio. Some artists are known to have made copies, or autograph replicas of their own works. The majority of Caravaggio scholars are not of the opinion that Caravaggio himself painted copies of his own works. However, Sir Denis Mahon, the new owner, claimed that this was precisely what Caravaggio had done in this case, and that this was an autograph replica of the Caravaggio painting *The Cardsharps*, which is on display in the Kimbell Art Museum, in Fort Worth, Texas, USA, and depicts a very similar scene.

E Sotheby's contends that any resemblances between *The Cardsharps* and the painting it sold on behalf of Mr Thwaytes are insufficient to attribute the latter as a genuine Caravaggio. It presented to the court a record of about 30 versions of the card game scene which had changed hands at auction, none of which were described as being by Caravaggio. An image of men cheating at cards was, they argued, popular subject matter at the time, and by no means unique to one artist.

F Unsurprisingly, auction houses such as Sotheby's go to great lengths not to misrepresent what is known about a painting's authorship in their catalogues. A set of phrases are employed to describe the degree of certainty as to the identity of the artist, such as 'Attributed to Giovanni Bellini', which means that the auctioneers consider that the work is probably by Bellini, but that they cannot be absolutely positive. 'Circle of Giovanni Bellini' would indicate that, in the considered opinion of Sotheby's experts, the work in question was produced by someone closely associated with Bellini, but almost certainly not by Bellini himself. 'After Giovanni Bellini' would mean that the work is considered to be a copy of a Bellini painting. In this case, Sotheby's attributed the work being sold by Mr Thwaytes to a 'follower' (and that does not necessarily mean someone who was a pupil) of Caravaggio.

G The court heard much discussion over the degree of artistic skill shown in the painting. The judge drew the conclusion that the quality was not up to that of the rest of the artist's known body of work, and as such ultimately ruled against Mr Thwaytes, who now faces substantial costs, but who still has the right to appeal, and may yet do so. While acknowledging many remarkable features of the picture, the judge instinctively felt that something was not quite right, and that Sotheby's were justified in being reluctant to label the painting a Caravaggio. She made a comparison with *The Cardsharps*, which is known to be a genuine Caravaggio, and pointed out how a feather in that picture looked lifelike, soft and fluffy, whereas the one in the painting in question was far less convincing and three-dimensional.

H Mr Thwaytes's legal team also put forward the case that changes had been made to the picture. It can be seen that the artist had repainted a ribbon which hangs from the elbow of one of the card players, making it shorter than it had been before. This, they argued, would not have been necessary had someone simply been producing a duplicate from the original painting. Yet the judge did not accept that this suggested evidence of a creative mind at work rather than a copyist, or that this repainting should have alerted Sotheby's to any need to investigate further.

Questions 1–8

Reading Passage 3 has eight paragraphs, **A–H**. Which paragraph contains the following information?

Write the correct letter, **A–H**, in boxes 1–8. **NB** You may use any letter more than once.

Q	Statement	
1	accusations against Sotheby's	
2	the result of Mr Thwaytes's court case	
3	a widespread belief about Caravaggio's practice	
4	an area in which legal professionals have limited knowledge	
5	an explanation of the way a painting is credited to a certain artist affects its value	
6	language used to convey any doubts about who a painting is attributed to	
7	whether conclusions can be drawn from changes which were made to Mr Thwaytes's painting	
8	evidence of sales of paintings by other artists which are similar to Caravaggio's work	

Questions 9 and 10

Which **TWO** of these beliefs are expressed by the writer?

OPTIONS

- A** It is possible that Mr Thwaytes might not accept the court's verdict.
- B** The painting which belonged to Mr Thwaytes is probably a genuine Caravaggio.
- C** When Sir Denis Mahon was purchasing the painting, he already knew it was by Caravaggio.
- D** Judges frequently make decisions about whether paintings are by particular known artists.
- E** Greater certainty is required when attributing work to artists than would be needed in other civil cases.

Answer: 9 10

Questions 11–16

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

It can be very difficult to say who an old painting is by, especially when there is little information as to its previous (11) ————, or when the artist worked in a (12) ————. It is important that vendors accurately state in their (13) ———— what is or isn't known about who the artist was. Sotheby's said that Mr Thwaytes's picture was by a (14) ———— of Caravaggio. The judge believed that the (15) ———— of Mr Thwaytes's painting was not that which would be expected of Caravaggio. The judge drew particular attention to the way a (16) ———— had been portrayed in the picture and argued that Caravaggio could and would have painted it differently.

Question 17

Choose the correct letter, **A**, **B**, **C** or **D**.

- (17) What would be the best title for the article?
- A** Sotheby's wins case over 'Caravaggio'.
 - B** Caravaggio forgery discovered by art collector.
 - C** Many great artworks are actually fakes, claims expert.
 - D** Judge clarifies misunderstood law on attribution of artworks.

Passage 63

Try It and See

In the social sciences, it is often supposed that there can be no such thing as a controlled experiment. Think again

A In the scientific pecking order, social scientists are usually looked down on by their peers in the natural sciences. Natural scientists do experiments to test their theories or, if they cannot, they try to look for natural phenomena that can act in lieu of experiments. Social scientists, it is widely thought, do not subject their own hypotheses to any such rigorous treatment. Worse, they peddle their untested hypotheses to governments and try to get them turned into policies.

① **B** Governments require sellers of new medicines to demonstrate their safety and effectiveness. The accepted gold standard of evidence is a randomized control trial, in which a new drug is compared with the best existing therapy (or with a placebo, if no treatment is available). Patients are assigned to one arm or the other of such a study at random, ensuring that the only difference between the two groups is the new treatment. The best studies also ensure that neither patient nor physician knows which patient is allocated to which therapy. Drug trials must also include enough patients to make it unlikely that chance alone may determine the result.

② **C** But few education programs or social initiatives are evaluated in carefully conducted studies prior to their introduction. A case in point is the 'whole-language' approach to reading, which swept much of the English-speaking world in the 1970s and 1980s. The whole-language theory holds that children learn to read best by absorbing contextual clues from texts, not by breaking individual words into their component parts and reassembling them (a method known as phonics). Unfortunately, the educational theorists who pushed the whole-language notion so successfully did not wait for evidence from controlled randomized trials before advancing their claims. Had they done so, they might have concluded, as did an analysis of 52 randomized studies carried out by the US National Reading Panel in 2000, that effective reading instruction requires phonics.

③ **D** To avoid the widespread adoption of misguided ideas, the sensible thing is to experiment first and make policy later. This is the idea behind a trial of restorative justice which is taking place in the English courts. The experiment will include criminals who plead guilty to robbery. Those who agree to participate will be assigned randomly either to sentencing as normal or to participation in a conference in which the offender comes face-to-face with his victim and discusses how he may make emotional and material restitution. The purpose of the trial is to assess whether such restorative justice limits re-offending. If it does, it might be adopted more widely.

④ **E** The idea of experimental evidence is not quite as new to the social sciences as sneering natural scientists might believe. In fact, randomized trials and systematic reviews of evidence were introduced into the social sciences long before they became common in medicine. An apparent example of random allocation is a study carried out in 1927 of how to persuade people to vote in elections. And randomized trials in social work were begun in the 1930s and 1940s. But enthusiasm later waned. This loss of interest can be attributed, at least in part, to the fact that early experiments produced little evidence of positive outcomes. Others suggest that much of the opposition to experimental evaluation stems from a common philosophical malaise among social scientists, who doubt the validity of the natural sciences, and therefore reject the potential of knowledge derived from controlled experiments. A more pragmatic factor limiting the growth of evidence-based education and social services may be limitations on the funds available for research.

⑤ **F** Nevertheless, some 11,000 experimental studies are known in the social sciences (compared with over 250,000 in the medical literature). Randomized trials have been used to evaluate the effectiveness of driver-education programs, job training schemes, classroom size, psychological counselling for post-traumatic stress disorder and increased investment in public housing. And where they are carried out, they seem to have a healthy dampening effect on otherwise rosy interpretations of the observations.

⑥ **G** The problem for policymakers is often not too few data, but what to make of multiple and conflicting studies. This is where a body called the Campbell Collaboration comes into its own. This independent non-profit organization is designed to evaluate existing studies, in a process known as a systematic review. This means attempting to identify every relevant trial of a given question (including studies that have never been published), choosing the best ones using clearly defined criteria for quality, and combining the results in a statistically valid way. An equivalent body, the Cochrane Collaboration, has produced more than 1,004 such reviews in medical fields. The hope is that rigorous review standards will allow Campbell, like Cochrane, to become a trusted and authoritative source of information.

Questions 1–6

Reading Passage 3 has seven paragraphs, A–G.

Choose the correct heading for paragraphs B–G from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Why some early social science methods lost popularity
- ii The cost implications of research
- iii Looking ahead to an unbiased assessment of research

- iv A range of social issues that have been usefully studied
- v An example of a poor decision that was made too quickly
- vi What happens when the figures are wrong
- vii One area of research that is rigorously carried out
- viii The changing nature of medical trials
- ix An investigative study that may lead to a new system
- x Why some scientists' theories are considered second-rate

Questions 7–10

Complete the summary below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

FIGHTING CRIME

Some criminals in England are agreeing to take part in a trial designed to help reduce their chances of (7) _____. The idea is that while one group of randomly selected criminals undergoes the usual (8) _____, the other group will discuss the possibility of making some repayment for their crime by meeting the (9) _____. It is yet to be seen whether this system, known as (10) _____, will work.

Questions 11–14

Classify the following characteristics as relating to

- A Social Science
- B Medical Science
- C Both Social Science and Medical Science
- D Neither Social Science nor Medical Science

Q	Statement	
11	a tendency for negative results in early trials	
12	the desire to submit results for independent assessment	
13	the prioritization of research areas to meet government needs	
14	the widespread use of studies that investigate the quality of new products	

Passage 64

Understanding the Lessons of the Past

Can historians really help us understand the present and plan for the future?

The historian and philosopher R.G. Collingwood likened the difference between those who understood history and those who did not to that between 'the trained woodsman' and 'the ignorant traveller' in a forest. While the latter marches along unaware of their surroundings, thinking 'Nothing here but trees and grass', the woodsman sees what lurks ahead. 'Look,' he will say, 'there's a tiger in that grass.' What Collingwood meant was that, through their familiarity with people, places and ideas, historians are often equipped to see how a situation might turn out. He had a wide-ranging view of the role historians might play in society, believing that their understanding of human behaviour and long-term economic or cultural processes meant that they could be more than just specialists in the past. By being able to spot the tiger in the grass, historians could also advise on contemporary and future challenges faced by the world.

For around 2,500 years, the notion of the historian as a commentator on the present day was well established. Thucydides, a historian in ancient Greece, believed that his History of the Peloponnesian War was not just an account of the war between Greece and Sparta, but a document revealing the basic causes of military conflict in every period and place, and so would have lasting value. Historians writing in later periods saw themselves as not only piecing together the details of specific events, but also helping their readers to understand current social and political situations.

Of course, historians cannot predict the future precisely and their assessments can be wrong. Yet the idea of history providing a valuable guide for present and future action was seldom questioned. This makes sense. After all, the past is our only source of information about what works and what does not; there is nothing else to draw upon. In their everyday lives, people generally look back and review their past experience and use this to help them determine their future actions. While two situations may not be perfectly alike, nevertheless they see patterns and lessons in the past that can help them make better choices.

In recent decades, the longstanding view that history is something which can help people understand the present and plan for the future has fallen out of favour with historians. Studying a series of historical events and using this knowledge to help society deal with current problems makes many of today's historians feel uneasy. There are two reasons for this. One is that historians no longer look at the big picture but work in ever-narrower fields, becoming specialists in topics which are of interest to only a small number of colleagues. The other is that because no two situations are exactly the same, attempting to compare two events or situations runs the risk of oversimplification.

However, other historians continue to insist that history can help those who are responsible for making important policy decisions, such as government officials. Most fundamentally, they believe that history teaches us to investigate the roots of a situation and search for the underlying, long-term causes of problems. If historians can pinpoint the factors that brought a situation about, they can make helpful observations about how likely it is that a proposed course of action will succeed.

Perhaps the most accessible tool offered by history is analogy — a comparison between similar events. It is conventional for people in all walks of life, not only professional historians, to liken a historical event to present circumstances and to spot similarities. People in government often use historical analogies in order to justify their policies. Using analogies is a valuable way of making a point, yet it is all too easy to misuse them or for analogies to be misunderstood. Historians can examine analogies and judge whether they are appropriate and can also identify the most relevant analogies and employ them to enrich public debate about important issues in society. Currently, the range of analogies utilised in public debate is depressingly small — almost any event of significance is related to either the international crises of the 1930s or the economic problems of the 1970s. The role of the historian should be to provide public speakers with a new and more appropriate range of alternatives.

So can we ever learn from history? Although history does not provide rules for action that are consistently reliable, or state definitively what should be done in any given scenario, we should remember Thucydides' view that 'the present, while never repeating the past exactly, must inevitably have some similarities to it. Hence, so must the future'. We should also be aware that if history teaches anything, it is that radical plans to completely change the status quo rarely work out as intended, so policy makers need to move cautiously.

The scepticism among the majority of historians about whether it is worthwhile applying the lessons of the past to the present is a shame. If they make a claim to expertise in cause and effect, they should have faith in their ability to see patterns and future trends. Furthermore, the public expect historians to be willing to tackle big problems. The apparent anxiety among historians about whether history has practical value may be one reason why history graduates worry that their history degree might prove 'useless'.

What makes history so stimulating is that it offers deep insights into the human condition which are of enduring value and impossible to acquire in other ways. Collingwood believed that the past is contained in the present and so 'lives on' because the present is nothing more than the accumulated decisions and actions of the past. History is 'alive and active' and stands 'in the closest possible relation to practical life'.

Questions 30–34

Choose the correct letter, **A, B, C or D**.

- 30 What is the point of the comparison made in the first paragraph?
- A It suggests that historians exaggerate potential problems in modern society.
 - B It explains the difficulties historians have in assessing unexpected events.
 - C It emphasises that historians have the skill to predict future developments.
 - D It describes the best way for historians to analyse unfamiliar situations.
- 31 What did the historian Thucydides see as the main purpose of his work?
- A to discuss what made a particular conflict unique
 - B to ensure there was an accurate record of military campaigns
 - C to focus on how to win wars quickly and easily
 - D to identify the underlying reasons for all armed conflict
- 32 What is the main idea the writer expresses in the third paragraph?
- A We tend to make decisions based on what happened to us in the past.
 - B We are unable to imagine a future life that is different from the past.
 - C We wrongly assume there are many connections between past events.
 - D We find it hard to know whether our past mistakes can teach us anything.
- 33 In the fourth paragraph, the writer suggests that historians feel concerned about
- A missing important details when analysing past events.
 - B failing to communicate clearly with the public.
 - C struggling to work with colleagues in related fields.
 - D making complex situations sound too straightforward.
- 34 In the fifth paragraph, the writer says that historians can help people in positions of power by
- A pointing out the mistakes that they have made.
 - B evaluating how practical their proposed measures are.
 - C reminding them of important moments in recent history.
 - D suggesting a specific course of action.

Questions 35–40

Do the following statements agree with the views of the writer in Reading Passage 3?

Write **YES, NO or NOT GIVEN**.

- 35 Using analogies is common practice among individuals working in a range of professions.
- 36 Politicians tend to avoid using historical analogies when discussing the measures they are proposing.
- 37 Analogies are a useful way of ensuring that people will understand the point being made.

- 38 Historians disagree about which historical periods have most in common with the present day.
- 39 In public debates, people refer too frequently to the events of the 1930s and 1970s.
- 40 People who engage in public debates are reluctant to make use of historians as a source of new analogies.

Questions 41–46

Complete the summary using the list of words, A–J, below.

LIST OF WORDS

- A influence
- B confidence
- C approach
- D resemblance
- E relevance
- F guidance
- G consequence
- H transformation
- I difference
- J behaviour

LEARNING FROM HISTORY

We cannot always get precise 41 — from history on what action should be taken to deal with current problems. However, there will always be some 42 — between the present and the past which we can learn from. One important lesson from history is that trying to bring about a comprehensive 43 — is usually a mistake. Unfortunately, some historians don't have 44 — in their skills and as a result can sometimes appear to doubt the 45 — of their subject to contemporary society. The insights that history provides regarding people's 46 — are always valuable and cannot necessarily be obtained from studying other subjects.

Passage 65

Unlocking the Mystery of Dreams

Dreams have captivated thinkers since ancient times, but their mystery is now closer than ever to resolution, thanks to new technology that allows scientists to watch the sleeping brain at work.

A Thousands of years ago, dreams were seen as messages from the gods, and in many cultures, they are still considered prophetic, foretelling things to come. In ancient Greece, sick people slept at the temples of Asclepius, the god of medicine, in order to receive healing dreams. Modern dream science really begins at the end of the 19th century with Sigmund Freud, who theorized that dreams were the expression of unconscious desires often from childhood. He believed that exploring these hidden emotions through analysis could help cure mental illness. After Freud, the most important event in dream science was the discovery in the early 1950s of a phase of sleep characterized by intense brain activity and rapid eye movement (REM).

B Adult humans spend about a quarter of their sleep time in REM, much of it dreaming. People awakened in the midst of REM sleep reported vivid dreams, which led researchers to conclude that most dreaming took place during REM. Using a machine called the electroencephalograph (EEG), researchers were able to see that brain activity during REM resembled that of the brain when the body is awake. The mystery of REM sleep is that even though it may not be essential, it is universal – at least in mammals and even birds. Some researchers think REM may have evolved for physiological reasons. 'One thing that's unique about mammals and birds is that they regulate body temperature,' says neuroscientist Jerry Siegel, director of UCLA's Center for Sleep Research. 'There's no good evidence that any coldblooded animal has REM sleep.' REM sleep heats up the brain and non-REM cools it off, Siegel says, and that could mean that the changing sleep cycles allow the brain to repair itself. 'It seems likely that REM sleep is filling a basic physiological function and that dreams are a kind of a side-effect, or by-product of this.'

C There is great disagreement about the psychological function of dreams and researchers have come up with some differing theories. On one side are scientists like Harvard's Allan Hobson, who believes that dreams are essentially random. In the 1970s, Hobson and his colleague Robert McCarley proposed what they called the 'activation-synthesis hypothesis,' which describes how dreams are formed by nerve signals sent out during REM sleep from a small area at the base of the brain called the pons. These signals, the researchers said, activate the images that we call dreams. That raised questions about dream research. If dreams are insignificant night-time images created by the brain, what is the point of studying them?

D But more recently, new theories have made some scientists take dreams more seriously. In 1997, Mark Solms of the University of Cape Town in South Africa found that there was more than one mechanism in the brain for activating dreams. Since then, Solms has argued that medical diagnostic equipment like functional magnetic resonance imaging (fMRI) and positron emission tomography (PET) that helps researchers watch dreaming brains might actually lend new support to Freud's ideas because the parts of the brain that are most active during dreaming control emotion. Further research has supported Solms's findings. Scientists using PET and fMRI technology to watch the dreaming brain have found that one of the most active areas during REM is the limbic system, which controls our emotions.

E Much less active during REM sleep is the prefrontal cortex, which is associated with logical thinking. That could explain why dreams in REM sleep often lack a coherent story line. Some researchers have also found that people dream in non-REM sleep as well, although those dreams generally are less vivid. Another active part of the brain in REM sleep is the anterior cingulate cortex, which detects differences or inconsistencies. Eric Nofzinger, director of the Sleep Neuroimaging Program at the University of Pittsburgh Medical Center, thinks that could be why people often solve tricky problems in their dreams.

F Deirdre Barrett, assistant professor at Harvard Medical School, would agree. In her book 'The Committee of Sleep,' she describes how painters like Jasper Johns and Salvador Dali found inspiration in their dreams. In her own research on problem solving through dreams, Barrett has found that even ordinary people can solve simple problems in their lives (like how to fit old furniture into a new apartment) if they focus on the dilemma before they fall asleep. There is also evidence that dreaming helps certain kinds of learning. Some researchers have found that dreaming about physical tasks, like a gymnast's floor routine, enhances performance.

G Whatever the function of dreams at night, they clearly can play a role in therapy during the day. The University of Maryland's Clara Hill, who has studied the use of dreams in therapy, says that dreams are a "back door" into a patient's thinking. "Dreams reveal stuff about you that you didn't know was there," she says. The therapists she trains to work with patients' dreams use dream imagery to uncover hidden emotions and feelings. Rosalind Cartwright from the university medical center in Chicago has been studying depression in divorced men and women, and she is finding that "good dreamers," people who have vivid dreams with strong story lines, are less likely to remain depressed. She thinks that dreaming helps diffuse powerful emotions. "Dreaming is a mental-health activity," she says.

Questions 27–31

Reading Passage 3 has seven paragraphs, A–G.

Which paragraph contains the following information?

Write the correct letter, **A–G**, in boxes 27–31.

Q	Statement	
27	a reference to the significance of dreams on artists' work	
28	a concern about the usefulness of dream research	
29	the types of living creatures that have REM sleep	
30	research results linking dreams to psychological well-being	
31	an account of how modern research tools have strengthened Freud's theory	

Questions 32–35

Choose the correct letter, **A, B, C** or **D**.

- 32 In ancient times, people thought that dreams
- A sent messages to the gods.
 - B helped resolve conflict.
 - C were a sign of physical illness.
 - D predicted future events.
- 33 According to the passage, which of the following happens during REM sleep time?
- A People rarely dream.
 - B People's dreams become confused.
 - C The temperature of the brain increases.
 - D The brain behaves differently than when you are awake.
- 34 What explanation is suggested in paragraph E for lack of a clear narrative in dreams?
- A Some dreams occur in non-REM sleep.
 - B Some dreams are generated in different areas of the brain.
 - C The part of the brain in control of reasoning is less involved.
 - D The part of the brain responsible for feelings is more involved.
- 35 According to the passage, which area of the brain helps people find solutions to difficult situations through their dreams?
- A the anterior cingulate cortex.
 - B the pons.
 - C the limbic system.
 - D the prefrontal cortex.

Questions 36–40

Look at the following statements (Questions 36–40) and the list of people below.

Match each statement with the correct person or people, **A–G**.

List of People

A Jerry Siegel

B Allan Hobson and Robert McCarley

C Mark Solms

D Eric Nofzinger

E Deirdre Barrett

F Clara Hill

G Rosalind Cartwright

Q	Statement	
36	Technology shows there is a link between dreams and the areas of the brain that deal with feelings.	
37	Dreams are meaningless pictures created by the brain.	
38	Dreaming is a method of calming strong feelings.	
39	Our dreams can show us unexpected things about ourselves.	
40	Dreams may be a result of maintaining an essential body function.	

Passage 66

A closer examination of a study on verbal and non-verbal messages

David Lapakko looks at limitations in Albert Mehrabian's famous 1967 study

A study of non-verbal communication carried out in 1967 continues to be widely quoted today. David Lapakko looks at limitations in the original study.

The findings of a study on verbal and non-verbal messages in communication by Albert Mehrabian and his colleagues at UCLA in 1967 have been quoted so frequently that they are now often regarded as a self-evident truth.

In the first experiment, subjects were asked to listen to a recording of a female saying the word 'maybe' in three tones of voice to convey liking, neutrality, and disliking. The subjects were then shown photos of female faces expressing the same three emotions and were asked to guess the emotions in the recorded voice and the photos. It was found that the photos got more accurate responses than the voices.

In the second experiment, subjects listened to nine recorded words spoken in different tones of voice. Three words had positive meanings (e.g., honey), three were neutral (e.g., oh), and three were negative (e.g., terrible). Again, the subjects had to guess the speaker's emotions. It was found that tone of voice carried more meaning than the individual words.

From these experiments, the researchers concluded that 7% of our feeling towards a speaker is based on the actual words they use, 38% on their tone of voice, and 55% on their body language (e.g., facial expression).

However, a closer look at the study reveals several limitations. The first is that the entire study involved only 62 subjects. Of these, 25 were used to select the word for the first experiment, while the key issue—comparing verbal and non-verbal communication—was determined by only the 37 remaining subjects. All were female undergraduates who participated as part of their introductory psychology course, and their ages and academic qualifications seem remarkably uniform. Thus, the findings may simply be a product of the nature of the sample.

Critics have also pointed out that the 7-38-55 formula is flawed since it was pieced together from two different experiments, neither of which involved all three channels (verbal, vocal, and facial). In addition, in the first experiment, the single word maybe was used throughout, so it was impossible for the effects of changes in verbal input to be assessed. The researchers intentionally used a "neutral" word so naturally, the subjects found little

meaning there. Clearly, such a methodology lacks validity. In the real world, people communicate in a particular context and speak in phrases and full-blown sentences, making extensive use of the multi-faceted vehicle of language.

My concern is that interpretations of this study have gained such prominence in our pedagogical literature. This 7-38-55 formula appears in many basic texts used for training in public speaking, interpersonal communication, and organizational communication.

Clearly, one appealing aspect of the Mehrabian study is its numerical precision. Communication is a complex phenomenon, but it seems less so when we can rely on these three magical numbers. In contrast to the ambiguities of language, numbers seem to possess exactness. And the popular appeal of the study has given the 7-38-55 formula enormous credibility. There is a certain mystique about non-verbal communication, and the continued references to this research sustain it, encouraging people to believe in the overwhelming importance of the non-verbal message compared with the verbal one. Yet we know that even one well-chosen word to a colleague or friend can make or break a communicative effort. Words do matter. Bradley (1991), one of the few textbook writers to criticize the Mehrabian study, makes the same point when he observes, "If we could communicate 93% of information and attitudes with vocal and facial cues, it would be wasteful to spend time learning a language."

Mehrabian himself believes his research should not be interpreted to devalue the role of language in communication, saying: "Please remember that all my findings dealt with communications of feelings and attitudes. It is absurd to imply or suggest that the verbal portion of all communication constitutes only 7% of the message... anytime we communicate abstract relationships (e.g., $x = y^2$) clearly 100% of the entire communication is verbal" (Mehrabian, 1996).

To be fair, many textbook writers attempt to be faithful to the context of Mehrabian's research. For example, Stewart and D'Angelo (1988) write: "Mehrabian argues that when we're uncertain about what someone's feeling, or about how much we like him or her, we rely... only 7% on the words that are spoken." Others try to play down the specific percentages, saying that an understanding of the general importance of non-verbal cues is more important. Nonetheless, other textbook authors simply use the numbers without placing any limits on their meaning.

Since this relatively small study was first published, it has achieved an influence far beyond its intended scope. We need to put it into its proper perspective and learn some important lessons from it regarding social science research, communication pedagogy, and the forces which have created widespread misunderstanding about communication.

Questions 27–30

Complete the summary using the list of words and phrases, **A–H**, below.

Albert Mehrabian and his colleagues carried out an influential study comparing the (27) _____ of verbal and non-verbal communication. This involved two experiments. In both experiments, subjects had to identify the (28) _____ being communicated by other people. The two main areas focused on in the first experiment were voice tones and (29) _____, while the second focused mainly on voice tones and (30) _____.

WORD BANK

- A** facial expressions
- B** purposes
- C** printed words
- D** effects
- E** word meanings
- F** gender differences
- G** feelings
- H** characteristics

Questions 31–35

Do the following statements agree with the views of the writer?

Write **YES** if the statement agrees with the views of the writer, **NO** if the statement contradicts the views of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- (31) One limitation of the study was that there were too few subjects involved.
- (32) The fact that the subjects in the study came from a similar background was an advantage.
- (33) The two experiments should have been carried out in a different order.
- (34) The researchers' choice of a neutral word was helpful in the context of the study.
- (35) The study would have been more valid if it had included a range of languages.

Questions 36–40

Choose the correct letter, **A**, **B**, **C** or **D**.

- 36 What does the writer say about the numerical precision of Mehrabian's study?
- A It makes the claims more attractive.
 - B It is the strongest point of the study.
 - C It will appeal to superstitious people.
 - D It allows comparison between languages.
- 37 What does the writer say about the popularity of the 7-38-55 formula?
- A It is unlikely to maintain its present status.
 - B It is leading to an undervaluing of language.
 - C It should be applied in a more practical way.
 - D It may help understanding of non-verbal messages.
- 38 What point is Bradley making about language learning?
- A Language could be learned more efficiently than it is.
 - B More research is needed into attitudes to communication.
 - C More should be spent looking at tone and body language.
 - D Language must be important since we make an effort to acquire it.
- 39 What does Mehrabian himself say about his findings?
- A They are relevant to only one area of communication.
 - B It is only in maths that 100% of communication is verbal.
 - C Feelings are more difficult to communicate than numerical facts.
 - D Non-verbal communication is the main part of the message.
- 40 What is the writer's purpose in the paragraph beginning *To be fair...*?
- A to justify the strong points of Mehrabian's study
 - B to outline other research on non-verbal behaviour
 - C to present varying interpretations of Mehrabian's study
 - D to show that textbooks tend to ignore non-verbal behaviour

Passage 67

Whale Culture

How the songs and dialects of humpback and killer whales reveal something remarkably close to culture

A Most social scientists stubbornly resist the idea that animals have culture. Even such advanced cetacean mammals as whales and dolphins clearly don't have art, literature, or architecture. But patient observation over many years has begun to reveal behaviours that can only be learnt from other whales. And that, say whale biologists, constitutes culture.

B So far, humpback and killer whales provide the best evidence of culture in cetaceans, and the song of male humpback is among the most striking examples. Humpback populations in different oceans sing different songs, but within the same ocean they all stick to the same one. However, during the breeding season the sounds change, as it appears that females are drawn to novel songs. One male might add an extra set of groans; another might drop a series of grunts. Soon all the other males have altered their own rendition to incorporate the changes until they are once again singing the same song. Since this occurs among thousands of whales spread across a vast part of the planet, the change cannot be in response to any factor in the animals' environment. The latest version of the song can be learnt only from other whales – almost certainly by imitation.

C Culture plays an even bigger part in the life of killer whales. Nowhere is this more obvious than along the north-west coast of America, where killer whales are split into two distinct populations – 'residents' and 'transients'. They live in the same stretch of water, but they don't mingle. In effect, they belong to two quite separate cultures. Residents live in stable groups, or 'pods', made up of two or three mothers and their offspring – perhaps 20 whales in all. Calves stay with their mothers throughout adulthood, and in many years of observation no one has ever seen a whale switch pods. Transients travel in smaller, more changeable groups of between three and six.

D One of the most obvious distinctions between the transient and resident societies is the way they impart information. Killer whales detect prey with a range of echo-locating clicks, but converse with a vocabulary of squeaks, whistles and whines. Transients have only a few such calls, and all transient societies share the same ones. Residents have a much more extensive repertoire, and each family group has its own unique and distinctive set of calls. Despite regular interaction between them, each resident pod sticks firmly to its own dialect. Research shows these dialects are maintained for at least 40 years.

E To qualify as part of killer whale culture, dialects must be learnt from other members of the pod. Animals with different dialects share the same waters, so the variation can't be a product of physical environment. 'And we can throw out the notion that dialects are

inherited,' says Lance Barrett-Lennard of the University of British Columbia. He has spent the past seven years analysing DNA from 270 whales. His paternity tests reveal the female killer whales invariably attract mates from outside their own pod – males with a very different dialect. If dialects were programmed by genetics, call patterns from both father and mother would be passed on to the calf. 'A calf uses the calls of its maternal pod very precisely. There's no input from the father,' says Barrett-Lennard.

F The question still remains – is this culture? It is, according to Frans de Waal of Emory University in Atlanta, who argues that culture is just another biological adaptation that has evolved in many creatures. One benefit of viewing culture in this way is that you can start to understand how and why it might have arisen in these creatures. Whales have several biological attributes that give them an advantage in social learning. Apart from their advanced mental abilities, they are adept at recognising sounds: ideal for communicating in the marine environment. Many species spend years rearing their offspring, and live in small, stable, multi-generational societies, a social system that provides ample opportunity for teaching and learning.

G But why have cetaceans evolved the ability to learn from other group members? Experts in whale biology believe that ecological factors and the need to adapt to sudden changes in the environment played a large part in the emergence of culture. Although the ocean is a relatively stable habitat in many ways, it is highly changeable in one crucial respect – the availability of food. One moment there might be a plentiful supply of fish, the next they've disappeared. When that happens, the past experience of senior members of the group – and the ability to share this knowledge – is a huge asset. The dialects of killer whales allow members of the group to identify each other, enabling them to share information about food hot spots. Among resident killer whales, it is also allow females to avoid inbreeding by picking out a mate with a strange dialect from outside their pod, says Barrett-Lennard.

H The importance of sharing information seems to have led to biological changes in at least some species of whale. Female killer whales, like humans, are very unusual in that they live up to a quarter of a century after they have had their descendants. And what whale matriarchs offer is the most important thing of all – cultural knowledge, vital for the group's survival, passed directly from one generation to the next.

Questions 1–5

*Do the following statements agree with the views of the writer in Reading Passage 3? Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① Resident killer whales appear to remain with their maternal group for life.
- ② Resident killer whales have a more restricted range of calls than transients.

- 3 There is a vocabulary of sounds which is common to all transient killer whales.
- 4 Resident killer whales share the dialects of other resident communities living in the same waters.
- 5 The dialects of transient killer whales remain constant over time.

Questions 6–8

Complete the summary below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

It has been observed that resident killer whales invariably live in fixed family groups, known as 6 _____. Each of these has its own unique set of calls, despite close contact with other family groups. As the same areas of ocean contain many different groups with widely varying dialects, it is clear that these differences could not have emerged as a result of the whales' 7 _____.

According to tests conducted by Lance Barrett-Lennard, a calf communicates exclusively with the dialect of the group to which its 8 _____ belongs. Barrett-Lennard also rejects the idea that the call patterns are inherited.

Questions 9–11

Choose **THREE** letters, **A–F**.

Which **THREE** of the following features of whales are mentioned in the passage?

OPTIONS

- A intelligence
- B physical strength
- C sensitivity to sound
- D prolonged life span
- E lengthy period of fertility
- F adaptability to variety of foods

Answer: 9 10 11

Questions 12–14

Reading Passage 3 has eight paragraphs, **A–H**. Which paragraph contains the following information?

Write the correct letter, **A–H**, in boxes 12–14.

Q	Statement	
12	an example of the kind of information passed by whales to each other	
13	a reference to variations in communication styles between different cultures within one species	
14	ways in which the skills of whales are favourable for development of culture	

Passage 68

What Should Companies Do to Survive?

Ten rules for good management in a world of uncertainty and constant upheaval

A In the world of business, where today's fashionable management theory soon becomes a stale cliché or a quaint archaism, it is essential to have a clear idea about the enduring rules of good management. The company that follows these is the company that is in the strongest position amid the uncertainties and violent upheavals of the world as we know it.

B A company is the sum of what its people understand and know how to do well. Value lies increasingly in creative ideas and knowledge. But ideas have value only if people share and develop them in ways that increase revenue; knowledge is useful only if people can find what they need to know. Getting intelligent people to share what is in their heads is vital, and takes more than mere money or clever software. Ideas must flow in every direction through a company — not merely from the top down. And knowledge is worth storing only if senior staff set careful rules to filter and structure it. What goes into a database determines the value of what comes out. So setting central rules and standards is key to good knowledge management.

C Good judgement is a key skill. Managers constantly blitzed with new information need to build in the data that matters and set aside the rest. Big-bang decisions are generally best avoided — or implemented in small incremental moves that leave room for flexibility and for altering course, if circumstances change.

D Customers matter — but some matter more than others. Acquiring new customers often costs more than making extra sales to existing ones. So companies must build loyalty by providing reliability and good service. Given the welter of product information reaching customers, memorable brands will grow more important. Companies need not only to widen their reach by finding new markets, but also to deepen existing relationships. They have more information than ever before about their customers, and must use this to offer their most profitable customers special deals and to make them feel part of an elite club. Some companies even seek to 'fire' unprofitable customers by charging them higher rates than others.

E Like its customers, some of a company's people matter more than others, and not just at the top: managing talent is also about capturing innovative ideas from middle managers and those further down the line. At every level, managers must identify where most value lies. In some cases, a few stars will encapsulate much of a company's value; in others, teams of employees will matter more. Some companies will want to hire the talents of 'free agents' as and when they are required; others, to employ directly their best brains.

F Companies need to collaborate more, in alliances that allow them to outsource production or spread risk or to enter new markets. Effective collaboration calls for trust and shared understanding, rather than the top-down, command-and-control approach of hierarchical structures. Successful collaboration also requires excellent communication, and incentives that reward sharing information and working for common goals. As costs of handling information in a company decline, new opportunities open for redefining corporate shape, and companies are becoming less hierarchical, with more ways to arrange and rearrange structure. Managers must think through from scratch which activities should be kept in-house and which outsourced, and normally a company should keep those activities it does better than its competitors.

G Given the pace of change, bosses need more than ever to be able to communicate persuasively through any channels, with their staff and the outside world. They must also listen: the most valuable communications will frequently be bottom-up, and the people nearest to the customer are best placed to explain what they see and hear. Managers must listen to them.

H Ironically, internet technologies, tools of freedom and decentralization, call for discipline and standard processes. Only by setting standards and insisting that everyone abide by them will companies reap their potential savings. Companies need to insist on common practices in areas such as purchasing and information technology in order to harvest real productivity gains. As a result, some aspects of centralisation will increase: a key task of top managers is to provide structures and standards, and to insist that they are observed.

I Once standards have been set, openness and freedom should reign. Centralisation of standards makes possible decentralisation of decision making. In addition, internet technologies increase the need for a culture of openness, to foster the sharing of knowledge and effective collaboration. Increasingly, companies will allow their suppliers and customers 'inside the machine,' as it were, by giving them extraordinary access to their databases and inner workings in order to integrate their operations and to make collaboration effective.

J The key to success lies much less in technical know-how than in excellent leadership to push through and build upon organisational change. At some points in a company's life, it will need a leader who can rally staff to push through the trauma of disruptive change. At other times, the right style will be the manager-as-coach, a selfless talent scout who specialises in assembling and motivating great teams. Always, the people at the top will set the tone in a firm. Their skills will determine whether it is a good company to work in and do business with.

Questions 1–8

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 1 Too few companies understand how to evaluate fashionable management theories.
- 2 Certain practices should always be followed when coping with unstable situations.
- 3 It is important that the development of ideas should lead to improvements in the company's financial position.
- 4 The structuring of a database should be decided by all its users.
- 5 More training needs to be provided to help managers assess the usefulness of new information.
- 6 Any major change should be introduced all at once.
- 7 Companies can benefit from using different strategies with different customers.
- 8 The activities which are common to all companies are best kept in-house.

Questions 9–14

Reading Passage 3 has ten paragraphs, **A–J**. Which paragraph contains the following information?

Write the correct letter, **A–J**, in boxes 9–14. **NB** You may use any letter more than once.

Q	Statement	
9	examples of qualities required when co-operating with other companies	
10	examples of business activities where all staff should follow the same practices	
11	examples of alternative employment patterns within a company	
12	a basis for deciding whether a company should carry out an activity itself or ask an outside company to do it	
13	an example of how information can be shared by a company and its customers	
14	a reason why customers might continue to buy from the same companies	

Passage 69

You Are What You Speak

Does your mother tongue really affect the way you see the world? Alison Motluk looks at some of the findings.

Does the language you speak influence the way you think? Does it help define your world view? Anyone who has tried to master a foreign tongue has at least thought about the possibility.

At first glance the idea seems perfectly plausible. Conveying even simple messages requires that you make completely different observations depending on your language. Imagine being asked to count some pens on a table. As an English speaker, you only have to count them and give the number. But a Russian may need to consider the gender and a Japanese speaker has to take into account their shape (long and cylindrical) as well, and use the number word designated for items of that shape.

On the other hand, surely pens are just pens, no matter what your language compels you to specify about them? Little linguistic peculiarities, though amusing, don't change the objective world we are describing. So how can they alter the way we think?

Scientists and philosophers have been grappling with this thorny question for centuries. There have always been those who argue that our picture of the Universe depends on our native tongue. Since the 1960s, however, with the ascent of thinkers like Noam Chomsky, and a host of cognitive scientists, the consensus has been that linguistic differences don't really matter, that language is a universal human trait, and that our ability to talk to one another owes more to our shared genetics than to our varying cultures. But now the pendulum is beginning to swing the other way as psychologists reexamine the question.

A new generation of scientists is not convinced that language is innate and hard-wired into our brain and they say that small, even apparently insignificant differences between languages do affect the way speakers perceive the world. 'The brain is shaped by experience,' says Dan Slobin of the University of California at Berkeley. 'Some people argue that language just changes what you attend to,' says Lera Boroditsky of the Massachusetts Institute of Technology. 'But what you attend to changes what you encode and remember.' In short, it changes how you think.

To start with the simplest and perhaps subtlest example, preparing to say something in a particular language demands that you pay attention to certain things and ignore others. In Korean, for instance, simply to say 'hello' you need to know if you're older or younger than the person you're addressing. Spanish speakers have to decide whether they are on intimate enough terms to call someone by the informal *tu* rather than the formal *Usted*. In Japanese, simply deciding which form of the word 'I' to use demands complex calculations

involving things such as your gender, their gender and your relative status. Slobin argues that this process can have a huge impact on what we deem important and, ultimately, how we think about the world.

Whether your language places an emphasis on an object's shape, substance or function also seems to affect your relationship with the world, according to John Lucy, a researcher at the Max Planck Institute of Psycholinguistics in the Netherlands. He has compared American English with Yucatec Maya, spoken in Mexico's Yucatan Peninsula. Among the many differences between the two languages is the way objects are classified. In English, shape is implicit in many nouns. We think in terms of discrete objects, and it is only when we want to quantify amorphous things like sugar that we employ units such as 'cube' or 'cup'. But in Yucatec, objects tend to be defined by separate words that describe shape. So, for example, 'long banana' describes the fruit, while 'flat banana' means the 'banana leaf' and 'seated banana' is the 'banana tree'.

To find out if this classification system has any far-reaching effects on how people think, Lucy asked English- and Yucatec-speaking volunteers to do a likeness task. In one experiment, he gave them three combs and asked which two were most alike. One was plastic with a handle, another wooden with a handle, the third plastic without a handle. English speakers thought the combs with handles were more alike, but Yucatec speakers felt the two plastic combs were. In another test, Lucy used a plastic box, a cardboard box and a piece of cardboard. The Americans thought the two boxes belonged together, whereas the Mayans chose the two cardboard items. In other words, Americans focused on form, while the Mayans focused on substance.

Despite some criticism of his findings, Lucy points to his studies indicating that, at about the age of eight, differences begin to emerge that reflect language. 'Everyone comes with the same possibilities,' he says, 'but there's a tendency to make the world fit into our linguistic categories.' Boroditsky agrees, arguing that even artificial classification systems, such as gender, can be important.

Nevertheless, the general consensus is that while the experiments done by Lucy, Boroditsky and others may be intriguing, they are not compelling enough to shift the orthodox view that language does not have a strong bearing on thought or perception. The classic example used by Chomskians to back this up is colour. Over the years many researchers have tried to discover whether linguistic differences in categorising colours lead to differences in perceiving them. Colours, after all, fall on a continuous spectrum, so we shouldn't be surprised if one person's 'red' is another person's 'orange'. Yet most studies suggest that people agree on where the boundaries are, regardless of the colour terms used in their own language.

Questions 1–5

Do the following statements agree with the claims of the writer in the Reading Passage? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 1 Learning a foreign language makes people consider the relationship between language and thought.
- 2 In the last century cognitive scientists believed that linguistic differences had a critical effect on communication.
- 3 Dan Slobin agrees with Chomsky on how we perceive the world.
- 4 Boroditsky has conducted gender experiments on a range of speakers.
- 5 The way we perceive colour is a well-established test of the effect of language on thought.

Questions 6–10

Look at the following features (Questions 6–10) and the list of languages below.

Match each feature with the correct language, **A–E**.

List of Languages

- A** Russian
- B** Japanese
- C** Korean
- D** Spanish
- E** Yucatec Maya

Q	Statement	
6	the importance of the relative age of speakers	
7	the use of adjectives to distinguish the names of objects or things	
8	a need to use some numbers with the correct gender	
9	a relationship between form and number	
10	the need to know how friendly your friendship is with the person you are addressing	

Questions 11–14

Complete the summary using the list of words, **A–J**, below.

WORD LIST

- A method
- B language
- C cardboard
- D design
- E purpose
- F gender
- G box
- H material
- I similarity
- J wood

LUCY'S EXPERIMENTS

In the likeness task, Lucy gave his subjects three combs. Two of these were made of the same **11** _____ and two were alike in that they had the same **12** _____. In another experiment, plastic and **13** _____ items were used. The **14** _____ that English and Yucatec speakers used to group these objects helped him show that speakers of different languages think about things differently.

Passage 70

Youth Homelessness

A look at the problems in assessing the scale of homelessness among young people in Britain, and the explanations that have been put forward to account for it

Youth homelessness did not emerge as a public problem in Britain until the second half of the 20th century. In 1975, a television programme called Johnny Go Home shocked the public by showing the risks faced by a runaway child in London, and placed the problem of youth homelessness firmly on the public agenda, although policy initiatives were not forthcoming. Behind the rise in homelessness among young people were the problems of youth unemployment, a reduction in state benefits, and a shortage of suitable housing.

Government responses to youth homelessness generally involve encouraging voluntary agencies to set up schemes. The proportions of young people affected vary from region to region, as do the underlying economic structures, making comparison difficult. However, homeless youngsters all over Britain face many of the same risks as they move between relatives, institutions and the street.

Statistics are an integral part of accounts of social issues, whether they be in the press or in the reports of agencies and academic commentators. The reaction of policy makers to a social problem also depends largely on the number of people it is thought to affect. However, there are very few agreed statistics on homelessness.

One reason for this is that there are a number of practical difficulties with measuring it. Young people who are staying temporarily with friends, for example, may not contact homelessness agencies, so their housing problem goes unrecorded. In addition, some may be homeless for just a night or two, others for months or years.

There is also no globally accepted definition of homelessness. For example, in Britain someone is deemed homeless by a local authority if they have no accommodation that they are entitled to occupy, but they must also fulfil a number of other criteria: they must have a local connection of some sort, they must not be intentionally homeless, and they must show that they are in "priority need" of accommodation. Official statistics therefore exclude large numbers of homeless people.

Explanations of the causes of youth homelessness also vary considerably. Structural explanations account for youth homelessness by reference to societal structures, such as changes in the housing market. This is crucial since, by definition, homelessness is a housing issue. In particular, there has been a substantial decline in rented housing in recent years. The economic power of young people is also significant. Both unemployment

and low pay are frequently viewed as being integral to homelessness and both factors affect young people in Britain. However, unemployment is not directly the issue – rather it is a question of the subsequent low income that unemployment incurs.

As such, the state benefit system is also seen to be fundamental to the rise in youth homelessness, because young people's benefits have been heavily restricted in a number of ways. As a result, many young people may be heavily in debt. These various difficulties have been further compounded by a number of demographic changes, such as the expanding number of single-person households. Consequently, while there has been a huge reduction in the supply of affordable accommodation, there has been an increase in the demand for such housing.

However, the problem of youth homelessness is often explained in more personal terms, which relate it to the individual behaviors of homeless people. A wide variety of such personal factors are often introduced to help explain youth homelessness, such as leaving home, mental illness, crime, and drug abuse, although there is usually disagreement over how these factors should be interpreted.

Four approaches can be taken towards the explanation of youth homelessness. One is the 'political' model, which sees homelessness as stemming primarily from the way society is organized. This is a view widely adopted in academic circles, but there are also a number of criticisms that can be made of it. Firstly, only by looking in detail at the individual features of the situation is it possible to assess the impact of these structural features on the homeless and to identify appropriate provision. In addition, this approach tends to treat young homeless people as powerless victims, which patronizes them and ignores the fact that they are still capable of self-determination.

The 'individual culpability' model sees homelessness as having little or nothing to do with societal structures but as resulting from the individual behavior of young homeless people themselves. Young homeless people, it is argued, do not have to leave home. As individuals in a free society they are accountable to themselves for their decisions. The advantage of this perspective is that it gives some credibility to the self-determination of young homeless people. However, because such an approach allocates blame to an individual, it also carries punitive implications.

In the 'pathology' model, the individual's situation is perceived to be a consequence of personality disorders that may result from early childhood experiences. According to this model, young homeless people have specific problems and they therefore require specialist support. This approach can present a misleading picture of the young homeless – many of whom have no problems other than one of accommodation.

Finally, according to the 'child' model, young people become homeless because they are simply too immature to cope with life. However, there is evidence to suggest that many homeless youngsters show considerable resilience and maturity in learning to cope with and survive homelessness.

Questions 1–5

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 How did the TV programme *Johnny Go Home* affect attitudes to youth homelessness in Britain?
 - A It raised awareness of the issue on a wide scale.
 - B It led to official measures to deal with the situation.
 - C It discouraged many young people from leaving home.
 - D It made people aware of the real causes of the problem.
- 2 Comparison of homelessness in different regions is difficult, partly because of variations in
 - A the places homeless children go to.
 - B government responses.
 - C economic structures.
 - D the work done by voluntary agencies.
- 3 Why is the lack of agreed statistics on homelessness a problem?
 - A Accounts of the problem by the press are ineffective.
 - B Policy makers may not respond appropriately to the situation.
 - C Academic commentary on the issue becomes unclear.
 - D Agencies may not be able to provide funding.
- 4 The lack of statistics on homelessness may partly be caused by
 - A disagreement over the type of data needed.
 - B reluctance of the homeless to identify themselves.
 - C lack of time to carry out surveys.
 - D difficulty in identifying the homeless.
- 5 The writer describes the criteria for classifying someone as homeless in Britain to illustrate
 - A the different ways of collecting statistics.
 - B the fact that too many people are defined as homeless.
 - C the variation there is in different parts of the country.
 - D the fact that official statistics may be misleading.

Questions 6–10

Complete the summary using the list of words, **A–J**, below.

WORD LIST

- A demographic explanations
- B government support
- C bank loans
- D solutions
- E money owed
- F economic explanations
- G personal explanations
- H structural explanations
- I income
- J property to rent

EXPLANATIONS FOR YOUTH HOMELESSNESS

There are various explanations for youth homelessness. Firstly, the reduction in (6) _____ was caused by changes in the housing market and is one of a number of (7) _____ for youth homelessness. There may also be (8) _____ for developments, such as the increased demand for single-person accommodation. Thirdly, some young people may find it difficult to get state benefits, and do not have jobs. These are (9) _____ for homelessness. Lastly, factors such as crime and mental illness are examples of (10) _____.

Questions 11–14

*Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- (11) The 'political' model is unpopular among academics.
- (12) The 'political' model may lead to young homeless people being regarded as having hardly any control over their lives.
- (13) The disadvantages of the 'individual culpability' model far outweigh the advantages.
- (14) The 'child' model is an accurate representation of the situation of many young homeless people.

Answer Key & Explanations

All 70 Passages

Passage 1 — 200 Years of Australian Landscapes at the Royal Academy in London

27. **NO**

narrow interpretation contradicts para 1: the range of artists and styles "results in a **flexible, wide-ranging notion of landscape**" — the opposite of narrow. Answer: **NO**.

28. **NOT GIVEN**

rejected Australian suggestions — the passage never mentions Australians submitting suggestions for the exhibition's theme, so there's nothing to confirm or deny. Answer: **NOT GIVEN**.

29. **YES**

Para 3: "Because of colonial ties, it was inevitable... that **it would reflect Britain's devotion to the beloved landscape** before its own character... took shape." Answer: **YES**.

30. **YES**

Para 3, Soriano: "What I wanted to show was how **Australian art develops a real distinctiveness**, associated with the landscape and the light." Answer: **YES**.

31. **NO**

Contradicts para 5: "there can be little denying that **British landscape painting is still relevant to a current generation of Australian practitioners**, however indirectly." Answer: **NO**.

32. **C**

Para 2 centres on Mendelssohn's view that "**British artistic values have directed this exhibition rather than allowing Australia the freedom to demonstrate its maturity**", and that the British "insisted on looking at the genre of landscape painting" expecting colonies to remain like Britain. Answer: **C**.

33. **B**

Para 2: "What Mendelssohn found surprising... was that **the underlying rules for the selection of works seemed to have been so conservative**." Answer: **B**.

34. **A**

Para 4, Gladwell's own words: "I wanted to make it **personal**," bringing the landscape tradition "to his experience and his world through skateboarding and beach culture" — a subjective, individual take. Answer: **A**.

35. C

Para 6: "I felt it was important that **it was seen as part of Australian art history, rather than a separate exhibition on its own.**" Answer: C.

36. A

Her China example illustrates the passage's framing statement that "**London is increasingly less important to today's generation of artists**" — young artists now "want to go everywhere," not specifically London. Answer: A.

37. F

Para 3: "the Academy is nevertheless **embracing the peculiarities of Australian art** from the mid-19th century onward." Answer: F.

38. E

Soriano, para 3: "the influence of English, French, or German art is **much more evident in the early periods, in the early 1800s to mid-1800s.**" Answer: E.

39. A

Gladwell's Storm Sequence shows him skateboarding as "one of Sydney's **signature brutal storms lingers offshore**" — he was "interested in Turner and the idea of **atmosphere affecting vision.**" Answer: A.

40. C

Para 6: "landscape struck me as being the right theme because **Aboriginal art started in and on the landscape**" — a deep, foundational connection to the land. Answer: C.

Passage 2 — A New Stage in the Study and Teaching of History

1. C

Para C: **historians are faced with a variety of forms of communication, ranging from simple audiotape to the promising complexities of videodiscs linked with computers.** Answer: C.

2. E

Para E: **the barrier seems to be that the profession is structured around the medium of the written word, and is somewhat insulated in its academic setting.** Answer: E.

3. F

Para F: **oral communication is not limited to words: it also includes body language, expression and tone, and is embedded in a context. little of this is evident in a written transcript.** Answer: F.

4. **A**

Para A: **film and video, especially as broadcast on television, are probably the major influence on the public's consciousness of history.** Answer: A.

5. **B**

Para B: **many schools, too, produce video yearbooks.** Answer: B.

6. **G**

Para G: **the many benefits of using moving images as historical evidence easily outweigh worries about cost, technical skills, or the effect of a camera.** Answer: G.

7. **C**

Para C: **the use of moving images to record current events for the benefit of future historians does not even have a commonly agreed name.** Answer: C.

8. **F**

Para F: **the narrator gives up some control and has less need to give explanations, while the viewer becomes involved in the process of interpreting and understanding history.** Answer: F.

9. **E**

Para E: **historians have rejected the training, the institutions, the motivations and the professional structures that would be needed in order to use moving images effectively.** Answer: E.

10. **NOT GIVEN**

the passage only says **many schools, too, produce video yearbooks** — nothing about student needs driving improvements in teaching. Answer: **NOT GIVEN.**

11. **YES**

mainstream academics who continue to use only the written word, while film and video... are probably the major influence on the public's consciousness of history. Answer: **YES.**

12. **NO**

professional historians have tended to avoid involvement in television programmes about history. Answer: **NO.**

13. **YES**

research into **films with a broad circulation, using them as a source of information on the social and intellectual history of the twentieth century. journals such as american history review have played a significant role in this field.** Answer: **YES.**

14. NOT GIVEN

the passage only discusses the effect of new technology on the discipline of **history** — no other academic subjects are mentioned. Answer: **NOT GIVEN**.

Passage 3 — Are We Born Able to Imitate?

1. A

surely such an achievement could only be due to clever genes; after all, it's impossible to imagine how babies could learn to produce the expressions they see — the writer lays out WHY Meltzoff and Moore concluded what they did. Answer: **A**.

2. A

worrying cracks began to appear in Meltzoff and Moore's picture... and these cracks have been spreading ever since — "cracks" = flaws/problems becoming more visible over time. Answer: **A**.

3. D

evidence that older infants don't copy any facial gestures, not even tongue protrusion, supported that interpretation — the interpretation being that tongue protrusion is a reflex, not real imitation. Answer: **D**.

4. C

without mirror experience, people **do no better on their tenth attempt than on their first... in order to learn imitation, we need to see, as well as feel, what we are doing**. Answer: **C**.

5. B

child psychologists have watched the flaws in Meltzoff and Moore's work becoming more apparent, without ever abandoning their belief in it. One of the reasons for this is... — "this" = continuing to believe despite the flaws. Answer: **B**.

6. H

Meltzoff portrays *Homo imitans* **in beautiful, compelling prose**. Answer: **H writing**.

7. C

every loving parent is drawn to the idea that if you stick out your tongue at a newborn, the baby likes to return the compliment. Answer: **C appeal**.

8. F

researchers might have realized that, if the imitation instinct was discredited, other treasured ideas would fall apart — implying reluctance to question it. Answer: **F reluctance**.

9. **B**

when small babies are not asleep, they are often inattentive... only a small number of babies stayed attentive for long enough to be analyzed. Answer: **B challenge.**

10. **A**

small variations in experimental procedure, could have produced misleading results. Answer: **A consistency.**

11. **YES**

the study was so large and thorough... following gold-standard procedures. Answer: **YES.**

12. **NOT GIVEN**

they found evidence for just one: tongue protrusion — stated as a finding, with no mention of the team's reaction or surprise. Answer: **NOT GIVEN.**

13. **NO**

the study finally undermined this argument... we have no good reason to believe that newborns can copy gestures — the opposite of unconvincing. Answer: **NO.**

14. **NO**

parents who copy babies' smiles make the infants better at imitating smiles, but not at copying hand movements. This specificity is important — the effect does NOT generalise to a range of actions. Answer: **NO.**

15. **YES**

through games, songs and rhymes that synchronize actions, by reacting in the same way and at the same time (for example, 'Ooh, what's that?'). Answer: **YES.**

16. **NOT GIVEN**

in most cultures, by allowing the child to spend time in front of actual mirrors, while building up the range of movements that they can copy in future — mentions cultural variation in mirror use, but not which specific gestures come first. Answer: **NOT GIVEN.**

Passage 4 — Assessing the Risk

1. **TRUE**

as a title for a supposedly unprejudiced debate... did not bode well — the writer signals the title's claimed neutrality is doubtful. Answer: **TRUE.**

2. FALSE

the 40 scientists' survey covered **no aeroplanes, bicycles or high-voltage power grids; no pasteurisation, pesticides or biotechnology; no quantum mechanics; no wheel** — far beyond medicine alone. Answer: **FALSE**.

3. FALSE

in short, their message was: no risk, no gain — the opposite of "shouldn't take risks"; their point is that risk IS necessary for progress. Answer: **FALSE**.

4. FALSE

all the technologies listed... were innately risky at their inception, as all technologies are — ordinary risk, not exceptional. Answer: **FALSE**.

5. TRUE

of course, the risks, such as they could be perceived, were worth taking. Answer: **TRUE**.

6. FALSE

in contrast, **GM crops stand out as an example of a technology whose benefits are far from clear** — unlike the other listed items, it is NOT shown clearly passing the same test. Answer: **FALSE**.

7. consumer's choice

7: a crucial issue is the consumer's choice. 8: the consumer's right to choose should be considered alongside considerations of risk and benefit. 9: people who ski choose to do so — skiing... is exhilarating. 10: promoters believe the future population cannot be fed without GM crops. 11: there is no GM research that will seriously affect the yield of wheat and rice. 12: GM is used to make production cheaper. 13: people continue to mistrust science. Answers: **consumer's choice / risk and benefit / skiing / GM crops / wheat and rice / production / mistrust.**

8. risk and benefit

7: a crucial issue is the consumer's choice. 8: the consumer's right to choose should be considered alongside considerations of risk and benefit. 9: people who ski choose to do so — skiing... is exhilarating. 10: promoters believe the future population cannot be fed without GM crops. 11: there is no GM research that will seriously affect the yield of wheat and rice. 12: GM is used to make production cheaper. 13: people continue to mistrust science. Answers: **consumer's choice / risk and benefit / skiing / GM crops / wheat and rice / production / mistrust.**

9. skiing

7: a crucial issue is the consumer's choice. 8: the consumer's right to choose should be considered alongside considerations of risk and benefit. 9: people who ski choose to do so — skiing... is exhilarating. 10: promoters believe the future population cannot be fed without GM crops. 11: there is no GM research that will seriously affect the yield of wheat and rice. 12: GM is used to make production cheaper. 13: people continue to mistrust science. Answers: **consumer's choice / risk and benefit / skiing / GM crops / wheat and rice / production / mistrust.**

10. GM crops

7: a crucial issue is the consumer's choice. 8: the consumer's right to choose should be considered alongside considerations of risk and benefit. 9: people who ski choose to do so — skiing... is exhilarating. 10: promoters believe the future population cannot be fed without GM crops. 11: there is no GM research that will seriously affect the yield of wheat and rice. 12: GM is used to make production cheaper. 13: people continue to mistrust science. Answers: consumer's choice / risk and benefit / skiing / GM crops / wheat and rice / production / mistrust.

11. wheat and rice

7: a crucial issue is the consumer's choice. 8: the consumer's right to choose should be considered alongside considerations of risk and benefit. 9: people who ski choose to do so — skiing... is exhilarating. 10: promoters believe the future population cannot be fed without GM crops. 11: there is no GM research that will seriously affect the yield of wheat and rice. 12: GM is used to make production cheaper. 13: people continue to mistrust science. Answers: consumer's choice / risk and benefit / skiing / GM crops / wheat and rice / production / mistrust.

12. production

7: a crucial issue is the consumer's choice. 8: the consumer's right to choose should be considered alongside considerations of risk and benefit. 9: people who ski choose to do so — skiing... is exhilarating. 10: promoters believe the future population cannot be fed without GM crops. 11: there is no GM research that will seriously affect the yield of wheat and rice. 12: GM is used to make production cheaper. 13: people continue to mistrust science. Answers: consumer's choice / risk and benefit / skiing / GM crops / wheat and rice / production / mistrust.

13. mistrust

7: a crucial issue is the consumer's choice. 8: the consumer's right to choose should be considered alongside considerations of risk and benefit. 9: people who ski choose to do so — skiing... is exhilarating. 10: promoters believe the future population cannot be fed without GM crops. 11: there is no GM research that will seriously affect the yield of wheat and rice. 12: GM is used to make production cheaper. 13: people continue to mistrust science. Answers: consumer's choice / risk and benefit / skiing / GM crops / wheat and rice / production / mistrust.

14. C

they have absolutely missed the point. The precautionary principle is a subtle idea — the whole article is a correction of a widespread misunderstanding. Answer: C.

Passage 5 — Australia's Megafauna Controversy

27. YES

Para 2: Field and Wroe "claim that it provides unequivocal evidence of a long overlap of humans and megafauna". Answer: YES.

28. **NOT GIVEN**

smaller megafauna species survived — the passage only says aridity "brought about their eventual demise" (megafauna generally); no distinction is made by species size. Answer: **NOT GIVEN**.

29. **NO**

Para 3: "**The dating of these layers is accurate**: ages for the sediments were obtained through radiocarbon dating... and luminescence dating..." — the writers call it accurate, not unreliable. Answer: **NO**.

30. **YES**

Para 3: "some of the stone tools show surface features indicating their use for processing plants, and a few even have **well-preserved blood and hair residues suggesting they were used in butchering animals**". Answer: **YES**.

31. **B (stone)**

large quantity of charcoal, ____ and artefacts ≈ para 4: "we estimate there are more than 3 tonnes of charcoal and **more than 300 tonnes of stone** buried there". Answer: **B – stone**.

32. **F (water)**

impossible if the area had been covered with ____ ≈ para 4: the tool count is "hard to reconcile with a site that was only available for occupation when the lake was dry" — implying it is otherwise normally covered by water. Answer: **F – water**.

33. **D (cooking)**

man-made structures... used for ____ ≈ para 4: "no cultural features such as **oven pits** have been discovered". Answer: **D – cooking**.

34. **C (sand)**

material... 36,000 years old, the ____ is much more recent ≈ para 5: "the charcoal samples are all roughly 36,000 years old. Second, **sand in the two upper levels is considerably younger than charcoal from the same levels**". Answer: **C – sand**.

35. **G (fossil bones)**

absence of tools and ____ a short distance away ≈ para 5: "a deep drill core made a mere 60 metres from the site recovered **no stone artefacts or fossil bones** whatsoever". Answer: **G – fossil bones**.

36. **A**

Para 5: "These points suggest strongly that **the sediments have been moved about** and some of the old charcoal has been re-deposited in younger layers." Answer: **A**.

37. **B**

Para 6: "Flood events more likely explain the accumulation of megafauna remains, and **could have mixed old bones with fresh deposits**." Answer: **B**.

38. **D**

Para 6: "we speculate that **the graziers made sure it was protected from the damage caused by cattle hooves by lining the surface with small stones collected from further afield**". Answer: **D**.

39. **B**

Para 7: "a group of 10 people killing only one juvenile Diprotodon each year would be sufficient to bring about **the extinction of that species within 1,000 years**". Answer: **B**.

40. **D**

Para 4: "If the sediment layers have remained undisturbed... as Field and Wroe contend, then the ages... should increase with depth. However, **our analysis revealed a number of inconsistencies**." Answer: **D**.

Passage 6 — Business Innovation

1. **NO**

as blockbuster novel products become more elusive, **big companies would do well to focus instead on making lots of small things better** — not on finding genuinely original products. Answer: **NO**.

2. **YES**

disruptive innovations **often herald the rapid downfall of thriving, long-standing businesses**. Answer: **YES**.

3. **NOT GIVEN**

sustaining innovations are described only as **the sort which simply improve on existing products for existing markets** — no mention of researching competitors. Answer: **NOT GIVEN**.

4. **YES**

rival products can be rushed onto the market at speeds which would have been unimaginable in those days. Answer: **YES**.

5. **YES**

the ultimate outcome of any one innovation may still be unpredictable. Answer: **YES**.

6. **D**

Vishwanath: companies need to adjust to **the ongoing fragmentation of markets... customised for a small group of consumers**. Answer: **D**.

7. **E**

Brynjolfsson: **the roots of the USA's productivity surge originate in a genuine revolution in how American companies are using information technology**. Answer: **E**.

8. **B**

Hammer cites the PC-maker case: **its head of manufacturing, who feared for his job, and its head of marketing, who did not want to upset his existing retail outlets.** Answer: **B**.

9. **C**

Baumol: large companies **have both cut back and redirected their Research and Development spending in recent years.** Answer: **C**.

10. **processes / business processes**

10: operational innovations = creativity in their business **processes**. **11:** Wal-Mart's pioneering idea of '**cross-docking**'. **12:** Southwest Airlines/Tetra Pak paragraph opens: "Companies are also being encouraged to embrace **strategic** innovations". **13:** Slywotsky and Wise's book recommends what they call **demand** innovations (Air Liquide). **14:** Air Liquide became a supplier not only of gas, but also of the **management services** to accompany it. Answers: **processes / cross-docking / strategic / demand / management services**.

11. **cross-docking**

10: operational innovations = creativity in their business **processes**. **11:** Wal-Mart's pioneering idea of '**cross-docking**'. **12:** Southwest Airlines/Tetra Pak paragraph opens: "Companies are also being encouraged to embrace **strategic** innovations". **13:** Slywotsky and Wise's book recommends what they call **demand** innovations (Air Liquide). **14:** Air Liquide became a supplier not only of gas, but also of the **management services** to accompany it. Answers: **processes / cross-docking / strategic / demand / management services**.

12. **strategic**

10: operational innovations = creativity in their business **processes**. **11:** Wal-Mart's pioneering idea of '**cross-docking**'. **12:** Southwest Airlines/Tetra Pak paragraph opens: "Companies are also being encouraged to embrace **strategic** innovations". **13:** Slywotsky and Wise's book recommends what they call **demand** innovations (Air Liquide). **14:** Air Liquide became a supplier not only of gas, but also of the **management services** to accompany it. Answers: **processes / cross-docking / strategic / demand / management services**.

13. **demand**

10: operational innovations = creativity in their business **processes**. **11:** Wal-Mart's pioneering idea of '**cross-docking**'. **12:** Southwest Airlines/Tetra Pak paragraph opens: "Companies are also being encouraged to embrace **strategic** innovations". **13:** Slywotsky and Wise's book recommends what they call **demand** innovations (Air Liquide). **14:** Air Liquide became a supplier not only of gas, but also of the **management services** to accompany it. Answers: **processes / cross-docking / strategic / demand / management services**.

14. **management services / the management services**

10: operational innovations = creativity in their business **processes**. **11:** Wal-Mart's pioneering idea of '**cross-docking**'. **12:** Southwest Airlines/Tetra Pak paragraph opens: "Companies are also being encouraged to embrace **strategic** innovations". **13:** Slywotsky and Wise's book recommends what they call **demand** innovations (Air Liquide). **14:** Air Liquide became a supplier not only of gas, but also of the **management services** to accompany it. Answers: **processes / cross-docking / strategic / demand / management services**.

Passage 7 — Children's Literature Studies Today

27. **A**

Para 2: **adult perceptions of what children are and of what they could and should be become woven into the literature**. Answer: **A**.

28. **C**

Para 3: some are **less interested in literary values than in the kinds of lessons it can teach**. Answer: **C**.

29. **B**

Para 5: new media **create definitional issues for scholars** — questioning where the limits of 'literature' lie. Answer: **B**.

30. **NO**

Para 5: **children themselves undoubtedly perceive their print literature as part of a broader continuum** — a continuum, not a clear distinction. Answer: **NO**.

31. **NO**

Para 6: **at least three disciplinary frameworks (English, education, and librarianship) fragment the focus**. Answer: **NO**.

32. **NOT GIVEN**

Para 8 links exploring children's perceptions to **a better understanding of how children read and how they learn to read, is a prerequisite to improved approaches to teaching** — nothing about parents choosing books. Answer: **NOT GIVEN**.

33. **YES**

Para 8: interpreting children's responses **frequently lead(s) to an understanding of how picture books appeal to children**. Answer: **YES**.

34. **NOT GIVEN**

Para 10 says the skill gap is why games feel harder — **a particular set of skills is absent** — but never suggests asking children about desired game features. Answer: **NOT GIVEN**.

35. **F**

Para 9: young people's accounts **enable a more sophisticated description of many of the complex processes involved in reading**. Answer: **F**.

36. **C**

Para 9: **asking other readers how they read, however, reduces that risk**. Answer: **C**.

37. **H**

Para 9: I may judge books by **their capacity to evoke a vivid visual response**. Answer: **H**.

38. **A**

Para 10: **understanding the significance of print literature as one aspect of literary culture**. Answer: **A**.

39. **E**

Para 10: **many adults will probably never develop the automatic skills to process a game as readily as they can read a book** — not that books are better, but that a skill is missing. Answer: **E**.

40. **C**

The article surveys multiple challenges (adult authorship, didacticism, commercialism, new media, disciplinary fragmentation) then, from para 7 onward, proposes a way forward: **how is the value of the imaginative encounter with the work of literature sustained and honored** — asking child readers for help. Answer: **C**.

Passage 8 — Conscious and Unconscious Thought

1. **D**

Matthew Walker: **rem awakenings provided a 32 per cent advantage in the number of puzzles solved, compared with non-rem awakenings**. Answer: **D**.

2. **F**

Howard Gardner suggests **that we free ourselves from the straitjacket of always doing things the same way**. Answer: **F**.

3. **E**

Daniel Goleman urges us to **discount... self-censorship** — the voice that says 'they'll think I'm foolish'. Answer: **E**.

4. **A**

Whitehead meant that **some of the greatest breakthroughs in creativity occur when a problem is consciously shelved for a while**. Answer: **A**.

5. **incubation**

this is a process which psychoanalysts call incubation. Answer: **incubation**.

6. **irrelevant task**

they were distracted by an irrelevant task whilst making a decision. Answer: **irrelevant task**.

7. **posters**

participants were presented with five posters and asked to choose one to take home. Answer: **posters**.

8. **TRUE**

the more complex a problem is, the less likely it becomes that conscious thought can contribute much... you should let the unconscious deal with it. Answer: **TRUE**.

9. **NOT GIVEN**

the passage discusses REM sleep and dreaming, but never mentions resting before sleep specifically. Answer: **NOT GIVEN**.

10. **FALSE**

rem awakenings provided a 32 per cent advantage in the number of puzzles solved, compared with non-rem awakenings — REM (dreaming) sleep helped more, not dreamless sleep. Answer: **FALSE**.

11. **FALSE**

subjects were asked to solve crossword puzzles, not given a memory test. Answer: **FALSE**.

12. **NOT GIVEN**

the passage says daydreams are so useful in the quest for creativity, but never compares them against sleep-dreams for relative effectiveness. Answer: **NOT GIVEN**.

13. **TRUE**

a key obstacle that the conscious mind imposes on the unconscious: that of self-censorship... keeping you imprisoned within the boundaries of what is deemed acceptable. Answer: **TRUE**.

Passage 9 — Designed to Last

1. **D**

tools serve 'conscience time', gathering dust on a shelf... people are reluctant to admit that they have wasted their money. Answer: **D**.

2. **B**

people **made the objects themselves, or family members passed them on... relied on expert manufacturers living close by, whom they probably knew personally** — gave objects a narrative. Answer: **B**.

3. **B**

when the novelty fades, they simply renew the excitement by buying more. Answer: **B**.

4. **D**

consumerist culture idolizes novelty instead... the chance to remake themselves with glossy, box-fresh products seems irresistible. When the novelty fades, they simply renew the excitement by buying more. Answer: **D**.

5. **C**

jeans **do not have the right feel until they have been worn and washed a hundred times**; a teddy bear — **no matter how ragged and worn... becomes, we don't rush out and buy another one**. Answer: **C**.

6. **TRUE**

power tools are **thrown away having been used, on average, for just ten minutes**. Answer: **TRUE**.

7. **TRUE**

to understand why we have become so wasteful, we should look to the underlying motivation of consumers. Answer: **TRUE**.

8. **NOT GIVEN**

we will buy less and rent a lot more — a future prediction, not a claim about the present. Answer: **NOT GIVEN**.

9. **NOT GIVEN**

at the moment, price competition between retailers makes it cheaper for consumers to replace rather than repair — no statement about future repair pricing. Answer: **NOT GIVEN**.

10. **D (laundry)**

installing energy-saving light bulbs, more efficient washing machines. "Washing machines" = **D laundry**.

11. **F (food)**

choosing locally produced groceries with less packaging. "Groceries" = **F food**.

12. **H (cars)**

instead of buying a second car, for example, we might buy into a car-sharing network.
Answer: **H cars**.

13. **C (tools)**

why own things such as tools that you use infrequently. Answer: **C tools**.

14. **A (mobile phones)**

electronic goods such as mobile phones will be designed to be recyclable. Answer: **A mobile phones**.

Passage 10 — Digital History

1. **A**

reduces the advantages once enjoyed by prestigious and well-funded international universities and centres for historical research. Answer: **A**.

2. **C**

there is a stereotype of the scholar alone in his or her workshop... this is how many in the profession probably still think of themselves. The solitary researcher. Answer: **C**.

3. **D**

these pieces of technology will undoubtedly have significant implications for future study and research in history — and indeed in most other academic subjects as well. Answer: **D**.

4. **B**

virtual meetings can be recorded to create an archive... especially important for future historians trying to validate or replicate the original research. Answer: **B**.

5. **YES**

the isolation felt by many, especially PhD students, will probably be reduced. Answer: **YES**.

6. **NOT GIVEN**

the passage only mentions video-conferencing for **interchanges among students from different universities** and joint classes — no mention of student-teacher consultations specifically. Answer: **NOT GIVEN**.

7. **NO**

the same website tool now **enables an advanced research group like ours to plan activities and share documents** — the writer extends a student tool INTO serious research use, the opposite of "more suitable for students". Answer: **NO**.

8. **NOT GIVEN**

the passage only describes what a **wiki allows multiple users to edit shared documents, whilst significantly also retaining all previous versions** — never discusses academics' opinion of its reliability. Answer: **NOT GIVEN**.

9. **D (transformation)**

our old methods of research and teaching may well be reshaped by collaborative technologies. "Reshaped" = **D transformation**.

10. **G (ownership)**

to whom does the product of a collaboration belong? = **G ownership**.

11. **E (contributions)**

with many scholars working together, how can a level of excellence be maintained? "Many scholars working together" = **E contributions**.

12. **A (cautious)**

conservative scholars may be uncomfortable using the website or video-conferencing. "Conservative" = **A cautious**.

13. **J (confident)**

the idea of being recorded or filmed can be disconcerting. "Disconcerting" (not feeling at ease) = not feeling **J confident**.

14. **K (guidelines)**

a set of informal social rules is necessary when discussing issues with people with whom you cannot make direct eye contact. "Social rules" = **K guidelines**.

Passage 11 — Dressed to Dazzle

1. **D**

made a prototype shirt that shortens its sleeves when room temperature rises. Answer: **D**.

2. **G**

manufacturing spider silk from the milk of genetically engineered goats — imitating a natural fibre. Answer: **G**.

3. **C**

make fabrics more durable, comfortable, wrinkle-free and stain-resistant — stain-resistant fabric needs washing less often. Answer: **C**.

4. **A**

uses nanotechnology to create fabrics that can store or release heat. Answer: A.

5. **B**

Sportswear could be impregnated with anti-perspirant. Answer: B.

6. **F**

the first man-made fibre derived from a 100% annually renewable resource. Answer: F.

7. **Ingeo**

7: Ingeo's impact on fashion will derive instead from its emphasis on using natural sustainable resources. 8: Soya bean fibre is also gaining ground... the fabric is a better absorber and ventilator than silk. 9: attaching electronic components to the fabrics **after the** normal weaving process. 10: attaching electronic components to the fabrics. 11: powered by a small battery. 12: Costumes made of the fabric wowed audiences at a production of the opera Aida. 13: electronic textiles capable of such wizardry are still too fragile to wear. 14: Accessories with this chameleon-like capacity — for instance a handbag — are more likely to appear first. Answers: Ingeo / soya bean / weaving / electronic components / battery / costumes / fragile / accessories.

8. **soya bean**

7: Ingeo's impact on fashion will derive instead from its emphasis on using natural sustainable resources. 8: Soya bean fibre is also gaining ground... the fabric is a better absorber and ventilator than silk. 9: attaching electronic components to the fabrics **after the** normal weaving process. 10: attaching electronic components to the fabrics. 11: powered by a small battery. 12: Costumes made of the fabric wowed audiences at a production of the opera Aida. 13: electronic textiles capable of such wizardry are still too fragile to wear. 14: Accessories with this chameleon-like capacity — for instance a handbag — are more likely to appear first. Answers: Ingeo / soya bean / weaving / electronic components / battery / costumes / fragile / accessories.

9. **weaving**

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10. **electronic components**

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11. **battery**

7: Ingeo's impact on fashion will derive instead from its emphasis on using natural sustainable resources. 8: Soya bean fibre is also gaining ground... the fabric is a better absorber and ventilator than silk. 9: attaching electronic components to the fabrics **after the normal weaving process**. 10: attaching electronic components to the fabrics. 11: powered by a small battery. 12: Costumes made of the fabric wowed audiences at a production of the opera Aida. 13: electronic textiles capable of such wizardry are still too fragile to wear. 14: Accessories with this chameleon-like capacity — for instance a handbag — are more likely to appear first. Answers: Ingeo / soya bean / weaving / electronic components / battery / costumes / fragile / accessories.

12. **costumes**

7: Ingeo's impact on fashion will derive instead from its emphasis on using natural sustainable resources. 8: Soya bean fibre is also gaining ground... the fabric is a better absorber and ventilator than silk. 9: attaching electronic components to the fabrics **after the normal weaving process**. 10: attaching electronic components to the fabrics. 11: powered by a small battery. 12: Costumes made of the fabric wowed audiences at a production of the opera Aida. 13: electronic textiles capable of such wizardry are still too fragile to wear. 14: Accessories with this chameleon-like capacity — for instance a handbag — are more likely to appear first. Answers: Ingeo / soya bean / weaving / electronic components / battery / costumes / fragile / accessories.

13. **fragile**

7: Ingeo's impact on fashion will derive instead from its emphasis on using natural sustainable resources. 8: Soya bean fibre is also gaining ground... the fabric is a better absorber and ventilator than silk. 9: attaching electronic components to the fabrics **after the normal weaving process**. 10: attaching electronic components to the fabrics. 11: powered by a small battery. 12: Costumes made of the fabric wowed audiences at a production of the opera Aida. 13: electronic textiles capable of such wizardry are still too fragile to wear. 14: Accessories with this chameleon-like capacity — for instance a handbag — are more likely to appear first. Answers: Ingeo / soya bean / weaving / electronic components / battery / costumes / fragile / accessories.

14. accessories

7: Ingeo's impact on fashion will derive instead from its emphasis on using natural sustainable resources. 8: Soya bean fibre is also gaining ground... the fabric is a better absorber and ventilator than silk. 9: attaching electronic components to the fabrics after the normal weaving process. 10: attaching electronic components to the fabrics. 11: powered by a small battery. 12: Costumes made of the fabric wowed audiences at a production of the opera Aida. 13: electronic textiles capable of such wizardry are still too fragile to wear. 14: Accessories with this chameleon-like capacity — for instance a handbag — are more likely to appear first. Answers: Ingeo / soya bean / weaving / electronic components / battery / costumes / fragile / accessories.

Passage 12 — Facial Expression

28. misidentified

28: both fear and surprise were constantly misidentified. 29: certain facial expressions correspond to particular emotions. 30–32: can not be covered, regardless of cultural background, and regardless of whether or not the culture has been isolated or exposed to the mainstream. Answers: misidentified / emotions / cultural background / isolated / exposed.

29. emotions

28: both fear and surprise were constantly misidentified. 29: certain facial expressions correspond to particular emotions. 30–32: can not be covered, regardless of cultural background, and regardless of whether or not the culture has been isolated or exposed to the mainstream. Answers: misidentified / emotions / cultural background / isolated / exposed.

30. cultural background

28: both fear and surprise were constantly misidentified. 29: certain facial expressions correspond to particular emotions. 30–32: can not be covered, regardless of cultural background, and regardless of whether or not the culture has been isolated or exposed to the mainstream. Answers: misidentified / emotions / cultural background / isolated / exposed.

31. isolated

28: both fear and surprise were constantly misidentified. 29: certain facial expressions correspond to particular emotions. 30–32: can not be covered, regardless of cultural background, and regardless of whether or not the culture has been isolated or exposed to the mainstream. Answers: misidentified / emotions / cultural background / isolated / exposed.

32. exposed

28: both fear and surprise were constantly misidentified. 29: certain facial expressions correspond to particular emotions. 30–32: can not be covered, regardless of cultural background, and regardless of whether or not the culture has been isolated or exposed to the mainstream. Answers: misidentified / emotions / cultural background / isolated / exposed.

33. C

difficult to interpret even in familiar individuals. For instance, disgust and fear can be tough to tell apart. Answer: C.

34. A

Facial expressions and their significance in the perceiver can, to some extent, vary between cultures with evidence from descriptions in the works of Charles Darwin — the first mention of culture's role. Answer: A.

35. D

He supports his data with statistics on the relation between the blink rates of presidential candidates and their success in their races. Answer: D.

36. H

none of these emotions has a definitive social component, such as shame, pride, or schadenfreude. Answer: H.

37. D

Blink rate can reveal how nervous or at ease a person maybe. Answer: D.

38. B

because expressions are closely tied to emotion, they are more often involuntary. Answer: B.

39. "B

B: a person's eyes **reveal much about how they are feeling, or what they are thinking**. **D:** the facial expressions are largely related to the mind and **each part of the face can express specific emotion**. (A is contradicted — expressions "also occur in most other mammals and some other animal species"; C is the opposite of Ekman's actual finding — expressions are universal, not culture-specific; E is never mentioned.) Answers: **B, D**.

40. "B

B: a person's eyes **reveal much about how they are feeling, or what they are thinking**. **D:** the facial expressions are largely related to the mind and **each part of the face can express specific emotion**. (A is contradicted — expressions "also occur in most other mammals and some other animal species"; C is the opposite of Ekman's actual finding — expressions are universal, not culture-specific; E is never mentioned.) Answers: **B, D**.

Passage 13 — First Memories

1. YES

we should welcome enquiries about our first memories - it is a compliment if others are interested in our lives and experiences. Answer: **YES**.

2. **NO**

their range of memories is also predictably diverse, from holidays and injuries to routine activities. Answer: **NO**.

3. **NOT GIVEN**

the passage only compares the proportion of negative and positive memories (around 25% vs nearly 50%), not which is more interesting. Answer: **NOT GIVEN**.

4. **NO**

our earliest memories were largely overlooked by researchers, which turns out to have been an important omission. Answer: **NO**.

5. **NO**

numerous experiments studying the memories held by witnesses to an event have shown conclusively that recollections of what happened can vary dramatically between individuals, and even in the same person over time. Answer: **NO**.

6. **B**

based on the assumption that american society places greater emphasis on individualism, this implies that earliest memories are influenced by social priorities. Answer: **B**.

7. **A**

who we are now appears to have some bearing on what we choose to recall from long ago. Answer: **A**.

8. **D**

our earliest memories may be pieced together from different elements of reality... information that actually comes from a photograph, a family story, a memory that we've heard from somebody else, or some combination of all of these. Answer: **D**.

9. **C**

does this really matter? they are still much-loved memories and perhaps we can learn about ourselves from them... such factors represent important elements of our humanity, whether or not the memory itself is completely accurate. Answer: **C**.

10. **F**

semantic memory... is the ability to recall what might loosely be called data, such as dates, times or places in which a particular incident occurred. Answer: **F**.

11. **C**

for the majority of adults, their earliest memories begin once the period of childhood amnesia has come to an end. Answer: **C**.

12. **G**

many people construct a narrative of their lives and select memories that contribute most forcefully to that narrative. Answer: **G**.

13. **D**

some memories stand out from the rest as much clearer and more colourful. these are usually formed during what's known as the 'reminiscence bump'. Answer: **D**.

14. **A**

flashbulb memories... are memories that derive from events about which we have particularly strong feelings. Answer: **A**.

Passage 14 — How Green is your PlanIT Valley?

1. **C**

a working model that will inspire the next generation of low carbon cities. Answer: **C**.

2. **A**

make Living PlanIT's plans sound more like an admirable experiment rather than a viable construction project. Answer: **A**.

3. **B**

the construction industry remains the last sector of the economy to resist the IT revolution... Their existing techniques are inadequate. Answer: **B**.

4. **D**

the essential 'anything is possible' mentality that is a feature of such companies. Answer: **D**.

5. **C**

it is hard not to admire a project that will put so many theories about smart cities to the test all in one go. Answer: **C**.

6. **YES**

the company's marketing language **requires a degree in Public Relations** to understand. Answer: **YES**.

7. **YES**

it becomes possible to believe that they might just deliver on their absurdly ambitious promises. Answer: **YES**.

8. **NOT GIVEN**

the passage says lessons were taken from aerospace, automotive and shipbuilding *manufacturing processes* — it never confirms the actual buildings have features matching those vehicles. Answer: **NOT GIVEN**.

9. **NO**

such techniques are not new, but they are rarely put into practice. Answer: **NO**.

10. **NOT GIVEN**

the technology **may sound like something out of a science-fiction novel, but we are increasingly living in a science-fiction age** — a comparison, not a claim that sci-fi inspired it. Answer: **NOT GIVEN**.

11. **B**

the IT industry has been complaining for a long time that the construction sector has failed to make adequate (if any) use of IT in its buildings. Answer: **B**.

12. **A**

almost all of the technology the company is planning to deploy in PlanIT Valley either already exists or is viable from a technical point of view. Answer: **A**.

13. **D**

cost concerns surrounding green developments are ill-founded. Answer: **D**.

14. **G**

The PlanIT Valley project presents many problems in terms of project management and coordination. Answer: **G**.

Passage 15 — How a Prehistoric Predator Took to the Skies

27. **D**

the least understood of all the extinct reptiles that once spent their lives in the skies. Answer: **D (reptiles)**.

28. **L**

at least part of their bodies was covered in hair, not unlike bats. Answer: **L (hair)**.

29. **F**

whether pterosaurs travelled on the ground by walking on their hind legs. Answer: **F (hind legs)**.

30. **J**

did such a heavy creature ever manage to take off? Answer: **J (weight)**.

31. **I**

how could a wing that appears to have been supported by fine, hollow bones. Answer: **I (strength)**.

32. **B**

a fish-eating sub-species of pterosaur that ruled the skies in the cretaceous period. Answer: **B (fish)**.

33. **YES**

the probability of finding three-dimensional pterosaur fossils anywhere is slim — 'their bones are usually crushed to dust.' Answer: **YES**.

34. **NO**

always been interested in how they managed to launch themselves, so i thought the real test would be to actually build one and fly it — the goal was take-off/locomotion, not distance. Answer: **NO**.

35. **NOT GIVEN**

The forward-facing pteroid **may have been instrumental in the evolution of large size by the pterosaurs** — overall body size, not wing size specifically. Answer: **NOT GIVEN**.

36. **NOT GIVEN**

The passage only says **we've had to take it flying without the head so far** — it never says several heads were built and failed. Answer: **NOT GIVEN**.

37. **D**

After describing the wings, the writer says it "did not answer the most pressing question: **how did such heavy animals manage to launch themselves into the sky?**" Answer: **D**.

38. **A**

to the surprise of many scientists, fossils from the araripe basin showed the pteroid possibly faced the opposite way — scientists had thought it pointed to the shoulder. Answer: **A**.

39. **B**

the model is stabilised by its head, which means it can start spinning around — 'we've had to take it flying without the head so far.' Answer: **B**.

40. **C**

Wilkinson wants huge pterosaurs but admits it's more likely they'll **start off with one of the smaller, flapping pterosaurs** — the writer hopes he is content to leave it at that, i.e. to stop at building small models. Answer: **C**.

Passage 16 — Improving Patient Safety

1. **C**

Mawle: **older people, especially those with arthritis, may have the same level of strength as a child**. Answer: **C**.

2. **D**

van der Waarde: **it is not known how much visually impaired patients will benefit nor how much the reading of visually able patients will be impaired**. Answer: **D**.

3. **B**

Trust statistic: **although child-resistant closures have reduced the number of incidents, they are not fully child-proof**. Answer: **B**.

4. **A**

Swayne: **a parent giving a cough medicine to a child in the middle of the night... packaging should be designed for moments such as these**. Answer: **A**.

5. **D**

van der Waarde: **many regulatory authorities do not have the resources to handle packaging information properly... they only see one box at a time and not several together**. Answer: **D**.

6. **C**

Mawle: **why are company details listed towards the beginning of a leaflet when what might be more important for the patient is that the medicine should not be taken with alcohol**. Answer: **C**.

7. **B**

a marketing team will prepare a brief. Answer: **B marketing teams**.

8. **D**

OTC designs are **usually commissioned from design agencies... the designers will come up with perhaps six or seven designs**. Answer: **D external designers**.

9. **A**

designs **might be tested on a consumer group**. Answer: **A consumers**.

10. **E**

most designs for prescription-only products are created in-house. Answer: E in-house designers.

11. **F**

the chosen design is then handed over to design engineers who work out how the packaging will be produced. Answer: F design engineers.

12. **B**

investigations following these tragedies have attributed some blame to poor typescript. Answer: B.

13. **B**

if all packaging is white with black lettering, people would have no choice but to read every box carefully. Answer: B.

14. **A**

these are innovative objects aiming to make things more user-friendly and safer, presented as positive developments, not as remaining problems. Answer: A.

Passage 17 — Inside the Mind of a Fan

1. **F**

Para F: we watch sport for the feeling, the human drama... they are directly connected to the amygdala, one of the main brain regions involved in emotion. Answer: F.

2. **B**

Para B: most novices swing with their consciousness, using an area of brain next to the premotor cortex. Answer: B.

3. **E**

Para E: the best athletes are those with a premotor cortex capable of imagining the movements of victory, together with the physical properties to make those movements real. Answer: E.

4. **C**

Para C: humans have an even more elaborate mirror neuron system [than monkeys]. Answer: C.

5. **D**

Para D: this phenomenon of neural mirroring was first discovered in 1954. Answer: D.

6. **E**

Para E: **watching a great golfer makes us better golfers, and watching a great sprinter actually makes us run faster.** Answer: E.

7. **C**

groups of cells that 'store vocabularies of motor actions'. Just as there are grammars of movement. These networks of cells are the bodily 'sentences' — a network of neurons, described using a language metaphor. Answer: C.

8. **A**

the expert's motor action **has become memorized... he hits the ball with the tranquility of his perfected autopilot.** Answer: A.

9. **C**

we watch sport for the feeling, the human drama. Answer: C.

10. **YES**

most novices swing with their consciousness. Answer: YES.

11. **NO**

humans have an even more elaborate mirror neuron system — the opposite. Answer: NO.

12. **NO**

if we are staring at a photograph, a fixed image of a runner mid-stride, our mirror neurons are totally silent. Answer: NO.

13. **NOT GIVEN**

they are introduced only as **two French physiologists** — nothing about them playing sport. Answer: NOT GIVEN.

14. **YES**

the mu system **is active when our bodies are still, and disappears whenever we do something active** — so it IS at rest during activity. Answer: YES.

Passage 18 — Kinesics

1. **v**

Psychology is a well-established subject, but one area of it — kinesics grew out of the older field of psychology. Answer: v.

2. **viii**

she has set the lead for the family action — the mother's body language, not words, marks her as the dominant figure. Answer: **viii**.

3. **vii**

we become aware of how that person feels, although we are conscious neither of seeing the signal nor of giving it meaning. Answer: **vii**.

4. **ii**

detect the effect of a television advertisement — a commercial (business) use of kinesics. Answer: **ii**.

5. **vi**

kinesics experts often have to take into consideration cultural and environmental differences — an untrained person easily misreads these signals. Answer: **vi**.

6. **ix**

such a touch must come at the right moment and in the right context — otherwise it is easily misread. Answer: **ix**.

7. **xi**

will touch others even though they are bombarded with body-language requests not to — they ignore the signals. Answer: **xi**.

8. **iv**

We lift one eyebrow for disbelief, shrug our shoulders for indifference, tap our fingers for impatience. The gestures are numerous. Answer: **iv**.

9. **YES**

was not identified until 1952, and research only began in earnest in the 1960s — roughly a decade gap before serious research started. Answer: **YES**.

10. **NO**

her words deny her leadership as she asks her husband or children for advice — her SPEECH contradicts, rather than shows, her leadership. Answer: **NO**.

11. **NOT GIVEN**

when there is any unconscious, positive response to the advertisement — the passage discusses detecting a positive response, never actual sales figures. Answer: **NOT GIVEN**.

12. **YES**

the other person may take it as an intrusion to his or her personal space. Answer: **YES**.

13. **YES**

from those that apply only in one culture to those that cut across cultural barriers — paired in the same sentence with the deliberate→unconscious spectrum, implying the cross-cultural ones tend to be the unconscious ones. Answer: **YES**.

14. **B**

A classic example is the young woman who told her psychiatrist that she loved her boyfriend very much while shaking her head from side to side in subconscious denial — an accessible, illustrative anecdote typical of writing for a general audience, not a specialist paper. Answer: **B**.

Passage 19 — Learning to be Bilingual

1. **B**

They understand that the family setting they create is central to this. Answer: **B**.

2. **C**

this creates the myth that mixing languages should always be avoided — the study suggests this "received wisdom" is a myth. Answer: **C**.

3. **A**

English dominated outside the home, so **the mothers were careful to carve out additional space for more sustained Japanese language learning with their children.** Answer: **A**.

4. **D**

they adopted a broader behavioural repertoire than that which is usually associated with a language. Answer: **D**.

5. **A**

parents feel it's problematic to keep correcting children when they mix languages. This is particularly the case when children show annoyance at being corrected. Answer: **A**.

6. **C**

the child said 'aisukurimuzu' — **the child had created something in between.** Answer: **C**.

7. **H**

Following firm rejections by her brother, the girl drew on her communicative repertoire. "Firm rejections" = **H refusal**.

8. **C**

the girl drew on her communicative repertoire to convince him. "Repertoire" = **C strategies**.

9. **A**

She rephrased the question by using various polite forms. "The question" = **A request**.

10. **E**

an assumed obligation to care on the part of the brother. "Obligation to care" = **E responsibility**.

11. **YES**

Whether bilingual or not, we all constantly select from our repertoire anything that will best serve our purpose. Answer: **YES**.

12. **NOT GIVEN**

the passage never discusses what these children do as parents themselves in later life. Answer: **NOT GIVEN**.

13. **NO**

the popularity of OPOL amongst parents rests on the simplicity of its message, which is that it should be applied consistently — not because it encourages switching. Answer: **NO**.

14. **NOT GIVEN**

the passage never discusses children eventually preferring one language over the other. Answer: **NOT GIVEN**.

Passage 20 — Left, or Right?

1. **D**

researchers started uncovering evidence of lateralisation in all sorts of animals — contradicting the belief that it was uniquely human. Answer: **D**.

2. **C**

strongly-lateralised individuals were able to feed twice as fast as weakly-lateralised ones when there was a threat of a predator — feeding while watching for danger. Answer: **C**.

3. **E**

the parrots with the strongest foot preferences worked out the puzzle far more quickly than their ambidextrous peers. Answer: **E**.

4. **B**

a group of fish survives best if the majority turn together in one direction while a very small proportion of the group escape in the direction that the predator is not expecting. Answer: **B**.

5. environmental

5: Genetics plays a part in determining lateralisation, but environmental factors have an impact too. 6: a chick's lateralisation depends on whether it is exposed to light before hatching from its egg. 7: Rogers tested the advantages of brain bias in chicks faced with the challenge of multitasking. 8–9: the birds incubated in the light looked for food mainly with their right eye, while using the other to check out the predator. Answers: environmental / light / multitasking / food / predator.

6. light

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7. multitasking

5: Genetics plays a part in determining lateralisation, but environmental factors have an impact too. 6: a chick's lateralisation depends on whether it is exposed to light before hatching from its egg. 7: Rogers tested the advantages of brain bias in chicks faced with the challenge of multitasking. 8–9: the birds incubated in the light looked for food mainly with their right eye, while using the other to check out the predator. Answers: environmental / light / multitasking / food / predator.

8. food

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9. predator

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10. E

Similar results probably hold true for many other animals — Bisazza's fish study extends Rogers' chick findings. Answer: E.

11. **H**

left-handers may have the advantage over the right-handed majority. Answer: **H**.

12. **B**

researchers have built up an impressive catalogue of animal lateralisation. Answer: **B**.

13. **B**

For many years it was assumed that lateralisation was a uniquely human trait — Rogers' 1970s chick study was among the first to contradict this. Answer: **B**.

14. **A**

This does, on some occasions, let the animal down — such as a toad failing to spot a snake with its weaker eye. Answer: **A**.

Passage 21 — Let's All Be Friends

27. **nature**

what mattered was the nature of the network binding its team together. Answer: **nature**.

28. **creative vision**

their 'weak' networks meant a lack of creative vision. Answer: **creative vision**.

29. **flops**

teams that had worked together successfully also tended to produce flops. Answer: **flops**.

30. **right mix**

groups with exactly the right mix of new and old participants reliably produced hits. Answer: **right mix**.

31. **density**

this variation in the 'density' of the ties allowed easy communication. Answer: **density**.

32. **NO**

the writer suggests governments should consider using the science of social networks as a tool — a proposal, not a description of current effective use. Answer: **NO**.

33. **NO**

Bring Back Birdie, its 1981 sequel, closed after just four nights. Answer: **NO**.

34. **NO**

he investigated **musicals that were launched between 1945 and 1989** — mostly the second half of the century. Answer: **NO**.

35. **YES**

teenagers are affected by their friends' friends' habits **even if they don't personally know those individuals**. Answer: **YES**.

36. **NOT GIVEN**

the passage says this **could** trigger a cascade effect — a possibility, not a guarantee. Answer: **NOT GIVEN**.

37. **exercise**

the same principle could be applied to **encouraging physical exercise or quitting smoking**. Answer: **exercise**.

38. **cascade**

the authority could **trigger a cascade effect, influencing the entire group**. Answer: **cascade**.

39. **harmful**

the same mechanisms **can also amplify harmful ones**. Answer: **harmful**.

40. **structure**

understanding the precise structure of these networks is critical. Answer: **structure**.

Passage 22 — Life on Mars?

1. **NO**

contrary to photographic evidence, Mars may be more promising than it appears — the photos actually make it look LESS promising than it really is. Answer: **NO**.

2. **YES**

you would notice some remarkable parallels between the atmosphere on Earth then and Mars today. Answer: **YES**.

3. **YES**

Three terraforming methods have been suggested, with the first already under development, albeit for a different purpose. Answer: **YES**.

4. **NO**

enough oxygen on the planet for humans to breathe using only special apparatus similar to that used by mountain climbers — special apparatus would still be needed. Answer: **NO**.

5. **YES**

it will stretch human ingenuity to its limits. Answer: **YES**.

6. **D**

it does have some things in common with our own planet — rotation rate, axial tilt, elements for life. Answer: **D**.

7. **A**

Terraforming would initially create a greenhouse effect that would heat the planet. Answer: **A**.

8. **C**

the factories would produce oxygen by mimicking the natural process of plant photosynthesis. Answer: **C**.

9. **C**

the passage covers both WHY (population pressure, long-term survival) and HOW (three terraforming methods) — not just one, and it stresses difficulty, not ease. Answer: **C**.

10. **I**

firing large, ammonia-bearing asteroids at the planet. "Firing" = **I shoot**.

11. **A**

it would take each asteroid about ten years to reach its destination. "Reach" = **A cover**.

12. **C**

one asteroid slamming into Mars' surface. "Slamming into" = **C hit**.

13. **B**

the energy produced by one asteroid. "Produced" = **B create**.

14. **D**

would raise the temperature of the planet by three degrees Celsius. "Raise" = **D increase**.

Passage 23 — Life on Mars?

27. **D**

Para D: ALH84001, a meteorite that was blasted into space when a large body struck Mars... and which spiralled across the solar system to crash into the Antarctic — a rock from Mars found on Earth. Answer: **D**.

28. **G**

Para G: **experiments aimed at conducting... to detect the presence of a substance called carbon 13 in soil and rock samples on Mars.** Answer: G.

29. **E**

Para E: **Mars lost its magnetic field several billion years ago... Without that protection, Mars's gases and water boiled off. That is why they are so different today.** Answer: E.

30. **F**

Para F: **much of the ground in the southern hemisphere of Mars is mixed with large quantities of ice. And where there is ice, there is likely to be liquid water.** Answer: F.

31. **A**

Para A: **These canals... had been built by technologically advanced Martians to transport water.** Answer: A.

32. **H**

Para H: **If DNA is located, it would mean that life on Mars and on Earth may spring from the same source.** Answer: H.

33. **B**

Para A: Herschel **declared that 'its inhabitants enjoy a situation similar to our own'.** Answer: B.

34. **C**

Para B: **taken up by the British writer H.G. Wells in 1898 in his masterpiece The War of the Worlds.** Answer: C.

35. **D**

Para C: probes sent back pictures **not of a world covered in vegetation, as many scientists still expected, but of vistas of cratered, arid deserts.** Answer: D.

36. **B**

Para D: Friedmann **believes that these could only have been created by living organisms. Others, however, argue that the crystals could have been laid down by simple chemical processes.** Answer: B.

37. **YES**

Para A: Lowell said **'Martian folk are possessed of inventions of which we have not dreamed'.** Answer: YES.

38. **NO**

Para C: Viking probes found Mars "as dead as the Sahara Desert" and **Mars was declared officially dead – for a while** — "for a while" signals this conclusion was later overturned by the evidence in paras E–F. Answer: **NO**.

39. **YES**

Para E: **the Martian surface, although hostile and arid today, was once lush and welcoming... Mars's gases and water boiled off**. Answer: **YES**.

40. **NOT GIVEN**

Para H discusses what finding DNA would mean, but says nothing about Spanish scientists launching any space probes. Answer: **NOT GIVEN**.

Passage 24 — Literacy and Rural Communities in Australia

1. **YES**

little is known about the incidence of literacy in rural areas of Australia, but anecdotal evidence suggests. Answer: **YES**.

2. **NO**

it is important, however, to remind the reader of the considerable diversity of rural areas. Answer: **NO**.

3. **NOT GIVEN**

following the father into **the family business** is mentioned only as ONE possible cause among several for males leaving school — never a claim that most rural businesses are family-run. Answer: **NOT GIVEN**.

4. **NO**

women's employment options are limited generally — the limitation applies to women overall, not just those with low literacy. Answer: **NO**.

5. **YES**

shift work and relatively high wages make such work relatively attractive. Answer: **YES**.

6. **YES**

more children in rural areas than urban areas tend to leave school without completing a full secondary education. Answer: **YES**.

7. **D**

Para D: **in times of hardship in farming areas... inadequate literacy skills add to the normal stresses**. Answer: **D**.

8. **G**

Para G: **some students travelled long distances to another town in order to avoid having to reveal their 'problem'**. Answer: G.

9. **B**

Para B: **reading directions for (often dangerous) sprays, the mixing of animal feed, the mixing of fuel for machinery**. Answer: B.

10. **A**

Para A: **for young people, inadequate literacy skills can affect their transition from school to work; for older people lack of literacy skills can reduce their employment options**. Answer: A.

11. **B**

Para B: **peer-group pressure from other young males who have decided to work on their family farms**. Answer: B.

12. **"B"**

B: Paras A/C/E show **reduced employment options and few alternatives** tied to poor literacy. **D:** Para D shows **a rural economic down-turn adds to the normal stresses** caused by poor literacy. **E:** Para F shows **the low value placed on literacy** within some families. (A: tutoring is mentioned only as something people avoid, never rated for effectiveness; C: no discussion of families relocating because of literacy; F: men and women's situations are described separately, never directly compared; G: no discussion of future trends.) Answers: **B, D, E**.

13. **"B"**

B: Paras A/C/E show **reduced employment options and few alternatives** tied to poor literacy. **D:** Para D shows **a rural economic down-turn adds to the normal stresses** caused by poor literacy. **E:** Para F shows **the low value placed on literacy** within some families. (A: tutoring is mentioned only as something people avoid, never rated for effectiveness; C: no discussion of families relocating because of literacy; F: men and women's situations are described separately, never directly compared; G: no discussion of future trends.) Answers: **B, D, E**.

14. **"B"**

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Passage 25 — Living Dunes

27. **iv**

their insidious creep can have serious consequences if there is an oil installation or a railway line in their path. Answer: **iv**.

28. **viii**

over half of that is classified as 'linear' sand dunes. Answer: **viii**.

29. **vi**

Barchan dunes only tend to form where you have one-directional winds on the edge of sandy deserts near coastal areas. Answer: **vi**.

30. **vii**

The grain tumbles, then stops on the face until subsequent mini-avalanches bury it. Eventually, it reappears at the back of the dune, ready to repeat the process. Answer: **vii**.

31. **v**

Even when two dunes collide, they quickly take on their distinctive shapes again. Answer: **v**.

32. **ix**

Local legends attributed the sounds to dangerous spirits which were trying to trap unwary travellers. Douady is more pragmatic. Answer: **ix**.

33. **ii**

they were able to take samples of the singing sand back to France and replicate the sound at ENS. Answer: **ii**.

34. **C**

These form in a long curving wave, as a result of wind blowing strongly from several quarters, flipping them from side to side. Answer: **C**.

35. **B**

his answer hinges on the fact that dunes aren't solid, but granular. Answer: **B**.

36. **D**

the dune can regulate its own shape, and maintain it as it moves. Answer: **D**.

37. **barchan dunes**

37: some **barchans** actually **sing**. **38:** the grains are a uniform **shape**, well-rounded from years of striking each other. **39:** the variations in size affect the **tone**. **40:** made from a precise combination of **minerals** from surrounding rocks. Answers: **barchan dunes / shape / tone / minerals**.

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38. shape

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39. tone

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40. minerals

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Passage 26 — Memory Champions...or just good at tricks?

27. D

Ed Cooke has memorized a specific human being, verb, and object that he associates with every card in the deck. Answer: D.

28. G

During two years of experimentation, S.V. increased his ability to recall numbers from seven to over eighty. Answer: G.

29. A

34 contestants gathered at the World Memory Championships. Answer: A.

30. E

most researchers agree there is no proof that such a memory truly exists. Answer: E.

31. B

reportedly originating in 477 BC with the Greek poet Simonides of Ceos. Answer: B.

32. F

Shereshevski also experienced synesthesia, a rare sensory condition where the senses are interconnected, which significantly improved his memorization ability. Answer: F.

33. specific human being

33: Ed Cooke has memorized a **specific human being**, verb, and object. **34:** he does it **three cards** in one go. **35:** he places those images along a **specific familiar path** in his mind. **36:** Cooke takes a **mental walk** along his route. Answers: **specific human being / three cards / specific familiar path / mental walk.**

34. three cards

33: Ed Cooke has memorized a **specific human being**, verb, and object. **34:** he does it **three cards** in one go. **35:** he places those images along a **specific familiar path** in his mind. **36:** Cooke takes a **mental walk** along his route. Answers: **specific human being / three cards / specific familiar path / mental walk.**

35. specific familiar path

33: Ed Cooke has memorized a **specific human being**, verb, and object. **34:** he does it **three cards** in one go. **35:** he places those images along a **specific familiar path** in his mind. **36:** Cooke takes a **mental walk** along his route. Answers: **specific human being / three cards / specific familiar path / mental walk.**

36. mental walk

33: Ed Cooke has memorized a **specific human being**, verb, and object. **34:** he does it **three cards** in one go. **35:** he places those images along a **specific familiar path** in his mind. **36:** Cooke takes a **mental walk** along his route. Answers: **specific human being / three cards / specific familiar path / mental walk.**

37. B

these regions are known to be involved in visual memory and spatial navigation. Answer: **B.**

38. C

the loci method became widely embraced across ancient Greece, serving as a fundamental technique for orators and scholars. Answer: **C.**

39. E

naturals exhibit strong memory across many types of tasks and can retain those memories more easily. Answer: **E.**

40. D

their remarkable abilities were actually based on the use of specific techniques. Answer: **D.**

Passage 27 — Metropolis

1. C

this mass of dials, levers and gauges symbolically stands for all machines and all industry — the lack of definition makes the machines universal symbols, reinforcing the film's central idea. Answer: C.

2. D

power, control, and the continuance of the system from one 10-hour shift to the next, are all that counts... their aims are to sustain the momentum of their own growth. Answer: D.

3. A

it is enlightening, I think, to relate the film to the modern global economy... Frederesen is certainly still the most potent movie symbol of the handful of elusive corporate figureheads. Answer: A.

4. C

when the images are allowed to speak for themselves, the film is impeccable... The problem with Metropolis is its sentimental storyline — superb imagery let down by a weak storyline. Answer: C.

5. NOT GIVEN

in 1924 Lang's inspiration was a night-time view of the New York skyline from the deck of an ocean liner — a city skyline, not a specific industrialist. Answer: NOT GIVEN.

6. YES

when the images are allowed to speak for themselves, the film is impeccable... The problem with Metropolis is its sentimental storyline. Answer: YES.

7. NOT GIVEN

in 1933 he fled Germany... She stayed in Germany and continued to make films — no mention of divorce or its cause anywhere. Answer: NOT GIVEN.

8. E (graphic)

Lang's graphic depiction of the future. "Graphic" = E graphic.

9. A (machines)

the workers as slave-like extensions of the equipment they have to operate. "Extensions of the equipment" = A machines.

10. K (organisation)

multi-national corporations... their aims are to sustain the momentum of their own growth. "Corporations" = K organisation.

11. **I (power)**

whose sole satisfaction seems to lie in the exercise of power. "Exercise of power" = **I power**.

12. **C (functions)**

the function of the individual machines is never defined. "Function" = **C functions**.

13. **D (abstract)**

Lang's graphic depiction of the future is conceived in almost totally abstract terms. "Abstract terms" = **D abstract**.

Passage 28 — The Value of Research into Mite Harvestmen

1. **C**

the animals respond by turning motionless, making them impossible for even a trained eye to pick out. 'They look like grains of dirt'. Answer: **C**.

2. **D**

they carry a record of hundreds of millions of years of geological history, chronicling the journeys that continents have made around the Earth. Answer: **D**.

3. **B**

fossils of similar species existed on both sides of the Atlantic, but **the ocean was too big for the species to have traveled across it on their own**. Answer: **B**.

4. **A**

armadillos are in S. America and Africa because **their ancestors evolved when the continents were joined**; later, **when South America and North America connected... armadillos spread north, too**. Answer: **A**.

5. **D**

having colonized Hawaii, it branched off from that species, losing its ability to fly. Answer: **D**.

6. **A**

it is difficult to say much about where cockroaches evolved... because they can move quickly from continent to continent. Answer: **A**.

7. **NO**

species spread **for other reasons, such as by crossing great stretches of water** — the Hawaiian goose is exactly this alternative case, not continental drift. Answer: **NO**.

8. YES

harvestmen are not found on young islands like Hawaii, as these types of islands emerged long after the break-up of Pangea. Answer: **YES**.

9. NOT GIVEN

by calculating how quickly the DNA mutated, the scientists could estimate when lineages branched off — a method, not a comparison of rates between species. Answer: **NOT GIVEN**.

10. NOT GIVEN

Dr Boyer then began investigating the geological history of Florida and found recent research to explain the mystery — no mention of wide acceptance. Answer: **NOT GIVEN**.

11. H (ancestors)

their ancestors were among the first land animals. Answer: **H ancestors**.

12. D (DNA)

from which they extracted DNA. Variations in the genes helped the scientists build an evolutionary tree. Answer: **D DNA**.

13. A (branches)

by calculating how quickly the DNA mutated, the scientists could estimate when lineages branched off. Answer: **A branches**.

14. I (continents)

they then compared the harvestmen's evolution to the movements of the continents. Answer: **I continents**.

Passage 29 — Musical Maladies

1. B

i had high expectations of musicophilia... and i confess to feeling a little guilty reporting that my reactions to the book are mixed. Answer: **B**.

2. C

sacks himself is the best part of musicophilia. he richly documents his own life in the book and reveals highly personal experiences. Answer: **C**.

3. B

he wants to convey the insights gleaned from the enormous and rapidly growing body of work on the neural underpinnings of musical perception and imagery, and the complex and often bizarre disorders to which these are prone. Answer: **B**.

4. **A**

there are now more sensitive tests, but cicoria has declined to undergo them; he does not want to delve into the causes of his musicality. what a shame. Answer: **A**.

5. **YES**

i confess to feeling a little guilty reporting that my reactions to the book are mixed. Answer: **YES**.

6. **NOT GIVEN**

the sonata is only mentioned in connection with the cover photo of sacks listening to it; the passage never discusses it as a treatment. Answer: **NOT GIVEN**.

7. **NO**

he knows that he needs to take heed of the latter, but his heart lies with the former. Answer: **NO**.

8. **NOT GIVEN**

the passage only says music therapy is underappreciated, without explaining why or claiming this is hard to understand. Answer: **NOT GIVEN**.

9. **YES**

he tends to be rather uncritical in accepting scientific findings and theories. Answer: **YES**.

10. **NO**

he conveys no sense of urgency about the pursuit of new avenues in the diagnosis and treatment of music-brain disorders. Answer: **NO**.

11. **F**

the many specific dissociations among components of music comprehension, such as loss of the ability to perceive harmony but not melody, indicate that there is no music center in the brain. Answer: **F**.

12. **B**

a drug can alleviate a symptom in one patient and aggravate it in another, or can have both positive and negative effects in the same patient. Answer: **B**.

13. **A**

the patient with music-brain symptoms is reported to have "normal" eeg results. Answer: **A**.

14. **D**

he does call for both approaches, though — observation and new technology together. Answer: **D**.

Passage 30 — Neanderthal Technology

27. **v**

west European cave-dwellers vs the Russian steppe, where **open-air sites with some sort of constructed shelter were the only option**. Answer: **v**.

28. **iv**

Caves, moreover, **aid the survival of archaeological material and can preserve the records of remote millennia** — explaining why we can examine so much more from cave sites. Answer: **iv**.

29. **iii**

the limestone caves of the Périgord region made ideal homes for the Neanderthal people. Answer: **iii**.

30. **vi**

the region's abundant food chain meant **men never needed to travel far from home to find abundant supplies of meat** — a steady, vital source of sustenance for the group. Answer: **vi**.

31. **i**

From Moldova, for example, comes evidence that has been interpreted as the remains of wind-break structures, or even a large tent. Answer: **i**.

32. **E**

A charred piece of birch **from Krapina in Croatia, is thought to be the remains of a fire-making twirl stick**. Answer: **E**.

33. **D**

The cave of Baume-Bonne in the Basses-Alpes region of France... boasts ten square metres of cobbles brought up from the local river and laid down, as though to take care of a puddle area in the cave. Answer: **D**.

34. **C**

the ash patches at Lazaret **are edged with small marine mollusc shells**. Answer: **C**.

35. **B**

a ring, up to about 8 x 5m in size, of mainly mammoth bones enclosing a dense concentration of stone tools, animal bones and ash. Answer: **B**.

36. **C**

at Lazaret, **finds of wolf and fox foot bones, without the rest of the skeletons, inside these 'tents' have been thought to indicate the use of animal pelts as bed coverings**. Answer: **C**.

37. sub-arctic

37: survival in the sub-arctic conditions faced by the Neanderthals is inconceivable without control of fire. 38: this fire was only of a short duration. 39: Fire gave warmth, light, heat for cooking and defence against predatory animals. Answers: sub-arctic / short / animals.

38. short

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40. C

the whole passage surveys cave and open-air dwelling evidence, site selection, and fire use across many regions — a broad discussion of what is currently known about Neanderthal settlements, not an argument, migration account, or developmental timeline. Answer: C.

Passage 31 — Pacific Navigation and Voyaging

1. YES

had no human population until ancestors of today's islanders sailed from Southeast Asia. Answer: YES.

2. NO

the romantic vision of some early twentieth-century writers of fleets of heroic navigators came before Sharp; no considered assessment... was forthcoming until 1956 when... Sharp published his research — he was first with a considered assessment, not the first to write at all. Answer: NO.

3. YES

the settlement of the islands was not systematic, being more dependent on good fortune by drifting canoes. Answer: YES.

4. YES

Sharp's theory was widely challenged, but... it did spark renewed interest in the topic and precipitated valuable new research. Answer: YES.

5. **NOT GIVEN**

Doran **followed a quite different approach**, focused on canoe performance data — no mention of agreeing or disagreeing with Lewis. Answer: **NOT GIVEN**.

6. **A**

he accompanied traditional navigators **in their traditional canoes, on test voyages** — his own yacht was only used to make first contact. Answer: **A**.

7. **C**

navigators would have known which course to steer by memorizing stars, **this gave them fixed directions by which to steer their boats**. Answer: **C**.

8. **B**

Doran **employed the latest electronic instrumentation... using his instruments to record canoe speeds in different wind strengths**. Answer: **B**.

9. **D**

he concluded that paddles, or perhaps long oars, could indeed have propelled for long distances what were primarily sailing vessels. Answer: **D**.

10. **B**

Wall Garrard's unusual method... **clever adaptation of computer simulation techniques pioneered in other disciplines**. Answer: **B**.

11. **C**

none of the researchers tried to use their findings to prove one theory or another; experiments such as these cannot categorically confirm or negate a hypothesis. Answer: **C**.

12. **A**

the strength of this research lay in the range of methodologies employed. Answer: **A**.

13. **D**

it was this adaptability which was their greatest accomplishment. These navigators observed the conditions... and altered their techniques accordingly. Answer: **D**.

14. **E**

the Pacific peoples were able to view the ocean as an avenue, not a barrier, to communication before any other race on Earth. Answer: **E**.

Passage 32 — Playing Psychological Games

1. **x**

'Psychological games' is an approach to relationships that developed around the 1960s — games are presented as an analytical framework for interaction. Answer: **x**.

2. **i**

Games fall into a number of categories, ranging from the harmless to the destructive. Answer: **i**.

3. **vi**

Some people set out to manipulate others for their own reasons. But others may not realize that they are being manipulative. Answer: **vi**.

4. **ix**

A number of potentially damaging games have been identified — the Corner Game and It's Your Decision Game. Answer: **ix**.

5. **iv**

an unwillingness to communicate with them directly. Answer: **iv**.

6. **iii**

needs to win a game in order to have a sense of self-esteem — by harming someone else's self-confidence. Answer: **iii**.

7. **vii**

Some game players have so many psychological needs that fulfilling their desires overshadows everything else in a relationship. Answer: **vii**.

8. **NO**

insisting they are not the game player forces the other person to take all responsibility for the consequences of the decision — the opposite: they push the decision onto someone else. Answer: **NO**.

9. **YES**

They can make it impossible for an atmosphere of trust to be created — even well-intentioned games. Answer: **YES**.

10. **YES**

Protector (who is over-supportive or over-protective) — listed as a type of manipulator. Answer: **YES**.

11. **YES**

A young couple that accepts that the husband needs to defend his wife against criticism by his family, or that he will automatically make the decisions about minor repairs on her car — they fall into the expected roles. Answer: **YES**.

12. **B**

places them in a situation where anything they do is wrong. Answer: **B**.

13. **A**

the whole passage **identifies what seem to be fixed scripts in a seemingly spontaneous conversation** — treating the theory as a genuinely useful analytical framework, not dismissing it. Answer: **A**.

Passage 33 — Rebranding Art Museums

27. **C**

what is the writer doing in paragraph 1 ≈ para A poses the question: "**Do any of these newly developed spaces indicate a fundamental shift in the ways we engage with art**" — this is framing/considering design's influence on engagement, before the NGV comparison begins in later paragraphs. Answer: **C**.

28. **B**

criticises the old NGV ≈ para D: the windowless wall and moat "**creates a rather unwelcoming expression of the museum as a temple of culture set apart from the everyday world**". Answer: **B**.

29. **C**

Para E, after describing the cloakroom, info office, ticket booths, cafe and shop: "**All of this is certainly impressive, but where has the art gone?**" Answer: **C**.

30. **B**

Para G: entrance is free, so "**the NGV needs something to keep the paying public coming back. The answer lies in the international 'blockbuster' exhibition**". Answer: **B**.

31. **NO**

Para B: the Great Hall "represented an unprecedented emphasis on the ancillary roles of **corporate functions and public performances in the contemporary museum**" — Grounds clearly did recognise this. Answer: **NO**.

32. **NO**

Para E: "**his design reflects what the new, improved, twenty-first century global museum feels it should be emphasising**" — it follows trends, not rejects them. Answer: **NO**.

33. **NOT GIVEN**

Bellini's subsequent museum projects — the passage discusses only his NGV work, never any other project of his. Answer: **NOT GIVEN**.

34. **NOT GIVEN**

monitors first-time visitors to permanent collections — the only visitor figure given is for the temporary "Impressionists" blockbuster (380,000); no monitoring of the permanent collection is mentioned. Answer: **NOT GIVEN**.

35. **YES**

Para H: **"the effect of too much innovation in museums can be disastrous"**. Answer: **YES**.

36. **F**

large space available ≈ para C: museums place increasing emphasis **"on corporate sponsorship, and on the need to generate additional funds via room hire"**. Answer: **F**.

37. **E**

children allowed to move freely ≈ para C: as an unthreatening introduction to visiting art museums, **"its impact on attendance cannot be overestimated"**. Answer: **E**.

38. **C**

too much emphasis on the museum building itself ≈ para F: new buildings can supplant the artworks inside them to **"become the major attractions themselves"**. Answer: **C**.

39. **A**

over-reliance on blockbuster exhibitions ≈ para G: **"they clearly narrow the options for other worthy but less glamorous projects"**, giving the programme an "off the rack" feeling. Answer: **A**.

40. **D**

Para H: the NGV **"needs to be able to juggle numerous, often conflicting priorities"**. Answer: **D**.

Passage 34 — Robert Louis Stevenson

27. **'D – elevated Stevenson's role as a writer'**

KF and CP **"chose not to criticise aspects of Stevenson's personal life. Instead, they focused on his writing, and gave high praise to his writing style and skill"** — this raises his role as a writer above the controversy. Answer: **D**.

28. **'D – lack of ethical nature'**

Para 2 ends by asserting that **"despite his skill, his literary works still lack moral value"** — the paragraph's throughline is that ethical/moral shortcoming is Stevenson's persistent flaw. Answer: **D**.

29. **'B – cannot be used to show moral values'**

Para 3: adventure stories are exciting but **"ultimately can't teach moral principles"** — plot points are one-dimensional rather than morally meaningful. Answer: **B**.

30. **'A – describe life in Scotland'**

Para 4: unlike other British authors, Stevenson's works **"captured the unique day to day life of average Scottish people"**. Answer: **A**.

31. **'C – gained more attention than his works'**

Para 5: **"interest in Stevenson's life had taken the place of interest in his works"** — his eccentric lifestyle overshadowed his writing in public attention. Answer: **C**.

32. **'TRUE'**

Para 3: **"Stevenson's works have deeply influenced Oscar Wilde"** (admiration), yet **"Wilde often joked that Stevenson would have written better works if he wasn't born in Scotland"** (could have written greater works). Answer: **TRUE**.

33. **'NOT GIVEN'**

encouraged... to start writing — para 3 only says **"Stevenson's works have deeply influenced Oscar Wilde"**; influence on an existing writer is not the same as encouraging him to begin writing. Answer: **NOT GIVEN**.

34. **'FALSE'**

Reversed from para 3: **"Galsworthy who claimed that Stevenson is a greater writer than Thomas Hardy"** — the opposite ranking. Answer: **FALSE**.

35. **'FALSE'**

Contradicts para 5: **"his writing was not the only source of attention for critics. Stevenson's personal life often attracted a lot of attention"**. Answer: **FALSE**.

36. **'F – technical control'**

Scott is more proficient in ____ ≈ para 4: critics **"exalt Sir Walter Scott as a literary genius because of his technical ability"**. Answer: **F – technical control**.

37. **'A – natural ability'**

Stevenson has better ____ ≈ para 4: Stevenson deserves recognition for his **"natural ability to capture stories and characters in words"**. Answer: **A – natural ability**.

38. **'H – depth'**

Scott's books illustrate ____, especially...tragedy ≈ para 4: Scott's works were **"taken more seriously as literature for their depth due to their tragic themes"**. Answer: **H – depth**.

39. **'G – storytelling'**

readers prefer Stevenson's ____ ≈ para 4: fans praise his "unique style of story-telling". Answer: G – storytelling.

40. **'I – human nature'**

Stevenson's understanding of ____ ≈ para 4: praised for his "capture of human nature". Answer: I – human nature.

Passage 35 — Science and Filmmaking

27. **D**

they were part of a growing cohort of computer scientists that has become fundamental to moviemaking. Answer: D.

28. **A**

the effect is less convincing when seen close up — i.e. better from further away. Answer: A.

29. **A**

the scientists, with admirable tenacity, continued their search for perfection. Answer: A.

30. **B**

Like classical radiology it uses X-rays. Answer: B.

31. **B**

a small group of actors can be made to look like hundreds of people gathered together. Answer: B.

32. **D**

the technology might have an application in online retailing. Answer: D.

33. **YES**

In 2004 a notable event took place, when researchers received an Oscar. Answer: YES.

34. **NOT GIVEN**

the passage explains CGI can make actors look older or younger, but never discusses whether ageing actors particularly favour it. Answer: NOT GIVEN.

35. **NO**

Marschner and his colleagues received the Oscar for their work in successfully producing a CGI model that simulates translucency — the (single) award was for skin, not the garment work. Answer: NO.

36. **NO**

for the moment the movement of a virtual cloak or the lifting of a computerized hat should look far more realistic — already usable now, not a long wait. Answer: **NO**.

37. **A**

this is when **light penetrates skin and then scatters below the skin's surface before re-emerging**. Answer: **A light**.

38. **E**

the mathematics for the model goes back many decades to the time when it was used in **astrophysics**. Answer: **E astrophysics**.

39. **D**

all the major special-effects studios had incorporated it into their digital rendering systems. Answer: **D use**.

40. **G**

Cinematography will benefit from this. Answer: **G improvement**.

Passage 36 — Scientific Research Reveals Hidden Benefits of Regular Exercise

1. **YES**

there is evidence to suggest that this lack of exercise is not only having a negative physiological effect on them, but is also adversely affecting their academic performance at school. Answer: **YES**.

2. **NO**

He also supports the long-held conviction that vigorous physical activity is much better than moderate activity — "long-held" contradicts "only recently". Answer: **NO**.

3. **NO**

Schools and parents should devise ways of increasing physical activity in and out of school time — a clear, explicit statement. Answer: **NO**.

4. **YES**

For those who are sceptical about this, and no doubt there are many. Answer: **YES**.

5. **NOT GIVEN**

the passage reports the Swedish IQ finding but never says researchers were "surprised" by it. Answer: **NOT GIVEN**.

6. **B**

This increases levels of chemicals like endorphin in the brain, which decrease stress and improve mood. "Stress" = **B anxiety**.

7. **I**

there is also evidence that certain parts of the brain become larger. "Evidence" = **I proof**.

8. **E**

children who are fit also tend to be better at multi-tasking and performing difficult mental tasks than their unfit friends. "Difficult mental tasks" = **E exercises**.

9. **C**

their findings could have important implications for improving children's performance at school. "Findings" = **C discoveries**.

10. **D**

it could help people **combat memory loss and retain problem-solving skills in old age**. "Old age" = **D elderly**.

11. **C**

Aerobic exercise is best for this [improving cognition]. Answer: **C**.

12. **A**

neuroimaging work **provided visual evidence to show that brain networks and structures actually change with exercise**. Answer: **A**.

13. **D**

Dr Mattson found that **restricting people's diets to just 500 calories every other day** increased protective proteins — "every other day" = certain days, not every day (rules out B). Answer: **D**.

Passage 37 — Sea Change for Salinity

1. **E**

Para E: **the waters of Australia's Murray River will regularly exceed the World Health Organisation's salt limits for drinking water**. Answer: **E**.

2. **H**

Para H: **individually, these technologies only gave clues to what was going on underground. Together they provide a far more revealing picture**. Answer: **H**.

3. **B**

Para B: **the past two years have seen a major development in its precision and powers of detection.** Answer: B.

4. **C**

Para C: **Howell and his fellow farmers had decided it was time for action and established a government-funded 'Landcare' group.** Answer: C.

5. **E**

Para E: **defeating this assault may take centuries, not decades.** Answer: E.

6. **H**

Para H: **this in turn will give the salt warriors time to model various ways of containing or curbing the menace, see what works best and then try it out on the ground.** Answer: H.

7. **B**

Para B: **the hard work now underway of planning and tree-planting on the surface may be rendered ineffective.** Answer: B.

8. **B**

electromagnetic surveys measure the electrical conductivity of soil to reveal the distribution of salt and the nature and variability of the regolith – the weathered rock and sediment that may lie above the bedrock. Answer: B.

9. **A**

these factors influence the mobility of salt through the soil profile and help determine where to plant particular crop species. Answer: A.

10. **C**

enables scientists to take 'slices' of the landscape at depths of five metres, ten metres, fifteen metres and so on, to determine where salt may be stored at depth. Answer: C.

11. **C**

a suite of high technologies used by geologists to see underground and prospect for gold and minerals is now being used to pinpoint the presence of salt. Answer: C.

12. **B**

unless this process is clearly understood... salt can still sneak past and erupt, following one of the ancient river channels formed millions of years ago. Answer: B.

13. C

how do you introduce the land-use changes that are needed when people still need to make a living. Answer: C.

14. A

salt is described throughout as **stealthily killing... salt can still sneak past and erupt... mount ambushes, emerging somewhere new, sometimes unexpected and unexplained.** Answer: A.

Passage 38 — Seeing the Colour of Sounds, Hearing the Colour of Numbers

27. C

Baron-Cohen's 1995 scan found that **areas of the brain lit up that are normally only active in response to vision and colour.** Answer: C (Simon Baron-Cohen).

28. C

gave the same answers in 92 per cent of instances when tested again a week later. Answer: C (Simon Baron-Cohen).

29. B

Locke **wrote about a man who experienced bright red as the sound of the trumpet.** Answer: B (John Locke).

30. A

Smilek: **Synesthesia, because it's abnormal, can give us new insights into normal perception.** Answer: A (Daniel Smilek).

31. F

A brain scanning experiment by Baron-Cohen in 1995 found that when synesthetes were listening to words, areas of the brain lit up. Answer: F.

32. I

the novelist Nabokov, who perceived the English 'a' as dark brown and the French 'e' as black. Answer: I.

33. A

have synesthesia — meaning literally 'joined sensation'. Answer: A.

34. F

pruning theory: **connections are pruned, so our senses become separated and the synesthetic mechanism is no longer intact;** plus the drug-repression counter-theory. Answer: F.

35. **B**

Others learn to keep it a secret for fear of people laughing at them. Answer: **B**.

36. **TRUE**

Kandinsky **believed synesthetes were like good, much-played violins, which vibrate in all their parts and fibres** — an admiring comparison. Answer: **TRUE**.

37. **TRUE**

Early claims that the multi-sensory experiences of synesthesia were linked with the hallucinations of mental illness have long been disproved. Answer: **TRUE**.

38. **FALSE**

Women are between two and eight times more likely than men to have the condition. Answer: **FALSE**.

39. **NOT GIVEN**

The passage never compares synesthetes' logical abilities to those of other people. Answer: **NOT GIVEN**.

40. **TRUE**

we may all have synesthesia at birth, when many parts of the brain are linked. Answer: **TRUE**.

Passage 39 — Shoemaker-Levy 9

1. **vii**

2. **iv**

3. **iii**

4. **ix**

5. **ii**

6. **B**

Para B: **serendipity played a large part in the discovery... they had a box of old film to hand which had been partially exposed by accident.** Answer: **B**.

7. **H**

Para H: **caused a massive plume or fireball up to 2000 km in diameter to rise... this material manifested itself as a dark scar surrounding the impact site.** Answer: **H**.

8. **F**

Para F: **Each fragment was assigned an identity letter A-W.** Answer: **F**.

9. **C**

Para C: **Other, more powerful, telescopes revealed that the comet was, in fact, composed of 21 cometary fragments, strung out in a line.** Answer: C.

10. **iron**

Jupiter is composed of a relatively small core of iron and silicates surrounded by hydrogen. Answer: **iron.**

11. **silicates**

core of iron and silicates. Answer: **silicates.**

12. **molecular form**

further from the centre, the pressure is lower and the hydrogen is in its normal molecular form. Answer: **molecular form.**

13. **cloud tops**

the Jovian cloud tops visible from Earth consist primarily of methane and ammonia. Answer: **cloud tops.**

14. **elements and compounds**

relatively small amounts of other elements and compounds which are thought to be responsible for the colours seen in the atmosphere. Answer: **elements and compounds.**

Passage 40 — Some views on the use of headphones

27. **NOT GIVEN**

easily persuaded by surveys — para 1 only says people "**report with confidence**" that music helps them; it never explains why they believe this or that surveys are what convinced them. Answer: **NOT GIVEN.**

28. **YES**

different studies share the same conclusions — para 2 cites multiple separate findings (the Taiwanese study, other research on lyrics), all converging on the same point: "**as silence has the best overall performance, it would still be advisable that people work in silence**". Answer: **YES.**

29. **NOT GIVEN**

doctors recommend — para 3 says music "**may... reduce blood pressure, heart rate and anxiety**", but this is only a possible effect, not a recommendation, and no doctors are mentioned anywhere in the passage. Answer: **NOT GIVEN.**

30. **NO**

respected government researcher contradicts para 4: Baldwin was an "**eccentric inventor and repairman**" working alone "in his kitchen" — not a researcher, and the Navy only adopted his invention afterward. Answer: **NO**.

31. **YES**

comparable to the invention of writing ≈ para 5: "**headphones have done what writing and literacy did for language**". Answer: **YES**.

32. **A**

a service economy ≈ para 3: office jobs "**require higher levels of concentration, reflection and creativity**" — i.e. mentally demanding. Answer: **A**.

33. **C**

The Sumerian notation and 44,000-year-old bone flute are presented as evidence that "**music evolved as a social glue for the species**" (para 5) — i.e. that music has mattered to humans throughout history. Answer: **C**.

34. **D**

Para 5: headphones "**made music private**", reversing music's earlier social/bonding purpose described in the same paragraph. Answer: **D**.

35. **B**

Para 6: "**wealth can buy... personal independence**" — independence is explicitly framed as something money can purchase. Answer: **B**.

36. **A**

The quote — "**when our minds are at ease, we're more likely to direct... attention inward**" — directly illustrates the writer's own point in that paragraph about the reflective benefits headphones provide. Answer: **A**.

37. **B**

changed the ____ wearers have with public spaces ≈ para 7: personal music devices "**change how we relate to public spaces**". Answer: **B — relationship**.

38. **E**

less keen on living in the ____ ≈ para 7: people are "**moving from the edges of cities to live in urban centres**" — "edges of cities" = suburbs. Answer: **E — suburbs**.

39. **G**

form ____ that isolate people ≈ para 7: headphones "**create shields for wearers, separating them from other people**". Answer: **G — barriers**.

40. **A**

treated with ____ that other people do not receive ≈ para 8: headphones are like a "'Do not disturb' sign" that makes people "respect their privacy". Answer: **A — courtesy**.

Passage 41 — Strictly English

1. **NOT GIVEN**

these are reasonably rare — the writer says the mistakes are RARE (infrequent), never says whether they are minor or serious in nature. Answer: **NOT GIVEN**.

2. **YES**

being a matter for debate, is less straightforward — explaining WHY correct English matters is harder than explaining WHAT it is. Answer: **YES**.

3. **NO**

the logic that drove it and that was common to all the other languages I knew — the writer says grammar's logic is COMMON (shared) across languages, not different. Answer: **NO**.

4. **YES**

was the ambiguity in a certain order of words deliberate or accidental — word order is discussed alongside word choice as equally rooted in communication. Answer: **YES**.

5. **D**

both will continue to evolve — English is always changing, never fixed. "Fixed" = **D permanent**.

6. **E**

learned men, many of whom contributed to the OED. "Learned" = **E knowledgeable**.

7. **B**

a standard that people recognize and are comfortable with — a widely-recognized, GENERAL standard. Answer: **B general**.

8. **H**

help people towards a capable grasp of the English tongue — the book aims to be USEFUL. Answer: **H useful**.

9. **C**

English grammar shouldn't be a matter for debate — not a CONTROVERSIAL subject. Answer: **C controversial**.

10. **C**

the academics feel they have to write in a stilted way **in order to be taken seriously by their peers** — to impress fellow group members. Answer: **C**.

11. **A**

they can rediscover the knack of writing reasonably plain English — for outside publications. Answer: **A**.

12. **D**

considered ugly and barbaric — repetitious, long-winded, abstract and abstruse. Answer: **D**.

13. **C**

unless what is considered acceptable within their discipline's changes — only changes if forced by discipline norms. Answer: **C**.

14. **B**

having a harmful effect on the language and causing it to be corrupted. Answer: **B**.

Passage 42 — Sun, Sand and Seawater

1. **YES**

sometimes ideas just pop up out of the blue — the idea for the seawater greenhouse came to Paton randomly, on a bus, not through deliberate research. Answer: **YES**.

2. **NOT GIVEN**

the passage only says **the bus was full of hot, wet people** and the windows steamed up — nothing is said about ventilation specifically. Answer: **NOT GIVEN**.

3. **YES**

when I woke, the thing was soaking wet, I had to wring it out. Answer: **YES**.

4. **NOT GIVEN**

no fan is mentioned anywhere in the passage. Answer: **NOT GIVEN**.

5. **NO**

after talking with Davies, Paton **started rigging up his own equipment** back home in London — the Persian Gulf greenhouse only appeared **today, a decade on**, not immediately. Answer: **NO**.

6. **hot dry air**

the front of the greenhouse faces into the prevailing wind so that hot dry air blows in through a front wall. Answer: **hot dry air**.

7. **moist**

the wall is made of perforated cardboard kept moist by a constant trickle of seawater. Answer: **moist**.

8. **heat**

heat from the infrared radiation is trapped in the space between the layers. Answer: **heat**.

9. **condenser**

the condenser is a metal surface kept cool by still more seawater. Answer: **condenser**.

10. **pure distilled water**

drops of pure distilled water form on the condenser and flow into a tank for irrigating the crops. Answer: **pure distilled water**.

11. **fans**

on windless days, fans ensure a constant flow of air through the greenhouse. Answer: **fans**.

12. **solar panels**

we could make it entirely independent of the grid, powered from a few solar panels. Answer: **solar panels**.

13. **construction costs**

critics point out that construction costs of around \$4 a square foot are quite high. Answer: **construction costs**.

14. **environmentally-friendly**

it really suggests an environmentally-friendly way of providing air conditioning on a scale large enough to cool large greenhouses. Answer: **environmentally-friendly**.

Passage 43 — Telepathy

27. **E**

Sceptics and advocates alike do concur on one issue, however: that the most impressive evidence so far has come from the so-called 'ganzfeld' experiments. Answer: **E**.

28. **B**

such signals might be more easily detected by those experiencing meditation-like tranquillity in a relaxing 'whole field' of light, sound and warmth. Answer: **B**.

29. **A**

the problem stems at least in part from the **lack of any plausible mechanism for telepathy**. Answer: A.

30. **F**

researchers trying to identify people who are particularly successful in autoganzfeld trials – **creative and artistic people do much better than average**. Answer: F.

31. **sender**

31–33: a person acting as a '**sender**' would attempt to beam the image; test involved **identification of a picture chosen from a random selection of four**; the '**receiver**' relaxing in the sealed room. Answers: **sender / picture / receiver**.

34–35: positive results ranged from '**sensory leakage**' – **where clues about the pictures accidentally reach the receiver** – to outright fraud. Answers: **sensory leakage / fraud**.

32. **picture / image**

31–33: a person acting as a '**sender**' would attempt to beam the image; test involved **identification of a picture chosen from a random selection of four**; the '**receiver**' relaxing in the sealed room. Answers: **sender / picture / receiver**.

34–35: positive results ranged from '**sensory leakage**' – **where clues about the pictures accidentally reach the receiver** – to outright fraud. Answers: **sensory leakage / fraud**.

33. **receiver**

31–33: a person acting as a '**sender**' would attempt to beam the image; test involved **identification of a picture chosen from a random selection of four**; the '**receiver**' relaxing in the sealed room. Answers: **sender / picture / receiver**.

34–35: positive results ranged from '**sensory leakage**' – **where clues about the pictures accidentally reach the receiver** – to outright fraud. Answers: **sensory leakage / fraud**.

34. **sensory leakage**

31–33: a person acting as a '**sender**' would attempt to beam the image; test involved **identification of a picture chosen from a random selection of four**; the '**receiver**' relaxing in the sealed room. Answers: **sender / picture / receiver**.

34–35: positive results ranged from '**sensory leakage**' – **where clues about the pictures accidentally reach the receiver** – to outright fraud. Answers: **sensory leakage / fraud**.

35. **fraud / outright fraud**

31–33: a person acting as a '**sender**' would attempt to beam the image; test involved **identification of a picture chosen from a random selection of four**; the '**receiver**' relaxing in the sealed room. Answers: **sender / picture / receiver**.

34–35: positive results ranged from '**sensory leakage**' – **where clues about the pictures accidentally reach the receiver** – to outright fraud. Answers: **sensory leakage / fraud**.

36. computers

36–37: autoganzfeld tests **used computers to perform many of the key tasks such as the random selection of images; by minimising human involvement, the idea was to minimise the risk of flawed results.** Answers: **computers / human involvement.**

38: results were **studied by Honorton in a 'meta-analysis'.** Answer: **meta-analysis.**

39–40: some parapsychologists remain disturbed by the **lack of consistency between individual ganzfeld studies** — a typical study involving around 40 people means **the group is just not big enough.** Answers: **lack of consistency / big enough.**

37. human involvement

36–37: autoganzfeld tests **used computers to perform many of the key tasks such as the random selection of images; by minimising human involvement, the idea was to minimise the risk of flawed results.** Answers: **computers / human involvement.**

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39–40: some parapsychologists remain disturbed by the **lack of consistency between individual ganzfeld studies** — a typical study involving around 40 people means **the group is just not big enough.** Answers: **lack of consistency / big enough.**

38. meta-analysis

36–37: autoganzfeld tests **used computers to perform many of the key tasks such as the random selection of images; by minimising human involvement, the idea was to minimise the risk of flawed results.** Answers: **computers / human involvement.**

38: results were **studied by Honorton in a 'meta-analysis'.** Answer: **meta-analysis.**

39–40: some parapsychologists remain disturbed by the **lack of consistency between individual ganzfeld studies** — a typical study involving around 40 people means **the group is just not big enough.** Answers: **lack of consistency / big enough.**

39. lack of consistency

36–37: autoganzfeld tests **used computers to perform many of the key tasks such as the random selection of images; by minimising human involvement, the idea was to minimise the risk of flawed results.** Answers: **computers / human involvement.**

38: results were **studied by Honorton in a 'meta-analysis'.** Answer: **meta-analysis.**

39–40: some parapsychologists remain disturbed by the **lack of consistency between individual ganzfeld studies** — a typical study involving around 40 people means **the group is just not big enough.** Answers: **lack of consistency / big enough.**

40. big enough / large enough

36–37: autoganzfeld tests **used computers to perform many of the key tasks such as the random selection of images; by minimising human involvement, the idea was to minimise the risk of flawed results.** Answers: **computers / human involvement.**

38: results were **studied by Honorton in a 'meta-analysis'.** Answer: **meta-analysis.**

39–40: some parapsychologists remain disturbed by the **lack of consistency between individual ganzfeld studies** — a typical study involving around 40 people means **the group is just not big enough.** Answers: **lack of consistency / big enough.**

Passage 44 — Termite Mounds

27. **iii**

the writer moves from termites' destructive reputation to the fact that **these pests may serve a useful purpose for us after all**. Answer: **iii**.

28. **vii**

termite mounds are impressive for their size alone — three to eight metres high. Answer: **vii**.

29. **iv**

this complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound — ventilation, temperature and moisture control. Answer: **iv**.

30. **ix**

part of their digestive process is carried out by a fungus, which they farm inside the mound — the termites feed the fungus, and the fungus feeds them back. Answer: **ix**.

31. **ii**

we may ultimately need to redefine our understanding of what constitutes a 'living' organism. Answer: **ii**.

32. **i**

the research team then carves the plaster of paris into half-millimetre-thick slices, and photographs them sequentially. Answer: **i**.

33. **vi**

the findings could be invaluable in informing future architectural design. Answer: **vi**.

34. **tunnels**

34: **this complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound**. 35: **ensuring them a continuous provision of fresh, breathable air**. 36: **the mound also automatically regulates moisture in the air, by means of both its underground 'cellar'**. 37: **evaporation from the top of the mound**. Answers: **tunnels / air / moisture / evaporation**.

35. **air**

34: **this complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound**. 35: **ensuring them a continuous provision of fresh, breathable air**. 36: **the mound also automatically regulates moisture in the air, by means of both its underground 'cellar'**. 37: **evaporation from the top of the mound**. Answers: **tunnels / air / moisture / evaporation**.

36. **moisture**

34: this complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound. 35: ensuring them a continuous provision of fresh, breathable air. 36: the mound also automatically regulates moisture in the air, by means of both its underground 'cellar'. 37: evaporation from the top of the mound. Answers: tunnels / air / moisture / evaporation.

37. **evaporation**

34: this complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound. 35: ensuring them a continuous provision of fresh, breathable air. 36: the mound also automatically regulates moisture in the air, by means of both its underground 'cellar'. 37: evaporation from the top of the mound. Answers: tunnels / air / moisture / evaporation.

38. **YES**

although the termites must generate waste, none ever leaves the structure, indicating that there is also some kind of internal waste-recycling system. Answer: YES.

39. **NO**

the team **carves the plaster of paris into half-millimetre-thick slices, and photographs them sequentially** — many photographs of many slices, not one photograph of the whole mound. Answer: NO.

40. **YES**

termite-inspired **buildings that are self-sufficient, environmentally, and cheap to run** would ease the burden described in "there will not be enough fuel to drive air conditioners around the world". Answer: YES.

Passage 45 — The Algebra Debate

1. **C**

competence in the stem fields... will keep the united states economically competitive in the ever-developing and accelerating world economy. Answer: C.

2. **B**

most errors occurred during the processing stage (57.8%), followed by comprehension error (21.9%), encoding error (15.6%), and verification error (4.7%). Answer: B.

3. **C**

algebra's value may not be as readily apparent as arithmetic's. Answer: C.

4. **B**

real-world mathematics, namely arithmetic, was the main focus of school mathematics in the former countries [Nordic/English-speaking]. Answer: **B**.

5. **D**

these questions are not easily answered, but are incredibly important and deserve more attention from both educators and researchers. Answer: **D**.

6. **B (source)**

the root of the problem must be identified. "Root" = **B source**.

7. **G (concepts)**

most people are able to grasp the major tenets. "Tenets" = **G concepts**.

8. **A (objects)**

numbers are tied to concrete things that surround us. "Things" = **A objects**.

9. **E (difficulties)**

past research relating to the teaching and learning of algebra was mostly concentrated on the issues of secondary school students. "Issues" = **E difficulties**.

10. **D (divergence)**

errors in school-level algebra stem from a fundamental difference in the approach to algebraic and arithmetic problem solving. "Difference" = **D divergence**.

11. **YES**

real-world mathematics, namely arithmetic, was the main focus of school mathematics in the former countries, and there has been a shift away from the more formal aspects of mathematics, such as algebra. Answer: **YES**.

12. **NOT GIVEN**

the passage only says these countries **place far less importance on the application of algebraic problem-solving** — nothing about student performance or ability being compared. Answer: **NOT GIVEN**.

13. **NO**

if teachers feel that they are not competent in algebra, it makes sense that they do not make a point of teaching it, or at least put off including it in their syllabi — expertise directly determines emphasis. Answer: **NO**.

14. **YES**

a central question in determining the importance of algebra is how to find a balance between what students need in their everyday lives and their requirements for this knowledge in pursuing higher education. Answer: YES.

Passage 46 — The Art of Deception

27. **C**

was addressing a group of young psychiatrists in training when he was asked a question — people in training being addressed by an expert are students. Answer: C.

28. **B**

he had already filmed a series of 12-minute interviews with psychiatric patients — re-watching one in slow motion is how he found the first micro-expression, showing where his theory started. Answer: B.

29. **A**

the ways in which we express rage, disgust, contempt, fear, surprise, happiness and sadness are universal — the same facial muscles fire regardless of language and culture. Answer: A.

30. **B**

most people will fail to spot these fleeting signals of inner torment — only 50 of 15,000 untrained people ('naturals') could do it. Answer: B.

31. **B**

he has held workshops for defense and prosecution lawyers, health professionals, even jealous spouses — plus the FBI, CIA and countless more agencies. Answer: B.

32. **C**

the show would exaggerate the effectiveness of his techniques and create the quite inaccurate impression among audiences. Answer: C (false beliefs).

33. **E**

he was concerned about unfair convictions, that one day someone not properly trained in his techniques might wrongly convict someone. Answer: E (justice).

34. **H**

that a commercial tv drama has been based on the work of just one scientist. Answer: H (research).

35. **D**

manifestly serious and well-intentioned reasons for making the programme. Answer: **D (motives)**.

36. **F**

per cent of the show is based on fact — and that's good enough for him. Answer: **F (accuracy)**.

37. **NOT GIVEN**

The passage only says Ekman **professes to have been a terrible liar ever since he was a small boy** — he was bad AT lying; nothing about regret. Answer: **NOT GIVEN**.

38. **NO**

the ability to detect a lie and the ability to lie successfully are completely unrelated. Answer: **NO**.

39. **NOT GIVEN**

Poker players are only mentioned as an example of people who aren't 'true' liars — nothing about Ekman training them. Answer: **NOT GIVEN**.

40. **YES**

he prefers to focus on the kinds of lies where the liar would be in grave trouble if they were found out, and the target would feel properly aggrieved. Answer: **YES**.

Passage 47 — The Benefits of Being Bilingual

1. eye movements / eye movement

1: the most compelling evidence comes from studying **eye movements**. **2:** this phenomenon is called 'language co-activation'. **3:** in the classic **Stroop Task**, people name the colour of a word's font. **4:** bilingual people often perform better on tasks that require **conflict management**. **5:** reflecting better **cognitive control** when having to make rapid changes of strategy. Answers: **eye movements / language co-activation / Stroop Task / conflict management / cognitive control**.

2. language co-activation / co-activation

1: the most compelling evidence comes from studying **eye movements**. **2:** this phenomenon is called 'language co-activation'. **3:** in the classic **Stroop Task**, people name the colour of a word's font. **4:** bilingual people often perform better on tasks that require **conflict management**. **5:** reflecting better **cognitive control** when having to make rapid changes of strategy. Answers: **eye movements / language co-activation / Stroop Task / conflict management / cognitive control**.

3. Stroop Task / the Stroop Task

1: the most compelling evidence comes from studying **eye movements**. **2:** this phenomenon is called '**language co-activation**'. **3:** in the classic **Stroop Task**, people name the colour of a word's font. **4:** bilingual people often perform better on tasks that require **conflict management**. **5:** reflecting better **cognitive control** when having to make rapid changes of strategy. Answers: **eye movements / language co-activation / Stroop Task / conflict management / cognitive control**.

4. conflict management

1: the most compelling evidence comes from studying **eye movements**. **2:** this phenomenon is called '**language co-activation**'. **3:** in the classic **Stroop Task**, people name the colour of a word's font. **4:** bilingual people often perform better on tasks that require **conflict management**. **5:** reflecting better **cognitive control** when having to make rapid changes of strategy. Answers: **eye movements / language co-activation / Stroop Task / conflict management / cognitive control**.

5. cognitive control

1: the most compelling evidence comes from studying **eye movements**. **2:** this phenomenon is called '**language co-activation**'. **3:** in the classic **Stroop Task**, people name the colour of a word's font. **4:** bilingual people often perform better on tasks that require **conflict management**. **5:** reflecting better **cognitive control** when having to make rapid changes of strategy. Answers: **eye movements / language co-activation / Stroop Task / conflict management / cognitive control**.

6. YES

such children were considered to be at a disadvantage... thereby identifying several clear benefits of being bilingual. Answer: **YES**.

7. NOT GIVEN

the passage describes how **the brain's language system begins to guess what that word might be** for both groups, but never compares guessing **ACCURACY** between bilinguals and monolinguals. Answer: **NOT GIVEN**.

8. NO

knowing more than one language can cause speakers to name pictures more slowly — the opposite. Answer: **NO**.

9. NO

without any intervening background noise, they show highly similar brain stem responses — only better WITH background noise, not in all situations. Answer: **NO**.

10. NOT GIVEN

the passage compares symptom-onset timing and disease severity among affected patients, but never states the overall **NUMBER** or rate of people who develop the disease. Answer: **NOT GIVEN**.

11. **D**

Para D: **bilingual listeners' neural response is considerably larger** when hearing speech sounds in background noise. Answer: **D**.

12. **G**

Para G: **seven-month-old babies... only the bilingual babies were able to successfully learn the new rule**. Answer: **G**.

13. **B**

Para B: **the sounds arrive in sequential order... the brain's language system begins to guess what that word might be**. Answer: **B**.

14. **C**

Para C: **knowing more than one language can cause speakers to name pictures more slowly, and can increase 'tip-of-the-tongue states'**. Answer: **C**.

Passage 48 — The Cause of Linguistic Change

27. **sound laws**

Para 1: **such regular changes are often called sound laws**. Answer: **sound laws**.

28. **fashion**

Para 3: **it is also possible that fashion plays a part in the process of change**. Answer: **fashion**.

29. **children**

Para 5: **the imitation of children is imperfect**: they copy their parents' speech, but never reproduce it exactly. Answer: **children**.

30. **principle of ease**

Para 6: one such force **is known as the principle of ease**. Answer: **principle of ease**.

31. **FALSE**

Para 1: various sources of evidence **can often give us a very good idea of the pronunciation of an earlier age**, though absolute certainty is never possible. Answer: **FALSE**.

32. **NOT GIVEN**

Para 1: **there are no universal sound laws, but simply particular sound laws for one given language at one given period** — no comparison of degree of change across languages is made. Answer: **NOT GIVEN**.

33. **TRUE**

Para 3: their steady infiltration upwards into the ranks of the landed gentry, **probably carried elements of middle class dialect into upper class speech**. Answer: **TRUE**.

34. **NOT GIVEN**

Para 5 says **the imitation of children is imperfect** in general, but makes no claim about specific sounds being hard for all children. Answer: **NOT GIVEN**.

35. **TRUE**

Para 6: an economy of effort has thereby been achieved, because **n and t are articulated in the same place**. Answer: **TRUE**.

36. **NOT GIVEN**

Para 7 only notes that the g is no longer pronounced **in words like gnat and gnaw** — nothing is said about future spelling changes. Answer: **NOT GIVEN**.

37. **TRUE**

Para 7: **temporary, which in Britain is pronounced as if it were tempory** — the spelling keeps a syllable the pronunciation drops. Answer: **TRUE**.

38. **A**

Para 1: **all the b sounds in a certain position (for example, at the end of a word) may change into p sounds** — a large-scale regular sound law. Answer: **A**.

39. **B**

Para 2: **the very characteristic melody of Welsh English has been carried over from Welsh**. Answer: **B**.

40. **C**

Para 7: **at one time the t was pronounced in words like castle and Christmas** — loss of a sound to reduce effort. Answer: **C**.

Passage 49 — The Conservation of Rare Species

1. **A**

the concept of rarity can refer to one of three characteristics: geographic range, habitat specificity, or local population size. Answer: **A**.

2. **C**

species with large ranges at low density **are ecologically interesting but almost never appear on lists of endangered species**. Answer: **C**.

3. **D**

helpers may wait many years before they acquire breeding status. Answer: **D**.

4. **B**

Walters and colleagues **artificially constructed cavities in trees at a number of sites... breeding groups were formed only in areas where artificial cavities were drilled**. Answer: **B**.

5. **TRUE**

others may be found in several habitats, but only in small numbers and in a geographically restricted area. Answer: **TRUE**.

6. **NOT GIVEN**

we can observe a species declining over time: for example, the African elephant population since 1950 — no mention of the rate slowing. Answer: **NOT GIVEN**.

7. **FALSE**

not all species that have declined to rarity are so well understood. Answer: **FALSE**.

8. **TRUE**

most of this woodland has been destroyed for agriculture and timber production. Answer: **TRUE**.

9. **FALSE**

red-cockaded woodpeckers compete for breeding vacancies in existing groups, rather than forming new groups and occupying abandoned territories, or starting at a new site — the opposite of preferring to leave. Answer: **FALSE**.

10. **TRUE**

because of the energy and time needed, typically several years, birds are better off competing for existing territories than building new ones. Answer: **TRUE**.

11. **NOT GIVEN**

the passage describes **the rescue of the red-cockaded woodpecker** as a completed example of successful conservation biology, without stating whether the project continues today. Answer: **NOT GIVEN**.

12. **insects**

the birds feed on insects and nest in cavities in old pine trees. Answer: **insects**.

13. **incubation and feeding**

helpers do not breed but assist in incubation and feeding. Answer: **incubation and feeding**.

14. C

the passage opens with general rarity concepts, then uses **one example that illustrates some of these factors is the red-cockaded woodpecker** as a detailed case study — the woodpecker illustrates the broader discussion, it isn't the sole topic. Answer: C.

Passage 50 — The Dark Side of the Technological Boom

1. B

most of us are too busy struggling to keep pace with ongoing innovations to question the implications. Answer: B.

2. C

over-reliance on electronic tools could also have serious repercussions on our ability to think creatively and develop new ideas. Answer: C.

3. E

service people... curtly terminate attempts at idle conversation because their performance is being electronically monitored. Answer: E.

4. B

Dennett: the obligation to make accurate predictions — the right decisions — becomes more onerous. Answer: B.

5. A

Brod: they need complete isolation to recover from the effects of the technology. Answer: A.

6. A

Brod: we need to avoid the temptation to replace informal gatherings for bouncing ideas off colleagues with electronic networking. Answer: A.

7. D

Nicholson: he likens the psychic numbing of electronic information overload to symptoms of post-traumatic stress syndrome. Answer: D.

8. C

Boyett and Conn: millions of people now charged with analyzing information and making routine decisions will be replaced by less skilled workers using 'intelligent' software. Answer: C.

9. D

Nicholson: if we are to survive the challenges of information-driven, hardwired offices... we need to provide psychological support systems. Answer: D.

10. **NOT GIVEN**

mass exposure to technological innovations in the workplace has come too recently for psychologists to reach a consensus — a lack of consensus is stated, but the passage never characterises the underlying knowledge itself as "limited". Answer: **NOT GIVEN**.

11. **TRUE**

one of the first signs of the struggle to adapt to the electronic office is often 'technostress'... people become accustomed to the patterns set by electronic tools – accelerated time and yes/no logic. Answer: **TRUE**.

12. **TRUE**

it increasingly difficult to convince ourselves that we're doing an adequate job... we are tormented by the knowledge that the world of information is limitless. Answer: **TRUE**.

13. **FALSE**

for executives near the top of the office pyramid, the benefits of the electronic revolution... may outweigh the disadvantages — executives fare **BETTER**, not worse. Answer: **FALSE**.

14. **NOT GIVEN**

the passage never discusses promotion or career advancement criteria at all. Answer: **NOT GIVEN**.

Passage 51 — The Fluoridation Controversy

27. **A**

Para 1: the ideal concentration is set "**depending on the regional temperature and hence the amount of water people are likely to drink**" — hotter regions mean people drink more water, so less fluoride per litre is needed. Answer: **A**.

28. **C**

Para 2: fluoridation is seen as "**an infringement on individual rights because it is compulsory medication of all members of a community**". Answer: **C**.

29. **D**

Para 5: Kuhn argued the shift between scientific ways of thinking "**can include social factors**", not just formal scientific rules. Answer: **D**.

30. **A**

Para 7: Sutton's study "**found that each showed significant methodological shortcomings**" — flaws in how the studies were conducted. Answer: **A**.

31. **B**

Para 9: **"this is not an issue on which declarations by scientific experts ought to be considered the final word"** — scientists shouldn't have the sole say on policy. Answer: **B**.

32. **NO**

Contradicts the writer's sociological view in paras 4–6: scientific knowledge **"is socially negotiated, and inevitably linked to the values of the relevant parties"** — the opposite of being kept separate. Answer: **NO**.

33. **YES**

Para 7: **"sociologists have also downplayed an important part of the debate by ignoring the number of eminent scientists who have questioned aspects of fluoridation"**. Answer: **YES**.

34. **NOT GIVEN**

scientists outside of North America — para 7 only says Sutton's book **"is not cited in a single analysis of the fluoridation issue by any sociologist"**; nothing is said about scientists specifically outside North America. Answer: **NOT GIVEN**.

35. **YES**

Para 8: the difference between supporters and opponents **"is not between rationality and irrationality but is a legitimate difference in values"**. Answer: **YES**.

36. **B**

the traditional view of science is that ≈ para 4: **"science is a special kind of knowledge, which is established through scientific methods and objectively applied"**. Answer: **B**.

37. **G**

a sociological view of science argues that ≈ para 4: scientific knowledge is **"socially negotiated, and inevitably linked to the values of the relevant parties"**. Answer: **G**.

38. **A**

Collins is of the opinion that ≈ para 5: the outcome of experiments **"was not something whose meaning could be immediately comprehended, but rather something for interpretation"**. Answer: **A**.

39. **D**

a supporter of fluoridation may conclude that ≈ para 7: **"the evidence for harm is limited and dubious... the likely benefits outweigh the possible dangers"** — i.e. not enough proof of harm to justify caution. Answer: **D**.

40. C

an opponent of fluoridation may conclude that ≈ para 7: "I oppose fluoridation of water supplies and favor the voluntary use of fluoride tablets by those who want to take them". Answer: C.

Passage 52 — The Ingenuity Gap

1. D

consists not only of ideas for new technologies... but, more fundamentally, of ideas for better institutions and social arrangements. Answer: D.

2. F

an easy climate or a punishing one. Answer: F.

3. A

depends not only on an adequate supply of useful ideas but also on the availability of other, more conventional factors of production, like capital and labor. Answer: A.

4. B

prosperity, stability and justice usually depend on the resolution, or at least the containment, of major political struggles over wealth and power. Answer: B.

5. C

Because these changes have accumulated slowly, it's often hard for us to recognize how profound and sweeping they've been. Answer: C.

6. B

these changes have sharply increased the density, intensity, and pace of our interactions with each other. Answer: B.

7. A

they are often very sensitive to the smallest of changes and perturbations, and their behavior can flip from one mode to another suddenly and dramatically. Answer: A.

8. YES

Over the last 100 years... the institutions and technologies we use to manage them must become more complex too, which further boosts our requirement for ingenuity. Answer: YES.

9. NO

we should not jump to the conclusion that the supply of ingenuity always increases in lockstep with our ingenuity requirement. Answer: NO.

10. **YES**

we can't always rely on the right kind of ingenuity appearing when and where we need it.
Answer: YES.

11. **YES**

Not many of us have more than a rudimentary understanding of how these systems work.
Answer: YES.

12. **NO**

the crush of information in our everyday lives is shortening our attention span, limiting the time we have to reflect on critical matters of public policy — the opposite. Answer: NO.

13. **NOT GIVEN**

the passage never discusses how future generations will judge today's governments. Answer: NOT GIVEN.

14. **YES**

some phenomena are intrinsically harder to understand than others, so the production of useful new knowledge in these areas can be very slow. Answer: YES.

Passage 53 — The Knowledge Society

1. **C**

the overwhelming majority of people in developed countries worked with their hands.
Answer: C.

2. **A**

their manual work is based on a substantial amount of theoretical knowledge which can be acquired only through formal education. Answer: A.

3. **D**

educational institutions to prepare knowledge technologists will grow rapidly in all developed and emerging countries. Answer: D.

4. **A**

most of it will be delivered in non-traditional ways, ranging from weekend seminars to online training programmes. Answer: A.

5. **B**

knowledge workers, whatever their sex, are professionals, applying the same knowledge, doing the same work, governed by the same standards and judged by the same results. Answer: B.

6. **NO**

knowledge **cannot be inherited or bequeathed from one generation to another. It has to be acquired anew by every individual.** Answer: **NO**.

7. **YES**

nowadays it is assumed that everybody will be a 'success'. Answer: **YES**.

8. **NOT GIVEN**

the passage discusses achievement through acquired knowledge and effort, but **never mentions success by accident.** Answer: **NOT GIVEN**.

9. **YES**

upward mobility is unlimited (good), but **the upward mobility of the knowledge society, however, comes at a high price: the psychological pressures and emotional traumas of the rat race** (bad). Answer: **YES**.

10. **NOT GIVEN**

schoolchildren in some countries may suffer sleep deprivation — described as a fact, with no recommendation about whether they SHOULD avoid it. Answer: **NOT GIVEN**.

11. **NO**

schools are becoming viciously competitive... indicates how much the fear of failure has already permeated the knowledge society — framed as a worrying symptom, not something to encourage. Answer: **NO**.

12. **YES**

knowledge workers need to develop **a non-competitive life and community of their own, and some serious outside interest.** Answer: **YES**.

13. **YES**

volunteering, an orchestra, or local government — **this outside interest will give them the opportunity for personal contribution and achievement**, all forms of contributing to others/community. Answer: **YES**.

Passage 54 — The Long Road to Artificial Intelligence

27. **D**

he allegedly said he sensed an 'alien intelligence' on the other side of the board when he played against deep blue — unlike previous programs, which "seemed predictable and mechanical". Answer: **D**.

28. **C**

intellectual prowess is what sets human beings apart from the rest of the animals, and chess is a supremely intellectual pursuit — unlike a car outrunning a sprinter, beating a human at chess is significant. Answer: C.

29. **A**

each of these activities requires the exercise of multiple sensorimotor skills. Answer: A.

30. **C**

the achievements of ai research in a variety of specialist domains (chess being just one among many success stories) contrast starkly with the field's failure to produce a machine with general purpose, adaptive intelligence. Answer: C.

31. **E**

the brain of an animal has evolved to prevent any harm from coming to that body and to perpetuate the genes that it carries. Answer: E (protection).

32. **G**

so that its actions can be made to react to the state of the environment. Answer: G (surroundings).

33. **I**

the brain shapes the animal's actions according to what it perceives. Answer: I (perception).

34. **A**

Same link established by **the brain shapes the animal's actions according to what it perceives** — perception feeding into action. Answer: A (results).

35. **D**

human intelligence, for all its glorious achievements, is fundamentally an extension of animal intelligence. Answer: D (qualities).

36. **C**

the human capacities for language, reason, and creativity all rest on a sensorimotor foundation. Answer: C (basis).

37. **NO**

the endeavor to create artificial general intelligence might succeed without much that is essential to biological life, such as metabolism and reproduction. Answer: NO.

38. **YES**

the need to engage with a messy, dynamic, physical environment full of complex and varied objects, both animate and inanimate, is at the root of intelligence. Answer: **YES**.

39. **NOT GIVEN**

the multi-specialist robot is going to get stuck as soon as it has to face a problem that is outside its areas of expertise — nothing about danger or unfixable faults. Answer: **NOT GIVEN**.

40. **NO**

general intelligence needs the ability to adapt **without recourse to trial and error or to training by a third party**. Answer: **NO**.

Passage 55 — The New Zealand Writer Margaret Mahy

27. **NOT GIVEN**

Para A calls Knox Mahy's "clearest heir" via "sheer singularity" — nothing about criticism between them either way. Answer: **NOT GIVEN**.

28. **NO**

Para A: **Mahy, too, has ventured into imaginative territory unknown to other local writers**, set against "our solidly realist tradition" — she departs from tradition, not explores it. Answer: **NO**.

29. **NOT GIVEN**

Para A mentions both her children's fiction and teenage fiction as achievements, but never compares their relative popularity. Answer: **NOT GIVEN**.

30. **YES**

Para A: **she is generous to a fault with her time and attention**. Answer: **YES**.

31. **NO**

Para A: **I'm reminded she isn't truly comfortable making herself or her life the heart of any conversation**. Answer: **NO**.

32. **B**

Para B: **going out and finding inspiration from her immediate environment**: a bus, a spelling mistake, a cat and a fur hat — ordinary, everyday sources. Answer: **B – everyday**.

33. **E**

Para B: **adventures on the high seas or in Antarctica, and also in quite unassuming places like the library or even down the back of a chair at home**. Answer: **E – exotic and ordinary**.

34. **I**

Para B: **a young adult burdened with special powers, such as the ability to cast spells in order to transform their world in supernatural and fantastical ways.** Answer: **I – magical.**

35. **H**

Para B: **the conclusions of her tales are usually predictable, they leave the reader feeling absolutely complete.** Answer: **H – unsurprising but satisfying.**

36. **B**

Para C: **The story ballooned at one point to 800 pages and has been through at least two substantial rewrites.** Answer: **B.**

37. **C**

Para D: **Bubble Trouble is the Mahy that New Zealand children and their parents know so well,** recited "at countless private and public functions". Answer: **C.**

38. **D**

Para E lists four generations reading Mahy — e.g. **my own children sat very still mouthing The Great White Man-Eating Shark in the 80s and 90s** — illustrating enduring, cross-generational popularity. Answer: **D.**

39. **C**

Para F: **a book is not properly finished, she has often said, until it has been read, because a reader brings something important to the book.** Answer: **C.**

40. **A**

Para H: David Hill's "This door is alarmed" anecdote illustrates **her peculiarly alert way of looking at the world, particularly the mad, slippery life of language.** Answer: **A.**

Passage 56 — The Peopling of Patagonia

1. **C**

it gives a baseline that all calculations regarding the rate of dispersion of humans throughout both continents must take into account — Patagonia's settlement date matters because it informs research across the whole region. Answer: **C.**

2. **C**

fundamental disagreements developed concerning the length of time involved... and the debate is still with us today — the writer lays out both sides of an ongoing scholarly disagreement. Answer: **C.**

3. **A**

the 'late' model is introduced only to set up the contrast with **many scholars now support the idea of an early model**. Answer: **A**.

4. **D**

this is surely the most realistic scenario as migrants slowly adapted to the diverse natural habitats they would have met while travelling through the continent. Answer: **D**.

5. **B**

evidence for human occupation of the centre of south america is now securely dated to around 12,500 years ago at the monte verde site, which casts doubt on the late model. Answer: **B**.

6. **NO**

the environment was **similar to that of early post-glacial europe** — not unique. Answer: **NO**.

7. **NOT GIVEN**

the passage only says **the evidence for this occupation remains relatively scant** — nothing about hunting specifically. Answer: **NOT GIVEN**.

8. **NO**

there is, however, reliable evidence from these sites to confirm the presence of humans by around 11,000 years ago — a clear conclusion was reached. Answer: **NO**.

9. **YES**

bones or other evidence may be deposited in caves by natural agency, in other words by other forces such as floods or predators. Answer: **YES**.

10. **D (good condition)**

the artefacts **were buried in layers of peat, which has the property of preserving wood remarkably well**. "Preserved well" = **D good condition**.

11. **B (random collection)**

dispersed flecks of carbon used in the test process were taken unsystematically from many different places in the site. "Unsystematically" = **B random collection**.

12. **C (similar properties)**

the artefacts from the earliest occupations were found at the same depth and have the same origins as those from los toldos. "Same origins" = **C similar properties**.

13. **H (reliable connection)**

there is no way of securely relating any of these remains with human occupation. "Securely relating" = **H reliable connection**.

14. **J (immediate surroundings)**

the utility of this site must be restricted to its direct vicinity... findings cannot be freely extrapolated further afield. "Direct vicinity" = **J immediate surroundings**.

Passage 57 — The Placebo Effect

1. **D**

each session with your patients should last at least half an hour. Answer: **D**.

2. **A**

describe your treatment in familiar words. Answer: **A**.

3. **G**

a good living if you are sufficiently convincing, or better still, really believe in your therapy. Answer: **G**.

4. **E**

if you are lucky and administer your treatment at just the right time you will get the credit. Answer: **E**.

5. **B**

many illnesses get better on their own. Answer: **B**.

6. **F**

conventional medicine **recognises but remains oddly ambivalent about** the placebo effect — implying it should be given more credit. Answer: **F**.

7. **A**

our state of mind can influence our physiology: anger opens the superficial blood vessels of the face; sadness pumps the tear glands. Answer: **A**.

8. **D**

pain research is favoured **because it's one of the commonest complaints and lends itself to experimental study**. Answer: **D**.

9. **C**

the placebo effect can be abolished by a drug, naloxone, which blocks the effects of endorphins. Answer: **C**.

10. **NO**

exactly how placebos work their medical magic is still largely unknown. Answer: **NO**.

11. **NO**

red dummy capsules made more effective painkillers than blue, green or yellow ones — the opposite of being taken off the market. Answer: **NO**.

12. **YES**

even branding can make a difference: if Aspro or Tylenol is what you like to take for a headache, their chemically identical generic equivalents may be less effective. Answer: **YES**.

13. **YES**

a doctor categorised his colleagues according to whether they were keen on it, openly skeptical of its benefits, or took a "let's try and see" attitude. Answer: **YES**.

14. **NO**

complementary practitioners are generally best at mobilising the placebo effect — the opposite of "seldom". Answer: **NO**.

Passage 58 — The Role of Accidents in Business

1. **YES**

the brothers left cooked wheat untended for more than 24 hours... they were surprised to discover that this time the rollers created flat flakes. Answer: **YES**.

2. **NO**

i believe that the record of my published papers conveys an impression of directedness and planning which does not at all coincide with the actual sequence of events. Answer: **NO**.

3. **NO**

the mistake that gave us cornflakes keeps repeating itself in the history of disruptive innovation... louis daguerre, for instance, discovered the technique that gave us photography. Answer: **NO**.

4. **YES**

saccharin, was the unintentional result of a medical scientist's work on a chemical treatment for gastric ulcers. Answer: **YES**.

5. **NOT GIVEN**

the passage only describes how art fry discovered the adhesive-note technique, not whether 3m should have funded him. Answer: **NOT GIVEN**.

6. **F**

in these sorts of environments, accidents are called failures and are discouraged — no distinction is made between different kinds of mistakes. Answer: **F**.

7. **G**

companies that were skilled at innovation were usually not that skilled when it came to commercialisation, and vice versa. Answer: **G**.

8. **C**

know how to take clever ideas from other firms and turn them into mass-market items. microsoft is a prime instance of this. Answer: **C**.

9. **A**

accidents, they say, come from unlikely mental associations such as memories and vague connections. Answer: **A**.

10. **B**

it means throwing together groups from diverse backgrounds, and combining ideas in unpredictable ways. Answer: **B**.

11. **D**

the cost of accidents that do not prove valuable are often of concern to people in business. Answer: **D**.

12. **A**

foster's used this lesson learned to go on and develop other brands instead. one of them, pure blonde, is now ranked as australia's fifth-largest beer brand. Answer: **A**.

13. **C**

unlike empire lager, there has been almost no promotion and its sales are generated more by word of mouth. Answer: **C**.

14. **B**

convincing employees and shareholders to tolerate failure and not play it safe is a big thing to ask. Answer: **B**.

Passage 59 — The Significant Role of Mother Tongue in Education

1. **C**

the cultural, linguistic, and economic resources of the nation are maximised... quite simply unintelligent from the point of view of national self-interest — framing mother-tongue development as a national economic resource. Answer: C.

2. **A**

more than 150 research studies conducted during the past 25 years strongly support what Goethe... once said — the quote backs up the research findings already presented. Answer: A.

3. **B**

children come to school well-prepared to learn the school language and succeed educationally. Answer: B.

4. **D**

worry that they take time away from the majority language. Answer: D.

5. **I (ability)**

how quickly bilingual children seem to 'pick up' conversational skills in the majority language at school. "Skills" = I ability.

6. **D (rate)**

the extent and rapidity of language loss will vary. "Rapidity" = D rate.

7. **J (area)**

concentration of families from a particular linguistic group in the neighbourhood. "Neighbourhood" = J area.

8. **F (family)**

speaking with their peers and siblings and in responding to their parents. "Peers/siblings/parents" = F family.

9. **C (dislocation)**

pupils frequently become alienated from the cultures of both home and school. "Alienated" = C dislocation.

10. **YES**

58% of kindergarten pupils come from homes where English is not the usual language of communication — so only 42% have English as their mother tongue, which is less than half. Answer: YES.

11. **NOT GIVEN**

research shows bilingual programmes work **without any negative effects on children's development in the majority language** — the reverse direction (majority language harming mother tongue via "research proof") is never stated. Answer: **NOT GIVEN**.

12. **NOT GIVEN**

the Foyer program in Belgium, which develops children's speaking and literacy abilities in three languages — no mention of the French education system accepting it. Answer: **NOT GIVEN**.

13. **NOT GIVEN**

pupils who know how to tell the time in their mother tongue understand the concept of telling time — only concept transfer is discussed, not an "earlier than monolingual children" comparison. Answer: **NOT GIVEN**.

14. **YES**

knowing how to distinguish the main idea from the supporting details of a written passage or story, and distinguishing fact from opinion transfers across languages. Answer: **YES**.

Passage 60 — The Strange World of Sight

27. **D**

This is still not generally recognized even now, 400 years later — the idea has existed since Locke and Newton's time. Answer: **D**.

28. **C**

it undermined the notion of humans as pre-eminently rational beings who could be held responsible for their actions. Answer: **C**.

29. **C**

mistaking seeing his clock for remembering it, and so failing to get up in the morning — Mr S is presented as a rare exception, since he had an exceptional memory. Answer: **C**.

30. **A**

that perceptions are not identical with what we perceive, and may be very different — the passage's central, recurring theme. Answer: **A**.

31. **NO**

It was his touch memories that enabled him to perceive them with his eyes — touch, not things he was told about. Answer: **NO**.

32. **NOT GIVEN**

Since his operation, his sight has gradually improved as he learns to see — improvement is shown, but the passage never confirms he reaches full parity with normally-sighted people. Answer: **NOT GIVEN**.

33. **YES**

he would have to shut his eyes while skiing to block out what he now found was a terrifying sight — skiing became harder despite his being a champion before. Answer: **YES**.

34. **NO**

now it seems that half the brain is occupied with seeing — not just one particular area. Answer: **NO**.

35. **NOT GIVEN**

A dramatic and discomfoting example is looking at the two sides of a face-mask — illustrates perception generally; no mention of training the newly-sighted. Answer: **NOT GIVEN**.

36. **YES**

even people with no visual impairment see what, at some level, they expect to see. Answer: **YES**.

37. **D**

the back of the mask will be seen as convex, though we know that it must be concave. Answer: **D convex**.

38. **F**

From the front it is a convex shape with the nose sticking out. Answer: **F nose**.

39. **I**

impossible to sketch the back of a hollow mask to look as it is. "Sketch" = **I drawing**.

40. **J**

The passage's overall theme is that perception is governed by prior expectation — subjects fail to see the hollow shape because of their **preconception** that faces have protruding features. Answer: **J preconception**.

Passage 61 — The Tuatara – Past and Future

27. **C**

Para B: The naturalists who came with the explorers disregarded the tuatara. Answer: **C**.

28. **B**

Para B: **He and his contemporaries were products of their age, strongly influenced by Charles Darwin's theory.** Answer: **B**.

29. **B**

Para C: **what he did not realise was that New Zealand has two species of rat... Gunther failed to recognise the distinction, believing all rats to be a relatively recent introduction.** Answer: **B**.

30. **D**

Para D: **This should not be seen... as evidence that tuatara and kiore can coexist.** Answer: **D**.

31. **D**

Para E: **tuatara and many other species were already rare or extinct when permanent human inhabitants – the Maori – arrived around 1300.** Answer: **D**.

32. **YES**

Para E: **This hypothesis is still being debated, but the evidence continues to accumulate in its favour.** Answer: **YES**.

33. **YES**

Para F: **Eradication of rats... was believed to be virtually impossible until about 1980, but since then has become routine.** Answer: **YES**.

34. **NOT GIVEN**

Para D describes Crook's findings about tuatara on rat-free islands, and para F later discusses sanctuaries generally – but the passage never says Crook was the *first* to recognise this potential. Answer: **NOT GIVEN**.

35. **NO**

Para F: **for the first time in 1800 years the decline has been reversed** – the opposite of "continuing to fall". Answer: **NO**.

36. **creative thought (B)**

Para G: **the truly significant outcome of this research is that it liberates the imagination.** Answer: **B (creative thought)**.

37. **trout (D)**

Para G: **Our rivers... are full of surrogate rats, in the form of introduced species of fish called trout.** Answer: **D (trout)**.

38. **native fish (G)**

Para G: **can we once again establish those species of indigenous fish that used to live in our rivers?**. Answer: **G (native fish)**.

39. **restoration (F)**

Para H: **Conservation is thus not primarily about the past but about imagining and then creating the future we wish** — a future built by restoring species. Answer: **F (restoration)**.

40. **extinction (H)**

Para H: **Just as we now no longer tolerate extinction**. Answer: **H (extinction)**.

Passage 62 — The Work of Caravaggio

1. **A**

Para A: **Mr Thwaytes proceeded to sue the auctioneers for professional negligence**. Answer: **A**.

2. **G**

Para G: **ultimately ruled against Mr Thwaytes, who now faces substantial costs**. Answer: **G**.

3. **D**

Para D: **the majority of Caravaggio scholars are not of the opinion that Caravaggio himself painted copies of his own works**. Answer: **D**.

4. **B**

Para B: **most judges... are reluctant to rule on whether an artwork should or should not be attributed to a particular artist, as this question lies outside their field of expertise**. Answer: **B**.

5. **C**

Para C: **no one would pay full price for a painting that was more-likely-than-not... by the legendary artist Picasso**. Answer: **C**.

6. **F**

Para F: **a set of phrases are employed to describe the degree of certainty as to the identity of the artist**. Answer: **F**.

7. **H**

Para H: **yet the judge did not accept that this suggested evidence of a creative mind at work rather than a copyist**. Answer: **H**.

8. E

Para E: **it presented to the court a record of about 30 versions of the card game scene which had changed hands at auction, none of which were described as being by Caravaggio.** Answer: E.

9. "A

A: the judge ruled against Thwaytes, **but who still has the right to appeal, and may yet do so.** **E:** **the degree of proof required is more similar to that needed in criminal trials, which require 'proof beyond all reasonable doubt'** — a higher bar than ordinary civil cases. (B: the judge ruled against this view; C: Mahon only concluded it was genuine AFTER buying and researching it; D: judges are described as RELUCTANT to make such rulings, not frequent.) Answers: **A, E.**

10. "A

A: the judge ruled against Thwaytes, **but who still has the right to appeal, and may yet do so.** **E:** **the degree of proof required is more similar to that needed in criminal trials, which require 'proof beyond all reasonable doubt'** — a higher bar than ordinary civil cases. (B: the judge ruled against this view; C: Mahon only concluded it was genuine AFTER buying and researching it; D: judges are described as RELUCTANT to make such rulings, not frequent.) Answers: **A, E.**

11. owners

11: namely records of all prior **owners**, is limited or non-existent. **12:** the artist had a **studio**, where pupils may have been engaged to make copies. **13:** not misrepresent what is known about a painting's authorship in their **catalogues**. **14:** Sotheby's attributed the work... to a '**follower**' of Caravaggio. **15:** the judge drew the conclusion that the **quality** was not up to that of the rest of the artist's known body of work. **16:** pointed out how a **feather** in that picture looked lifelike. Answers: **owners / studio / catalogues / follower / quality / feather.**

12. studio

11: namely records of all prior **owners**, is limited or non-existent. **12:** the artist had a **studio**, where pupils may have been engaged to make copies. **13:** not misrepresent what is known about a painting's authorship in their **catalogues**. **14:** Sotheby's attributed the work... to a '**follower**' of Caravaggio. **15:** the judge drew the conclusion that the **quality** was not up to that of the rest of the artist's known body of work. **16:** pointed out how a **feather** in that picture looked lifelike. Answers: **owners / studio / catalogues / follower / quality / feather.**

13. catalogues

11: namely records of all prior **owners**, is limited or non-existent. **12:** the artist had a **studio**, where pupils may have been engaged to make copies. **13:** not misrepresent what is known about a painting's authorship in their **catalogues**. **14:** Sotheby's attributed the work... to a '**follower**' of Caravaggio. **15:** the judge drew the conclusion that the **quality** was not up to that of the rest of the artist's known body of work. **16:** pointed out how a **feather** in that picture looked lifelike. Answers: **owners / studio / catalogues / follower / quality / feather.**

14. follower

11: namely records of all prior **owners**, is limited or non-existent. **12:** the artist had a **studio**, where pupils may have been engaged to make copies. **13:** not misrepresent what is known about a painting's authorship in their **catalogues**. **14:** Sotheby's attributed the work... to a '**follower**' of Caravaggio. **15:** the judge drew the conclusion that the **quality** was not up to that of the rest of the artist's known body of work. **16:** pointed out how a **feather** in that picture looked lifelike. Answers: **owners / studio / catalogues / follower / quality / feather**.

15. quality

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16. feather

11: namely records of all prior **owners**, is limited or non-existent. **12:** the artist had a **studio**, where pupils may have been engaged to make copies. **13:** not misrepresent what is known about a painting's authorship in their **catalogues**. **14:** Sotheby's attributed the work... to a '**follower**' of Caravaggio. **15:** the judge drew the conclusion that the **quality** was not up to that of the rest of the artist's known body of work. **16:** pointed out how a **feather** in that picture looked lifelike. Answers: **owners / studio / catalogues / follower / quality / feather**.

17. A

the article opens with the case **at the centre of a case at the High Court in London** and closes with the judge ruling against Thwaytes — the whole piece is framed around this specific court case and its outcome, not a general legal explainer or a claim of forgery. Answer: **A**.

Passage 63 — Try It and See

1. vii

The accepted gold standard of evidence is a randomized control trial, in which a new drug is compared with the best existing therapy. Answer: **vii**.

2. v

the educational theorists who pushed the whole-language notion so successfully did not wait for evidence from controlled randomized trials before advancing their claims. Answer: **v**.

3. ix

This is the idea behind a trial of restorative justice which is taking place in the English courts, which "might be adopted more widely" if successful. Answer: **ix**.

4. **i**

This loss of interest can be attributed, at least in part, to the fact that early experiments produced little evidence of positive outcomes. Answer: **i**.

5. **iv**

Randomized trials have been used to evaluate the effectiveness of driver-education programs, job training schemes, classroom size, psychological counselling for post-traumatic stress disorder and increased investment in public housing. Answer: **iv**.

6. **iii**

This is where a body called the Campbell Collaboration comes into its own — evaluating existing studies via systematic review. Answer: **iii**.

7. **re-offending**

7: The purpose of the trial is to assess whether such restorative justice limits **re-offending**. **8:** assigned randomly either to **sentencing** as normal. **9:** a conference in which the offender comes face-to-face with his **victim**. **10:** the idea behind a trial of **restorative justice**. Answers: **re-offending / sentencing / victim / restorative justice**.

8. **sentencing**

7: The purpose of the trial is to assess whether such restorative justice limits **re-offending**. **8:** assigned randomly either to **sentencing** as normal. **9:** a conference in which the offender comes face-to-face with his **victim**. **10:** the idea behind a trial of **restorative justice**. Answers: **re-offending / sentencing / victim / restorative justice**.

9. **victim**

7: The purpose of the trial is to assess whether such restorative justice limits **re-offending**. **8:** assigned randomly either to **sentencing** as normal. **9:** a conference in which the offender comes face-to-face with his **victim**. **10:** the idea behind a trial of **restorative justice**. Answers: **re-offending / sentencing / victim / restorative justice**.

10. **restorative justice**

7: The purpose of the trial is to assess whether such restorative justice limits **re-offending**. **8:** assigned randomly either to **sentencing** as normal. **9:** a conference in which the offender comes face-to-face with his **victim**. **10:** the idea behind a trial of **restorative justice**. Answers: **re-offending / sentencing / victim / restorative justice**.

11. **A**

This loss of interest can be attributed, at least in part, to the fact that early experiments produced little evidence of positive outcomes — discussed specifically about social science's history. Answer: **A**.

12. **C**

The hope is that rigorous review standards will allow Campbell, like Cochrane, to become a trusted and authoritative source of information — both Campbell (social science) and Cochrane (medicine) do independent review. Answer: **C**.

13. **D**

the passage never discusses research priorities being set to meet government needs, for either field. Answer: **D**.

14. **B**

Governments require sellers of new medicines to demonstrate their safety and effectiveness. Answer: **B**.

Passage 64 — Understanding the Lessons of the Past

30. **C**

historians are often equipped to see how a situation might turn out — spotting 'the tiger in the grass' means seeing what is coming. Answer: **C**.

31. **D**

a document revealing the basic causes of military conflict in every period and place. Answer: **D**.

32. **A**

people generally look back and review their past experience and use this to help them determine their future actions. Answer: **A**.

33. **D**

attempting to compare two events or situations runs the risk of oversimplification. Answer: **D**.

34. **B**

they can make helpful observations about how likely it is that a proposed course of action will succeed. Answer: **B**.

35. **YES**

it is conventional for people in all walks of life, not only professional historians, to liken a historical event to present circumstances. Answer: **YES**.

36. **NO**

people in government often use historical analogies in order to justify their policies — the opposite of avoiding them. Answer: **NO**.

37. **NOT GIVEN**

Analogies are "a valuable way of making a point, yet it is all too easy to misuse them or for analogies to be misunderstood" — never claimed to guarantee understanding. Answer: **NOT GIVEN**.

38. **NOT GIVEN**

The passage says which decades are over-used in public debate, but never reports historians disagreeing about this. Answer: **NOT GIVEN**.

39. **YES**

the range of analogies utilised in public debate is depressingly small — almost always the 1930s or 1970s. Answer: **YES**.

40. **NOT GIVEN**

The writer only recommends that historians "provide public speakers with a new and more appropriate range of alternatives" — never says speakers are unwilling. Answer: **NOT GIVEN**.

41. **F**

although history does not provide rules for action that are consistently reliable. Answer: **F (guidance)**.

42. **D**

must inevitably have some similarities to it. Answer: **D (resemblance)**.

43. **H**

radical plans to completely change the status quo rarely work out as intended. Answer: **H (transformation)**.

44. **B**

they should have faith in their ability to see patterns and future trends — implying they currently lack it. Answer: **B (confidence)**.

45. **E**

the apparent anxiety among historians about whether history has practical value. Answer: **E (relevance)**.

46. **J**

it offers deep insights into the human condition which are of enduring value and impossible to acquire in other ways. Answer: **J (behaviour)**.

Passage 65 — Unlocking the Mystery of Dreams

27. **F**

Para F: **how painters like Jasper Johns and Salvador Dali found inspiration in their dreams.**
Answer: **F**.

28. **C**

Para C: **If dreams are insignificant night-time images created by the brain, what is the point of studying them?.** Answer: **C**.

29. **B**

Para B: **There's no good evidence that any coldblooded animal has REM sleep** — discusses which creatures have REM sleep (mammals, birds). Answer: **B**.

30. **G**

Para G: **people who have vivid dreams with strong story lines, are less likely to remain depressed.** Answer: **G**.

31. **D**

Para D: **might actually lend new support to Freud's ideas because the parts of the brain that are most active during dreaming control emotion.** Answer: **D**.

32. **D**

Para A: **they are still considered prophetic, foretelling things to come.** Answer: **D**.

33. **C**

Para B: **REM sleep heats up the brain and non-REM cools it off.** Answer: **C**.

34. **C**

Para E: **Much less active during REM sleep is the prefrontal cortex, which is associated with logical thinking.** Answer: **C**.

35. **A**

Para E: Nofzinger **thinks that could be why people often solve tricky problems in their dreams** — referring to the anterior cingulate cortex. Answer: **A**.

36. **C**

Para D: **one of the most active areas during REM is the limbic system, which controls our emotions** — Solms's findings, supported by PET/fMRI technology. Answer: **C**.

37. **B**

Para C: Hobson **who believes that dreams are essentially random.** Answer: **B**.

38. **G**

Para G: **She thinks that dreaming helps diffuse powerful emotions.** Answer: **G.**

39. **F**

Para G: Clara Hill: **Dreams reveal stuff about you that you didn't know was there.** Answer: **F.**

40. **A**

Para B: Siegel: **filling a basic physiological function and that dreams are a kind of a side-effect, or by-product of this.** Answer: **A.**

Passage 66 — A closer examination of a study on verbal and non-verbal messages

27. **D**

comparing the ____ of verbal and non-verbal communication ≈ para 5: **"the researchers concluded that 7% of our feeling towards a speaker is based on the actual words they use, 38% on their tone of voice, and 55% on their body language"** — the study compares how much impact each channel has. Answer: **D – effects.**

28. **G**

Para 3: subjects **were asked to guess the emotions in the recorded voice and the photos**; para 4: subjects **"had to guess the speaker's emotions"**. Answer: **G – feelings.**

29. **A**

Para 3: subjects were **shown photos of female faces expressing the same three emotions.** Answer: **A – facial expressions.**

30. **E**

Para 4: **it was found that tone of voice carried more meaning than the individual words.** Answer: **E – word meanings.**

31. **YES**

Para 6: **the entire study involved only 62 subjects**, explicitly listed as a limitation. Answer: **YES.**

32. **NO**

Para 6: **their ages and academic qualifications seem remarkably uniform. Thus, the findings may simply be a product of the nature of the sample** — framed as a weakness, not an advantage. Answer: **NO.**

33. **NO**

The passage never criticises the sequence of the two experiments — the first (selecting a suitable neutral word) necessarily had to come before the second (the main comparison). No objection to the order itself is raised anywhere in the text. Answer: **NO**.

34. **NO**

Para 7: **The researchers intentionally used a "neutral" word so naturally, the subjects found little meaning there. Clearly, such a methodology lacks validity.** Answer: **NO**.

35. **NOT GIVEN**

a range of languages — the passage never mentions language variety as a factor in the study's design or validity. Answer: **NOT GIVEN**.

36. **A**

Para 9: **one appealing aspect of the Mehrabian study is its numerical precision** — precision makes the study attractive/appealing. Answer: **A**.

37. **B**

Para 9: the formula's popularity is **encouraging people to believe in the overwhelming importance of the non-verbal message compared with the verbal one. Yet we know that even one well-chosen word... Words do matter.** Answer: **B**.

38. **D**

Para 9: Bradley says **"If we could communicate 93% of information and attitudes with vocal and facial cues, it would be wasteful to spend time learning a language"** — since we do spend time learning language, it must matter. Answer: **D**.

39. **A**

Para 10: **all my findings dealt with communications of feelings and attitudes** — Mehrabian restricts his findings to that one domain (feelings/attitudes), not communication in general. Answer: **A**.

40. **C**

Para 11 lists several different ways textbook writers treat the study: some are faithful to context, some play down the percentages, and **other textbook authors simply use the numbers without placing any limits on their meaning** — a survey of varying approaches. Answer: **C**.

Passage 67 — Whale Culture

1. **TRUE**

calves stay with their mothers throughout adulthood, and in many years of observation no one has ever seen a whale switch pods. Answer: **TRUE**.

2. **FALSE**

residents have a much more extensive repertoire than transients — the opposite of restricted.
Answer: **FALSE**.

3. **TRUE**

transients have only a few such calls, and all transient societies share the same ones. Answer: **TRUE**.

4. **FALSE**

despite regular interaction between them, each resident pod sticks firmly to its own dialect.
Answer: **FALSE**.

5. **NOT GIVEN**

research shows these dialects are maintained for at least 40 years — stated only for RESIDENT dialects, nothing said about transient dialects over time. Answer: **NOT GIVEN**.

6. **pods**

residents live in stable groups, or 'pods', made up of two or three mothers and their offspring. Answer: **pods**.

7. **genetics**

if dialects were programmed by genetics, call patterns from both father and mother would be passed on to the calf — but they aren't, ruling out genetics. Answer: **genetics**.

8. **mother**

a calf uses the calls of its maternal pod very precisely. There's no input from the father.
Answer: **mother / maternal pod**.

9. **"A"**

A: Para F mentions whales' **advanced mental abilities**. **C:** Para F says whales are **adept at recognising sounds**. **D:** Para H notes female killer whales **live up to a quarter of a century after they have had their descendants** — a long post-reproductive lifespan. (B: no mention of physical strength; E: it's post-reproductive SURVIVAL that's highlighted, not a lengthy fertile period; F: no mention of diet variety.) Answers: **A, C, D**.

10. **"A"**

A: Para F mentions whales' **advanced mental abilities**. **C:** Para F says whales are **adept at recognising sounds**. **D:** Para H notes female killer whales **live up to a quarter of a century after they have had their descendants** — a long post-reproductive lifespan. (B: no mention of physical strength; E: it's post-reproductive SURVIVAL that's highlighted, not a lengthy fertile period; F: no mention of diet variety.) Answers: **A, C, D**.

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A: Para F mentions whales' **advanced mental abilities**. **C:** Para F says whales are **adept at recognising sounds**. **D:** Para H notes female killer whales **live up to a quarter of a century after they have had their descendants** — a long post-reproductive lifespan. (B: no mention of physical strength; E: it's post-reproductive SURVIVAL that's highlighted, not a lengthy fertile period; F: no mention of diet variety.) Answers: **A, C, D**.

12. **G**

Para G: **the dialects of killer whales allow members of the group to identify each other, enabling them to share information about food hot spots**. Answer: **G**.

13. **D**

Para D: **one of the most obvious distinctions between the transient and resident societies is the way they impart information**. Answer: **D**.

14. **F**

Para F: **whales have several biological attributes that give them an advantage in social learning**. Answer: **F**.

Passage 68 — What Should Companies Do to Survive?

1. **NOT GIVEN**

the passage only says **it is essential to have a clear idea about the enduring rules of good management** — it never claims most companies fail to evaluate management theories. Answer: **NOT GIVEN**.

2. **YES**

it is essential to have a clear idea about the enduring rules of good management — in the strongest position amid uncertainties and violent upheavals. Answer: **YES**.

3. **YES**

ideas have value only if people share and develop them in ways that increase revenue. Answer: **YES**.

4. **NO**

knowledge is worth storing only if senior staff set careful rules to filter and structure it — senior staff decide, not all users. Answer: **NO**.

5. **NOT GIVEN**

managers constantly blitzed with new information need to build in the data that matters and set aside the rest — no mention of training. Answer: **NOT GIVEN**.

6. **NO**

big-bang decisions are generally best avoided — or implemented in small incremental moves. Answer: NO.

7. **YES**

offer their most profitable customers special deals... 'fire' unprofitable customers by charging them higher rates. Answer: YES.

8. **NO**

a company should keep those activities it does better than its competitors — the criterion is doing it BETTER, not being common to all companies. Answer: NO.

9. **F**

Para F: **effective collaboration calls for trust and shared understanding... requires excellent communication, and incentives.** Answer: F.

10. **H**

Para H: **companies need to insist on common practices in areas such as purchasing and information technology.** Answer: H.

11. **E**

Para E: **some companies will want to hire the talents of 'free agents' as and when they are required; others, to employ directly their best brains.** Answer: E.

12. **F**

Para F: **normally a company should keep those activities it does better than its competitors [in-house, the rest outsourced].** Answer: F.

13. **I**

Para I: **companies will allow their suppliers and customers 'inside the machine'... giving them extraordinary access to their databases.** Answer: I.

14. **D**

Para D: **companies must build loyalty by providing reliability and good service.** Answer: D.

Passage 69 — You Are What You Speak

1. **YES**

Anyone who has tried to master a foreign tongue has at least thought about the possibility. Answer: YES.

2. **NO**

the consensus has been that linguistic differences don't really matter — the opposite of the statement. Answer: **NO**.

3. **NO**

The brain is shaped by experience, says Slobin — siding with the new generation who disagree with Chomsky's view. Answer: **NO**.

4. **NOT GIVEN**

Boroditsky agrees, arguing that even artificial classification systems, such as gender, can be important — her opinion, not a description of her own experiment. Answer: **NOT GIVEN**.

5. **YES**

The classic example used by Chomskians to back this up is colour. Answer: **YES**.

6. **C**

In Korean, **you need to know if you're older or younger than the person you're addressing**. Answer: **C**.

7. **E**

In Yucatec, **objects tend to be defined by separate words that describe shape**. Answer: **E**.

8. **A**

a Russian may need to consider the gender [when counting]. Answer: **A**.

9. **B**

a Japanese speaker has to take into account their shape (long and cylindrical) as well, and use the number word designated for items of that shape. Answer: **B**.

10. **D**

Spanish speakers have to decide whether they are on intimate enough terms to call someone by the informal tu rather than the formal Usted. Answer: **D**.

11. **H**

but Yucatec speakers felt the two plastic combs were [alike] — same MATERIAL (plastic). Answer: **H material**.

12. **D**

English speakers thought the combs with handles were more alike — same DESIGN feature (handle). Answer: **D design**.

13. C

Lucy used a plastic box, a cardboard box and a piece of cardboard. Answer: C cardboard.

14. A

Americans focused on form, while the Mayans focused on substance — a different grouping METHOD. Answer: A method.

Passage 70 — Youth Homelessness

1. A

the programme placed the problem of youth homelessness firmly on the public agenda, although policy initiatives were not forthcoming — wide awareness, but no official action. Answer: A.

2. C

the proportions of young people affected vary from region to region, as do the underlying economic structures, making comparison difficult. Answer: C.

3. B

the reaction of policy makers to a social problem also depends largely on the number of people it is thought to affect. However, there are very few agreed statistics. Answer: B.

4. D

young people who are staying temporarily with friends, for example, may not contact homelessness agencies, so their housing problem goes unrecorded. Answer: D.

5. D

they must fulfil strict criteria, so official statistics therefore exclude large numbers of homeless people. Answer: D.

6. J

there has been a substantial decline in rented housing in recent years. Answer: J property to rent.

7. H

structural explanations account for youth homelessness by reference to societal structures, such as changes in the housing market. Answer: H structural explanations.

8. A

these various difficulties have been further compounded by a number of demographic changes, such as the expanding number of single-person households. Answer: A demographic explanations.

9. **F**

young people's benefits have been heavily restricted... both unemployment and low pay are frequently viewed as being integral to homelessness. Answer: **F economic explanations.**

10. **G**

a wide variety of such personal factors are often introduced to help explain youth homelessness, such as leaving home, mental illness, crime, and drug abuse. Answer: **G personal explanations.**

11. **NO**

the political model is a view widely adopted in academic circles — the opposite of unpopular. Answer: **NO.**

12. **YES**

this approach tends to treat young homeless people as powerless victims, which patronizes them. Answer: **YES.**

13. **NOT GIVEN**

the writer lists the advantage of this perspective... however... it also carries punitive implications — both are stated, but no claim that one "far outweighs" the other. Answer: **NOT GIVEN.**

14. **NO**

there is evidence to suggest that many homeless youngsters show considerable resilience and maturity — contradicting the model's premise that they are simply too immature. Answer: **NO.**